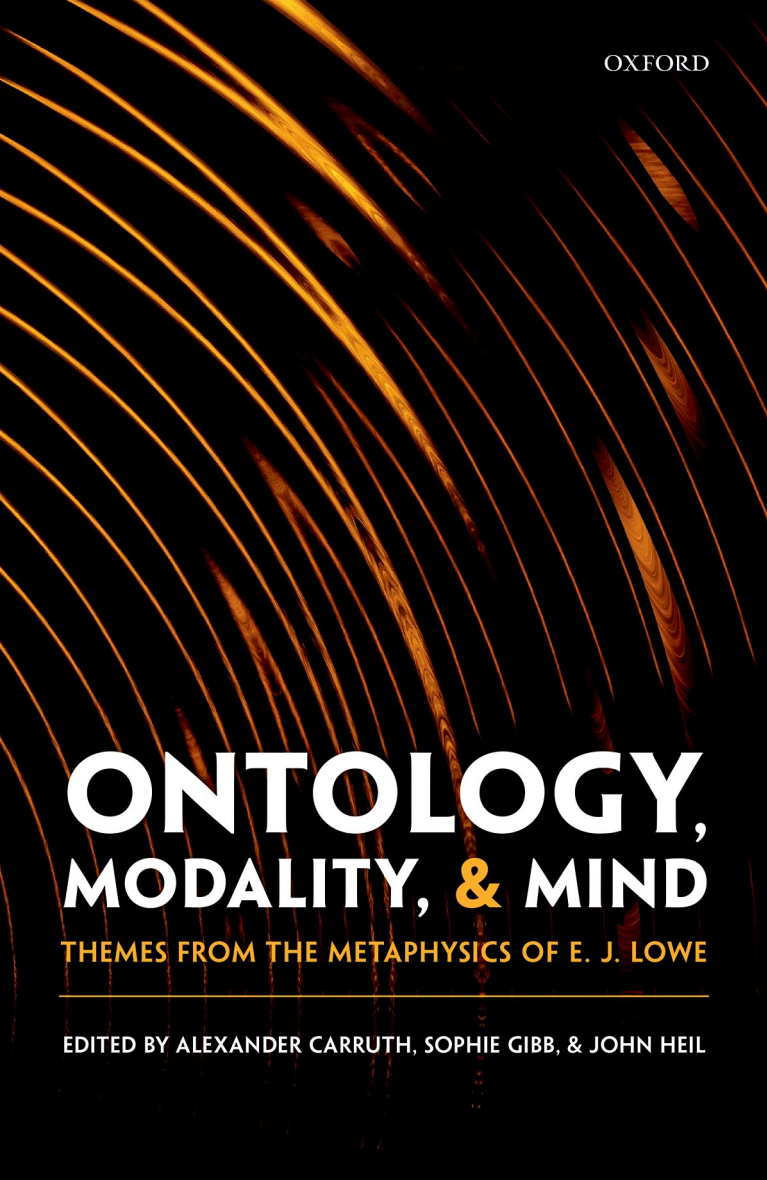




OXFORD

ONTOLOGY, MODALITY, & MIND

THEMES FROM THE METAPHYSICS OF E. J. LOWE

EDITED BY ALEXANDER CARRUTH, SOPHIE GIBB, & JOHN HEIL

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Introduction

John Heil

1. E. J. Lowe

This volume is dedicated to the memory of E. J. Lowe, late Professor of Philosophy at Durham University.¹ It has fallen to me to say something about Lowe's philosophical work. Before setting out to do so, I cannot forbear a comment on the man.

Jonathan Lowe was a vital and ongoing influence on the philosophical development of many. His departing the scene, like the departure of David Lewis, C. B. Martin, J. J. C. Smart, and David Armstrong has contributed to a progressive thinning of the ranks in serious ontology. Contemporary metaphysics more and more reflects social trends in which a few are elevated to the status of stars, sometimes on the strength of cleverness and an ability to game the system without making any sort of substantive contribution likely to withstand the crucible of shifting trends—philosophical counterparts of hedge fund managers. Lowe, in contrast, was the real deal. His modesty and natural reticence masked a powerful and fearless intellect immune to the vicissitudes of philosophical fashion.

Lowe combined an instinct for the big picture with an analytical temperament that equipped him to master the details: a foxy hedgehog. He was a sensitive reader of history prepared to learn from earlier philosophers—most especially Aristotle and Locke—and not simply invoke them as argumentative props. Like the best philosophers, he led by example. I recall his masterful soliloquies during the fourth week of a six-week NEH Summer Seminar I directed in June and July 2009, in which he mesmerized a roomful of young philosophers lucky enough to observe a master craftsman at work.

In the summer of 2013 I directed another NEH Seminar. Lowe was scheduled to attend, and in fact I had built the seminar around the expectation that he would be our first visitor and set the tone for everything that followed. Tragically it never happened. By May of 2013, he had become too ill to travel abroad. He died on 5 January of the following year.

¹ The author is grateful to A. D. Carruth, S. C. Gibb, and Matthew Tugby for comments on an earlier draft of this chapter.

Jonathan Lowe was a peerless philosopher, colleague, teacher, husband, and father—a man whose absence is keenly felt by many. In my own case, not a day goes by when I haven't wished that I could sound him out on a particular topic or argument. Now I wonder what he would make of my description of his views that follows.

2. You Have to Do What You Have to Do

After agreeing to write this introduction, I procrastinated for more than a year, paralyzed by the thought of attempting a summary of Lowe's views. The problem is that he wrote and thought deeply about countless subjects, many of which fall outside my comparatively impoverished philosophical range. With deadlines looming, I decided not to attempt the impossible, but to address three topics that hold a central place in the Lowe corpus: the four-category ontology, essence and modality, and what Lowe called 'non-Cartesian substance dualism.' I settled on these topics because they illustrate the breadth of his work in metaphysics, and because they bear the marks of the evolution of some of his most penetrating views.

Although Lowe gave the impression of a philosopher who carried around a fully developed Big Picture on which he drew in addressing smaller, more narrowly circumscribed issues, I believe he was in fact more or less continuously evolving philosophically, often in unexpected ways. Thus, some readers will be surprised at my description of Lowe's conception of universals, but I am reasonably confident—based on face-to-face discussions with the man himself—that my description is accurate. Perhaps this is because I am in agreement with him on the topic and on his account of essences.² In contrast, I have always found the account of mental causation that emerges in Lowe's defense of non-Cartesian substance dualism harder to fathom. Harder to fathom, but intriguing and for that reason eminently worthy of exploration.

A caveat. My aim in what follows is not to provide full-scale elucidations of Lowe's positions but merely to say enough to convey the flavor of those positions and to provide a feel for how he was thinking about central topics in metaphysics. I focus on conclusions and say little about details of Lowe's arguments supporting those conclusions. My hope is that readers who find the conclusions provocative will track down the arguments in the many papers and books devoted to them.

3. The Four-Category Ontology

Lowe is well known for defending a 'four-category' ontology reminiscent of Aristotle's *Categories*.³ The basic entities are individual substances—particular horses, particular statues, particular electrons. Individual substances are themselves instances of

² Readers who know me might be astonished to hear this, but, as I note in my contribution to this volume, I am convinced that our positions were converging—or, at any rate I was discovering that my qualms about his views were based largely on my having misjudged them.

³ Discussion in this section is based on Lowe 2006, 2011, 2012a, as well as Armstrong 1997.

substantial universals or kinds. In addition, individual substances are various ways, the ways being ‘modes’, particularized properties. A particular horse is brown and possesses a particular size and shape. A particular electron has a negative charge, a particular mass, a particular spin. These are all ways the horse and the electron are, modes. Modes, in turn, are instances of non-substantial universals, ‘attributes’.⁴

So we have four categories of entity: individual substances, substantial universals (or kinds), modes, and non-substantial universals (or attributes). What of relations? Relations are not a fundamental ontological category. Lowe subscribed to the Leibnizian thesis that relational truths have non-relational truthmakers (Lowe 2016).

Modes are non-transferrable ways particular substances are. A mode owes its identity to the substance it modifies. Socrates’s paleness is *Socrates’s* paleness. Socrates’s paleness might be exactly similar to Simmias’s paleness, but the two are distinct palenesses. This means that the relation between a mode and the substance it characterizes is ‘internal’: you could not have Socrates’s paleness without Socrates’s being pale. Modes, then, are not ‘glued’ onto substances, substances are not propertyless substrata, ‘bare’ or ‘thin’ particulars. To be a substance is to be something that is various ways, and to be a way, to be a mode, is to be a way some substance is.

Both individual substances and modes are particulars. What of substantial and non-substantial universals? Like David Armstrong and unlike the Platonists, Lowe holds that universals must have instances. Instances of substantial universals are individual substances. Bucephalus is an instance of the kind horse. Instances of non-substantial universals are not, as the Platonists would have it, individual substances. If Socrates is pale, Socrates is not an instance of paleness. Instances of non-substantial universals are modes. Socrates’s paleness, not Socrates himself, is an instance of paleness. The relationship between Socrates and paleness, the non-substantial universal is ‘exemplification’. Socrates exemplifies paleness by virtue of being characterized by a paleness mode, by virtue of being a pale way. Putting all this together yields Lowe’s ‘ontological square’, as depicted in Figure 0.1.

The four-category ontology is not a ‘relational’ ontology. That is, a substance’s possessing a property—Socrates’s being pale—is not a matter of the substance standing in a relation to a universal. The relation between Socrates’s paleness and Socrates is internal, hence ontologically recessive, ‘no addition of being’. Does this mean that Lowe belongs in Armstrong’s camp?

For Armstrong, the basic entities are ‘states of affairs’, where a state of affairs is a substance’s instantiating a universal (at a time, a qualification I henceforth omit): Socrates’s being pale, where paleness is a universal that can have other instances. If Socrates and Simmias are both pale, they share something, they literally have something in common: paleness, a universal. There is but one multiply located paleness capable of being ‘wholly present’ in each of its instances. Lowe rejects this conception of universals, regarding it as unintelligible: Lowe’s universals are not Armstrong’s universals.

⁴ Lowe preferred ‘mode’ to ‘trope’ in part, I suspect, because of its associations with scholastic and early modern philosophers, especially Locke.

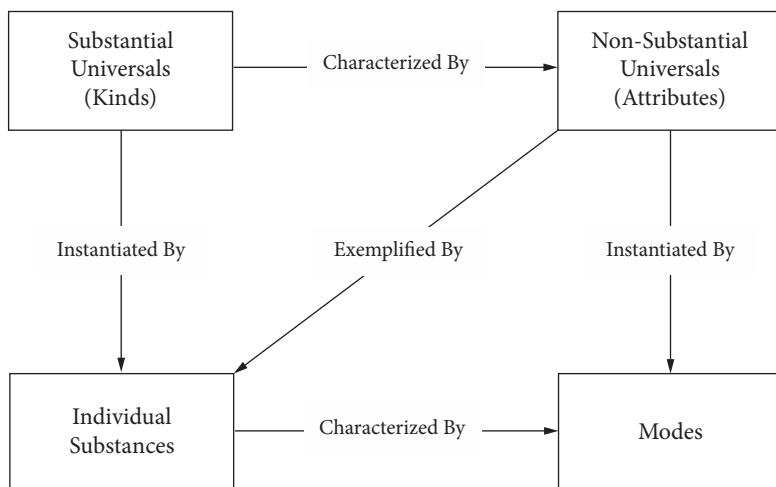


Figure 0.1. The Ontological Square

Before saying what Lowe's universals are, however, it is worth noting another point on which Lowe and Armstrong differ. Armstrong's universals correspond to properties, Lowe's non-substantial universals, attributes. Armstrong has no place for substantial universals. For Armstrong a kind—horsehood, say—is not a distinct ontological category but something closer to a collection of universals definitive of instances of the kind—properties any horse must possess.

As noted earlier, Armstrong's basic entities are states of affairs: particular substances' instantiating universals. God does not create particular substances—'thin particulars'—and universals, then assemble them into states of affairs. If God wants to create a universal or a substance, God must create a particular state of affairs: the 'victory of particularity'. States of affairs are non-mereological composites. Universals and particular substances are alike abstractions, 'aspects' of states of affairs. Every state of affairs has dual aspects: a particular aspect and a universal aspect.

Lowe's conception of universals and his reasons for thinking that universals—both substantial and non-substantial universals—must have instances are very different. For Lowe a universal is an abstraction from a particular. Locke characterized abstraction as 'partial consideration'. Think of Socrates's being pale. You can consider Socrates without considering his paleness, and you can consider Socrates's paleness without considering Socrates. (If this seems odd, imagine considering the color of a color chip in a paint store without considering the chip itself or its shape.) This gets you to Socrates's paleness, a mode. Now consider just the paleness, abstract from its being *Socrates's* paleness. When you do this, when you abstract from its particularity, you have arrived at a universal, something that could have many instances.⁵

⁵ I discuss this conception of universals and a related conception advanced by D. C. Williams in my contribution to this volume.

The same reasoning applies to Socrates himself and his kind—humanness. You can consider the individual substance, Socrates, and you can consider Socrates as a human being abstracting from his particularity.

If universals are abstractions in this sense, it is clear why they must have instances. Just as Socrates's paleness requires Socrates, so paleness, the universal, requires some paleness or other. Socrates's paleness depends 'rigidly' on Socrates, paleness, the universal, depends 'non-rigidly' on there being some particular paleness or other.

On this conception of universals, a universal is not a general entity, not an entity capable of being wholly present in distinct spatio-temporal locations at once, and certainly not a resident of a Platonic heaven. Universals are a species of abstract entity, where an abstract entity is understood, not as being transcendent, outside spacetime, but as being dependent for its identity on its instances. Substantial universals depend on individual substances, non-substantial universals depend on modes. Lowe's realism, then, is, like that of D. C. Williams, a species of what Keith Campbell calls 'painless realism' (Campbell 1990).

Before moving on, two points require emphasis. First, universals are in no sense language- or mind-dependent. Although abstraction—Locke's 'partial consideration'—is a mental operation, abstraction merely reveals what is there to be abstracted. A particular electron has a particular charge. If the electron is an individual substance, its charge is a mode, one way the electron is. In considering that mode, in considering that way independently of the electron, you are considering a way many things, many electrons are. More generally, two individual substances are 'the same' way, they 'share' a characteristic, if they are exactly similar in some respect, where the 'respects' are modes.

Second, this conception of universals differs from a conception embraced by some trope theorists who identify universals with classes or collections of exactly resembling tropes. For Lowe a universal is what it is for an individual substance to be a particular way, what it is for an electron, for instance, to have a particular charge. And this is something many electrons have in common.

4. Metaphysics: The Science of Essence

Lowe's conception of essences is closely related to his conception of universals.⁶ Every entity of whatever category, he thinks, has an essence, or more precisely a 'general essence', what it is to be an entity of that kind, and an 'individual essence', what it is to be this particular entity of that kind. In this he follows Aristotle, who describes essences as 'the what it is to be', or 'the what it would be to be' an entity of a particular kind (*Metaphysics* Z, 4), and Locke for whom 'the proper and original signification' of 'essence' is 'the very being of a thing whereby it is what it is' (1690/1978, III, iii, 15).

⁶ See Lowe 2008a, 2012a, 2012b, 2012d. Kit Fine's work on essences has sparked a vast and growing literature on the topic; see, for instance, Fine 1982, 1994 and many subsequent papers.

Thus conceived, essences are not entities, not constituents of ‘hylomorphic compounds’ as Aristotelians suppose, nor are they microconstitutions or hidden structures of the kind favored by ‘scientific essentialists’. It might turn out that the essence of something, water, for instance, is to have a particular kind of microconstitution—what it is to be water is to be constituted by H_2O —but it would be a category mistake to identify this microconstitution with water’s essence. (For reasons to doubt that it is essential to water that it be constituted by H_2O , see Lowe 2008a.) In any case, were essences entities, they themselves would have to have essences, and those essences have essences, and a regress would ensue.

Associated with an essence is a ‘real definition’, a specification of what it is to be an entity of a particular kind, what it is to be a horse, or a statue, or an electron, or more generally, what it is to be a substance, a mode, or a universal. A grasp of essence is, Lowe believes, required to ‘think comprehendingly’ about a given kind of entity or a particular entity of a given kind. Thus to ‘think comprehendingly’ about a statue, you must have a grasp of what it is to be a statue. If you know what it is to be a statue, then you know the identity and persistence conditions for statues. You know, for instance, what kinds of change a statue could and could not undergo. A statue could be damaged and repaired, some of its matter replaced, but a statue could not survive a dramatic change in shape or a dispersal of its parts.

Modal truths about entities—what is or is not possible for a given entity, for instance—stem from essences of the entities in question. You know what could or couldn’t be true of an electron if you know what it is to be an electron. Part of what it is to be an electron is to possess a negative charge, and it is of the essence of negative charges that they empower entities so charged to repel similarly charged entities and attract positively charged entities.

Lowe characterizes metaphysics as the ‘science of essence’. God aside, essence, he thinks, ‘precedes existence’, at least in the sense that, before you could know whether an entity of a particular kind exists, you must at least know what it is to be an entity of that kind. This does not mean that you must know everything about entities of the kind in question. To discover new species of fish, an ichthyologist must know what it is to be a fish. Once located, empirical study can reveal unexpected characteristics of a new species. Physicists unsure of the existence of black holes knew what they were looking for, even though some characteristics of black holes awaited subsequent empirical investigation.

Essences provide a basis for distinguishing objects’ essential properties from their ‘accidental’ properties. An essential property of an object ‘flows’ from the essence of the object in the sense that the object could not continue to exist as an object of that kind were it to lack that property. You could think of an essential property as a property an object must have if the object is to satisfy a given real definition. An accidental property is one an object could lack while remaining an object of that kind. Socrates’s rationality is essential to his humanity, but Socrates’s paleness is not, although Socrates is both rational and pale.

Lowe's conception of essences might be usefully compared with his thoughts on universals. Substantial universals, for instance, are revealed by abstracting from individual substances, something possible only if you know the essence of the individual in question. You can abstract statuehood from a particular statue—a statue of Buchephalus, for instance—only if you understand what it is to be a statue. And indeed you might conclude that kinds—substantial universals—just *are* essences. Instances of kinds share essences in the sense that they all satisfy the same real definition, the 'what it is to be' an instance of a given kind is the same in each case.

5. Subjects of Experience

I conclude my sketch of Lowe's philosophical contributions with an account of 'non-Cartesian substance dualism'.⁷ Imagine that minds—or better, following Descartes, *selves*—were individual substances distinguishable from, but dependent on, the material substances in which they were embodied. Lowe distinguishes selves from their bodies in the way you might distinguish a statue from the lump of bronze that makes it up. A self *has* a body, a complex material substance, on which it depends for its existence. When you identify yourself, you are identifying a substance that has, and depends on, a body, but which is not identical with that body. Nor are you to be identified with any part of your body (your brain, for instance). At this point the statue analogy breaks down. Although the self shares some properties with the body, a self is not made up of the body or the body's parts as a statue is, at a particular time, made up of a portion of bronze.

Bodies and selves have very different essences, very different identity and persistence conditions, so you are not identical with your body. Similar considerations lead to the conclusion that you are not identical with any part of your body, your brain, for instance. Your body is a complex biological substance that includes complex substances as parts. Your brain is one of these substantial parts. Your brain could exist when you do not. Further, you have a particular height and mass. These you share with your body, not with your brain and not with any other part of your body.

One important difference between your body and you is that, unlike your body, you are a simple substance, one that lacks parts that are themselves substances. For consider: what would parts of you be? If you grant that the self is not the body or a part of the body, then parts of the body could not be parts of the self, unless the self has, in addition, other, non-bodily parts. But, again, what might these parts be? There are no obvious candidates.

Might the self be said to have *psychological* parts? Minds could be thought to include distinct 'faculties', for instance. You have various sensory faculties as well as a faculty for memory, and a faculty of imagination. Might these faculties be regarded as *parts* of you? Whatever the faculties are, they are not substances in their own right, entities

⁷ See Lowe 1996, 2003, 2010, 2012c. Much of the discussion to follow is based on Heil 2004, ch. 4.

capable of existence independently of the self in the way parts of a body—a brain or a heart, for instance—are capable of existing independently of the body of which they are parts. Mental faculties are something like capacities, modes dependent on the substances they modify.

So the self is a simple substance distinct from the body and from any substantial part of the body. What characteristics *do* selves possess? You—the self that is you—possesses some characteristics only *derivatively*. Your having a nose, for instance, stems from your having a body that has a nose. But you also have a particular height and mass. These characteristics are, in addition to being characteristics of your body, characteristics of you, your self. This is the point at which Lowe and Descartes part company. According to Descartes, selves, but not bodies, possess mental characteristics; bodies, but not selves, possess physical characteristics. For Lowe, a self can have physical as well as mental characteristics.

What accounts for the distinction between physical characteristics you have and those you have only by virtue of having a body that has them? If the self is simple, then it can possess only characteristics capable of possession by a simple substance. Having two arms is possible only for a complex substance. You have two arms only derivatively, only by virtue of having a body that has two arms. In contrast, being a particular height or having a particular mass does not imply substantial complexity, so these are characteristics you could be said to possess non-derivatively.⁸

In addition to possessing a range of physical characteristics, selves possess mental characteristics. Your thoughts and feelings belong, not to your body, or to a part of your body (your brain), but to you. More generally, selves, but not their bodies, possess mental characteristics—or perhaps your body ‘possesses’ them only derivatively, only by virtue of *your* possessing them.

Because selves, on a view of this kind, are not regarded as immaterial substances, the Cartesian problem of causal interaction between selves and physical substances does not arise. Still, we are bound to wonder how a self, which is not identical with a body or with any part of a body, could act so as to mobilize a body. You decide to take a stroll and subsequently move your body in a characteristic manner. How is this possible? The causal precursors of your strolling apparently include only bodily events and various external causes of bodily events.

Lowe argues that the model of causation in play in discussions of mental causation is inappropriate. A Cartesian imagines selves initiating causal sequences in the brain and thereby bringing about bodily motions. One worry about such a view is that it apparently violates principles central to physics such as the conservation of mass–energy and the idea that the physical universe is ‘causally closed’. Perhaps such a worry is, in the end, merely the manifestation of a prejudice or, more charitably, a well-confirmed but fallible presumption that could be undermined by empirical evidence. Until such evidence turns up, however, we should do well to remain suspicious of those who

⁸ One assumption here is that a simple object could be extended. Thanks to Matthew Tugby.

would deny closure solely in order to preserve a favored thesis. What is required, Lowe thinks, is a way of understanding mind–body causal interaction that is at least consistent with our best physical theories as they now stand.

Lowe argues that there is, in any case, a more telling difficulty for the Cartesian model. Consider your decision to take a stroll, and your right leg’s subsequently moving as a consequence of that decision. A Cartesian supposes that your decision, a mental event, initiates a causal chain that eventually issues in your right leg’s moving, a bodily event. This picture is captured in Figure 0.2 (M_1 is your deciding to stroll, a mental event; B_1 is your right leg’s moving, a physical event; E_1 and E_2 , intervening physical events in your nervous system; t_0 is the time of the decision; and t_1 , the time at which your right leg moves).

The Cartesian picture, Lowe thinks, incorporates a distortion. Imagine tracing the causal chain leading backwards from the muscle contractions involved in the motion of your right leg. That chain presumably goes back to events in your brain, but it goes back beyond these to earlier events, and eventually to events occurring prior to your birth. Further, and more significantly, when the causal chain culminating in B_1 is traced back, it quickly becomes entangled in endless other causal chains issuing in a variety of quite distinct bodily motions, as depicted in Figure 0.3.

Here, B_1 is your right leg’s moving, and B_2 and B_3 , are distinct bodily motions. B_2 might be your left arm’s moving as you greet a passing acquaintance, and B_3 might be a non-voluntary motion of an eyelid. The branching causal chains should be taken to extend up the page indefinitely into the past.

Now, although your decision to stroll is presumed to be responsible for B_1 , and not for B_2 and B_3 , the causal histories of these bodily events are inextricably entangled. Prior to t_0 , there is no identifiable event sequence causally responsible for B_1 , but not for B_2 or B_3 . It is hard to see where in the complex web of causal relations occurring in your nervous system a mental event might initiate B_1 .

Lowe advocates the replacement of the Cartesian model of mental causation with something like a model reminiscent of one proposed by Kant. The self affects the physical universe, although not by initiating or selectively intervening in causal chains. Indeed, in one important respect—and excluding events such as the decay of a radium atom—nothing *in* the universe *initiates* a causal chain. Rather, to put it somewhat mysteriously, the self makes it the case that the universe contains a pattern of causal sequences issuing in a particular kind of bodily motion. A mental event (your deciding to stroll, for instance) brings about a physical event (your right leg’s moving in a particular way), not by instigating a sequence of events that culminates in your right leg’s so moving, but by bringing it about that a particular kind of causal pattern exists.

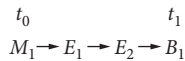


Figure 0.2. The Cartesian Model of Mental Causation

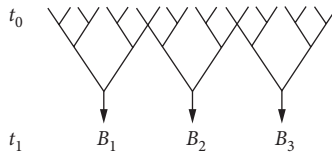


Figure 0.3. Complex Causal Chains

To see how this might work, imagine a spider scuttling about on its web. Although the web is causally dependent on the spider, it is a substance in its own right, not identifiable with the spider's body or a part of the spider. Moreover, the web affects the spider's movements, not by initiating them, but by 'enabling' or 'facilitating' them. The web, you might say, makes it the case that the universe contains motions of one sort rather than another. In an analogous way, the self might be regarded as a product of complex physical and social processes, a product not identifiable with its body or a part of its body. The self accounts for the character of bodily motions, not by initiating causal chains, but by making it the case that those causal chains have the particular 'shape' they have.⁹

6. Looking Ahead

These brief sketches of themes central to Lowe's metaphysics fall well short of capturing their power and scope. The hope is that they will steer readers to Lowe's extensive body of published work, a comprehensive listing of which appears on pages 177–91. Various issues arising in that work occupy the contributors to this volume.

Peter Simons discusses Lowe's commitment to the primacy of metaphysics in our understanding the universe, ourselves, and our scientific endeavors, and takes up the basis of categorical distinctions, such as those featuring in the four-category ontology. Simons' preferred categorical scheme differs from Lowe's, but both authors accept that categories are not themselves categorized entities, and distinctions among categories are internal, requiring no appeal to a category of relations.

I have mentioned already that my contribution to the volume concerns universals as understood by D. C. Williams and by Lowe. It was my reading of a posthumously published paper by Williams that led me finally to understand how Lowe could have thought that an ontology such the one to which I am attracted, one in which universals are absent, could in the end be consistent with his.

Anna Marmodoro takes up Lowe's objections to hylomorphism—the doctrine that objects are compounds of form and matter. Although Lowe placed himself in

⁹ The position calls to mind one defended by F. I. Dretske (1988); but see Gibb 2015 for discussion. A. D. Carruth and S. C. Gibb, in their contribution to this volume, note that Lowe distinguishes between event and fact causation, and takes mental causation to be a species of fact causation. Lowe provides a more conventional treatment of mental causation in his 2003; for discussion see D. M. Robb's contribution to this volume.

the neo-Aristotelian camp, he sharply distinguished his four-category ontology—which he associated with Aristotle’s *Categories*—from the matter–form ontology Aristotle develops in the *Metaphysics*. Marmodoro thinks that there are many ways to unpack hylomorphism and proceeds to articulate a version she thinks brings Lowe much closer to the Aristotle of the *Metaphysics*.

David Oderberg tackles Lowe’s essentialism and what he sees as unwarranted concessions to the prevailing conception of laws of nature as contingent. Lowe, unlike Aristotle, distinguishes something like natural necessity from metaphysical necessity, and espouses epistemological modesty concerning natural essences. This, Oderberg thinks, introduces tensions in Lowe’s essentialism best resolved by moving toward ‘real essentialism’ and a univocal notion of necessity.

Tuomas Tahko is interested in the epistemology of essences. He has in mind essences as characterized by Lowe and Kit Fine, but also more robust conceptions such as the conception developed by Oderberg. If an entity’s essence is what it is to be that entity or an entity of that kind, how do we come to know an essence? Straightforward empirical investigation is ruled out because empirical investigation of entities seems to presuppose knowledge of their essences. But, as Lowe recognized, gaining access to essences cannot be a purely a priori matter either. Tahko develops a suggestion of Lowe’s meant to clarify both what essences are and how we could know them.

Extending Lowe’s essentialism to the normative realm, Antonella Corradini advances a robustly nonnaturalist normative ontology according to which an object’s essence can include a ‘telic structure’, so it can be of the nature of an object—part of what makes it what it is—that it has certain ends. If these are included in the supervenience base of normative properties and truths—and so metaphysically necessitate the properties and truths—the result is a nonnaturalist normative ontology sharply contrasting with more familiar ‘nonreductive’ ontologies with exclusively naturalistic supervenience bases.

Peter van Inwagen addresses Lowe’s ‘new modal version of the ontological argument’. One of the premises of the argument—that abstract beings are dependent beings—was touched on in my earlier discussion of Lowe’s conception of universals. After setting out Lowe’s argument in some detail, van Inwagen advances reasons for rejecting the dependence thesis, thereby calling into question the soundness of Lowe’s ontological argument.

A. D. Carruth and S. C. Gibb discuss in depth an attempt by Lowe to accommodate mental causation to the apparent ‘causal completeness’ of the physical domain, an attempt sketched briefly in my discussion of non-Cartesian substance dualism. Physical causation is event causation—one definite event brings about another definite event—whereas mental causation is ‘fact causation’—your decision to wave making it the case that a certain physical fact obtains. Carruth and Gibb question whether a suitably robust distinction between facts and events can be drawn in a way that would allow this position to be maintained, examining typical approaches to this distinction before going on to discuss whether Lowe’s four-category ontology has the resources to distinguish facts from events.

David Robb takes up a somewhat different picture of mental-to-physical causation advanced elsewhere by Lowe. Lowe hopes to ward off concerns about ‘causal closure’ and the completeness of physics, by suggesting that certain mental events might be required to produce certain physical events—certain bodily movements, for instance—but that this could occur in a manner that would make the mental event and its contribution to the physical effect ‘invisible’. This would be the case if a physical event produced a simultaneous mental event, and the two together conspired to produce a subsequent physical event. The physical cause would suffice for the effect, but only by producing a mental event. Robb argues that, owing to the complexity of the pertinent physical powers, there are reasons to think that it would be possible to factor out the contribution of constituent physical powers and thereby establish empirically the need for a mental power. This means that the thesis could be empirically tested and potentially falsified.

The philosophical diversity of these contributions illustrates what I earlier described as the range and power of Lowe’s metaphysical vision, a reminder that philosophy is like art: as middling art isn’t art, middling philosophy isn’t philosophy. (The corollary: most philosophers aren’t philosophers.) Edward Jonathan Lowe is most definitely a philosopher, indeed a philosopher’s philosopher.

7. References

- Armstrong, D. M. 1997. *A World of States of Affairs*. Cambridge: Cambridge University Press.
- Campbell, K. 1990. *Abstract Particulars*. Oxford: Basil Blackwell.
- Dretske, F. I. 1988. *Explaining Behavior: Reasons in a World of Causes*. Cambridge, MA: MIT Press.
- Fine, K. 1982. ‘Acts, Events, and Things.’ *Language and Ontology: Proceedings of the 6th International Wittgenstein Symposium*. Vienna: Holder-Pichler-Tempsky: 97–105.
- Fine, K. 1994. ‘Essence and Modality.’ *Philosophical Perspectives* 8: 1–16.
- Gibb, S. C. 2015. ‘The Causal Closure Principle.’ *Philosophical Quarterly* 65: 626–47.
- Heil, J. 2004. *Philosophy of Mind: A Contemporary Introduction* 2nd edn. London: Routledge.
- Locke, J. 1690/1978. *An Essay Concerning Human Understanding*. Ed. P. H. Nidditch, Oxford: Clarendon Press.
- Lowe, E. J. 1996. *Subjects of Experience*. Cambridge: Cambridge University Press.
- Lowe, E. J. 2003. ‘Physical Causal Closure and the Invisibility of Mental Causation.’ In S. Walter and H.-D. Heckmann, eds. *Physicalism and Mental Causation: The Metaphysics of Mind and Action*. Exeter: Imprint Academic: 137–54.
- Lowe, E. J. 2006. *The Four-Category Ontology: A Metaphysical Foundation for Natural Science*. Oxford: Clarendon Press.
- Lowe, E. J. 2008a. ‘Essentialism, Metaphysical Realism, and the Errors of Conceptualism.’ *Philosophia Scientiae* 12: 9–33.
- Lowe, E. J. 2008b. *Personal Agency: The Metaphysics of Mind and Action*. Oxford: Oxford University Press.
- Lowe, E. J. 2010. ‘Why My Body is not Me: The Unity Argument for Emergentist Self–Body Dualism.’ In A. Corradini and T. O’Connor, eds. *Emergence in Science and Philosophy*. London: Routledge: 127–48.

- Lowe, E. J. 2011. 'Ontological Categories: Why Four are Better than Two.' In J. Cumpa and E. Tegtmeier, eds. *Ontological Categories*. Frankfurt: Ontos Verlag: 100–26.
- Lowe, E. J. 2012a. 'A Neo-Aristotelian Substance Ontology: Neither Constituent nor Relational.' In T. E. Tahko, ed. *Contemporary Aristotelian Metaphysics*. Cambridge: Cambridge University Press: 229–48.
- Lowe, E. J. 2012b. 'Essence and Ontology.' In L. Novak, D. D. Novotny, P. Sousedik, and D. Svoboda, eds. *Metaphysics: Aristotelian, Scholastic, Analytic*. Frankfurt: Ontos Verlag: 93–111.
- Lowe, E. J. 2012c. 'Non-Cartesian Substance Dualism.' In B. P. Göcke, ed. *After Physicalism*. Notre Dame, IN: Notre Dame University Press: 48–71.
- Lowe, E. J. 2012d. 'What Is the Source of Our Knowledge of Modal Truths?' *Mind* 121: 919–50.
- Lowe, E. J. 2016. 'There Are (Probably) No Relations.' In A. Marmodoro and D. Yates, eds. *The Metaphysics of Relations*. Oxford: Oxford University Press: 100–12.

1

Metaphysics as the Science of Essence

E. J. Lowe

1. Introduction

What is metaphysics?¹ And how is it to be pursued? That is, by what method of inquiry can we hope to acquire metaphysical knowledge, if indeed there is any distinctive kind of knowledge that deserves to go by that name? Elsewhere, I have defended the view that the central task of metaphysics is to chart the *possibilities of being*, with a view to articulating the structure of reality as a whole, at its most fundamental level (Lowe 1998, ch. 1). A key thought here is that knowledge of what is *actual* presupposes and rests upon knowledge of what is *possible*—that is, of what is *really* or *metaphysically* possible—and hence that every empirical science requires some sort of metaphysical foundation. Moreover, this foundation had better be, at bottom, the same for all such sciences, since each empirical science has the pursuit of truth as its objective and truth itself is unitary and indivisible.

According to this conception of the aim and content of metaphysical theory, metaphysics is above all concerned with identifying, as perspicuously as it can, the fundamental *ontological categories* to which all entities, actual and possible, belong. This it does by articulating the *existence and identity conditions* distinctive of the members of each category and the relations of *ontological dependency* in which the members of any given category characteristically stand to other entities, either of the same or of different categories. The proper conduct of this task, as I conceive of it, is an *a priori* exercise of the rational intellect, so that pure metaphysics should be thought of as a science whose epistemic basis and status are entirely akin to those of pure mathematics

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and logic, differing from both of the latter primarily in having formal ontological questions at its heart.

If metaphysics is, for the foregoing reason, centrally concerned with charting the domain of the possible, it is incumbent upon metaphysicians to explain what it is that *grounds* metaphysical possibility—and to do so in a way that allows our knowledge of metaphysical possibility to be something that is itself *possible*, given a metaphysically defensible account of our own nature as rationally cognisant beings occupying a distinctive place in the fundamental structure of reality as a whole.

My own belief—which I shall endeavour in this chapter to explain and justify—is that the only coherent account of the ground of metaphysical possibility and of our capacity for modal knowledge is to be found in a version of *essentialism*: a version that I call *serious* essentialism, to distinguish it from certain other views which may superficially appear very similar to it but which, in fact, differ from it fundamentally in certain crucial respects. Above all, my preferred version of essentialism eschews any appeal whatever to the notion of *possible worlds* in its account of the nature and ground of metaphysical possibility, for reasons that I shall try to make clear in due course. I am at most prepared to allow that the language of possible worlds may *sometimes* function as a useful *façon de parler*, albeit one that carries with it the constant danger of misleading those who indulge in it.

2. Serious Essentialism

It is vital for my purposes in this chapter that the doctrine of *essentialism* be suitably understood. Many possible-worlds theorists happily describe themselves as essentialists and propose and defend what *they* call essentialist claims, formulated in terms of the language of possible worlds. They will say, for instance, that an essential property of an object is one that that object possesses in every possible world in which it exists, or, alternatively, that is possessed by the ‘counterpart(s)’ of that object in every possible world in which that object has a ‘counterpart’. And they will typically claim that some, but not all, of an object’s actual properties are essential to it in this sense. A doctrine of this sort is not serious essentialism in my sense, because it attempts to characterize essence in terms of antecedently assumed notions of possibility and necessity and thus—in my view—puts the cart before the horse. It is at best *ersatz* essentialism.

So what is *serious* essentialism? To begin to answer this question, we need to ask *what essences are*. However, this question is potentially misleading, for it invites the reply that essences are *entities* of some special sort. And, as I shall argue, it is simply incoherent to suppose that essences are entities. According to serious essentialism, as I understand it, all entities *have* essences, but their essences are certainly not further entities related to them in some special way.

What do we or, rather, what *should* we mean by the ‘essence’ of a thing—where by ‘thing’, in this context, I just mean any sort of entity whatever? We can, I suggest, do no better than to begin with John Locke’s perceptive words on this matter, which go right

to its heart. Essence, Locke said, in the ‘proper original signification’ of the word, is ‘the very being of any thing, whereby it is, what it is’ (Locke 1690, III, III, 15). In short, the essence of something, *X*, is *what X is*, or *what it is to be X*.²

In another locution, *X*’s essence is the very *identity* of *X*—a locution that I am happy to adopt, provided that it is clearly understood that to speak of something’s ‘identity’ in this sense is quite different from speaking of the identity *relation* in which it necessarily stands to itself and to no other thing. In order to avoid potential confusion about the meaning of locutions such as these, it is important to draw, from the very start, a distinction between *general* and *individual* essence.³ Any individual thing, *X*, must be a thing of *some general kind*—because, at the very least, it must belong to some *ontological category*.

Remember that by ‘thing’ here I just mean ‘entity’. So, for example, *X* might be a material object, or a person, or a property, or a set, or a number, or a proposition, or whatnot—the list goes on, in a manner that depends on what one takes to be a full enumeration of the ontological categories to be included in it (see Lowe 2006, pt. 1 for my own account of what ontological categories we should recognize and which we should regard as fundamental). This point being accepted, if *X* is something of kind *K*, then we may say that *X*’s *general* essence is *what it is to be a K*, while *X*’s *individual* essence is *what it is to be the individual of kind K that X is*, as opposed to any other individual of that kind.

Before I proceed, an important complication must be dealt with. It should be evident that we cannot simply assume that there is only ever a *single* appropriate answer to the question ‘What kind of thing is *X*?’. For instance, if ‘a cat’ is an appropriate answer to this question, then so will be the answers ‘an animal’ and ‘a living organism’. So too, of course, might be the answer ‘a Siamese cat’. It is important to recognize, however, that some, but not all, of these answers plausibly announce the fact that *X* belongs to a certain ontological category. In my view, ‘*X* is a living organism,’ *does* announce such a fact, but ‘*X* is a cat’ does not. I take it that the substantive noun ‘cat’ denotes a certain *natural kind* and consider that such kinds are a species of *universal*. Thus, natural kinds, such as the kind *cat*, are themselves things belonging to a certain ontological category—the category of universals—but such a kind is not *itself* an ontological category, because ontological categories are not *things* at all, to be included in a complete inventory of what there is (see Lowe 1998, ch. 8 and Lowe 2006, ch. 2). One upshot of all this is that a certain sort of ambiguity attaches to questions concerning a thing’s general essence, as I shall now try to explain.

² The historical source of this view lies with Aristotle, whose phrase τὸ τί ἦν εἶναι is standardly translated as ‘essence’: see Aristotle, *Metaphysics* Z, 4. Its more literal meaning is ‘the what it is to be’ or ‘the what it would be to be’.

³ I do not attempt to offer here a *semantic analysis* of expressions such as ‘what *X* is’, ‘what it is to be *X*’ or ‘the identity of *X*’, though that is no doubt an exercise that should be undertaken at some stage in a full account of what I am calling *serious essentialism*. I assume that our practical grasp of the meaning of such expressions is adequate for a preliminary presentation of the approach of the sort that I am now engaged in.

An implication of what I have said so far is that if ‘a cat’ is an appropriate answer to the question ‘What *kind* of thing is *X*?’, then we may say that *X*’s general essence is *what it is to be a cat*. But, while I don’t want to retreat from this claim, I do want to qualify it. I should like to say that if *X* is a cat, then *X*’s *fundamental* general essence is *what it is to be a living organism*, because that—in my view—is the most narrow (or ‘lowest’) ontological category to which *X* may be assigned. The reason for this is that it is part of the *individual* essence of the natural kind *cat*—of which *X* is *ex hypothesi* a member—that it is a kind of living organism. Now, there are, I believe, certain essential truths concerning *X* that do not issue from its fundamental general essence but only from the fact that it belongs to this particular natural kind. These are essential truths concerning *X* that are determined solely by the individual essence of that natural kind.

I maintain that *X*’s fundamental general essence determines what is *absolutely* metaphysically necessary for *X*, whereas the individual essence of the natural kind *cat* determines only what is metaphysically necessary for *X qua* member of that kind. Thus, *being a cat* is not an absolute metaphysical necessity for any individual living organism that is, in fact, a cat. To put it another way: it is *metaphysically* possible—even if not *biologically* or *physically* possible—for any individual cat to survive ‘radical’ metamorphosis, by becoming a member of *another* natural kind of living organism (see Lowe 1998, 54–6).

Accordingly, *what it is to be a cat*, while it is not *X*’s fundamental general essence, is nonetheless what we might appropriately call *X*’s *specific* general essence, on the grounds that the kind *cat* is the most specific (or ‘lowest’) natural kind to which *X* may be assigned.⁴ However, I acknowledge that the distinction that I am now trying to draw between ‘fundamental’ and ‘specific’ general essence in the case of individual members of natural kinds is a controversial one that needs much fuller justification than I am able to give it here. Hence, in what follows, I shall try as far as possible to prescind from this distinction, hoping that the simplification involved in doing so will cause no damage to the overall thrust of my arguments.

One consequence of this simplification is that I shall continue to speak of ‘the’ kind to which a thing belongs, without discriminating between ‘kind’ in the sense of *ontological category* and ‘kind’ in the sense of *natural kind*, and without explicit acknowledgement of the fact that the question ‘What kind of thing is *X*?’ may be capable of receiving more than one appropriate answer.

3. Why Are Essences Needed?

I have just urged that all individual things—all entities—have both general and individual essences, a thing’s general essence being *what it is to be a thing of its kind* and its

⁴ I take it here, at least for the sake of argument, that there are ‘higher’ natural kinds to which *X* may be assigned, such as the kinds *mammal* and *vertebrate*, but that Siamese cats—for example—do not constitute a distinct natural kind of their own.

individual essence being *what it is to be the individual of that kind that it is*, as opposed to any other individual of that kind. But why suppose that things must *have* 'essences' in this sense and that we can, at least in some cases, know those essences? First, because otherwise it makes no sense—or so I believe—to say that we can talk or think comprehendingly about things at all. If we do not at least know *what a thing is*, how can we talk or think comprehendingly about it?⁵ How, for instance, can I talk or think comprehendingly about *Tom*, a particular cat, if I don't know what cats are and which cat, in particular, *Tom* is? I am not saying that I must know *everything* about cats or about *Tom* in order to be able to talk or think comprehendingly about that particular animal.⁶ But I must know enough to distinguish the kind of thing that *Tom* is from other kinds of thing, and enough to distinguish *Tom* in particular from other individual things of *Tom's* kind. Otherwise, my talk and thought cannot really fasten upon *Tom*, as opposed to something else.

It is fashionable at present to suppose that our talk and thought have, in general, their *referents* in the 'external' world secured through the existence of appropriate *causal* links between certain constituents of our talk and thought—certain of our linguistic and mental 'representations'—and various extra-linguistic and extra-mental entities belonging to that world: links that can, and mostly do, obtain without our needing to have any knowledge of them. On this sort of view, my talk and thought can fasten upon *Tom* because there is an appropriate causal link between the name '*Tom*,' as I have learnt to use it, and *Tom*—and an analogous causal link between a certain 'mental representation' of mine (perhaps a certain 'symbol' in the putative 'language of thought' supposedly utilized by my brain) and *Tom*. I will only say here that I cannot begin to understand how it might seriously be supposed that a linkage of this sort could genuinely suffice to enable me to *talk and think comprehendingly about Tom*, even if it is conceded that there is a (relatively anodyne) notion of 'reference' that could perhaps be satisfactorily accounted for by a causal theory of the foregoing sort.

I should emphasize that I am not presently concerned to challenge the so-called causal theory of reference, much less to defend in opposition to it some sort of neo-Fregean theory of reference as being mediated by 'sense'. I am not interested, at present, in semantic questions or rival semantic theories, but rather in the purely metaphysical question of *how it is possible to be acquainted with an object of thought*: my answer being that it is so through, and only through, a grasp of that object's *essence*—that is, through knowing *what it is*.

⁵ I ask only how we can *talk or think* comprehendingly about a thing if we do not know what it is—not how we can *perceive* a thing if we do not know what it is. I am happy to allow that a subject *S* might, for example, *see* an object *O* even though *S* does not know what *O* is. Seeing, however, is not a purely intellectual act. Even lower animals that cannot at all plausibly be said to understand *what* objects exist in their environment, may nonetheless be said to *see* or *feel* or *smell* some of those objects.

⁶ Perhaps, all I need to know about cats is that they are *animals* or *living organisms* and perhaps, likewise, all I need to know about *Tom* is which animal or living organism he is.

Denying the reality of essences doesn't only create an epistemological problem: it also creates an ontological problem. Unless Tom has an 'identity'—whether or not anyone is acquainted with it—there is nothing to make Tom the particular thing that he is, as opposed to any other thing. Anti-essentialism commits us to anti-realism, and indeed to an anti-realism so global that it is surely incoherent. It will not do, for instance, to try to restrict one's anti-essentialism to 'the external world', somehow privileging us and our language and thought. On the one hand, how could it be that there is a fact of the matter as to *our* identities, and the identities of *our words and thoughts*, but not as to the identities of the mind-independent entities that we try to capture in language and thought? On the other hand, how could there *not* be any fact of the matter as to our identities and the identities of our words and thoughts? Everything is, in Joseph Butler's memorable phrase, *what it is and not another thing*. That has sounded to many philosophers like a mere truism without significant content, as though it were just an affirmation of the reflexivity of the identity relation. In fact, Butler's dictum does not merely concern the identity relation but also identity in the sense of *essence*. It implies that there is a fact of the matter as to what any particular thing is—that is, as to its 'very being', in Locke's phrase. Its very being—its identity—is what makes it the thing that it is and thereby distinct from any other thing.

Essences are apt to seem elusive and mysterious, especially if talked about in a highly generalized fashion, as I have been doing so far. Really, I suggest, they are quite familiar to us. Above all, we need to appreciate that in very many cases a thing's essence involves *other things*, to which it stands in relations of *essential dependence*. Consider the following thing, for instance: the set of planets whose orbits lie within that of Jupiter. What kind of thing is that? Well, of course, it is a *set*, and as such an abstract entity that depends essentially for its existence and identity on the things that are its members: Mercury, Venus, Earth, and Mars. Part of *what it is to be a set* is to be something that depends in these ways upon certain other things—the things that are its members. Someone who did not grasp that fact would not understand *what a set is*. Furthermore, someone who did not know *which things* are this set's members, or at least what determined which things are its members, would not know *which particular set* this set is. So, someone who knew that its members are the planets just mentioned would know which set it is, as would someone who knew what it is to be a planet whose orbit lies within that of Jupiter.⁷

This is a simple example, but it serves to illustrate a general point. In many cases, we know what a thing is—both what kind of thing it is and which particular thing of that kind it is—only by knowing that it is related in certain ways to other things. In such

⁷ There are, broadly speaking, two different views of what a set is: one which takes a set simply to be the result of—as David Lewis puts it—'collecting many into one', and another which takes a set to be the extension of a property or of a concept. (For Lewis's remark, see his 1991, vii.) I see no compelling reason why, in principle, our ontology should not accommodate sets in *both* of these understandings of what they are. Because I am using the example of sets only for illustrative purposes, this is a matter on which I can afford to remain agnostic here.

cases, the thing in question depends essentially on these other things for its existence or its identity. To say that X depends essentially on Y for its existence and identity is just to say that it is *part of the essence* of X that X exists only if Y exists and *part of the essence* of X that X stands in some unique relation to Y (see Lowe 1998, ch. 6; Tahko and Lowe 2016). Knowing a thing's essence, in many cases, is accordingly simply a matter of understanding the relations of essential dependence in which it stands to other things whose essences we in turn know.

4. Essences Are Not Entities

I said earlier that it is wrong to think of essences as themselves being *entities* of any kind to which the things having them stand in some special kind of relation. Locke himself unfortunately made this mistake, holding that the 'real essence' of a material substance just *is* its 'particular internal constitution'—or, as we would now describe it, its atomic or molecular structure. Thus, at one point Locke remarks: '[W]e come to have the *Ideas of particular sorts of Substances*, by collecting such Combinations of simple *Ideas*, as are by Experience. . . taken notice of to exist together, and are therefore supposed to flow from the particular internal Constitution, or unknown Essence of that Substance' (1690, II, xxiii, 3).

This is a mistake that has been perpetuated in the modern doctrine, made popular by the work of Saul Kripke and Hilary Putnam, that the essence of water *consists* in its molecular make-up, H_2O , and that the essence of a living organism *consists* in its DNA—the suggestion being that we *discover* these 'essences' simply by careful scientific investigation of the things in question (see Putnam 1975; Kripke 1980). Now, as we saw earlier, it may well be part of the essence of a thing *that it stands in a certain relation to some other thing*, or kind of things. But *the essence itself*—the very being of a thing, whereby it is, what it is—is not and could not be some further entity. So, for instance, it might perhaps be acceptable to say that it is part of the essence of water *that it is composed of H_2O molecules* (an issue that I shall return to shortly). But the essence of water could not simply *be* H_2O —*molecules* of that very kind—nor yet the *property* of being composed of H_2O molecules.

For one thing, if the essence of an entity *were* just some further entity, then *it in turn* would have to have an essence of its own and we would be faced with an infinite regress that, at worst, would be vicious and, at best, would appear to make all knowledge of essence impossible for finite minds like ours. To know something's essence is not to be acquainted with some *further thing* of a special kind, but to understand *what exactly that thing is*. This is why knowledge of essence is possible, for it is a product simply of *understanding*—not of empirical observation, much less of some mysterious kind of quasi-perceptual acquaintance with esoteric entities of any sort. And, on pain of incoherence, we cannot deny that we understand what at least some things are, and thereby know their essences.

Here it might be objected that it is inconsistent of me to deny that essences are entities and yet go on, as I apparently do, to *refer to* and even *quantify over* essences. Someone who voices this objection probably has in mind W. V. Quine's notorious *criterion of ontological commitment*, encapsulated in his slogan 'to be is to be the value of a variable' (see, for instance, Quine 1969).

I reply, in the first place, that I could probably say all that I want to say about my version of essentialism while avoiding all locutions involving the appearance of reference to and quantification over essences, by paraphrasing them in terms of locutions involving only sentential operators of the form 'it is part of the essence of X that'—where 'the essence of X ' is not taken to make an independent contribution to the meaning of the operator, which might be represented symbolically by, say, ' E_x ' in a sentential formula of the form ' $E_x(p)$ '. The latter is a kind of locution that I certainly do want to use and find very useful. However, effort spent on working out such paraphrases in all cases would be effort wasted. If a paraphrase *means the same* as what it is supposed to paraphrase—as it had better do, if it is to be any good—it carries the same 'ontological commitments' as whatever it is supposed to paraphrase, so that constructing paraphrases cannot be a way of relieving ourselves of ontological commitments. We cannot discover those commitments simply by examining the syntax and semantics of our language, for syntax and semantics are very uncertain guides to ontology. In other words, I see no reason to place any confidence in Quine's famous criterion.

5. Essence Precedes Existence

Another crucial point about essence is this: in general, *essence precedes existence*. And by this I mean that the former precedes the latter both *ontologically* and *epistemically*. That is to say, on the one hand, it is a precondition of something's existing that its essence—along with the essences of other existing things—does not preclude its existence. And, on the other hand—and this is what I want to concentrate on now—we can in general *know* the essence of something X antecedently to knowing whether or not X exists. Otherwise, it seems to me, we could never find out *that* something exists. For how could we find out *that* something, X , exists before knowing *what* X is—before knowing *what it is* whose existence we have supposedly discovered?

(Notoriously, Descartes is supposed to have claimed, in the *Second Meditation*, to know *that* he existed before he knew *what* he was—that is, before he grasped his own essence. But it seems to me that any such claim must be construed as being either disingenuous or else intended non-literally, if it is not to be dismissed as being simply incomprehensible. It might, for instance, be taken to imply merely that Descartes was certain that the word 'I' *had a reference*, before knowing what that reference was. To be accurate, though, what Descartes actually says is 'But I do not yet have a sufficient understanding of what this "I" is, that now necessarily exists' (1641/1986, 17). That is consistent with saying that Descartes *does* already grasp his own essence, but needs to

clear his mind of confused thoughts concerning it. Query: might we not come to know what *X* is *neither before nor after* discovering that *X* exists, but *simultaneously* with that discovery? I see no reason to deny this possibility in some cases. But that concession need not be taken to undermine the claim that, in general, we *can* know the essence of something *X* before knowing whether or not *X* exists.)

Consequently, we know the essences of many things that, as it turns out, *do not* exist. We know what these things *would be*, if they existed, and we retain this knowledge when we discover that, in fact, they do not exist. Conceivably, there are exceptions. Perhaps it really is true in the case of God, for instance, that essence does not precede existence, although this could not quite generally be the case. However, saying this is perfectly consistent with acknowledging that, sometimes, we only come to know the essence of something after we have discovered the existence of certain *other* kinds of things.

This is what goes on in many fields of theoretical science. Scientists trying to discover the transuranic elements knew before they found them *what it was* that they were trying to find, but only because they knew that what they were trying to find were elements whose atomic nuclei were composed of protons and neutrons in certain hitherto undiscovered combinations. They could hardly have known what they were trying to find, however, prior to the discovery of the existence of protons and neutrons—for only after these sub-atomic particles were discovered and investigated did the structure of atomic nuclei become sufficiently well understood for scientists to be able to anticipate which combinations of nucleons would give rise to reasonably stable nuclei.

Here it might be objected that Kripke and Putnam have taught us that the essences of many familiar natural kinds—such as the kind *cat* and the kind *water*—have been revealed to us only *a posteriori* and consequently that in cases such as these, at least, it cannot be true to say that ‘essence precedes existence’, whatever may be said in the case of the transuranic elements.

The extent to which the Kripke–Putnam doctrine has become a commonplace of contemporary analytic philosophy is illustrated by the following remark of Frank Jackson’s, which he makes simply in passing and without acknowledging any need to justify it: ‘[W]e rarely know the essence of the things our words denote (indeed, if Kripke is right about the necessity of origin, we do not know our own essences)’ (Jackson 1998, 50). Yet, I would urge, it should strike one as being odd to the point of paradoxicality to maintain that we can talk or think comprehendingly about things without knowing *what it is* that we are talking or thinking about—that is, without grasping their essences.

The charitable conclusion to draw would be that philosophers like Jackson do not use the term ‘essence’ in what Locke called its ‘proper original signification’. Now, of course, Locke himself says that the ‘real’ essences of material substances are *unknown* to us—and the Kripke–Putnam doctrine is recognizably a descendant of Locke’s view, to the extent that it identifies the ‘real essences’ of material substances with their

'internal constitutions', many of which are certainly still unknown to us and may forever continue to be so. But Locke, at least, concluded—unlike modern adherents of the Kripke–Putnam doctrine—that 'the *supposition of Essences, that cannot be known*; and the making them nevertheless to be that, which distinguishes the Species of Things, is so wholly useless... [as] to make us lay it by' (1690, III, III, 17) and he accordingly appeals instead to what he calls *nominal* essences.

The correct position, I suggest, is neither Locke's nor that of the Kripke–Putnam doctrine, but rather (what I take to be) Aristotle's: the real essences of material substances *are* known to those who talk or think comprehendingly about such substances—and consequently that such essences are not to be identified with anything that is *not* generally known to such speakers and thinkers, such as the 'particular internal constitution' of a material substance, or a human being's (or other living creature's) 'origin' in the Kripkean sense.

The presupposition here is that Kripke and Putnam are *correct* in identifying the essence of water, for example, with its molecular make-up, H_2O . I have already explained why I think that such identifications are mistaken, to the extent that they can be supposed to involve the illicit *reification* of essences. It may still be urged against me that even if, more cautiously, we say only that it is part of the essence of water *that it is composed of H_2O molecules*, it still follows that the essence of water has only been revealed to us—or, at least, has only been *fully* revealed to us—a *posteriori*.

In point of fact, however, the Kripke–Putnam doctrine is even more obscure and questionable than I have so far represented it as being. Very often, it is characterized in terms of the supposed modal and epistemic status of identity-statements involving natural kind terms, such as 'Water is H_2O ', which are said to express truths that are at once necessary and *a posteriori*. In such a statement, however, the term ' H_2O ' is plainly not functioning in exactly the same way as it does in the expression ' H_2O molecule'. The latter expression, it seems clear, means 'molecule composed of two hydrogen ions and one oxygen ion'. But in 'Water is H_2O ', understood as an identity-statement concerning kinds, we must either take ' H_2O ' to be elliptical for the *definite description* 'the stuff composed of H_2O molecules' or else simply as being a *proper name* of a kind of stuff, in which case we cannot read into it any significant semantic structure. On the latter interpretation, 'Water is H_2O ' is exactly analogous to 'Hesperus is Phosphorus' and its necessary truth reveals nothing of substance concerning the composition of water. If we are inclined to think otherwise, this is because we slide illicitly from construing ' H_2O ' as a proper name to construing it as elliptical for the definite description 'the stuff composed of H_2O molecules'. Now, when 'Water is H_2O ' is understood on the model of 'Hesperus is Phosphorus', its necessary *a posteriori* truth may in principle be established in a like manner—namely, by appeal to the familiar logical proof of the necessity of identity, together with the *a posteriori* discovery of the co-reference of the proper names involved—but not so when it is construed as meaning 'Water is the stuff composed of H_2O molecules', for the latter involves a definite description and the logical proof in question notoriously fails to apply where

identity-statements involving definite descriptions are concerned (see Kripke 1971; compare Lowe 2005).

Thus far, we have been given no reason to suppose that 'Water is H_2O ' expresses an *a posteriori* necessary truth that reveals something concerning the essence of water. The appearance that we have been given such a reason is the result of mere sleight of hand.

It might be thought that 'Water is the stuff composed of H_2O molecules' follows unproblematically from the supposed empirical truth 'Water is H_2O ' (construed as an identity-statement involving two proper names) and the seemingly trivial, because analytic, truth ' H_2O is the stuff composed of H_2O molecules'. But the latter, when the first occurrence of ' H_2O ' in it is interpreted as a proper name, is no more trivial than 'Water is the stuff composed of H_2O molecules'—and this is how it must be interpreted for the inference to go through.

There is, in any case, another important consideration that we should bear in mind when reflecting on the frequently invoked analogy between 'Water is H_2O ' and 'Hesperus is Phosphorus'. It is all very well to point out that the discovery that Hesperus is Phosphorus was an empirical one. But it was not *purely* empirical. The identity was established because astronomers discovered that Hesperus and Phosphorus *coincide in their orbits*: wherever Hesperus is located at any given time, there too is Phosphorus located. However, spatiotemporal coincidence only implies identity for things of appropriate kinds. It is only because Hesperus and Phosphorus are taken to be *planets* and thereby *material objects of the same kind* that their spatiotemporal coincidence can be taken to imply their identity. The principle that distinct material objects of the same kind cannot coincide spatiotemporally is not an empirical one: it is an *a priori* one implied by *what it is* to be a material object of any kind—in other words, it is a truth grounded in *essence*. It is only because we know that it is *part of the essence* of a planet not to coincide spatiotemporally with another planet, that we can infer the identity of Hesperus with Phosphorus from the fact that they coincide in their orbits. One must already know *what a planet is*—know its essence—in order to be able to establish by *a posteriori* means that one planet is identical with another.

It might be asked: did astronomers know *which* planet Hesperus is—that is, know its *individual* essence—before knowing that it is identical with Phosphorus? It might seem that the answer must be 'No': for if they did, it may be wondered, how could they have been in any doubt as to its identity with Phosphorus?

We need to bear in mind, however, that it is clearly not part of the essence of any planet that it has the particular orbit that it does: a planet can certainly change its orbit, and indeed could have had a quite different one. What led to the discovery that Hesperus is the same planet as Phosphorus was simply that their orbits were plotted and found to coincide. And since one can know which planet a planet is without knowing what its orbit is, it is therefore perfectly explicable that astronomers should—and did—know which planet Hesperus is and which planet Phosphorus is without knowing that Hesperus is the same planet as Phosphorus. So how, in general, *does* one know which material object of kind *K* a certain material object, *O*, is?

One way one can know this is through *perceptual acquaintance* with *O* that is informed by knowledge of the *general* essence of objects of kind *K*. (Recall that perception of an object *O* does not *in itself* presuppose knowledge of what *O* is, so that the foregoing claim does not beg the very question at issue.) That is to say, it very often happens that one perceives an object *O* in circumstances that enable one to know that *what* one is perceiving, *O*, is a particular object of kind *K*. In such circumstances, one is thus in a position to know *which* object of this kind *O* is—namely, *that* one (the one that one is perceiving). And one can retain this knowledge by *remembering* which object it was that one perceived. I should emphasize, however, that this does not imply that it is part of *O*'s individual essence that it is the object of kind *K* that one perceived on a particular occasion—for, of course, it will in general be an entirely contingent matter that one happened to perceive it then, or indeed at all.

By the same token, one must already know *what a kind of stuff is*—know its essence—in order to be able to establish by *a posteriori* means that one kind of stuff is identical with another. It can hardly be the case, then, that we can *discover* the essence of a kind of stuff simply by establishing *a posteriori* the truth of an identity-statement concerning kinds of stuff—any more than we can be supposed to have discovered the essence of a particular planet by establishing *a posteriori* the truth of an identity-statement concerning that planet. Even granting that 'Water is H_2O ' is a true identity-statement that is both necessarily true and known *a posteriori*, it does not follow that it can be taken to reveal to us the essence of the kind of stuff we call 'water'.

Be all this as it may, we still have to address the question of whether, in fact, we ought to say that it is part of the essence of water that it is composed of H_2O molecules. So far, we have at best seen only that the Kripke–Putnam semantics for natural kind terms have given us no reason to suppose that we ought to. I am inclined to answer as follows.

If we are using the term 'water' to talk about a certain chemical compound whose nature is understood by theoretical chemists, then indeed we should say that it is part of the essence of this compound that it consists of H_2O molecules. At the same time, it should be acknowledged that the existence of this compound is a relatively recent discovery, which could not have been made before the nature of hydrogen and oxygen atoms and their ability to form molecules were understood. Consequently, when we use the term 'water' in everyday conversation and when our forebears used it before the advent of modern chemistry, we are and they were *not* using it to talk about a chemical compound whose nature is now understood by theoretical chemists. We are and they were using it to talk about a certain *kind of liquid*, distinguishable from other kinds of liquid by certain fairly easily detectable macroscopic features, such as its transparency, colourlessness, and tastelessness. We are right, I assume, in thinking that a liquid of this kind actually exists, but *not* that it is part of its essence that it is composed of H_2O molecules. At the same time, however, we should certainly acknowledge that empirical scientific inquiry reveals that, indeed, the chemical compound H_2O is very largely what bodies of this liquid are made up of. In fact, the natural laws governing this and other chemical compounds make it overwhelmingly unlikely that this

kind of liquid could have a different chemical composition in different parts of our universe. But the 'could' here is expressive of mere physical or natural possibility, not *metaphysical* possibility (for extended discussion of the need to distinguish between these two species of possibility, see Lowe 2006, ch. 9 and ch. 10). Only an illicit conflation of these two species of possibility could reinstate the claim that water is *essentially* composed of H_2O molecules.

What about our supposed 'intuitions' in so-called 'Twin-Earth' cases—for example, the supposed intuition that if, on a distant planet, a watery stuff was discovered that was not composed of H_2O molecules, then it would not be *water*? In answer to this question, I would remark only that these supposed intuitions need to be interpreted in the light of the fact, just mentioned, that the natural laws governing chemical compounds in our universe almost certainly render such scenarios physically impossible. The supposedly 'watery' stuff on Twin Earth would be like fool's gold (copper pyrites): it would at best be *casually* mistakable for water and *that* is why it would not be water. The *chemical* explanation for this would be that fool's water, as we could justly call it, is not composed of H_2O molecules. But we cannot turn this perfectly legitimate chemical explanation into a logico-cum-metaphysical argument that genuine water is of *metaphysical* necessity composed of H_2O molecules—unless, once again, we conflate physical with metaphysical necessity.

6. Essence as the Ground of All Modal Truth

I have urged that the following two principles must be endorsed by the serious essentialist: that *essences are not entities* and that, in general, *essence precedes existence*. But by far the most important principle to recognize concerning essences, for the purposes of the present chapter, is that *essences are the ground of all metaphysical necessity and possibility* (compare Fine 1994). One reason it can be the case that X is *necessarily* F is that it is part of the essence of X that X is F . For example, any material object is necessarily spatially extended because it is part of the essence of a material object that it is spatially extended—in other words, part of *what it is to be a material object* is to be something spatially extended.

This is not the only possible reason something could be necessarily F . X might be necessarily F on account of the essence of *something else* to which X is suitably related. Socrates is necessarily the subject of the following event—the *death of Socrates*—because it is part of the essence of that event that Socrates is its subject, even though it is not part of Socrates's essence that he is the subject of that event. It is not on account of *what Socrates is* that he is necessarily the subject of that event but, rather, on account of *what that event is*.⁸ This is not to say that Socrates could not have died a different death,

⁸ Analogously it could be conceded that H_2O molecules necessarily compose water without its being conceded that it is part of the essence of *water* to be composed of H_2O molecules—for the necessity may be explained instead as arising from the essence of H_2O molecules.

only that no one but Socrates could have died the death that he in fact died. What goes for necessity goes likewise, *mutatis mutandis*, for possibility.

I venture to affirm that all facts about what is necessary or possible, in the metaphysical sense, are grounded in facts concerning the essences of things—not only of existing things, but also of non-existing things. But, I repeat, facts concerning the essences of things are not facts concerning *entities* of a special kind, they are just facts concerning *what things are*—their very beings or identities. And these are facts we can therefore grasp simply in virtue of understanding what things are, which we must in at least some cases be able to do, on pain of being incapable of thought altogether. Consequently, all knowledge of metaphysical necessity and possibility is ultimately a product of the understanding, not of any sort of quasi-perceptual acquaintance, much less of ordinary empirical observation.

How, for example, do we know that two distinct things of suitably different kinds, such as a bronze statue and the lump of bronze composing it at any given time, can—unlike two *planets*—exist in the same place at the same time? Certainly not by *looking very hard* at what there is in that place at that time. Just by looking, we shall not see that two distinct things occupy that place. We know this, rather, because we know *what a bronze statue is* and *what a lump of bronze is*. We thereby know that these are *different* things and that a thing of the first sort must, at any given time, be composed by a thing of the second sort, since it is part of the essence of a bronze statue to be composed of bronze. We know that they are different things because, in knowing what they are, we know their *identity conditions*, and thereby know that one of them can persist through changes through which the other cannot persist—that, for instance, a lump of bronze can persist through a radical change in its shape whereas a bronze statue cannot. These facts about their identity conditions are not matters we could discover purely empirically, by examining bronze statues and lumps of bronze very closely, as we might in order to discover whether, say, they conduct electricity or dissolve in sulphuric acid (see Lowe 2002, a response to Olson 2001, and Lowe 2003). Rather, they are facts about them we must grasp *antecedently* to being able to embark upon any such empirical inquiry concerning them. We can only inquire empirically into something's properties if we already know *what it is* we are examining.

7. The Errors of Conceptualism

At this point I need to counter a rival view of essence that is attractive to many philosophers but is, I think, ultimately incoherent. I shall call this view *conceptualism*.⁹ It is

⁹ Who, it might be asked, is really a conceptualist in the sense that I am about to articulate? That is difficult to say with any assurance, since most conceptualists are understandably coy about proclaiming their position too explicitly. However, amongst major analytic philosophers of the twentieth century, Michael Dummett very plausibly counts as one, in virtue of his apparent endorsement of the view that reality is an 'amorphous lump' that can be 'sliced up' in indefinitely many different but equally legitimate ways, depending on what conceptual scheme we or other thinkers happen to deploy (see Dummett 1981, 563 and 577).

the view that facts about essences are really, in the end, just facts about certain of *our concepts*—for example, *our concept* of a bronze statue and *our concept* of a lump of bronze. This would reduce modal truths to conceptual truths or, if the old-fashioned term is preferred, *analytic* truths.

I have no objection to the notion of conceptual truth as such. Perhaps, as is often alleged, ‘Bachelors are unmarried’ indeed expresses such a truth. Let us concede that it is true in virtue of our concept of a bachelor, or in virtue of what we take the word ‘bachelor’ to mean. But notice that ‘Bachelors are unmarried’ has a quite different modal status from an *essential* truth such as ‘Statues are composed of matter’. In calling the former a ‘necessary’ truth, we cannot mean to imply that bachelors *cannot marry*, only that they cannot marry *and go on rightly being called ‘bachelors’*. The impossibility in question is only one concerning the proper application of a word. In calling ‘Statues are composed of matter’ a necessary truth, however, we can’t be taken to mean merely that statues cannot fail to be composed of matter *and go on rightly being called ‘statues’*—as though *the very same thing* that, when composed of matter, was properly called a ‘statue’ might exist as something immaterial. No, we must be taken to mean that statues cannot fail to be composed of matter *period*. Statues are things such that, *if they exist at all*, must be composed of matter. That is because it is part of the essence of a statue to be so composed. In contrast, it is not part of the essence of any bachelor to be unmarried, for a bachelor is just an adult male human being who happens to be unmarried, and any such human being undoubtedly *can* marry.

So, ‘Statues are composed of matter’ is certainly not a mere conceptual truth, and the same goes for other truths that are genuinely essential truths—truths concerning the essences of things. They have, in general, nothing to do with our concepts or our words, but with the nature of the things in question. Of course, since concepts and words are themselves things of certain sorts, there can be truths concerning *their* essences. Indeed, what we could say about ‘Bachelors are unmarried’ is that it is, or is grounded in, a truth concerning the essence of the concept bachelor, or of the word ‘bachelor’. We could say that it is part of the essence of the concept *bachelor* that only unmarried males fall under it, and part of the essence of the word ‘bachelor’ that it applies only to unmarried males.

I said that conceptualism is ultimately *incoherent*. For one thing, as we have just seen, the proper thing to say about ‘conceptual’ truths is, very plausibly, that *they are grounded in the essences of concepts*. That being so, the conceptualist cannot maintain that *all* putative facts about essence are really just facts concerning concepts. For this is to imply that putative facts about *the essences of concepts* are really just facts concerning *concepts of concepts*—and we have set out on a vicious infinite regress.

So might David Wiggins, who calls his position ‘conceptualist realism’ and acknowledges, as the only admissible notion of individuation, a cognitive one that takes this to be a *singling out of objects by thinkers* (see Wiggins 2001, 6). And so might Hilary Putnam, on the evidence of such papers as ‘Why There Isn’t a Ready-Made World’ (Putnam 1983), whose flavour seems distinctly different from that of earlier work of his cited previously.

Conceptualists will object, no doubt, that this complaint is question-begging. However, even setting that complaint aside, we can see that conceptualism is untenable. Conceptualists are at least committed to affirming that *concepts*—or, in another version, *words*—exist and indeed that concept-users do, to wit, *ourselves*. *These*, at least, are things conceptualists must acknowledge to have *identities*, independently of how we conceive of them, on pain of incoherence in his position. The conceptualist must at least purport to understand *what a concept or a word is*, and indeed *what he or she is*, and thus grasp the essences of at least some things. And if of *these* things, why not of *other* kinds of things? Once knowledge of essences is conceded, the game is up for the conceptualist. It must be conceded, even by conceptualists, on pain of denying that they know what *anything* is, including the very concepts that lie at the heart of their account. For recall, all that I mean by the essence of something is *what it is*.

Why is anyone ever tempted by conceptualism? I'm afraid that it is the legacy of scepticism, particularly scepticism concerning 'the external world'. Sceptics feel at home with themselves and with their words and concepts, but express doubt that we can ever really know whether those words and concepts properly or adequately characterize things in the external world. The sceptic thinks that we can know nothing about how or what those things are 'in themselves', or indeed even whether they are *many* or *one*. According to the sceptic, all that we can really know is how we *conceive* of the world, or *describe* it in language, not how *it is*. But by what special dispensation does the sceptic exclude *our concepts* and *our words* from the scope of his doubt? For are they not, too, things that exist? There is, in truth, no intelligible division that can be drawn between *the external world*, on the one hand, and *us and our concepts and our language* on the other.

Here it may be protested: but how, then, *can* we advance to a knowledge of what and how things are 'in themselves', even granted that the sceptic is mistaken in claiming a special dispensation with regard to the epistemic status of our concepts and our words? However, the fundamental mistake is to suppose, with the sceptic, that such an 'advance' would have to proceed *from* a basis in our knowledge of our concepts and words—*from* a knowledge of how we conceive of and describe the world—to a knowledge of that world 'as it is in itself', independently of our conceptual schemes and languages. This 'inside-out' account of how knowledge of mind-independent reality is to be acquired already makes such knowledge impossible and must therefore be rejected as incoherent.

What alternative is there, barring a retreat to some form of anti-realism? Again, *knowledge of essence* comes to the rescue. Because, in general, essence precedes existence, we can at least sometimes know *what it is* to be a *K*—for example, *what it is* to be a material object of a certain kind—and thereby know, at least in part, what is or is not *possible* with regard to *Ks*, in advance of knowing whether, or even having good reason to believe that, any such thing as a *K* actually *exists*. Knowing already, however, *what it is* whose existence is in question and that its existence is at least *possible*, we can intelligibly and justifiably appeal to empirical evidence to confirm or cast doubt upon existence claims concerning such things.

By 'empirical evidence' here I emphatically do *not* mean evidence constituted purely by the contents of our own perceptual states at any given time, as though all that we had to go on is how the world in our vicinity *looks* or otherwise *appears* to be. That, certainly, is not the conception of 'empirical evidence' operative in scientific practice, which appeals rather to the results of controlled experiments and observations, all of which are reported in terms of properties and relations of mind-independent objects, such as scientific instruments and laboratory specimens.

The growth of objective knowledge consists, then, in a constant interplay between an *a priori* element—knowledge of essence—and an *a posteriori* element, the empirical testing of existential hypotheses whose possibility has already been anticipated *a priori*. This process does not have a foundational 'starting point' and it is constantly subject to critical reappraisal, both with regard to its *a priori* ingredients and with regard to its empirical contributions.

Here we do not have a hopeless 'inside-out' account of objective knowledge, because our own subjective states as objective inquirers—our perceptions and our conceptions—are accorded no special role in the genesis of such knowledge. Those subjective states are merely some amongst many possible objects of knowledge, rather than objects of a special kind of knowledge that supposedly grounds knowledge of all other things. But, to repeat, it is crucial to this account that knowledge of *essences* is not itself knowledge of *objects or entities* of any kind, nor is it grounded in any such knowledge—such as knowledge of our own concepts.

8. The Redundancy of Possible Worlds

I want to conclude by looking at the language of possible worlds and its bearing upon the nature and ground of metaphysical modality. I have already made it clear that, in my opinion, all modal facts concerning what is metaphysically necessary or possible are ultimately grounded in the essences of things—hence *not* in facts concerning entities of any sort, because *essences are not entities*. But—it may perhaps be urged—this in itself does not necessarily prevent the language of possible worlds from casting at least some light on the nature and ground of metaphysical modality. Well, let us see.

First, consider non-fictionalist construals of the language of possible worlds, according to which, possible-worlds variables in that language range over a domain of *existing entities* of some kind, such as Lewisian 'parallel universes' or maximal consistent sets of propositions—the former conceived as being *concrete* and the latter as *abstract* entities of certain kinds.¹⁰ According to possible-worlds theorists adopting this approach, any modal statement in which the modal terminology involved is

¹⁰ For Lewis's approach, see Lewis 1986; the use of the expression 'parallel universes' to describe possible worlds as he conceives of them is mine rather than his. For the view that possible worlds are maximal consistent sets of propositions, see Adams 1974, and for a similar view that takes them to be maximal possible states of affairs—again conceived as being abstract entities—see Plantinga 1974.

expressive of metaphysical modality is *semantically equivalent* to one quantifying over existing entities of the favoured kind—as it might be, parallel universes or maximal consistent sets of propositions. Moreover, according to such an approach, the truth or falsehood of the modal statement in question is grounded in facts concerning those entities. For example, the truth or falsehood of the statement ‘Possibly, there are talking donkeys’ is, supposedly, grounded in facts concerning the inhabitants of certain parallel universes or facts concerning the membership of certain maximal consistent sets of propositions. But, I suggest, it should strike one as being obviously problematic to suppose that—where the metaphysical modalities are concerned—modal facts are grounded in facts concerning existing entities of *any* kind.

The salient point, again, is that *essence precedes existence*. An *existing* entity must at the very least be a *possible* entity—something whose essence does not preclude its existence. And what is *true* of an entity will likewise depend at least in part on *what it is*—its essence. It can only be the case, for example, that some parallel universe *does* in fact contain amongst its inhabitants such a thing as a talking donkey if there *could be* such things as parallel universes and such things as talking donkeys inhabiting them. The very facts that are being proposed as the grounds of modal truths already *presuppose* modal truths, simply because they are, supposedly, facts concerning *existing entities* of certain putative kinds.

The upshot is this. Suppose we grant that there could be such things as Lewisian parallel universes or maximal consistent sets of propositions because, understanding *what these entities are*—knowing their essences—we know that their essences do not preclude their existence. Let us go further and suppose that such things *do in fact exist*. *Even so*, facts concerning such entities could not constitute *the ground of all modal truths*. Why not? Because, first and foremost, such facts could not constitute the ground of modal truths concerning *those entities themselves*. If these entities exist, then there must indeed be modal truths concerning them, since there are modal truths concerning *any* existing entity. So, for example, if parallel universes exist, it must either be true, concerning them, that infinitely many of them could exist, or else be true, concerning them, that only finitely many of them could exist. Similarly, it must either be true, concerning them, that two or more of them could be qualitatively indiscernible, or else be true, concerning them, that any two of them must be qualitatively distinct. And so on.

Quite evidently, *concretists*—as we may call devotees of Lewis’s approach—cannot contend that, for example, ‘Possibly, there are infinitely many possible worlds’ is true or false for the same sort of reason that they contend that ‘Possibly, there are infinitely many electrons’ is true or false. For the latter is true, he maintains, just in case there is a possible world—a parallel universe—in which there are infinitely many electrons (or electron ‘counterparts’). But they cannot maintain that the former is true just in case there is a possible world in which there are infinitely many possible worlds. For, knowing what a ‘possible world’ is supposed to be according to concretists—to wit, a ‘parallel universe’, akin to our cosmos—we know already that it is not the sort of thing that

could have *another* such thing amongst its inhabitants, let alone infinitely many other such things.

Far from its being the case that facts concerning possible worlds—*whatever* these are conceived to be—are the ground of all modal facts, there must be modal facts not grounded in the existence of entities of *any* kind, including possible worlds. And if this must be so for *some* modal facts, why not for *all*, as serious essentialism contends?

Abstractionists—as we may call devotees of possible worlds conceived as maximal consistent sets of propositions—might protest at this point that they, at least, never intended to suggest that modal truths could be reduced, without remainder, to *non-modal* truths concerning possible worlds and that this exempts them from the foregoing strictures. Abstractionists openly acknowledge, for example, that they appeal to an unreduced notion of *consistency* in explaining what they take a ‘possible world’ to *be*—to wit, a maximal consistent set of propositions, or something like that. This might be an acceptable response if the *only* modal notion being relied upon by the abstractionist was that of consistency—the notion, that is, of the *possible joint truth* of two or more propositions. But my complaint does not focus on this well-known feature of abstractionism and its consequent repudiation of any aspiration to offer a reductive account of modality. Rather, my complaint focuses on the fact that abstractionism, just like concretism, appeals to *existing entities* of certain putative kinds in presenting its account of the semantics of modal statements. In this case, the entities in question are abstract objects such as *propositions* and *sets* thereof. But propositions and sets, if they exist, are just *further entities*, concerning which various modal truths must hold.

For example, it must either be true, concerning sets, that they could have contained different members, or else it must be true, concerning sets, that they could not have contained different members. Suppose it is true. Suppose, that is, that the following modal statement is true, where *S* is any given set whose actual members are certain objects: ‘Possibly, *S* has members that are different from its actual members.’ What is *this* supposed to mean, according to the abstractionist? Clearly, something like this: ‘Some maximal consistent set of propositions contains the proposition that *S* has members that are different from its actual members.’ But *S* was supposed to be *any set we like*. So what happens if we try to let *S* be the maximal consistent set of propositions whose actual members are all and only the propositions that are *actually* true—in other words, if we try to let *S* be the maximal consistent set of propositions that the abstractionist identifies as *the actual world*, W_α ? In that case, the abstractionist translates the putative modal truth ‘Possibly, W_α has members that are different from its actual members’ as meaning ‘Some maximal consistent set of propositions contains the proposition that W_α has members that are different from its actual members’—or, in the language of possible worlds, ‘In some possible world, the actual world is different from how it actually is.’ But it is very hard to see how the abstractionist could allow *this* to be true.

The implication is that abstractionists’ semantics for modal statements compels them to deny, after all, that any set whatever could have contained different members. Now, I am not quarrelling with that verdict as such, since I consider that it is part of the essence of any set that it has the members that it does—that *their* identities determine

its identity. However, it is plainly not a verdict that should be forced upon us merely by the machinery that we invoke to articulate the semantics of modal statements: rather, it is one that should emerge from an adequate understanding of what sets *are*—an understanding that carries modal implications and one abstractionists themselves must possess *prior* to constructing their preferred machinery for modal semantics.

That abstractionism runs into this and similar problems is just a symptom of the fact that abstractionism, like other possible-worlds accounts of metaphysical modality, has simply *mislocated* the meaning and grounds of modal truths, by trying to find them in facts concerning a special class of entities of an esoteric kind—in this case, maximal consistent sets of propositions.

What, finally, of *fictionalism* (as in Rosen 1990)? Fictionalism can be dismissed without more ado, I think, because in seeking to reap the advantages of theft over honest toil, it relies on the toil in question at least being *effective*. If the toil was wasted effort, no advantages can be got from it. But we have seen that both concretism and abstractionism fail on their own terms, whence there is no profit to be had in a theory that rests on a *pretence* that either of them is true. This would be like stealing the harvest of a farmer whose crops had failed.

I conclude that the language of possible worlds, whether or not it is interpreted in an ontologically serious manner and *whatever* possible worlds are taken to be, can throw no real light at all on the nature and ground of metaphysical modality. If possible worlds, whatever they are taken to be, exist at all, that is a fact that may hold some interest for the ontologist—who is, after all, concerned to provide as full and accurate an inventory of what there is as is humanly possible—but it is not one that can usefully be recruited for the purposes of modal metaphysics. For that, I suggest, we have no viable option but to turn to *serious essentialism*.

9. References

- Adams, R. M. 1974. 'Theories of Actuality'. *Noûs* 8: 211–31.
- Descartes, R. 1641/1986. *Meditations on First Philosophy*. Tr. J. Cottingham. Cambridge: Cambridge University Press.
- Dummett, M. 1981. *Frege: Philosophy of Language*, 2nd edn. London: Duckworth.
- Fine, K. 1994. 'Essence and Modality'. In Tomberlin 1994: 1–16.
- Jackson, F. C. 1998. *From Metaphysics to Ethics: A Defence of Conceptual Analysis*. Oxford: Clarendon Press.
- Kripke, S. A. 1971. 'Identity and Necessity'. In M. K. Munitz, ed. *Identity and Individuation*. New York: New York University, 135–64.
- Kripke, S. A. 1980. *Naming and Necessity*. Cambridge, MA: Harvard University Press.
- Lewis, D. K. 1986. *On the Plurality of Worlds*. Oxford: Basil Blackwell.
- Lewis, D. K. 1991. *Parts of Classes*. Oxford: Basil Blackwell.
- Locke, J. 1690/1978. *An Essay Concerning Human Understanding*. Ed. P. H. Nidditch. Oxford: Clarendon Press.
- Lowe, E. J. 1998. *The Possibility of Metaphysics: Substance, Identity, and Time*. Oxford: Clarendon Press.

- Lowe, E. J. 2002. 'Material Coincidence and the Cinematographic Fallacy: A Response to Olson.' *The Philosophical Quarterly* 52: 369–72.
- Lowe, E. J. 2003. 'Substantial Change and Spatiotemporal Coincidence.' *Ratio* 16: 140–60.
- Lowe, E. J. 2005. 'Identity, Vagueness, and Modality.' In J. L. Bermúdez, ed. *Thought, Reference, and Experience: Themes from the Philosophy of Gareth Evans*. Oxford: Oxford University Press, 290–310.
- Lowe, E. J. 2006. *The Four-Category Ontology: A Metaphysical Foundation for Natural Science*. Oxford: Clarendon Press.
- Olson, E. T. 2001. 'Material Coincidence and the Indiscernibility Problem.' *The Philosophical Quarterly* 51: 337–55.
- Plantinga, A. C. 1974. *The Nature of Necessity*. Oxford: Clarendon Press.
- Putnam, H. 1975. 'The Meaning of "Meaning".' In K. Gunderson, ed. *Language, Mind, and Knowledge (Minnesota Studies in the Philosophy of Science vol. 7)*. Minneapolis: University of Minnesota Press, 131–93. Reprinted in H. Putnam, *Mind, Language and Reality (Philosophical Papers vol. 2)*. Cambridge: Cambridge University Press, 215–71.
- Putnam, H. 1983. 'Why There Isn't a Ready-Made World.' In *Realism and Reason (Philosophical Papers vol 3)*. Cambridge: Cambridge University Press: 205–29.
- Quine, W. V. O. 1969. 'Existence and Quantification.' In *Ontological Relativity and Other Essays*. New York: Columbia University Press: 91–113.
- Rosen, G. 1990. 'Modal Fictionalism.' *Mind* 99: 327–54.
- Tahkko, T. and Lowe, E. J. 2016. 'Ontological Dependence.' *The Stanford Encyclopedia of Philosophy*. Ed. E. N. Zalta, <http://plato.stanford.edu>.
- Tomberlin, J. E., ed. 1994. *Logic and Language (Philosophical Perspectives vol. 8)*. Atascadero, CA: Ridgeview Press.
- Wiggins, D. 1983. *Sameness and Substance Renewed*. Cambridge: Cambridge University Press.

PART I

Categorical Ontology

2

Lowe, the Primacy of Metaphysics, and the Basis of Categorical Distinctions

Peter Simons

One can be a realist concerning the distinctions that are captured by a system of ontological categories without having to maintain that the categories themselves are elements of being.¹

The Four-Category Ontology, p. 43

1. The Primacy of Metaphysics

If you really want to know what you are *talking* about, you need to know *what* you are talking about. This seeming truism can be given a logico-epistemological slant, as the maxim ‘Define your terms!’ As such, it is clearly doomed to fail, since not everything can be defined. Some terms cannot be given definitions: but we can try to reach an understanding with others, what Bolzano calls a *Verständigung* (Bolzano 1837, § 668), about the meaning of the term in question, with all the pitfalls that ensue, and aspire down the line to capture prime aspects of that meaning in accepted principles or meaning postulates restricting and governing the term.

The slant I give to the truism is however ontological, one of which Jonathan strongly approved, namely the Aristotelian view that discourse which is not anchored in a proper ontology is not fully serious, and that metaphysics therefore enjoys a primacy among the sciences which is neither linguistic nor epistemological, but reflects a

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¹ I believe it was at the 2012 *Gesellschaft für analytische Philosophie* conference in Constance that I first heard Jonathan put the view forward. He was the first philosopher that I found to have independently reached the conclusion at which I had also arrived. Being both of us British, male and of a certain age, we did not hug after his talk, but we were both pleased. My own first published foray into the theme is Simons 1999, but, as made clear there, it originated outside academia in software design discussions at Ontek Corporation in the 1990s.

fundamentally realistic view. According to this, the world of things comes self-differentiated into different fundamental kinds, and these things combine and interact in their own ways, mostly without input from ourselves.

The critical turn in philosophy, starting with Descartes and Locke, and leading to the metaphysical deflationism of Hume, pushed metaphysics away from its regal position in philosophy. Kant strove to rescue metaphysics from Hume's criticisms by confining it to the sphere of our cognition, leaving mind-independent reality outside the pale. History repeated itself over a century later when Quine, Strawson, and Dummett strove to rehabilitate metaphysics after the positivist onslaught by making it an adjunct to semantics. The result was in each case the same: to relativize what exists to our linguistic or conceptual scheme, and so to approach that most evil of philosophical positions: idealism.

There is no such thing as a safe metaphysics or an innocent metaphysics, or an easy *a priori* metaphysics which can be established by armchair methods, or by reflection on pure reason alone. It makes bold, speculative, and shaky conjectures. Serious metaphysics risks refutation by empirical discovery; it courts inconsistency and incoherence; it is subject to historical, cultural, and personal biases; and it is bound to be perpetually provisional. Saving a beatific vision, this is both what we must expect and that to which we must be reconciled. Yet metaphysics is inescapably a philosopher's business, since it is universal: as Aristotle says, all the special sciences cut off a portion of being. When philosophers ignore metaphysics, others move in and do their job for them, usually badly. Jonathan's insistence on the primacy of metaphysics and its relative independence of the philosophy of language completes the turn away from the linguistic turn. The extent, interconnectedness, and quality of Jonathan's writings on metaphysics makes him the most influential British metaphysician since Strawson, and the most substantial since Whitehead.

2. The Tasks of Metaphysics

The scope and tasks of metaphysics were conceived and outlined by Aristotle, continued by medieval scholastics, and elaborated by Wolff, Husserl, Whitehead, and D. C. Williams. They consist in two parts. One part consists in the careful elaboration of a scheme of fundamental kinds or categories, their justification, connection, and governing principles or *archai*: an *ontologia sive metaphysica generalis* including a *Kategorienlehre*. Husserl called this *formal ontology*; Williams called it *analytic ontology*. The other part is a broader enterprise of showing how these categories and principles apply to a wide range of things in the world, what Williams called *speculative cosmology* and Husserl *regional ontology*, but which I prefer to call *systematics*. Without a view to its application, ontology remains a detached glass-bead game, but without a view to ontology, systematics remains at best a haphazard congeries of disconnected insights. Both sides of metaphysics require and reinforce one another. Metaphysical problems may up to a point be tackled piecemeal, but metaphysics as a whole must aspire to be

systematic. That requirement is part of what makes metaphysics difficult to do well, something that Jonathan rightly emphasized in the face of those easy critics who scorn metaphysics as little more than hot air or think it is something a ‘real’ scientist could knock out on a Sunday afternoon. Not many philosophers in recent times have succeeded in being truly systematic, but Jonathan is one of them.

Williams’s adjective ‘speculative’, emphasized also by Whitehead, reminds us of the unavoidably conjectural and fallible nature of metaphysics. We cannot avoid going beyond the evidence in postulating the applicability of metaphysical categories and principles, and because of their highly abstract nature, correction through experience is indirect, haphazard, and slow. Several of my own speculative conjectures are diametrically opposed to Jonathan’s: I subscribe to naturalism, anti-dualism, and nominalism, and the primacy of processes over continuants. That will in no way inhibit my ability to judge our convergence on other, perhaps deeper, matters.

3. Categories

The principal task of ontology is the provision of a *Kategorienlehre*. As long as analytic philosophy was beholden to the forms of predicate logic and its semantic partner set theory, the notion of category and the theory of categories languished. The assumption, whether implicit (in most cases) or explicit (in the case of Wittgenstein), of a thoroughgoing harmony or categorial and structural match between the forms of language and the forms of objects, left no room for consideration whether a system of classification of objects, including in its most general levels a system of categories, was anything other than something one could read off logical semantics, or in the case of ordinary-language philosophy, the meaning and use of vernacular expressions. Along with a select handful of others, including John Anderson, Roderick Chisholm, David Armstrong, and Reinhardt Grossmann, Jonathan recognized that in the absence of a methodological handrail from logic and linguistics, ontology needed to be more specific and deliberate about its selection, articulation, and justification of categories. In Jonathan’s case, this took the form of reverting to a well-tried model, that of Aristotle, in his four-category ontology (Lowe 2006), a reversion which was completed once Jonathan recognized the inescapable need for tropes, or modes, as following his favourite Locke he preferred to call them. The need for modes alongside properties and relations emerges when one asks the Aristotelian question *why* certain things are true. If it is true that Caius is seated, why is it true? Aristotle said: because Caius is seated. And in what does this consist? Not in Caius alone, since he was not and will not be and might not now have been seated. His being seated consists in his body being configured in a certain bent way and his backside resting on a support to secure his body against a force such as gravity. This *posture*—one of Aristotle’s minor categories—is peculiar to Caius and will cease to be when he stands up, falls, or kneels down, etc. It is, in other words, a mode, dependent on Caius, and in virtue of the temporary existence of this mode of Caius, the statement is true.

Recognizing the role of modes as truthmakers, Jonathan steers clear of the siren call of states of affairs, the logical atomists' (and others') choice of truthmaker. I consider he was right to do so.

4. Ontic Categories

In Kant's view, the categories (taken together with the forms of intuition) cannot be taken to structure the world as it is in itself, but only our experience. From a realist perspective this gives up the race before the starting gun is fired. Aristotle's categories, notwithstanding their role in thought and language, were taken to divide the world into its own fundamental kinds.

One lesson of logical atomism, especially in its strict Wittgensteinian version, is that there are concepts which are indispensable to us but which have no worldly representative, denotatum or counterpart. Wittgenstein's example in the *Tractatus* was the logical constants: negation, disjunction, quantification, etc. His fundamental thought (contra Frege) was that they do not represent (Wittgenstein 1921, 4.0312). He was half-heartedly followed by Russell, who did find worldly counterparts to negation and quantification in negative and general facts. The need for logical constants stems from our cognitive inadequacy: God does not need them. They do not structure reality, any more than does the indispensable ontological concept of existence, which stands for no quality and fails to divide entities. Call such non-representative concepts *Kantian* or *auxiliary*, reserving the terms *Aristotelian* and *ontic* for those concepts which in fact or in intent discern the kinds of things. Ontic categories are then those concepts of the highest level of generality which divide reality at its most fundamental joints. In Aristotle, there are eight or ten of them, in Williams just one, in Armstrong there are three, in Whitehead eight, and in Jonathan, four. The world divides, according to Lowe, into substances, kinds, modes, and properties, a division which traces its pedigree back to the second chapter of Aristotle's *Categories*. It is with ontic categories alone that I shall be further concerned.

5. Factors: Grounds of Categorical Distinctions

Jonathan's four categories do not simply sit alongside one another: they are related. Objects *instantiate* substantial kinds and modes *instantiate* modal kinds or attributes.² Modes *inhere in* or *characterize* substances. As a result, objects *exemplify* attributes, which comes about when a mode which instantiates an attribute characterizes an object. The colour-mode of this piece of paper characterizes the paper and instantiates whiteness, so the paper exemplifies whiteness. Instances are rigidly ontologically dependent on their respective universals; conversely, universals are only weakly dependent on their

² Properties and relations: latterly Jonathan came to the view, to which I am sympathetic, that relations as entities are not needed.

instances. Substantial kinds are identity-dependent on the properties that constitute that kind's essence. So the four corners of the ontological square are uniquely determined by the patterns of dependence among the members of the categories. The distinctness, and the distinctive nature of, each of the categories, more exactly, members of the categories, as belonging to this category rather than that, as well as their distinctive interrelations, need to be elucidated, or grounded.

At the beginning of chapter 3 of *The Four-Category Ontology*, Jonathan lists the following relations as contributing to this grounding; namely, *identity*, *instantiation*, *characterization*, *exemplification*, *constitution*, *composition*, and *dependence*. The last is not considered basic by him, and we have seen that exemplification is definable in terms of instantiation, characterization, and some logic. Whether one agrees with this precise list or not (I do not), the structural point is this. These relations are not themselves additional entities alongside what they relate, but relate their terms internally.

Talk of internal relations is apt to mislead: if two things stand in an internal relation, however that is defined, is that relation not something rather than nothing? Were it nothing, would they then not stand so to one another? To avoid such double-talk, I prefer to talk about *why* they so stand. Things stand in an internally related way to one another if their so standing is necessitated by their existing at all. Making a semantic ascent, we can say: the truthmaker for the proposition that they so stand is just the things themselves. No additional entity apart from the related things is required for the proposition to be true. Given the battles of Trafalgar and Waterloo, it cannot fail to be the case that Trafalgar occurred earlier than Waterloo. We need neither a relational mode of temporal precedence nor a state of affairs of Trafalgar's preceding Waterloo, nor any other thing, for this to be true. They are internally so related that Trafalgar precedes Waterloo.

Contrapositively, were the things not so related, at least one of them would not exist. Internal relations are the relational counterparts of essential properties. Where something is an essential property of a thing, were that property not to be exemplified by the thing, the thing would not exist; contrapositively, the existence of the thing necessitates that it have the property.

Whatever it is that differentiates the categories from one another and accounts for the interrelations, to have a name, I call *formal factors*. Like Jonathan, I consider that any serious metaphysics must employ them. But they are not entities in the ontology, and for that reason one might wish to avoid them. There are two ways to avoid being committed to formal factors. One is to subscribe to a single-category ontology, such as a reistic ontology of substances only, or a tropes-only ontology. Even then one will have need of something to ultimately account for how the various members of the one category stand to one another. That will bring us back to factors eventually. Or one could be a monist, holding there is only one entity, which may be consistent, but is certainly false. Alternatively, one could take the view that the categories are brutally or surdly different. I call this view *surdism*. Again this may be consistent, but it is highly unattractive. It also runs counter to the history of metaphysics, in which

such distinctions as those between particular and universal, one and many, substance and accident, property and relation, have turned out to be eminently discussable, even if not exactly definable. I conclude that every ontology, if carried through properly, will need to invoke some factors or other.

Not all metaphysicians have been explicit about their factors, but the more reflective ones have. Aristotle's ontological square, the inspiration for Jonathan's four categories, is structured by the two relational factors of *being said of* and *being in*. Aristotle's own eight or ten categories, which at first sight look 'piecemeal', 'haphazard', and merely 'picked up as they came his way', as Kant said, or simply derived from grammatical features of Greek, as Trendelenburg and Benveniste argued, have a more subtle justification which can be spelled out in terms of factors. This was argued in a doctoral dissertation of 1862 called *On the Several Senses of Being in Aristotle*, by the young Franz Brentano (Brentano 1862). Factors include being prior, being towards another, being inherent, being a movement, being contained. These phrases used by Aristotle certainly do not designate entities in any of the categories: they are factorial. The most explicitly and self-consciously factorial ontology of recent times is that of Roman Ingarden, who in his *Controversy over the Existence of the World* (Ingarden 1947–8) constructs his categories, which he calls 'modes of being', in terms of a handful of families of factors, which he calls 'existential moments'. It is perhaps no coincidence that Brentano and Ingarden were regarded by their respective contemporaries as 'scholastic', an epithet I also heard used about Jonathan, and while I would not myself want to be so called, it is not a label to which I personally attach any pejorative force.

6. Bedrock Regresses

Not all internal relations are at the level of instantiation or characterization. We need another way to separate the formal from the material, and to give us a clue that we are at or near fundamentality. A very useful diagnostic symptom is the presence or threat of an infinite regress. Fundamental ontology has left a trail of regresses behind it. Plato's regress of instantiation, known since Aristotle as the Third Man, but in Plato's own example from the *Parmenides* the less Viennese-sounding Third Large, is the most famous. A second is the Bradley Regress, intended to support Bradley's Spinozistic monism, but less extremely pointing out a relative of the instantiation regress, when relations are considered, and particularly acute when applied to the question of what unifies a state of affairs or fact in those ontologies which accept such things. Another regress discussed in the context of the debate about universals is Russell's regress of resemblance, raised in chapter 9 of *The Problems of Philosophy* (Russell 1912), and an alleged thorn in the side of nominalists. Russell's argument was stated some years earlier in §§ 3–4 of the second of Husserl's *Logical Investigations* (Husserl 1900–01), which like Russell endorses universals. Husserl had a good nose for regresses. He finds another one in Twardowski and discusses it in § 22 of the third Investigation. The regress-engendering thesis says that for a complex object (one consisting of several parts) to be unified as one object, there must be, besides the several parts, an additional part, a

moment of unity, which unifies the whole. Husserl rightly points out that this leads to an infinite regress of parts. We might call it the *Composition Regress*. Another regress argument applied in connection with a basic feature of reality occurs in McTaggart's attempt to show that time is unreal.

Not all regresses and regress arguments are ontological in nature: some concern logic or epistemology. But a good few are clearly ontological. Such regresses come into view whenever ontologists try to cheat by postulating a 'special' kind of relation or object. Strawson's non-relational tie of exemplification and Bergmann's nexus of instantiation, both deployed to block Bradley's Regress, are prominent examples of such cheats. Regresses rear up when we are getting close to fundamentals, so let us call them *bedrock regresses*.

Bedrock regresses readily come to mind in regard to distinguishing and relating categories. We already met regresses of instantiation and composition, concerning two of Jonathan's formal relations. We can multiply them. Here is a regress suggested by ideas of Meinong. Existence is pretty fundamental. Suppose an object exists *because* the state of affairs of its existing (what Meinong calls a *Seinsobjektiv*) exists. Then that state of affairs exists only because a second-order state of affairs saying that it exists itself exists. And so on. Meinong himself does not deny that there are infinitely many *Seinsobjektive* based on a single existing object, but he sensibly denies that the object's existing *consists in* or is explained by its personal *Seinsobjektiv*'s existing. There is an infinity, but not a vicious one. Here is a mereological regress. Let A be part of B. Suppose, as a nominalist, I say this is true because of the existence of a parthood-trope linking A to B. This would appear, qua individual, to be housed wholly within B, as another trope of B. So we have an infinity of parthood tropes, which, if tropes are parts, is a variant of Twardowski's Compositional Regress. Here is a locational regress. Let object O be located at L. Where is its being so located? What about identity? Is the identity of A with itself something distinct from A? (Meinong and Russell would say: yes!) Finally, here is a causal regress. If E causes F, what is the cause of E's causing F?

I am not saying all of these regresses are equally serious, but they all arise because we are approaching bedrock, the bedrock being signified by two features. One is the purported universality of the feature in question: everything is somewhere, everything exists, everything has a cause, everything is in some way, everything is itself, everything is related to everything else. The other is that because we are at bedrock, we cannot shift the burden of expression and explanation onto something more fundamental: we are there already. We are in the business of trying to explain something in a way which invokes that same something in the explanation.

7. What To Do About Them

The best thing to do about regresses is not to let them get started. Internal relatedness is often a good way to go about this, especially when combined with a judicious use of the notion of truthmaking. This strategy works well in many cases. Take existence. A true proposition that A exists is made true by A itself: nothing answering specifically to

'exists' is required: this is Hume's point. Existence, or A's existence, is not an additional item alongside A. And if the proposition is false, this is because there is nothing answering to 'A' to make it true. We do not need an additional truthmaker, a fact that A does not exist, *das Nichtsein des A*, to do the job. Or take similarity, specifically, the exact similarity of two modes M and N, where say M is the redness of tomato A and N the exactly similar redness of tomato B. For them to be exactly similar, nothing else is required than M and N themselves. Being modes, they have a 'thin' essence, so for them to exist is for them to be exactly as they are, and therefore they resemble one another exactly. The passing and contingent similarity of the tomatoes A and B is then explained by the contingent inherence of M in A and N in B.

To combat the compositional regress a two-stage strategy is required. Firstly, where A is composed of B, C, D, ... this may be for a variety of reasons, some of which concern non-formal relations. An isolated uncharged helium atom has its four nucleons and two electrons as parts, the former essentially, the latter accidentally. While the nucleons do not individually need to be together, for this helium atom to exist, they all do: they are individually essential parts of it. But what keeps them together is a physical force, not a metaphysical principle. The presence of electromagnetic attraction between electrons and nucleus and the proximity of these two electrons to the nucleus account for their being parts of the atom. At the level of the individual electrons, the features of mass, charge, energy, location, and spin are more tightly unified. Assuming these features exhaust the electron, as trope bundle theorists hold, the togetherness of the various mutually requisite tropes comprising the electron turn on direct and individual mutual and one-sided needs which Husserl called *foundation*. The exploration of the varieties of such needs takes up the majority of Husserl's third Logical Investigation. Husserl stresses over and over that nothing else is required than the parts themselves.

The immediate mutual requirement of tropes for one another in what I call the essential *nucleus* or kernel of an individual (Simons 1994), and the immediate but one-sided requirement of one trope for a more substantial bearer, are reasons why I do not agree with Jonathan that dependence is not a basic notion. In these cases, I think it is, and it is significant that Husserl opted to use the word *Fundierung* (foundation) rather than the broader *Unselbständigkeit* (dependence) for the basic case.

Consider another fundamental ontic distinction: that between one and many, more specifically, between one, two, three, and so on. Despite the claims of Frege and Russell, these are not purely logical notions. The reason is obscured by Frege's and Russell's methodological presuppositions as built into their logical languages. For them, every genuine name or denoting expression is singular. They do not countenance plural terms, and while Frege knows natural languages contain empty terms (Russell apparently does not), in science these are banned as they cannot contribute to truth. Modern free logics and plural logics however rightly dispense with these constraints. Therefore we are entitled to know why sentences of forms such as 'A is the same object as B', 'A is a different object from B', 'A is one of the objects B', 'A are the

same objects as B' are true when they are. Jonathan here takes individual identity as fundamental. I disagree with that, for two reasons. One is that identity applies to pluralities (and indeed masses) just as to individuals. We need to define the notion of individuality to narrow the notion down. The other and more crucial reason is that there is an ontological dimension to difference: $A \neq B$ *because* A and B both exist. If either fails to exist, ' $A \neq B$ ' is false. So we should distinguish two distinct identity predicates: one requiring the existence of A when $A = A$, the other not. This means there are at least three identity predicates: existence-entailing, existence-entailing and singular, and not existence-entailing. So contrary to Jonathan I consider the notion 'A is one of B' (existence-entailing in both positions) to be a primitive alongside a purely logical non-discernibility predicate.

The suggested solution to the problem of bedrock regresses then lies in either internal relations, or essential properties, or both, where we are understanding 'property' and 'relation' in an ontologically non-committal, light-touch way. To speak in a paradoxical Meinongian vein: there are ways that things are and ways they stand to one another such that there are no such ways. This rose is red. Its redness inheres in it, and is so related to it, but there is no such thing as inherence, or instances of inherence. This shows that quantifying is a less sure guide to what there is than truthmaking.

It might be thought that an ontology in which bedrock regresses do not threaten is less problematic, better crafted, than one where they do. Not so: it merely shows the ontology in question has not reached bedrock. Another thought is that such things are unsayable: they belong to what can be shown and cannot be said, to think with the author of the *Tractatus*. There are two problems with this. One is that it assumes a pre-established harmony between ways of speaking and ways of being. Wittgenstein was not the first to hold to such a view—it is fully present among the medieval speculative grammarians—but he was perhaps the first to issue such puritanical bans on disrupting the harmony by inappropriate chatter. Secondly—and whether there is such a harmony, or, as I believe, not, there is no reason why further chatter must be nonsense, even when it is not in accord with harmony assumptions. One simply needs to be clear what one is doing.

8. Factors and Categories, Form and Matter

Like Ingarden, I consider that factors come in families, but unlike Ingarden I do not think the families consist in a small finite number of alternatives. Take the mereology family for instance. One could consider objects just from the point of view of whether or not they are mereological atoms; that is, things lacking any proper parts, giving a binary division between atoms and non-atoms. That would be an impoverished approach, for two reasons. Firstly, it is not clear whether the real world actually contains any atoms. If it did not, the distinction would be idle. Secondly and more crucially, the notion of atom is embedded in mereology, which deals with the part-whole relation

and its many cognates. If anything is the basic factor here, it is part-whole, or it and its several interdefinable cognates, and the meaning postulates constraining it are the axioms of mereology, a conservative mereology confining itself to the explication of the terms alone. A mereology which ventures into unverifiable and sometimes outlandish hypotheses about if and or when several things compose another thing (e.g. always, or never) is overstepping that brief and becoming speculative. It then rashly and prematurely ventures into systematics rather than being simply ontological.

Other factor families include the multitude family, dealing with one, many, number, and membership. This is concrete membership, of those extensional collections Bolzano called *Mengen* (Bolzano 1837, § 84), a term that was appropriated by his admirer Cantor. For Bolzano, a *Menge* is a plurality: it has at least two members, and if these are all concrete, so is the *Menge*. What I call a *multitude* is simply one or more objects, an individual or a plurality: it is what Cantor himself called a *Vielheit*. Multitudes differs from sets in lacking any null or empty multitude, nor are there singleton multitudes distinct from their sole members.

A third family is the location family. This deals with the varieties of being at (Simons 2014). Exercised carefully, it should be metaphysically as neutral as possible between relational and substantival accounts of space, time, and space-time.

The dependence or essence family is one in which Jonathan invested considerable thought. It was his work and the convergent work of Kit Fine (Lowe 1994; Fine 1994) that convinced me that dependence is its own thing, not to be analysed in terms of standard alethic modality as I had done earlier (Simons 1987, ch. 7).

The final factor family about which I am fairly certain is that of determination, whose principal and perhaps sole primitive is causation, considered as holding between individual events and processes and their parts.

Putting these factor families together and letting them congruously interweave gives us categories, classes of beings. The job of a *Kategorienlehre* is not just to decide on a list and work with it, but also to uncover the factorial anatomy of those classes which for historical and systematic reasons have been taken as categories, such as substance, accident, property, relation, event, process, situation, place, time, and so on. In the interweaving there may be no single top-level list as there is in Aristotle or Jonathan: that needs to be considered.

In anatomizing ontic categories we inevitably employ the auxiliary categories of logic and we need to consider other cognitive operations that we regularly employ, such as abstraction and reification. In so doing, we can judiciously wield Ockham's Razor and clear away some of the excessive growth to which a too literal interpretation of the noun-phrases these operations throw up gives occasion.

Here again we enter into speculative and uncertain territory, so I need to stress again the provisional and error-prone enterprise that constitutes serious metaphysics. Jonathan Lowe was ever open-eyed about the risks and difficulties thus entailed, and we can be grateful that he nevertheless stuck to the enterprise with a tenacity that is hard to emulate and much, much harder to match. Once again, sadly not to his face, I salute him.

9. References

- Bolzano, B. 1837. *Wissenschaftslehre*. 4 vols. Sulzbach: Seidel. Translation: *Theory of Science*, tr. P. Rusnock and R. George. Oxford: Oxford University Press, 2013.
- Brentano, F. 1862. *Von der mannigfachen Bedeutung des Seienden nach Aristoteles*. Freiburg: Herder, 1862. Translation: *On the Several Senses of Being in Aristotle*, tr. R. George. Berkeley: University of California Press, 1975.
- Fine, K. 1994. 'Ontological Dependence'. *Proceedings of the Aristotelian Society* 95: 269–90.
- Husserl, E. 1900–01. *Logische Untersuchungen*. 2 vols. Halle/Saale: Niemeyer. Translation: *Logical Investigations*, tr. J. N. Findlay, 2 vols. London: Routledge & Kegan Paul.
- Ingarden, R. 1947–8. *Spór o istnienie świata*. Kraków: Polska Akademia Umiejętności, Volume I, 1947, Vol. II, 1948. Translation: *The Controversy over the Existence of the World*, tr. A. Szylewicz. Frankfurt/Main: Lang, 2013.
- Lowe, E. J. 1994. 'Ontological Dependency'. *Philosophical Papers* 23: 31–48.
- Lowe, E. J. 2006. *The Four-Category Ontology: A Metaphysical Foundation for Natural Science*. Oxford: Oxford University Press.
- Russell, B. 1912. *The Problems of Philosophy*. London: Home University Library.
- Simons, P. M. 1987. *Parts*. Oxford: Oxford University Press.
- Simons, P. M. 1994. 'Particulars in Particular Clothing: Three Trope Theories of Substance'. *Philosophy and Phenomenological Research* 54: 553–75.
- Simons, P. M. 1999. 'Die Urfaktoren des Seienden: eine anaximandrische Alternative'. In J. Nida-Rümelin, ed., *Rationalität, Realismus, Revision*. Berlin: de Gruyter, 399–406.
- Simons, P. M. 2014. 'Where it's At: Modes of Occupation and Kinds of Occupant'. In S. Kleinschmidt, ed. *Mereology and Location*. Oxford: Oxford University Press: 59–68.
- Wittgenstein, L. 1921. 'Logisch-Philosophische Abhandlung'. *Annalen der Naturphilosophie* 14:185–262. Translation: *Tractatus Logico-Philosophicus*, tr. D. Pears and B. F. McGuinness. London: Routledge & Kegan Paul, 1961.

3

Existents and Universals

John Heil

Abstract entities are not denizens of some ‘Platonic’ realm which is ‘separated’ from the world of things existing in space and time. According to this view, to say that abstract entities do not exist ‘in’ space and time is not to say that they somehow exist ‘elsewhere,’ a notion which is doubtfully coherent in any case. It is merely to say that when we speak of abstract entities we must ‘abstract away’ from all spatio-temporal determinations and distinctions. (Lowe 2002, p. 66)

It is clear that abstraction, which is common to all intellects, makes a form universal.

(Aquinas, *Quaestiones Disputatae de Veritate*, II, 6, ad 1)

Even if this individual is a human being and that individual is a human being, it is not necessary that both have numerically the same humanity, any more than it is necessary for two white things to have numerically the same whiteness. On the contrary, it is necessary [only] that the one resemble the other in having [an individual] humanity just as the other does. It is for this reason that the intellect—when it considers [an individual] humanity, not as belonging to this [or that] individual, but as such—forms a concept that is common to them all.

(Aquinas, *Sent* 2.17.1.1)

1. Personal Note

In the course of preparing this chapter, I found myself wondering how E. J. Lowe—Jonathan, as I knew him—would respond to this or that point, particularly in light of the fact that I address a matter concerning which he had thought long and hard, a matter concerning which he had definite, well-crafted views.¹ I recognize that whatever I have to say in what follows Jonathan would have wanted to query. He would have been two steps ahead of me, seeing what I could only have been brought to see with considerable effort. Absent Jonathan’s steadying hand, I am no longer sure who I can trust to set me straight.

Still, I can think of no better way to honor Jonathan’s memory than to dive into a vexed metaphysical topic, and few topics could be more vexed than that of universals. I chose the topic, not because I want to promote a new and exciting proposal, but

¹ Presented at a conference honoring E. J. Lowe 27–29 July 2014 at the University of Durham. I am especially grateful for extensive comments on an earlier draft of this chapter provided by Anthony Fisher and to Jeffrey Brower for correcting my hopeless Latin.

because it is one Jonathan and I discussed face to face on more than one occasion and over which we seemed utterly at odds. *Seemed* at odds, because I vividly recall a 2004 conversation with Jonathan in Durham in which he endeavored to assure me that, really, we were *not* at odds. Then, in 2010, at a conference on universals in Pisa, I attempted to spell out what I thought might effect a reconciliation of our apparently incompatible positions. Jonathan appeared to agree. (The paper was eventually published as Heil 2015a.)

This struck me as an important breakthrough, not least because in my dealings with Jonathan I had come round to the comforting view that, if Jonathan and I independently accepted *p*, *p*. (The principle is one I urge on you the reader as a potential time-saver when you engage in metaphysical deliberations.)

2. Williams's Tropes

My subject is a conception of universals that appears sensible, unmythical, and metaphysically inviting, a conception that I believe was once historically influential, but, owing to changes in philosophical fashion, became invisible in the twentieth century. With one or two important exceptions, you will find no mention of it in contemporary work devoted to universals. The conception was reinvented—or rediscovered—in the mid-twentieth century by D. C. Williams, a much underappreciated practitioner of ontologically serious metaphysics. Williams's conception belongs to a family of conceptions defended by others, including Aquinas and perhaps by Aristotle and by Lowe—or so it would appear.²

The universals I have in mind are a species of abstract entity in a traditional sense of 'abstract'. Abstract entities of the kind in question are not 'necessary beings' residing in an incorruptible Platonic realm, not entities situated 'outside' space-time, whatever that might mean. One way an entity can qualify as abstract is when its identity—its being the entity it is—depends on some other entity. Socrates's paleness, a particular way Socrates is, depends on Socrates. Socrates's paleness is *Socrates's* paleness. Socrates can survive the loss of his paleness, but if Socrates goes, if Socrates ceases to exist, his paleness thereby ceases to exist. The crucial point is that Socrates's paleness is separable from Socrates 'only in thought', only by way of a process of abstraction, what C. B. Martin, following Locke, liked to call 'partial consideration'.

So is Socrates's paleness a *trope*? Socrates's paleness is a property, a particular way Socrates is, and properties so conceived are nowadays *called* tropes. This unfortunate label is due to Williams, who borrowed the term from George Santayana, a predecessor at Harvard. Confusingly, as Williams himself notes, Santayana meant by 'trope' essence or universal. I once spent several weeks pondering the hermeneutical question, why would Williams embrace a term used by a philosopher whose work he respected and use it to mean the opposite of what that philosopher meant?

² G. F. Stout (1921, 1936) might have had something similar in mind as well. In recent years, Keith Campbell has done the most to articulate and develop Williams's doctrine; see his 1990.

I can think of a number of reasons Williams might have had for making the choice, but one in particular springs to mind. According to the *Oxford English Dictionary*, one meaning of ‘trope’ is ‘a word used in a sense different from its intended meaning.’ Williams’s ‘trope’ is a trope.³

But why bother? Why saddle philosophers with a perversely chosen technical term when others are available? One possibility is that Williams was banking on those conversant with Santayana’s philosophy to connect Santayana’s conception of universals. I think this unlikely, however, owing to Williams’s doubts concerning philosophers’ appreciation of Santayana’s metaphysics.

George Santayana was master of a solemn showmanship which is to blame for his being taken for a more trivial and alien philosopher than he was, both by those who like the trivial and the alien and by those who do not... The longer one peers, however, through the opalescent glass with which his style conceals the man, the more one discovers of a more solid philosophical citizen. (Williams 1954, 31–2)

Setting aside irony and historical associations, what might explain Williams’s decision to inflict ‘trope’ on the philosophical community? As Williams would have recognized, historical precursors of tropes (individual accidents, for instance, or modes) are at home in traditional two-category substance–attribute ontologies of the kind Williams rejected. Williams makes attributes the basic entities in an austere one-category ontology. Williams’s universe comprises a four-dimensional patchwork of qualities—the tropes—and their mereological sums. This reverses the traditional order of dependence between attributes, considered as accidents or modes, and objects. Williams’s objects, being sums of attributes, depend on the attributes.⁴ Under the circumstances, it would make sense to introduce a new term, ‘trope’, a neutral term that brings with it no unwelcome associations with individual accidents or modes.

Williams’s tropes are, I suspect, ontologically unstable in a way he occasionally acknowledged. Consider a historical precedent. Medieval philosophers, beginning in the eleventh century, posited what came to be called ‘real accidents’ in an effort to account for the continued appearances of bread and wine after the substances in which they inhered had been ‘converted’ into the body and blood of Christ in the sacrament of the Eucharist. Real accidents were accidents that could survive the subtraction of the substances to which they once belonged, properties now belonging to no substance. A well-rehearsed criticism of this view noted that the accidents in question were accidents in name only. A round white accident is a round white something, a something that is the round way and the white way, a *substance* (see Heil 2015b for discussion).

Although I find these historical matters endlessly fascinating, I introduce them only to make it clear why, in speaking of ‘particularized’ properties, I, like Lowe, prefer to

³ For discussion, see Heil 2012, ch. 5. In the end, given Williams’s account of universals, Williams’s tropes are in fact close to Santayana’s tropes, so there need be no irony in his use of the term.

⁴ You might think of Williams’s tropes as abstract entities, not because they depend on objects to which they belong—indeed the dependence goes the other way—but because they are compossible *parts* of those objects (which are sums of tropes), separable ‘only in thought’.

speak of modes, not accidents, and certainly not tropes. 'Mode'—from the Latin *modus*, way—is a traditional term that took on special prominence in the seventeenth century as a label for properties considered as particular modifications or characteristics of substances, properties but not universals. Tropes were meant by Williams to be self-standing entities from which objects are constructed: quite literally *parts* of objects, objects being mereological sums of tropes. Accidents might or might not be capable of trope-like feats. But modes are of necessity modifications of substances, particular ways those substances are, hence dependent on the substances they modify. Every mode is a mode of some substance, a way some substance is. Conversely, every substance is some way—or ways—or other. The universe includes substances that are various ways and their endless combinations and interrelations. Or at least this is one possibility.

Although philosophers who regard properties as fully particular differ in various respects, these differences have little or no bearing on the topic under discussion. For reasons of expository simplicity, then, I shall speak indifferently of tropes, modes, and accidents. It is not that these labels circumscribe a single category, but that their differences are irrelevant here.

My plan in what follows is to indicate how Williams's reflections on universals provide a way to make sense of the possibility of meaningful talk of universals in a universe of particulars. I would like to think that the resulting picture is close to Lowe's in important respects. Whether that is so or not, this conception of universals is I believe close to conceptions held by assorted medievals including Aquinas, and perhaps by Aristotle himself. A caveat: I recognize the inadequacy of my historical credentials. I could easily be wrong about Aquinas, wrong about Aristotle, and wrong about Williams. Even if that were so, however, the position I discuss and attribute to Williams—and find echoed in Aquinas, Aristotle, and Lowe—has an appeal independent of its pedigree.

3. Immanent Universals

For Williams—and Lowe as well—universals are taken to be immanent: respectable denizens of space–time. This places them in the same camp as David Armstrong, another exponent of immanent universals, although immanence, for Armstrong, falls out of a very different metaphysical picture. According to Armstrong, the universe comprises states of affairs. A state of affairs is a nonmereological structure, the instantiation of a universal by a particular at a time. States of affairs are Armstrong's elements of being. God does not make the particulars and the universals, then assemble them into states of affairs. God's creating a particular or a universal is a matter of God's creating a state of affairs incorporating the one or the other. Particulars and universals alike are abstract, in the sense that they are separable from states of affairs only in thought.

The basicness of states of affairs guarantees that Armstrong's universals are immanent. A universal lacking instances, would be a universal existing apart from any state of affairs, a manifest impossibility. You could think of states of affairs as having two 'aspects': a particular aspect and a general or universal aspect. Socrates's being pale on

Tuesday (I will omit the temporal reference henceforth) involves Socrates, a particular, and paleness, a shareable, repeatable universal. Paleness can participate in states of affairs that do not include Socrates—Simmias might be pale—and Socrates is a constituent of states of affairs that do not include paleness—Socrates, in addition to being pale, is snub-nosed. The fabric of Armstrong's universe is a tightly woven tapestry of intersecting particulars and universals.

Many philosophers, Williams and Lowe included, have found the idea of an entity's being at once wholly present in distinct locations unfathomable. Nevertheless, Williams takes the idea seriously, supposing it to be what proponents of immanent universals embrace, coupling it to the thought that universals differ from particulars in satisfying the principle of the identity of indiscernibles. Particulars can be indiscernible but distinct, not so universals. If *A* and *B* are indiscernible universals, then *A is B*. Peas in a pod could be indiscernible yet distinct; but their colors and shapes could not be distinct, at least not if their colors and shapes are indiscernible Armstrong-style universals. The difference here is marked by the difference between

- (1) one thing, *a*'s, being identical *to* *b*, and
- (2) *a*'s being identical *with* *b*.⁵

One pea is identical *to* another colorwise; their respective colors, however, are, *qua* universals, identical *with* one another, not merely exactly similar, but 'strictly identical'.

So, if two peas are the same color, the color of one *is*—is identical with—the color of the other. At least, says Williams, this is what proponents of universals *say* they mean. Indeed, they will insist that this is what they *do* mean, that this is the whole *point* of universals, their very *raison d'être*. Even so, Williams holds, they could not *really* mean this.

Williams bases this contention, not on an argument, but on kinds of consideration commonly advanced by critics of immanent universals. You start by thinking that one pea is identical to another colorwise. In considering their respective colors, however, identical *with* replaces identical *to*: this pea's color is identical *with*—*is*—that pea's color. The task, Williams thinks, is to explain how this illicit shift occurs and why it might seem natural, even inevitable.

One possibility, a possibility endorsed by Armstrong, is that similarity is 'grounded' in identity.⁶ Socrates and Simmias are similar by virtue of sharing a common element: paleness. However, as Williams notes, 'we are much less likely to explain similarity as due to the presence of a common or general constituent than to explain the notion

⁵ Although the distinction between 'identical to' and 'identical with' is not one to which Williams appeals explicitly, it provides a convenient way of representing Williams's argument and affords me the opportunity to mention the fact that the distinction increasingly eludes philosophers writing on identity, one more casualty of the ongoing erosion of grammatical norms and subsequent loss of precision in contemporary philosophical prose.

⁶ So if particulars *a* and *b* are similar, this is either because *a is b* (everything is similar to itself), or because *a is F* and *b is G*, and *F is G*, taking 'is' in each case to express strict identity. Problems arise in cases of less-than-perfect similarity, as when *a* is more similar to *b* than to *c*, for instance. See Heil 2003, ch. 14.

of a common or general constituent as somehow a resultant of similarity' (1959, 6). This will be a recurring theme in what follows.

4. Rules for Counting

Human beings have a capacity to direct their attention to substances—particular tomatoes, for instance—or to properties of those substances, ways those substances are—a tomato's color or shape. You 'abstract' the tomato's color or shape from the tomato by attending to or considering its color or shape without attending to what *has* the color or shape, the tomato. Although abstraction is a mental activity, *what* is abstracted is not something mental or mind-dependent. The tomato's color and shape are real, perfectly 'objective' features of the tomato, here and now qualitative characteristics that make a difference to what the tomato does or would do. The tomato would, for instance, look red owing to its color; the tomato would roll, would make a circular concave impression in the carpet, owing to its shape.

In this way, you arrive at a distinction between substances as bearers of properties, and properties borne by substances. The distinction leaves open the question whether properties are accidents, modes, tropes, or universals. You could think of Williams as defending the idea that properties—his tropes—are particulars, but wanting to make sense of the philosophical tendency to treat properties as universals, repeatable, multiply locatable *general* entities.

Start by abstracting this tomato's color, considering its color apart from the tomato as you might in the course of selecting tomatoes at the grocer's. You can, however, engage in a further level of abstraction, what Williams calls 'generization'. This leads you to universals. You consider the color of this tomato, this tomato's redness, then turn your attention to the color of a distinct tomato, recognizing that these are *the same*. In so doing, Williams thinks, you are abstracting from the color's immanent particularity—from its being the color of this tomato or the color of that tomato—and considering just its intrinsic qualitative nature. You are, although you did not know it, generizing, abstracting, as Lowe puts it in a passage quoted at the outset, 'from all spatio-temporal determinations and distinctions' (2002, 66). The fact is there are two distinct but indiscernible qualitative natures here, two modes, two accidents, two tropes. Once you generize, once you abstract from the tropes' respective particularities, you count the two as one. Here is redness, and here is redness again, the *same* redness.

In this way, the redness of a particular tomato can serve as a universal when it is considered purely qualitatively apart from its being the particular here and now redness of *this* tomato. The result is what Keith Campbell calls *painless realism*: 'the difference between a case and a kind, a particular and a universal, is not a difference of *category*, but a difference in *rule for counting*' (Campbell 1990, 44).

Consider a design inspired by Mondrian, and imagine that each shade of grey is a particular color (red, blue, yellow). How many colors occur in Figure 3.1? On one natural way of counting (and assuming that we count white as a color) the image includes four colors. The truthmaker for this claim is something that features three cases of

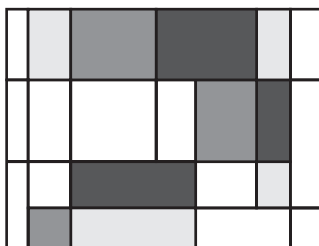


Figure 3.1. *Composition with Red, Blue and Yellow*, Piet Mondrian, 1930

red, three cases of blue, four cases of yellow, and a dozen cases of white. (Compare counting the words in a 10,000-word article with counting the words mastered by your two-year-old niece.)

As Campbell notes, this is a familiar way of introducing Peirce's distinction between types and tokens. The image includes three tokens of a single type, red. These different ways of counting reflect different interests on the part of counters. But they do not thereby render types or tokens mind-dependent. Given the universe as it is, you have all the types and tokens there are. Are cases considered as kinds universals? You make the call. Truthmakers for claims about types are modes or tropes. The universe does not include, in addition to these, general, non-particular, multiply locatable entities. As Locke insists, 'all things that exist are only particulars' (*Essay*, III, iii, 6).

This way of understanding—or finessing—universals differs from another that has devotees among those philosophers who regard properties as particulars. I have in mind the idea that universals are to be identified with sets, or classes, or pluralities of resembling properties. These would be coextensive with Armstrong's universals, amounting to what he calls *ersatz universals*. On such a view, to say that two tomatoes are the same color is to say that each tomato has a particular color, and that the two colors belong to a class of exactly resembling colors of particular objects.⁷

G. F. Stout puts it this way:

A character characterizing a concrete thing or individual is as particular as the thing or individual which it characterizes. Of two billiard balls, each has its own particular roundness separate and distinct from that of the other, just as the billiard balls themselves are distinct and separate. As Jones is separate and distinct from Robinson, so the particular happiness of Jones is separate and distinct from that of Robinson. What then do we mean when we say, for instance, that roundness is a character common to all billiard balls? I answer that the phrase 'common character' is elliptical. It really signifies a certain general kind or class of characters. To say that particular things share in the common character is to say that each of them has a character

⁷ Stout (1921, 1936) defends an alternative view, arguing that resemblance is itself 'founded' on a primitive 'distributive unity', which functions as a universal in something like Williams's sense.

which is a particular instance of this kind or class of characters. The particular instances are distributed amongst the particular things and so shared by them. (1921, 2–3)

Such views face well-known difficulties. Members of classes can increase in number and decrease, but redness—the universal—seems incapable of enlargement or diminution. And it might be true that were the color of that tomato red, it would be the same color as this tomato, but it makes no sense to think that a particular greenness might become a member of the class of rednesses.

Such worries are perhaps resolvable, but even if they were not, they do not apply to Williams's universals. Williams's universals are neither general entities nor classes. Regarding a property as a universal is a matter of regarding it as a representative kind, prescinding from its particularity.

5. Real Universals

Williams speaks of 'two modes or dimensions of reference', reference to cases and to kinds (1959, 9). I have suggested treating a particular case—the redness of this tomato, for instance—as a kind is at bottom a matter of eliding the distinction between something's being identical *to* and its being identical *with* something else. Presently I shall suggest a slightly different way of thinking about the relation of cases to kinds.

Meanwhile, you might worry that anything along these lines demeans universals. This would be so, however, only if you were convinced that universals must be *entia generalia*. On the view I am ascribing to Williams, what exists are particulars—modes or tropes—that do double duty as universals.

That universals are determined by a 'weaker' identity condition than particulars does not even mean that they have an inferior or diluted reality. A tabulation of universals is just one way of counting, as it were, the same world which is counted in a legitimately different and more discriminating way, in a tabulation of particulars. (1959, 9)

Does this turn universals into creatures of reason? Not according to Williams.

We are not saying that universals are '*entia rationis*', if this ancient phrase implies that they have their source or habitat in reason or discourse. It is by a twist of language that we designate them, it is true, singly or in troops, but it is only by twists of language that we designate anything, and none of these twists creates its own [designatum].⁸

If you think of universals in this way, you have a ready defense of the idea that universals must have instances, an idea embraced by Lowe—and by Armstrong, although for very different reasons. A universal is a mode (a trope), an 'instance' considered

⁸ Williams (1959, 9). The paper as it appears in the *Australasian Journal of Philosophy* has 'designatum', but Anthony Fisher has noted in correspondence that this is one of several typographical errors in the published version that do not occur in the original manuscript.

without regard to its particularity, the tomato's redness considered apart from its particularity.

Williams acknowledges that a conception of this kind is unlikely to satisfy a philosopher who persists in thinking of universals as general entities. It does, however, reflect an important feature of our thoughts about the universe. Truthmakers for those thoughts can turn out to be surprising ways the universe is, ways that cannot be deduced from the truths. Truthmakers for claims about moving bodies, for instance, could turn out to involve nothing that moves. This would be so if the truthmakers for truths about the motions of cricket balls and planets resembled the truthmakers for truths about the movement of a cursor across your computer screen—a genuine possibility if moving bodies turned out to be wavelike disturbances in fields.

In the same way judgments featuring general terms, including talk of shared or common properties, could have fully particular truthmakers. You could see this as a denial of universals or you could see it as encompassing the deep story about universals.

Whether previous immanent realists would recognize their view in this opinion that universals are immanent because they are, to speak crudely, the similarity roles (or 'adjectival identities') of abstract occurrents, I have some doubt. I am sure, from experience with myself, that an immanent realist begins by thinking he means more, but can bring himself to see, or think he sees, that he *couldn't* mean more – that every attempt to state an alternative results in something verbally but not significantly different from just redefining 'identity' by resemblance. (1959, 10)

You might regard Williams's 'painless realism' as an aberration, a philosophical outlier. I believe that the position, or something like it, is one that philosophers have struggled to articulate off and on at least since Aristotle. You can find precursors of the view among the philosophical giants. Aquinas, for instance, while denying that universals are real existents—not constituents of objects or states of affairs, and not Platonic forms—holds that universal concepts have application to particular things owing to those things' 'common' elements. Individual human beings, for instance, share a nature, an 'essence', in virtue of which all of them are human beings.

Human nature itself, indeed, has being in abstraction from all that individuates, and for this reason it has a content which is the same for all individuals which are outside the soul, as it is equally a likeness of all of them, leading to a knowledge of all men insofar as they are men.

(*De Ente et Essentia*, IV)

It is hard for me to read such passages without thinking of Williams. In fact I find this reading of Aquinas much more interesting than an alternative reading according to which Aquinas's universals are located just in the mind.

I like to think that Williams's and perhaps Aquinas's universals are close to Lowe's.⁹ These are universals even the most dedicated trope theorist could love. When you think about it, what else could universals be?

⁹ See below and see Lowe (1998, 155–6 and 2002, 66).

6. Postscript

The original version of this chapter ended here. After completing it, however, I discovered a paper of Lowe's published in 2012 in a collection to which I myself had contributed, criticizing—as I now think, erroneously—Lowe's inclusion of universals in his 'four-category' ontology. The paper, 'A Neo-Aristotelian Ontology: Neither Relational nor Constituent' (Lowe 2012), reveals that Lowe's conception of universals is indeed very close to the conception I attribute to Williams.¹⁰

Lowe's four-category ontology, which is inspired by the metaphysics of Aristotle's *Categories*, includes two species of universal: substantial universals or kinds, and attributes (see Lowe 2006). Objects—individual substances—are instances of kinds, substantial universals; modes—ways individual substances are—are instances of attributes, property universals, Aristotle's accidents. In each case the instances are 'ontologically prior' to the universals.

As noted earlier, in Armstrong's case, immanence follows from the fact that the fundamental things are not substances or universals but states of affairs: particulars instantiating universals at times. Universals and particulars alike are abstractions, 'aspects' of states of affairs. States of affairs are not concocted from stocks of particulars and universals. Universals—and particulars—are on the scene only via the obtaining of states of affairs. There is a clear sense in which Armstrong's universals are non-particular 'constituents' of states of affairs, items 'wholly located' everywhere states of affairs that include them are located.

Immanence for Lowe is, as it is for Williams, importantly different. A universal is an abstraction:

It is part of the overall picture offered by the four-category ontology that universals are *ontologically posterior* to particulars – which is why it takes universals to be incapable of existing uninstantiated. On this view, although universals are perfectly real, they are perhaps best seen as being *abstractions from*, or *invariants across*, particulars. (Lowe 2012, 245)

So the conception of universals I have associated with Williams fits Lowe nicely. You can engage in 'partial consideration' and attend to Socrates, an individual substance that is various ways, the ways being modes, and you can attend to Socrates's paleness, 'abstracting' it from Socrates. Socrates's paleness is an instance of paleness, a mode, what Williams calls a case. But you can also abstract from Socrates's paleness, a mode, and consider just the paleness independently of its being Socrates's paleness. In this way, Socrates's paleness affords an instance of what it is to be pale, a universal.

Where Lowe differs from Williams is in his allegiance to substance as an ontological category and his deployment of substantial universals. For both Williams and Lowe, Socrates's paleness is an instance of paleness, an instance of what it is to be pale. For Lowe, but not for Williams, Socrates himself is an instance of humanity, a kind, a

¹⁰ As noted earlier, my original discussion of Williams and Lowe on universals occurred in a paper presented in 2010 to an audience that included Lowe (Heil 2015a).

substantial universal. However, humanity is not in any sense a constituent of, or entity present in, Socrates—or in other human beings. A substantial universal constitutes the ‘essence’ of its instances, and an essence is—for Lowe as it is for Aristotle—*what it is to be* an individual substance of a particular kind—a human being, for instance. If you associated essences with Aristotelian ‘real definitions’, then instances of substantial forms are just those things that satisfy one or another real definition. Many things could satisfy a single definition, but this is not because they all include a common ingredient, identified with a substantial universal or an essence, it is because they are similar in the pertinent ways.

For Lowe and Williams all you need in order to have a universal is a single case—a particular paleness, for instance. Once you have that, you have a vehicle for ‘generalization’, you appreciate *what it is to be* pale and thus the possibility of other cases of paleness, palenesses exactly resembling this one. And, as Williams and Lowe insist, none of this makes universals mind- or language-dependent. Objects are what they are and are various ways whether we bother to consider those ways or not, whether or not we deign to assign them names.

Philosophers are trained to be hard on one another. This can lead to ill-conceived unproductive criticisms of a kind that is all too familiar. But it is also possible to err in another, rather less obviously pernicious manner by advancing ‘charitable’ interpretations of positions that, in effect, represent those positions in ways that reflect interpreters’ preconceptions. We hear what we expect to hear, we recast the unfamiliar so as to make it familiar, we make sense of the other by making the other one of us. This tendency, especially prominent when we read philosophers writing in earlier eras, can lead to the dumbing down of our forebears and thereby forgoing the opportunity to learn from them (see Melamed 2013 for discussion and examples).

This is how you might see many contemporary discussions of universals and their historical antecedents. If the position I have ascribed to Williams and Lowe strikes you as strange or revisionary, that could be due to your having taken on board a distinctively twentieth-century conception of what universals would have to be: multiply located general entities, perhaps, or other-worldly *abstracta*, colleagues of numbers and sets residing in an incorruptible Platonic heaven. I readily confess that my own relentless hostility toward universals stems from what I now regard as egregious misconceptions about what I imagined universals would have to be. What Williams and Lowe demonstrate is that this hostility is, as is much philosophy throughout its history, misguided and self-defeating.

7. References

- Campbell, K. 1990. *Abstract Particulars*. Oxford: Basil Blackwell.
 Heil, J. 2003. *From an Ontological Point of View*. Oxford: Clarendon Press.
 Heil, J. 2012. *The Universe as We Find It*. Oxford: Clarendon Press.

- Heil, J. 2015a. 'Universals in a World of Particulars.' In G. Galluzzo and M. J. Loux, eds. *The Problem of Universals in Contemporary Philosophy*. Cambridge: Cambridge University Press, 114–32.
- Heil, J. 2015b. 'Cartesian Transubstantiation.' In J. Kvanvig, ed. *Oxford Studies in the Philosophy of Religion*, vol. 6. Oxford: Oxford University Press, 139–57.
- Lowe, E. J. 1998. *The Possibility of Metaphysics: Substance, Identity, and Time*. Oxford: Clarendon Press.
- Lowe, E. J. 2002. 'Metaphysical Nihilism and the Subtraction Argument.' *Analysis* 62: pp. 62–73.
- Lowe, E. J. 2006. *The Four-Category Ontology: A Metaphysical Foundation for Natural Science*. Oxford: Clarendon Press.
- Lowe, E. J. 2012. 'A Neo-Aristotelian Substance Ontology: Neither Constituent nor Relational.' In T. E. Tahko, ed. *Contemporary Aristotelian Metaphysics*. Cambridge: Cambridge University Press, 229–48.
- Melamed, Y. 2013. 'Charitable Interpretations and the Political Domestication of Spinoza, or, Benedict in the Land of the Secular Imagination.' In E. Schliesser, M. Laerke, and J. Smith, eds. *The Methodology of the History of Philosophy*. Oxford: Oxford University Press, 258–79.
- Stout, G. F. 1921. 'The Nature of Universals and Propositions.' *Proceedings of the British Academy* 10 (1921–23): 157–72. Oxford University Press. Reprinted in G. F. Stout, *Studies in Philosophy and Psychology*. London: Macmillan and Co, 384–403.
- Stout, G. F. 1936. 'Universals Again.' *Proceedings of the Aristotelian Society, Supplementary Volume* 15: pp. 1–15.
- Williams, D. C. 1954. 'Of Essence and Existence in Santayana.' *Journal of Philosophy* 51: pp. 31–42.
- Williams, D. C. 1959/1986. 'Universals and Existents.' A paper delivered to the Yale Philosophy Club and published posthumously in *The Australasian Journal of Philosophy* 64 (1986): pp. 1–14.

4

Whole, but not One

Anna Marmodoro

1. Introduction

This chapter investigates the problem of the unity of objects from the point of view of an ontology that posits only instances of physical powers (e.g. mass, spin, charge) at the fundamental level of reality.^{1, 2} The account I put forward is a neo-Aristotelian one. One of the keystones of Aristotle's metaphysics is his hylomorphism, the view (as I interpret it) that objects are *unified wholes* analysable in their component matter and form. In my understanding of Aristotle, matter and form are holistically rather than mereologically composed in a hylomorphic composite.³ I submit that Aristotle had philosophically sound insights on holistic composition; on the other hand, he also assumed that a whole is *ipso facto* metaphysically one. It is with this assumption that I take issue in this chapter. By contrast with Aristotle, I distinguish being a *whole* and being *metaphysically one*; and I give in this chapter arguments for such distinction. I show how powers may combine in both ways, into wholes, and *further*, into metaphysical unities.

For the sake of building my neo-Aristotelian approach on firm foundations, in this chapter I also engage with E. J. Lowe's critique of Aristotle's hylomorphism. Lowe, while often declaring how inspirational Aristotle was for his own work, considers Aristotle's key doctrine of hylomorphism unable to meet, in his words, 'the challenge of explaining how *a new substance* is brought into existence' when matter and form

¹ The preparation of this chapter benefited from a fellowship at the Paris Institute for Advanced Studies (France), with the financial support of the French State, managed by the Agence Nationale de la Recherche, programme 'Investissements d'avenir' (ANR-11-LABX-0027-01 Labex RFIEA+). The chapter draws (*verbatim* in some sections) on two previously published papers of mine: 'Aristotle's hylomorphism without reconditioning', *Philosophical Inquiry*, Volume 37, Issue 1/2, 2013, pp. 5–22; and 'Power mereology: structural versus substantial powers', in M. P. Paoletti and F. Orilia (eds.), *Philosophical and Scientific Perspectives on Downward Causation*, Routledge, 2017, pp. 110–27.

² I have argued for such ontology in my 'Power mereology: structural versus substantial powers', in M. P. Paoletti and F. Orilia (eds.), *Philosophical and Scientific Perspectives on Downward Causation*, Routledge, 2017.

³ There is much controversy in the literature as to how to interpret Aristotle's view; for reasons of space here I can only state my own interpretation and refer the reader to my 'Aristotle's hylomorphism without reconditioning', *Philosophical Inquiry*, Volume 37, Issue 1/2, 2013, pp. 5–22, which includes also a review of some modern interpretations of Aristotle's views.

combine (2012, 236). In response, I show that Lowe had not captured the essence of Aristotle's theory of holistic composition, and that his 'reconditioned' version of the theory does not fare well in addressing the very question that Lowe thinks Aristotle failed to address successfully.

2. Lowe's Critique of Aristotle's Holism

In his analysis of Aristotle's hylomorphism Lowe concentrates on what he identifies as its main desideratum, which is 'to justify the judgement that a new concrete object—an "addition of being"—really has been brought into existence, rather than some previously existing things merely being re-arranged' (2012, 236). Lowe interprets Aristotle as accounting for the emergence of a new entity by positing that matter and form, the parts of an object, are (in a Fregean fashion) 'incomplete entities'. According to Lowe, what allows for the emergence of a new unified object is the mutual 'saturation' of its parts. No additional metaphysical glue is needed; it is the incomplete nature of the parts that secures that they get unified. Lowe characterizes the view thus:

The key point is that, on this view, individual substances exhibit 'internal' ontological complexity, being combinations of 'incomplete' entities that are completed by each other in the substance. (2012, 231)

Lowe explains this (supposedly) Aristotelian account of the unity of substance through the incompleteness of its parts thus:

unless we can see the new substance as being a combination of items neither of which can exist independently of the other in just such a combination, rather than as merely being composed of other independently existing things, each possessing their own features, we shall be unable to justify the judgement that a new concrete object... really has been brought into existence. (2012, 235)

Lowe finds this (putative) Aristotelian view problematic; I take it, on the ground that incompleteness is an unexplained concept:

What I don't understand is what it means to say that the completed house's form—the way in which its 'matter' is organized—is an 'incomplete' constituent of the house which 'combines' together with that equally 'incomplete' matter to constitute the house, a complete substance. (2012, 236)

I agree with Lowe that what he describes here would not offer an explanation of how a substance is a new entity, rather than a mereological sum of its parts arranged in a certain way. There is no reason anyone should expect that two incomplete entities could make up a complete whole over and above the sum of its parts; for instance, half a pear and half an orange do not make up a whole fruit. But the two parts of a drawbridge over a river do make up a complete bridge. Why? Because of their complementarity. This is the sense in which, each on its own, is incomplete. Complementary entities complete each other on account of what is achieved when they complement one another. For

Aristotle, the wholesomeness of the achievement is what licenses the description of the contributing entities as incomplete. Complementary entities may be either existentially independent of each other, like a peg and a bookshelf; or dependent on each other, like quarks (which cannot exist on their own); or both, like the parts of a computer. Their joint achievement may be either a function that can be performed only together (e.g. a calculation) or a new entity they jointly bring about (e.g. a bookshelf on a bookcase, a computer, or a baryon (consisting of three quarks)). The nature of the achievement will determine whether the result is a new function or a new entity.⁴ In the case of matter and form, if they are thought to be incomplete in a complementary way, the achievement of their mutual completion is a new entity. The difference between them and the case of the baryon is that the fusion between the quarks is *physical*, by means of a physical bond; but the fusion of matter and form is *metaphysical*, and this has to be explained.

Aristotle however, I argue, did not distinguish the two ways, physical and metaphysical, in which a compound object is formed. This conflation (which Lowe too did not see) is perpetuated in much contemporary metaphysics. On the one hand, Aristotle says that the substantial form is responsible for the physical formation and constitution of a substance; he writes that since ‘substances [...] are *formed naturally* and in accordance with their *nature*, [...] it would appear that this nature is their substance’ (namely their substantial form) (*Metaphysics* VII 17, 1041b, 29–30, my emphasis). On the other hand, the substantial form determines what a substance is, where all the constituents of the substance are defined by their functional role in the substance, assigned to them by the substantial form; Aristotle writes that ‘we shall define each part [of a substance], if we define it well, not without reference to its *function*’ (*ibidem*, 1035b, 16–18, my emphasis). On my reading of Aristotle, it follows that the substantial form is responsible *both*, for the constituents of the substance being *physically structured* as they are in the substance; but also for the constituents making up a whole whose parts are *defined* in terms of the nature of the whole.⁵ So a particular substance is both physically constituted the way it is in virtue of its substantial form; and it is a metaphysically united whole of a particular sort in virtue of the substantial form. The substantial form plays, for Aristotle, both, a physical and metaphysical unifying role; Aristotle runs these together as if one, as the quotation below illustrates; and since Aristotle, the two roles have *not* been teased apart in metaphysics:

[I]t would seem that this [what unifies the constituents of a substance] [...] is something, and not an element, and that it is the *cause* that makes this thing flesh and that a syllable [i.e. what sort they are]. And similarly in all other cases. And this is the substance [essence/nature] of each thing (for this is the primary cause of its being); and since [...] substances are *formed*

⁴ Note that these two types of emergence, that of a novel property (function) versus that of a novel entity, are often run together in the literature.

⁵ See my ‘Aristotle’s hylomorphism without reconditioning’, *Philosophical Inquiry*, Volume 37, Issue 1/2, 2013, pp. 5–22.

[physically] in accordance with a *nature of their own* and by a *process of nature*, their substance would seem to be this kind of 'nature' [i.e. form], which is not an element but a principle.

(*Metaphysics* VII 17, 1041b, 25–31; my emphasis)

Notwithstanding this ambiguity in Aristotle's claim that the substantial form unifies a substance into a whole, his account ofhylomorphic composition withstands philosophical scrutiny. As I argued elsewhere, following Scaltsas (1994), I take Aristotle to hold that being unified into a whole *re-identifies* the parts into being what they cannot be apart from the whole. The parts are re-identified according to the unifying principle of the whole, the substantial form. Once re-identified, they have *no distinctness* in the substance; they exist in it holistically.⁶ If they were severed from the whole, they would lose their functional identity, which is conferred to them by the form, on the basis of their role in the whole substance. Thus, if severed, the parts would lose their form and become like (originative) matter is to the substance it can constitute (e.g. like extracting marble from a statue).⁷

Does Aristotle's hylomorphism require any sort of 'reconditioning'? My view is that it does not; yet, there have been many recent attempts in current metaphysics to 'recondition' Aristotle's views by rejecting one or the other of the premises on which his hylomorphism is based. Lowe too, however, among others, offers a way of 'reconditioning' Aristotle's hylomorphism, by jettisoning matter from the ontology, on the ground that he considers matter an unintelligible notion.⁸ Lowe illustrates his view by means of the following example:

When... the electron is captured by the proton and occupies an orbit around it, then indeed we have a new concrete object of a very different kind: a hydrogen atom... In the newly created hydrogen atom, the proton remains exactly what it was before, just *a proton*, and the electron remains just *an electron*. A new form is instantiated... The form does not, in any sense I can understand, 'combine' with the proton and the electron as to constitute with them the atom. The only things that do any 'combining' are the proton and the electron.

(2012, 237, emphasis in the original)

It is clear that this reconditioned account has no room for the Aristotelian notion of matter; there is no metaphysical job for it to do. On Lowe's way of thinking about the given example, the proton and the electron are there before and after the creation of the hydrogen atom: there is no need for a material continuant that takes on a new form when a new substance is created, because the proton and the electron can do this by combining, when the composite is created.

⁶ What can be extracted from the whole exists in it only potentially, in the sense that it derives from it.

⁷ A key text for this conclusion is where Aristotle describes his *Homonymy Principle*, in the *Metaphysics*: [The parts of a substance] cannot even exist if severed from the whole; for it is not a finger in *any* state that is the finger of a living thing, but the dead finger is a finger only *homonymously*. (1035b, 24–5, last emphasis added)

⁸ Lowe writes: 'I have no serious need for the hylomorphist category of matter' (2012, p. 237).

This proposed reconditioning of hylomorphism is not however free of difficulties. The main one arises with the claim it makes that the generation of a new substance leaves the constituting components *as they were* (e.g. when the hydrogen atom forms, the proton and the electron remain what they were before making up the atom). For Aristotle, a hydrogen atom is, at best, an instance of a substance, but not a *model* for understanding the metaphysics of substance; reconditioned hylomorphism cannot help us understand the generation of a *new* entity (e.g. of flesh out of bread and water). How do bread and water come to constitute flesh when we feed ourselves? Aristotle's solution, as I have argued elsewhere and briefly indicated here above, is that bread and water undergo a radical qualitative and functional change when they make up flesh—a change such that they are no longer identifiable as bread and water once they constitute flesh. But what could one say in answer to the above question if one endorses Lowe's model? One might want to argue that ultimately bread and water are protons and electrons. (But do we have to assume atomism on *a priori* grounds because we assume all generation is combination?) This move however would not serve to recondition *Aristotelian* hylomorphism, as for Aristotle the ultimate level of reality comprises the fundamental powers (hot, cold, wet, and dry) and not a layer of particles (even conceding that protons and electrons are particles of a special kind).⁹

But independently of the issue of whether Lowe's account is genuinely a way to recondition Aristotle's hylomorphism or not, a further pressing question that Lowe's account gives rise to is this: how are we to understand the claim that the proton and the electron *combine* into a hydrogen atom? That is, what is the metaphysical difference between the proton and the electron on the one hand, and the combined proton and electron on the other? It is widely assumed that the difference is not primitive, but rather one that metaphysics can explain. Yet does it come down to just a difference of being related to each other or not?

3. Whole, but not One

Let us take an electron as an example of a unified entity constituted of parts: its elemental properties.¹⁰ In an ontology (like the one I endorse and I have argued for elsewhere) in which such properties are powers, this means that the electron is constituted by its elemental powers, such as the power of electric charge, of spin, etc. The question is: what is it, metaphysically, for an electron to be *constituted* by these powers? An electron *isn't* simply the compresence of powers of charge, spin, mass. These powers are arranged,

⁹ See my 'Potentiality in Aristotle's Metaphysics' in K. Engelhard and M. Quante (eds), *The Handbook of Potentiality* (Dordrecht, 2018).

¹⁰ Physics offers us a number of ways of understanding the constitution of an electron; but I submit that on any of them the electron is defined in terms of elemental properties. In the standard model, an electron is an elementary particle that is characterized by certain values of three elementary properties: mass, charge, and spin, and exhibits wave-particle behaviour. In quantum field theory, an electron, like all elementary particles, is an excitation state of an underlying physical field; different types (properties) of excitation of the field account for the different types of elementary particle. In string theory, an electron is a vibrating string, of the lepton type determining its core structure.

structured in a particular configuration, when making up an electron; there is at least composition, and more, as we have seen in the previous section of this chapter, which needs to be explained. I will argue that physical structure *unites*; while metaphysical structure *unifies*. The former brings about wholes, the latter unities. But wholes are not always unities. No degree of organization at the physical level entails metaphysical oneness, that it, being one entity. There are in nature aggregates of powers that endure and even evolve and develop over time (e.g. an organism); as well as aggregates whose component powers are functionally interconnected (e.g. a computer). Such aggregates of physically united powers (in enduring structures, and recurrently enduring ones in organic cases) are structurally united through physical causal means, such as attraction and repulsion of electrons at the fundamental level, etc. All these structures of powers exhibit physical continuity and connectedness, synchronically and often diachronically, too; but crucially, no particular instance of physical structure, here and now, can unify the structured powers metaphysically into one. A set of powers that are merely *physically united* are a *plurality* of many, just like the grains of sand on the beach. They can even be a one-something, a whole, and still be many, like the class of students who are rowdy.¹¹ By contrast, an electron is constituted of a structured plurality of powers, but it is further unified and one, and is not identical to the plurality of its powers, on any Aristotelian metaphysics. The powers of an electron (its charge, spin, and mass), which are physically united into a structure, are also *metaphysically unified*.

There are countless structures of powers in nature which are (merely) physically united. Although some structures are physically more strongly united than others, this in itself does not change their standing as merely physical structures. On the other hand, some of the physically united structures are *also* metaphysically unified; for instance, a tree, or an organism. What, then, differentiates a merely physically united structure of powers from one that is also metaphysically unified? Physical structures are ‘*numberless*’, in the sense that they do not bring with them a count principle. It is an *open question* how many entities a physical structure is, which is not determined by the structure, or even the structure’s being a whole. What is merely physically structured can be counted as one, or as many entities, or as either (e.g. as one can think of the 64 chess squares in various ways). I argue that number accrues to physical structures of powers on account of their *metaphysical unification*.

Let us begin with the question of what metaphysical unification is. How does a physically united structure become unified in such a way as to acquire number, and become one? Consider a squad of soldiers. One can think of it as a structured plurality of soldiers or a whole of many; but also, one can think of it/them as a single army unit. Similarly, a swarm of bees can be thought of as a structured plurality of insects, or as a single super-organism. What is the difference? My claim is that the difference between a structured plurality and a single individual it may constitute lies in the *dependencies* that develop between the components of the structure. Unification of structured components into a

¹¹ The reasoning is the same as W. V. O. Quine’s (1960) when he argues that what instantiates a *rabbit* instantiates just as much *un-detached rabbit parts*.

single individual involves more than ontological dependence of the parts on the whole; it involves ontological-cum-definitional dependence, namely holistic dependence.

How does a component in a physical structure (e.g. a power) become holistically dependent on the whole-structure-as-a-single-individual? The transformation demands a change in the criteria of individuation applied to the power, which results in a change in the power's ontological status. Consider the electric charge of an electron; the power is re-individuated as *a way of being of* the electron, and is no longer a discrete entity in the whole structure. It is not, anymore, an electric charge which is compresent with spin, mass, etc.; rather it is the electrically charged electron. An individual emerges from a structure of powers, as the structured powers cease being discrete parts in the structure and become qualifications of the individual.¹²

The view is then this: a physical structure of powers is a whole—an aggregate of many powers. When it is metaphysically unified, it becomes a single entity, which bears these powers as its own properties. The 'one' that emerges is a subject (e.g. an electron), characterized by its constituents as its qualifications (e.g. being charged). I call such unification *metaphysical unity*. The discrete constituents of a physical structure come to constitute a subject by becoming properties of the subject, and are henceforth individuated as such, namely, as 'of the subject', dependent on it, rather than as discrete entities.¹³ They are dependent on the subject for what each of them is (i.e. as the type of qualification of the subject each of them is). This holds of all types of constituents of a subject, whether abstract or concrete, particular or universal; they are not compresent in the subject, but unified as ways the subject is. In this sense, the subject is constituted of the physical structure of powers that makes it up, but acquires a new metaphysical structure, changing from being an aggregate of many powers (or other types of constituent) to being an individual subject, a substance, qualified by these powers.¹⁴

I submit that the metaphysical transition from a united physical *structure* of powers to a unified powerful *subject* is achieved by a holistic re-individuation of the constituents of the structure, in accordance with either sortal or mass individuation principles. The sortal or mass individuation principles dictate the type of entity that is individuated by the criteria they set for being that entity. Oneness of entity is generated by the dependence of the parts on the whole. Sortal and mass predicates enable science to 'carve' the world in ways that explain what happens in the world. Alternative carvings of the world deliver alternative numbers of entities in the world, and alternative

¹² Each qualification can be individuated *by abstraction* from the individual; this involves division of the individual by abstraction, which I have discussed elsewhere. [Reference omitted because the paper is under peer review.]

¹³ By 'discrete' here I mean that they are entities which can be individuated without reference to other individuals.

¹⁴ What follows is that the powers in the physical structure that constitutes the subject are not identical to the qualifications of powerfulness that come to thereby characterize the subject. The holism of the subject changes their ontological status from discrete components to its own qualifications.

explanations of what happens in the world. The ultimate alternatives are classified into two general types of ontology: extensional mereologists (such as David Lewis) see pluralities, where substantialists (myself included) see emergent unities. Where the mereologists see *structured compresence*, the substantialists see *holistic dependence*. These alternative ontologies derive from alternative individuation principles for entities in the world, and dependencies between them.

4. Structural versus Substantial Powers

The building blocks of all there is in nature are instances (tropes) of physical powers. They occur in nature in physical structures of dependencies. I have argued that such structures unite the powers together, but do not unify them metaphysically into individual entities. On the other hand, there exist a small number of physical structures of powers that are also metaphysically unified into individual entities, on account of the way their constituents are individuated. Thus, for instance, an electron is one entity, not on account of its being physically structured, but on account of the way the powers in the structure are holistically individuated into qualifications of the electron. The physical structure of the electron's powers *constitutes* the electron, but *is not* the electron. The electron is a single emergent (exercising) power, constituted of a physical structure of powers—mass, spin, charge, and space–time—which are holistically composed into one emergent entity, the electron, under the individuation principle of the sortal 'electron'. I call the structure of powers constituting an emergent entity a *structural power* that can be conceived of plurally; the electron is constituted of a structural power—the structure of mass, spin, charge, space–time. The unification of the structural power of electron into one is derivative from the metaphysical oneness of the electron. In itself, the structural power of electron is a structure of powers; it is a whole of many powers. Is it a power? Strictly, no; it is many powers; a powerful whole made of many powers. On the other hand, the single power of electron emerges from the unification of the physical structure of powers into a single individual; I call the emergent power of electron a *substantial power*. This is constituted of the structure of powers of the electron, but which are now re-individuated as qualifications of a subject according to the principle of the sortal 'electron'.

So why differentiate between a structural power and a substantial power? Isn't the substantial power of electron sufficient for understanding what type of power the electron is? The reason for differentiating them is twofold. On the one hand, the structural power is what *constitutes* the electron; whereas the substantial power *is* the electron. On the other, the differentiation between the structural power of electron and the substantial power of electron enables us to understand the relation between the *extensional mereological* conception of the electron (as a whole of many interrelated powers); and the *substantialist* conception of the electron (as an individual emergent power). The constitution relation turns the powers in the electron's physical structure into properties of the emergent substantial power. Although every structure of powers

could be thought of as a structural power, I reserve the term 'structural power' for those structures of powers which constitute substantial powers. (Thus, for instance, the winds around me are not a structural power, although they are structured.) Every science individuates its own individual subjects or substances, and hence its own substantial powers (i.e. the entities that the science needs to describe and investigate reality within its own domain of inquiry). These structural powers are holistically unified, top-down,¹⁵ according to the individuation principles of the respective concepts which figure in the scientific laws and explanations of the science in question.

Top-down unification gives rise to *downward emergence* of components. For the substantialist (and for Aristotle), the ultimate components individuated at the ground level are dependent for what they are on the whole they belong to. So even in the extreme case of a general who treats a squad of soldiers as a single superorganism, the soldiers become (for all intents and purposes in that general's army) properties of the squad-organism. In the case of a dog, the substantialist treats it as one organism constituted of the structural (causal) unitedness of its components. These same reasons do not, by contrast, incline the mereologist to treat the dog as one individual, rather than as a sum of components. For the substantialist, the individuation concepts facilitate the top-down unification of structural powers, imposing dependence relations of the powers on the emergent subject. The difference between the two ways of individuating a dog is that on both, there is the structural power of the dog. But while the work of the mereologist stops here, when she identifies the dog with the structural power, which is an aggregate of interrelated powers, the substantialist goes further; she turns structured aggregates into individuals; the structural power of dogness comes to constitute a substantial power, the dog, as a diachronic power.

The difference between structural powers and substantial powers has not been placed on the map in the history of metaphysics or in current metaphysics so far. As we have seen, on the contrary, Aristotle claimed that in substances, the *formal* cause (namely the substantial form, e.g. being a tiger) is the same as the *efficient* cause of the substance (e.g. the physically generative power of a substance; it's as if the substantial power of a substance were the same as the structural power of a substance. This has been held as an important insight of Aristotle, which has been repeated over the ages as an explanation of the physical oneness together with the metaphysical oneness of a substance. But Aristotle was (dare I say it?!) wrong to identify the formal and efficient causes of a substance. I have argued here that structural powers do not entail substantial powers, but only vice versa; and that structural powers constitute substantial powers, through the re-individuation of their components according to holistic sortal or mass principles (e.g. for dogs or lava).

¹⁵ In contrast with the bottom-up unification that structural powers bring about.

5. Conclusion

In this paper, I have examined the issue of the unity of objects from the viewpoint (which I endorse) that the ultimate constituents of reality are instances (i.e. tropes) of physical powers. Powers compose in two different ways, physically and metaphysically. When physically united they make up wholes, which are plural; when metaphysically unified they make up unities, which are one. The unification results from the sortal concepts we use to individuate structures of powers into entities, that is, single subjects that are qualified by the powers that constitute them as parts. I argued that structural powers are physically united, while substantial powers are metaphysically unified. No physical relation can unify a structure into one, but only into plural wholes. The unification into one is a further step which results from the holistic, top-down individuation of the constituents of a structural power into a single entity—downward emergence of components. Hence, whereas wholeness results from physical relations, oneness results from conceptual individuation. Conceptual individuation is justified either on scientific grounds, for the expediency of explanation, or on pragmatic grounds, for other kinds of expediency.

6. References

- Aristotle. 1995. *The Complete Works*. Princeton, NJ: Princeton University Press (6th printing).
- Jaworski, W. 2016. *Structure and the Metaphysics of Mind: How Hylomorphism Solves the Mind–Body Problem*. Oxford: Oxford University Press.
- Koons, R. C. 2014. ‘Staunch vs. Faint-hearted Hylomorphism: Toward an Aristotelian Account of Composition’, *Res Philosophica* 91: 1–27.
- Lowe, E. J. 2011. ‘A neo-Aristotelian substance ontology: neither relational nor constituent’ in *Contemporary Aristotelian Metaphysics*, ed. T. Tahko, 229–48. Cambridge: Cambridge University Press.
- Marmodoro, A. 2013. ‘Aristotle’s hylomorphism without reconditioning’, *Philosophical Inquiry*, Volume 37, Issue 1/2: 5–22.
- Marmodoro, A. forthcoming. ‘Power mereology: structural versus substantial powers’, in *Philosophical and Scientific Perspectives on Downward Causation*, ed. M. P. Paoletti and F. Orilia. New York: Routledge.
- Quine, W. V. O. 1960. *Word and Object*. Cambridge, MA: MIT Press.
- Rea, M. C. 2011. ‘Hylomorphism Reconditioned’, *Philosophical Perspectives* 25: 341–58.
- Scaltsas, T. 1994. *Substances & Universals in Aristotle’s Metaphysics*. Ithaca: Cornell University Press.
- Tahko, T. ed. 2012. *Contemporary Aristotelian Metaphysics*. Cambridge: Cambridge University Press.

PART II

Necessity and Essence

5

The Impossibility of Natural Necessity

David S. Oderberg

1. Introduction

The consensus among philosophers, it is fair to say, is that there is such a thing as physical or natural necessity, that it is distinct from metaphysical necessity, and that it is the necessity characterizing both the laws discovered by natural science and their instantiations.¹ Natural necessity, it is generally thought, is a ‘softer’ or less wide-ranging kind of necessity than logical necessity or metaphysical necessity or conceptual necessity. There are natural necessities, say that carbon has an auto-ignition temperature of 700°C or that planets move in an ellipse with a star at a focus (Kepler’s First Law), that are in no way logically guaranteed, do not flow from the very concept of carbon or of a planet, and might not obtain in some other possible worlds—including, crucially, worlds in which carbon or planets still exist.

By contrast, there is a minority view,² now usually called ‘scientific essentialist’ or sometimes ‘dispositional essentialist’, according to which natural necessity is a species of *metaphysical* necessity, the necessity that holds in virtue of the essences of things (Ellis 2001; Bird 2007; Shoemaker 1980, 1998; Bigelow, Ellis, and Lierse 1992). ‘Scientific’ essentialism tends to be associated with a belief in physical and perhaps chemical laws but a scepticism about laws in ‘messy’ fields such as biology (Ellis 2001, 170) and psychology. ‘Dispositional’ essentialism focuses on causal laws grounded in dispositions (or powers as I will call them), having little to say about the possibility of non-causal laws—laws not grounded in powers. Both kinds of metaphysical necessitarianism—necessitarianism *tout court* for my purposes—can be seen as subsets of a broader, more thoroughgoing Aristotelian necessitarianism about laws. According to Aristotelian necessitarianism, the laws of *nature* are the laws of *natures*: in other words a law is, ontologically speaking, the expression of the essences of the things to which

¹ See, for a few examples among many: Fine 2002; Lange 2009; Kment 2014; Lewis 1983; Armstrong 2010. I will use the term ‘natural necessity’ rather than ‘physical necessity’, but I implicitly take ‘physical’ in this context broadly to cover material objects in general, whether they be objects of physics as such or some other special science such as chemistry or biology.

² I do not mean to claim that these two views exhaust all those that are or could be held.

it applies. Speaking linguistically, laws describe the essential behaviour of things. On this conception, no privileged place is obligatorily granted to physics and/or chemistry over other domains, and there is no requirement that laws must all be grounded in the powers of the objects to which they apply. Geometrical and other mathematical laws, for instance, apply to abstract mathematical objects and are expressions of their natures but are not grounded in powers since purely abstract objects have none.

It is not hard to see why laws are necessary on the Aristotelian conception. If *Fs* have essential natures then they *must* behave in certain ways, so long as *Fs* exist. The descriptions of their behaviour *must* be true, so long as the description concerned entails that *Fs* exist. There is simply no room for contingency here, in the only sense that matters. If *Fs* do not exist in some world, then there will be no law that they behave in some kind of way in that world. In *that* sense, the laws are contingent if they concern any kind of object whose very existence is contingent. But this is not a sense of contingency that has any metaphysical interest. The interesting idea is that laws for *Fs* might be contingent in the sense that a law *L* for *Fs* might be true in w_1 and false in w_2 even though *Fs* exist in both worlds. Let us call this the ‘strong contingency claim.’ I submit that the central way of understanding natural necessity as something less than metaphysical necessity is as the kind of necessity laws have if the strong contingency claim is true. This is what the ‘natural necessitarian’ believes. Indeed, the supporter of natural necessity in this sense is committed to the biconditional: there is natural necessity if and only if there is strong contingency. According to Aristotelian necessitarianism, by contrast, natural necessity exists, but it is nothing other than a species of metaphysical necessity—the necessity possessed by laws for natural objects.

In what follows, I examine the strong contingency claim with particular reference to E. J. Lowe’s view of laws and natural necessity. Although Lowe developed a fairly well worked-out account of the philosophical logic of law statements,³ which will be part of my analysis, his work on the ontological status and modal force of laws never received the systematic and extended treatment the topic deserved. That said, what Lowe does claim is important and relatively clear, forming a key element in the defence of natural necessitarianism. Once we subject to scrutiny his own and related views, the case for natural necessitarianism looks at best shaky. I would go further and make the bold claim that natural necessity looks to be a sheer impossibility, perhaps the product of a myth that developed sometime during the ‘scientific revolution.’ It is beyond my task to uncover the origin of the idea, however; it is enough for me to make the cumulative case that it is an idea whose time for exit has come.

2. The Circularity Problem

Probably the most serious objection to the very idea of natural necessity is that it is impossible to define non-circularly if the definition appeals to the laws of nature. This long-standing worry is sometimes called the triviality problem, sometimes the

³ Lowe 2009, chs 9 and 11, drawing on his earlier 1980 and 1982.

circularity problem, and has been formulated in a variety of ways. It applies to any attempt to define natural necessity by relativization to worlds in which the laws of nature are the same. John Foster argues that if we construe natural necessity as a relative form of necessity obtaining in those worlds sharing the same ‘nomological organisation’ as ours, then the necessary regularities in our world will hold in all of these worlds but in a way that is ‘wholly trivial and of no help in illuminating the sense in which the regularities are necessary’ (Foster 2004, 90). Putting the problem in terms of showing how a law supports counterfactuals, Peter van Inwagen argues that in order to show that a given law does so we need to limit the worlds we assess to those in which the objects to which the law applies are physically possible; but ‘our analysis would be circular’ if we specified such worlds in terms of objects whose existence is consistent with the laws (van Inwagen 1979, 449).

The problem is a wholly general one for attempts to explain natural necessity in terms of relativization to worlds in which the laws are the same, *a fortiori* for all accounts according to which natural necessity is weaker than metaphysical necessity due to relativization to a subset of possible worlds. Take the law of reflection, that when light reflects off a mirror the angle of incidence equals the angle of reflection. Each instantiation of the law is naturally necessary; but to account for that particular necessity as being explained by the law itself is merely to invite the question of what explains the necessity of the law. We should, of course, accept that each instantiation is entailed by the law, but the relative necessity view seeks to *explain* the necessity of the former in terms of the latter. But then what explains the necessity of the law itself? Suppose, however, that we had an independent account of the necessity of the law: the natural necessitarian could avoid the triviality/circularity problem, but at what seems a big cost—having two completely different accounts of natural necessity, namely one for the laws and one for their instances. This seems wildly implausible, for surely whatever gives an instance of the law of reflection its necessity—that *this* particular beam of light, reflecting off *this* particular surface, *must* produce an angle of reflection equal to the angle of incidence—should be just the same as what gives the very law itself its necessity, and not because a law is a mere conjunction of instances but because the law in question is itself about light, reflection, and surfaces. So there must be something about the connection between these *kinds* of thing that *explains* the connection between any instances of them.

Moreover, it is highly doubtful whether any independent account of laws in the offing even preserves and explains the necessity of the laws themselves. Lange says, of David Lewis’s Best Systems Account (Lewis 1973, 73), that it ‘respects the laws’ contingency but fails to do justice to their necessity. Laws in Lewis’s picture retain nothing that we would ordinarily recognize as genuine necessity’ (Lange 2009, 46). The same goes for any Humean account of the laws, hence a trilemma for the natural necessitarian who appeals to relative necessity: either an independent account of the laws themselves explains their necessity or it does not. If it does not, then if it denies natural necessity it is of no use to the target of my critique, which is the natural necessitarian; but if it is consistent with natural necessity yet fails to explain it, it is patently inadequate. If it

does explain their necessity, then it divorces this from the necessity of the instances of the laws, resulting in an implausible dualism of natural necessity.

It is clear that Lowe supports the relative necessity analysis of natural necessity: 'My view is that there is such a thing as natural necessity . . . but that it is only a species of *relative* necessity. To say that a state of affairs is "naturally necessary" is merely to say that it is a state of affairs which must be the case given the laws of nature' (Lowe 2006, 147–8). One assumes he then takes on the circularity problem with full force, but as far as I can tell he does not. He recognizes the problem, to be sure: 'Plainly, one cannot non-circularly explain this notion of necessity in terms of truth in all worlds in which the actual laws of nature obtain, if one appeals to the very notion of necessity in question in explaining what constitutes a law' (2006, 148; see also Lowe 2009, 151). He implies that his own account avoids the problem, and states explicitly that Armstrong's theory (Armstrong 1983; 2010, ch. 5) does so as well. As for Armstrong, Lowe plausibly claims that the higher-order 'necessitation' relation between universals hardly deserves to be called necessity at all, and the natural necessity obtaining between particular states of affairs is mere relative necessity (Lowe 2006, 48–9). For my purposes, it does not matter whether Armstrong's higher-order relation is a genuine kind of necessity or not, since his theory falls to the trilemma above whatever the correct interpretation.

Of greater urgency is how, exactly, Lowe proposes to avoid the circularity objection to the relative necessity account. Although he does not say so explicitly, the idea is that by grounding laws of nature in *kinds* of thing, as his well-known theory holds, he provides the needed independent account of the necessity of the laws themselves (Lowe 1980; 1982; 2006; 2009). That a particular beam of light must reflect in accordance with the law of reflection *consists* in the entailment by the law of the particular state of affairs. That the law itself must hold consists in its being a relation between *kinds*, namely light and mirrors. This is the ontology behind Lowe's analysis of law statements as being of the form K is (K s are) F , or K is (K s are) characterized by the attribute F (Lowe 2006, 132, 133).

The problem with this way of avoiding circularity, however, is that the price is necessitarianism, by which I mean Aristotelian metaphysical necessitarianism. Now on my view this is not merely a price worth paying but a toll: you *must* pay it to avoid circularity. By grounding the necessity of the laws themselves in the necessity attaching to the *essences* of the kinds they involve, we are in the happy position of being able to account for the necessity of the instances precisely as instances of necessary laws without having to ground the necessity of the laws in the necessity of the instances. Rather, the necessity of the laws, being metaphysical, is grounded in the very natures of what they are about. It is of the *essence* of light and mirrors that they obey the law of reflection. Now this does not mean that obedience to the law of reflection is *part* of the essence of light or mirrors; that is to say, what you would include in a *real definition* of these kinds of thing. Rather, such obedience can be seen to be a *property*, what Fine calls a 'consequential essence' (Fine 1995, 5), what scholastics used to call a

proprium—a particular kind of feature that, in Locke's apt metaphor, 'flows' from the essence of a thing (Oderberg 2011).

Lowe, however, explicitly declines to pay the essentialist toll despite his appeal to kinds as an independent account of the ontology of laws. Before considering this directly, I want to highlight Lowe's opinion that if we hold the laws to be metaphysically necessary, we consider them to be true in all possible worlds (Lowe 2006, 142). This, he says, is unacceptable because 'many philosophers have strong intuitions that natural laws are *not* necessary in the strongest possible sense—that a natural law which obtains in this, the actual world, need not obtain in every possible world' (Lowe 2006, 142). Yet it is no part of the necessitarian position—nor should it be—that natural laws hold in every possible world. The necessitarian thesis is that, *given the existence* in any world *w* of the objects that obey a given law in the actual world, they also obey the same law in *w*. The necessitarian need not and should not deny the possibility of alien worlds in which things very similar to, but not of the same essence as, say, light and mirrors do not obey the law of reflection. Moreover, albeit less importantly, in worlds in which either light or mirrors do not exist, the necessitarian can happily accept either that the law of reflection is vacuously true *or* that there is no such law, or maybe even that the law is false or that it exists but has no truth value. Such questions are probably best settled on semantic grounds. For the necessitarian, the key idea is that in all worlds in which light and mirrors exist, light obeys the law of reflection. Now Lowe had earlier said, 'I believe that water—that very substance—could, very arguably, exist in a possible world in which there was not a natural law that water dissolves common salt' (Lowe 2006, 132). So it seems that this is what he means by his later objection to necessitarianism, but the potential for confusion is significant.

3. Kinds Are Not Enough

Lowe, then, explicitly declines to appeal to essences in the metaphysical grounding of natural necessity; and if he had, as I have argued, he would have been committed to Aristotelian necessitarianism about laws. He does not eschew the appeal to essence because he is anti-essentialist, but because he does not consider it to be 'part of the essence' of water that it dissolves salt. The essence of water, he says, is (in part) its chemical composition, but 'the laws governing hydrogen and oxygen atoms are quite conceivably different in different possible worlds' (Lowe 2006, 132–3).⁴

Now Lowe is aware of Alexander Bird's important challenge to the claim that salt contingently dissolves in water (Bird 2001), but his response is relatively desultory (Lowe 2006, 165–6). Recall, to put it as concisely as possible, that Bird argues for the

⁴ Note that Lowe holds chemical composition to be 'part of the essence' of water. He does not say in his (2006) what the other part or parts are. However, in his (2011) he reverses himself and denies that chemical composition is *any* part of the essence of water, which he holds to consist solely of 'macroscopic, observable' features (2011, 18). This change of view does not affect my argument, and evaluating Lowe's later position is neither possible nor necessary here.

necessity of salt's dissolution in water on the ground that the very law, Coulomb's Law, in virtue of which dissolution can occur, is required for the very existence of salt in the first place. There cannot be a world in which salt fails to dissolve but still exists, given the supposed failure of Coulomb's Law. There are two parts to Lowe's response to Bird. The first is that Bird's argument is speculative. Although a fair point, it does not establish much. On its face, as Lowe recognizes, the argument may well be correct. Moreover, whatever the truth of the scientific details, it gives the necessitarian a *template* for advancing further such arguments in other cases, as the basic metaphysical framework is itself relatively clear and plausible. Secondly, Lowe claims that Bird's argument, even if sound for Coulomb's Law and hence the specific case of salt and water, *presupposes* that the laws of physics (rather than chemistry) are themselves *contingent*. This is because the argument relies on there being electrons governed by different electrostatic laws in other possible worlds, worlds in which salt does not exist and ipso facto does not dissolve in water. The chemical law of dissolution might not be contingent on Bird's argument, says Lowe, but the laws governing electric charge have to be contingent for the argument to work.

In reply to Lowe, the argument does not *presuppose* the contingency of any electrostatic laws; it *assumes* it for the sake of argument. If we assume that salt both exists in a world and fails to dissolve in water in that world, we must also assume that Coulomb's law is both true and false in that world. But this is logically impossible, so there is no such world. The only relevant possible worlds are ones in which either salt does not exist, or salt does exist and dissolves in water. In the latter, the electrostatic laws are the same as in the actual world. But we have no reason to infer further, from Bird's argument, that in the former—a world where salt does not exist—there are still electrons governed by different electrostatic laws. In such a world, there would be no electrons at all: such is the only position a necessitarian should hold. Moreover, it is wholly consistent with Bird's argument, albeit not forming *part* of that argument.

Be that as it may, and whether or not Bird's argument is sound (which I think it is), the first thing to note is that had Lowe distinguished between essences and properties this might to some extent have assuaged his worries about dissolution's being 'part of the essence' of water. Indeed, the category of *propria* is conspicuously absent from his four-category ontology, which recognizes only 'attributes'—what an Aristotelian would call *accidents*, and which for Lowe are universals without further qualification bar one. The qualification is that if an attribute characterizes a *kind* (salt dissolves in water), the characterization is dispositional, whereas if this is not the case but instead the attribute merely⁵ has a mode (trope) that characterizes a particular (Fred is walking), the characterization is occurrent.

Had Lowe recognized properties in the Aristotelian sense—a special category of universal whose instantiation flows from, or is expressed by, the essence of the kind

⁵ 'Merely' in the sense that the particular does not have the attribute *because* the former instantiates a kind that is characterized by the attribute.

whose instances are characterized by its modes—he would have been able to say the following of light (to return to my example). He could have said that light obeys the law of reflection, and necessarily so, even though such behaviour is no part of the essence of light *in the sense* that it would not figure in a real definition of light as a kind of electromagnetic radiation within a given portion of the electromagnetic spectrum.⁶ But his ontological scheme does not have the resources to allow such a claim. Given that it distinguishes only between dispositional and occurrent characterizations, it falls prey to its own internal problem of how to characterize kinds in the right way—the way that yields extensional equivalence with the laws—without *presupposing* essences in the first place. Suppose there is a kind *K* characterized by an attribute *F* but where *F* has not *always* characterized *K*. (Think of heritable but relatively insignificant⁷ mutations in biology.) Is it a law that *K*s are *F*? Presumably Lowe would say no, on the grounds that *F* does not truly *characterize* the kind *K*. But why not? It must be because there have been *K*s that existed without being *F*. Why, though, should that matter? Why isn't it *now* a law that *K*s are *F*? After all, for Lowe laws are contingent. One way they could have been different is by *actually* varying over time. Yet suppose that the *K*s could, at some future time, *lose* the attribute of being *F*. Do we say that the law, which came into existence when *F*ness began to characterize the *K*s, would again cease to exist? We can iterate the problem to the point of absurdity: why even think that such characterizations amount to laws? (I will say more later about variation over time.)

The only way, it seems, for Lowe to resist this absurdity is to deny at the outset that in the case of the *K*s that become *F*, we have a true characterization in his technical sense. Yet the only possible ground on which he can base the denial is that for an attribute genuinely to *characterize* a kind, it must be true of all possible instances of the kind—subject to a normality clause that might apply in certain domains such as biology (Lowe 2009, 150). Since, in our example, there are possible *K*s—namely, the actual past ones—that are not *F*, Lowe can decline to regard *F*ness as characterizing *K*s. But to insist that characterization of a kind requires attribution to all possible members of the kind is precisely to rely on essentialism: we do not have true characterization of a kind, and hence a genuine law, unless it is grounded in the *essence* of the kind, not merely in the kind itself. And if the characterization is grounded in the essence of the kind, thus generating the law, then the kind cannot exist, given its essence, without the law's being true; which is just what the necessitarian claims. Perhaps Lowe could have produced another way of explaining what characterization in his technical sense amounts to—one that doesn't rely on essentialism—but it is hard to know what it could be and, more importantly, it would have to enable us to distinguish between characterization, albeit temporary, and merely universal attributes that are in no way law-generating. (Compare, say, the temporary hexadactyly of all humans due to a mutation and the temporary

⁶ Nothing, of course, hangs on this definition, which is in any case incomplete since the range is not specified. Maybe the real definition is quantum-theoretical, or something else.

⁷ That is, not significant enough to warrant the conclusion that there has been speciation.

sickness of all humans due to a pandemic. It would take a very subtle account of characterization in Lowe's sense to explain how the former might come out as a contingent law while the latter came out as an accidental universal generalization.)

In short, kinds are not enough to ground laws; we need *essential* kinds. But if we appeal to essential kinds, ipso facto we are grounding laws—explaining their ontological status—in terms of essentialism. And if laws track essences, laws must be metaphysically necessary, in the only sense Aristotelian necessitarians ever meant or should have meant—that if something is a law, and it concerns (i.e. is a law about) certain kinds of thing, then in every world in which those kinds of thing exist, that law is true.

4. An Unstable Position

Lowe strongly supports the contingency of some laws of nature, although he thinks other laws are metaphysically necessary, such as what he calls the 'non-causal law' that electrons have unit negative charge (Lowe 2006, 145). Since having unit negative charge is part of the essence of electrons, electrons cannot exist without having that attribute; the same idea, presumably, applies to any kind with an essential attribute.⁸ In which case, Lowe should hold it to be a law that cows are mammals or that oak trees have trunks. We ought to find 'law talk' odd here, if for no other than methodological reasons. Laws are the sorts of thing scientists establish by a combination of experiment and postulation in order to explain the data experiments yield. Bare observation might be the simplest kind of experiment, but it is stretching credulity to suppose that some large number of these, combined with some hypothesizing, yielded the conclusion that it was a *law* that cows are mammals or that oak trees have trunks. Indeed, experiment showed that electrons have unit negative charge, but did experiment show this to be a *law*? It takes some mix of common sense and philosophical thinking (even by scientists themselves) to work out that it is part of the *essence* of oak trees that they have trunks; no botanist ever discovered it. For this reason alone we should not call such truths *laws*.

Metaphysically, moreover, there is no gain in understanding from calling essentialist truths *laws*, where by 'essentialist truths' I mean truths about what is part of the real definition of a thing—what the thing *is*. Laws have a *relational* character: they connect things to their behaviour; qualities or quantities to other qualities or quantities; kinds of object to other kinds of object. These are *real* relations, by which I mean something like the following: for the domain in question, there is a domain-involving *process* or *traceable connection* between the law's relata. It is a law that electrons attract protons, and we can trace a process whereby this happens using quantum field theory. It is a law that plant metabolism requires light, which is underwritten by the process of

⁸ Tahko (2015, 523) says that, for Lowe, 'the simpler the kind in question, the more likely it is that all or most of its attributes are essential to it', which is supposed to explain Lowe's use of an example from subatomic physics; but there is no reason why examples should, on Lowe's position, be limited to these.

photosynthesis. Even mathematical laws, like the simple commutative law of arithmetic, involve a real relation between two kinds of thing (ordered addition), within which we can trace by the process of addition why the law holds.

I fully accept that the term 'law' is used promiscuously to cover many things this particular (and admittedly sketchy) view of laws would not admit, but it is not clear to me what clarity the promiscuity provides. We should hold distinct the real definitions of things from the ways in which things behave, or operate, or are connected to other things. There is no real relation between an electron and its charge, or a cow and its mammalian nature, no process or connection to trace or comprehend. Electrons just *are* and *must* be unit negatively charged, and cows are and must be mammals. History is, of course, littered with erroneous real definitions—mistaken conceptions of what things really are. But all that demonstrates is that essentialists must be *fallibilists*, using a combination of common sense, science and/or philosophical reflection to reach correct real definitions.

Given this approach, Lowe is wrong to conceive of what are actually truths given by real definitions as metaphysically necessary laws: for these truths are not laws at all. There remain, however, the truths he takes to be contingent laws, and here I think he is again mistaken. Lowe appears to adhere to a 'governing' conception of the contingent natural laws, according to which they determine kinds of object to behave in certain ways, but where the kinds themselves are ontologically independent of those laws in the sense that the identity of those kinds does not depend on the laws in which they figure.⁹ So, as we have already seen, Lowe believes that 'water—that very substance—could, very arguably, exist in a possible world in which there was not a natural law that water dissolves common salt' (Lowe 2006, 132), and a bit later: 'I consider that the laws governing hydrogen and oxygen atoms are quite conceivably different in different possible worlds' (Lowe 2006, 132–3).

Yet how, one might wonder, can laws 'govern' kinds, on Lowe's view, given that he also thinks the 'lawlike connection' to be no more than the 'familiar characterizing or predicative tie' between universals—that is, kinds and attributes (Lowe 2006, 132)? Either the laws govern kinds or they are *immanent* to kinds. If the latter, and—as I have argued—kinds are not enough, they must be immanent to the *essential* kinds of things; in other words, what 'ties' kinds to attributes in a law-like way is part of the very essence of those kinds, without which tie those kinds would be something other than what they are. If, on the other hand, the laws 'govern' the kinds, in what way do they do so? Other than interpreting governance in terms of objects' *obeying* their

⁹ This is not quite the governing conception usually discussed in the literature (Loewer 1996; Beebe 2000, for example) but it is close. The usual contrast is between the anti-Humean, for whom the laws metaphysically determine certain kinds of behaviour (e.g. certain privileged kinds of regularity) and the Humean, for whom the laws are mere descriptions of certain kinds of behaviour. For the anti-Humean, laws determine, in a metaphysically strong sense, objects' future behaviour; for the Humean, they merely logically entail their future behaviour. On my usage of 'governing', laws determine the behaviour of kinds whose identity does not depend on those laws, and this is implied by the usual governing conception.

essences, something they *must* do, metaphysically, for them to be the kinds of object they are, it is hard to know what governance can amount to. There is, of course, a strain of theological voluntarism, represented for example in Calvinism, which sees the laws of nature as *nothing but* the ordinances of God *imposed* upon nature (Witte and Alexander 2007, 76, quoting the Calvinist theologian Abraham Kuyper (1837–1920)). Again, one can hold a Platonist view of laws according to which the law-like relation between transcendent universals somehow governs that between their instances (such is the view of Tooley 1987). Lowe is no friend of either approach, nor is either attractive in its own right.

In sum, Lowe appears to be left in the following unstable position. There are laws that are metaphysically necessary, such as that electrons have unit negative charge, because it is part of the essence of electrons to be so characterized. This must be an *immanent* fact about electrons; that is, embedded in their very nature. But there are also metaphysically contingent laws, such as those ‘governing’ hydrogen and oxygen atoms, whereby kinds of object governed by them in some worlds exist in other worlds in which they are not so governed. These contingent laws cannot be immanent to the natures of what they govern: if they were, they would not be contingent. A less strong position, such as governance in terms of characterization of kinds by attributes, will not do since kinds, as we saw, are not enough to account for laws as a distinctive class of truths. There is no apparent and plausible alternative interpretation of governance in the offing; moreover, a view combining both a governing conception of contingent laws and an immanentist conception of necessary laws introduces another kind of unappealing dualism into the ontology of laws.

5. Against Contingency

We still have not seen *why* Lowe thinks that some laws are metaphysically contingent. To say that the contingent laws are not essentialist truths will not do, since (a) we have seen that truths about essences in the strict sense, such as that electrons have negative unit charge or that cows are mammals, are not laws at all, and (b) there might still be room for essentialist truths in the non-strict sense, such as that light obeys the law of reflection or planets travel in an ellipse with a star at a focus.¹⁰ Now (a) and (b) are precisely the position of the Aristotelian, for whom laws are ties between the essences of things and the way they operate (in the broadest sense). So what does Lowe have to say?

It seems that Lowe rests the entirety of his case for metaphysical contingency on the conceivability of worlds in which certain essential kinds exist that also exist in the actual world, but in which there is a relatively small change in a law governing the kinds. This might be due to a slight variation in a constant figuring in the law, such as the gravitational constant *G* in Newton’s law of universal gravitation, or a variation in some other

¹⁰ In other words, given (b) it would not be enough to hold that the laws divided into the strict essentialist truths about what things *are* (necessarily true) and all the rest (contingently true).

element, say from the square of the distance to some small, non-integral power. It would, he thinks, be highly implausible to insist that such inter-world variations entail that mass as it is in the actual world does not exist in those differing worlds. Moreover, it is epistemically possible that at least some of the very things we call 'constants of nature' vary over time in the *actual* world. Perhaps *G* is one of them; yet if this were discovered, we should not say that mass as it existed in the past when *G* had a given value is not the same property as mass existing now when *G* has some other value (Lowe 2006, 151).

Nor is Lowe without company in this view of the variability of the laws. Kit Fine, perhaps somewhat surprisingly, also appeals to the possible 'inverse cube law' against the metaphysical necessity of the actual laws.¹¹ He pictures a situation in which mass obeys the inverse square law in the actual world but 'schmass' obeys the inverse cube law in another world, *w*. Fine's clever move is to argue that the necessitarian is then committed to holding that the proposition that there is no schmass (call it *S*) is metaphysically contingent but naturally necessary. It is metaphysically contingent because the very metaphysical possibility of schmass is what enables the necessitarian to evade the putative counterexample involving the inverse cube law: for the necessitarian, in response to this alleged counterexample to their position, holds that there is no mass in a world governed by the inverse cube law, only some other kind of thing—call it schmass. The inverse square law remains metaphysically necessary since it governs mass, not schmass: it is true in all worlds in which there is mass and in no other worlds.¹² Yet *S*, adds Fine, is also naturally necessary. His reason: 'For our original judgment was that the inverse square law was a natural necessity, though not a metaphysical necessity. Now that we see that the metaphysically possible worlds in which it was taken to fail are ones with schmass rather than mass (and given that our universe is taken to be completely governed by the Newtonian Laws), we should take it to be a natural necessity that there is no schmass' (Fine 2002, 239–40).

I confess to finding this passage rather unconvincing. To the extent there is within it an argument at all, it is that *S* is a natural necessity because the inverse square law is a natural necessity. Yet why should we accept that the natural necessity of the inverse square law entails the natural necessity of *S*? It cannot be because whatever is entailed by a natural necessity is itself a natural necessity, for then the proposition that $2 + 2 = 4$ would be a natural necessity. Moreover, the natural necessity of the inverse square law entails the disjunction of the natural necessity of the inverse square law and the natural necessity of the inverse cube law, but that disjunction is not a natural necessity: it was no more a scientific discovery than was the proposition that $2 + 2 = 4$.

¹¹ Fine 2002, 238–9. He himself admits that on his neo-Aristotelian view of necessity (Fine 1994), '[n]atural necessities would simply be the special case of those essentialist truths that arise from the identity of natural kinds' (239). Yet he argues in Fine 2002 that natural necessity is *sui generis* and that at least some natural necessities are metaphysical contingencies.

¹² I leave aside whether there might be worlds containing mass-like stuff that is also governed by the inverse square law, for this is irrelevant to the main point, which is the denial by the necessitarian that *mass* can exist in a world in which *it* is not governed by the inverse square law.

Perhaps Fine's implicit assumption is that when a natural necessity N entails a proposition p , p is itself naturally necessary when it concerns those kinds governed by N either explicitly or implicitly. Now S does not concern mass explicitly; it only concerns schmass. But one might think it *implicitly* concerns mass, since S says that there is no schmass and we are all assuming, for simplicity's sake, that the existence of mass and the existence of schmass are the only metaphysical possibilities. Moreover, we are assuming, as Fine does explicitly, that our world is 'completely governed by the Newtonian Laws' (Fine 2002, 240), so the existence of mass and the existence of schmass in our world (indeed any world) are mutually exclusive. Yet this does not make Fine's argument any more plausible. Assuming immanentism about kinds (which seems implicit in Fine and is a given for the Aristotelian), that salt dissolves in water (of natural necessity) entails that there is water, but the existence of water is hardly a natural necessity. If it is, who discovered it?¹³ Similarly, who discovered that there is no schmass?

Now although Fine would deny the following proposal, since he holds S to be a 'different natural necessity' from the inverse square law (Fine 2002, 239; emphasis in original), perhaps one might advance on his behalf (and as a defence of Lowe's strong contingentism) the thought that S is logically equivalent to, maybe even just means, 'there is mass' (assuming, again, mutual exclusivity for mass and schmass as well as no other alternative). But who, exactly, discovered that there is mass?¹⁴ Not Newton, who discovered that mass is governed by the inverse square law—which is *not* the same as discovering that there is mass.¹⁵ That there is mass does not look like a natural necessity.¹⁶

Assume, on the other hand, that S is indeed naturally necessary and metaphysically contingent, just as Fine asserts. It is still no threat at all to necessitarianism because it does *not* entail the strong contingency thesis, namely that there can be an essential kind governed by one law in the actual world and a contradictory one in another, for example that there can be a world containing *mass* that is not governed by the inverse square law. All it entails is that there can be a world containing *schmass* that is not governed by the inverse square law; and this would be merely an example of *weak* contingency. So even if it were the case, as Fine thinks (and Lowe might have agreed) that S is naturally necessary, this would not remotely affect the thesis that the laws of nature are metaphysically necessary—true in all those worlds in which the essential kinds they are about exist.

¹³ Not 'who discovered water?' but 'who discovered the (putative) natural necessity that there is water?'

¹⁴ Let alone 'who discovered the (putative) natural necessity that there is mass?'

¹⁵ Note: not every scientific discovery must be a natural necessity, but every natural necessity has to be a scientific discovery—and that includes kitchen-table science and amateur sleuth science as much as white-coat science.

¹⁶ The related thought that S is just logically equivalent to 'the inverse square law is naturally necessary and metaphysically contingent', even though not strictly synonymous with it, is even less promising. It is question begging to say that S is naturally necessary but metaphysically contingent in virtue of being logically equivalent to 'the inverse square law is naturally necessary and metaphysically contingent', because the latter is precisely what necessitarianism denies.

Putting Fine's key argument to one side, what should we say about the variability of the so-called constants of nature? Needless to say, a constant that varied would not be a constant, so we would need another word. Let us call a quantity (in particular, one thought erroneously to be a constant) that varies within a world a *variant*. And call a quantity that can vary across worlds a *fluctuable*.¹⁷ Should the existence of variants or fluctuables make necessitarians worry about their position?

Let's start with fluctuables. The power in the law of universal gravitation that varied from a square in the actual world to a cube in *w* is a case in point. We are invited by the strong contingentist to think, as Lowe does, that we would still be dealing with *mass* in *w*, and this because the *only* thing varying was the power of the distance; the equation in each world is exactly the same but for that fluctuable. That this is the motivating idea should be evident from what we would naturally think were there to be nothing remotely like Newton's law in *w*, rather a wholly different equation governing attraction between whatever bodies existed in that world—say, one relating the volumes of two bodies by addition and multiplying the sum by the tenth root of their average temperature. Call this alien equation *A*. Would anyone, contingentist or not, be remotely tempted to think that *w* contained *mass* rather than *schmass*—some other weird kind of stuff, perhaps similar to mass in some respects¹⁸ but essentially different? Well, maybe not in advance of what we might discover, but should we foreclose on what physics might have to say about such a case? Perhaps some ingenious reasoning could show us that the putative *schmass* in *w* was mass after all? Necessitarians should reply that this is precisely what they deny on metaphysical grounds, with the sort of confidence with which they deny that physics could ever show us how an electron might transmute into a butterfly.¹⁹

The contingentist, then, is trading on the close similarity between Newton's Law in the actual world and the inverse cube law in *w*, as though the difference—between the square of the distance and the cube of the distance—were a mere matter of degree. After all, it's just a small change of number! But small differences of number, as we know, can make all the difference in the world to the way things behave.²⁰ We can be fairly certain that were mass to obey an inverse cube law²¹ there would be no stable,

¹⁷ Unfortunately, 'variable' is already taken.

¹⁸ Including, of course, the respect of manifesting mutual attraction just as massive objects do in the actual world.

¹⁹ It might be objected that this is too quick. After all, hasn't the contingentist *stipulated*, and hasn't the necessitarian accepted, that the inverse cube law in *w* involves mass? The law is, precisely, $F = Gm_1m_2/r^3$. In reply, it is *this* move that is too quick, just as it would be sleight of hand for the necessitarian to hold that the inverse cube law just involves *schmass* because we should really hold it to be $F = Gm_1^*m_2^*/r^3$. We simply cannot say, in advance of further argument, whether the variables $\{m_1, m_2\}$ in the first formulation of the inverse cube law range over values for the same kind of property as $\{m_1^*, m_2^*\}$ in the second formulation or whether the variables in either of them range over values for the same kind of property as $\{m_1, m_2\}$ in Newton's Law itself.

²⁰ Hence the need for scepticism over claims that a tiny percentage difference in coding DNA between phenotypically distinct species reveals a more significant underlying similarity: 1 or 2 per cent can make a huge difference.

²¹ We should not get hung up about metaphysically impossible antecedents in counterfactuals. They are not a problem for necessitarianism any more than for someone who reasons plausibly that if they

closed orbits and the planets would spiral into the sun.²² More important, though, is the question of what it is about mass that makes it subject to an inverse square law. The answer seems to be, as many physicists from Dumbleton in the fourteenth century to Hooke and Newton in the seventeenth surmised, that gravity as a force emanating from a point source (idealized) spreads over a larger area the further it is from the source, with the area describing a sphere, whose surface area is $4\pi r^2$. If the force decreases in concentration the further it spreads from the source, the force is weaker as the area is larger, and the area increases with r^2 . So as r^2 increases the force decreases, which is an inverse relationship.²³

What this shows, if correct, is that the inverse square relationship is explained in terms of the essence of massive objects as sources of central force. It is because we can idealize them as point sources, and because we think of gravitational force as emanating from those point sources in a spherical, radiating fashion, that we are able to explain the empirical phenomenon embodied in the inverse square relationship between force and distance. Varying the power of the distance would require us to conceive of mass in a completely different way.

Suppose, on the other hand, that this explanation of the inverse square law is wrong. Suppose, that is, that in w mass were in fact governed by an inverse cube law. Would that compel the necessitarian to abandon the view that the inverse square law was necessary? Not at all, because the most natural move to make would be to hold that *if* gravitational behaviour were closely tied (in a way we don't perhaps fully understand) to what mass is in its very nature, and *if* mass could be governed by an inverse cube law in w , then what existed in w would still be mass, only *generically* so. That is, the kind *mass* would turn out to be a *genus* rather than a species, and what existed in w would be merely a different species of the same genus. And *that* is a schmass that any Aristotelian necessitarian can take on board—objects that are specifically schmassive but generically massive. In which case, what exists in the actual world would also be mass, only a different species of mass for which we have no name. (Call it $\text{mass}_@$.) We know that what is a law for the species need not be a law for the genus: otherwise, for example, the laws governing springs would be the same as those governing fluids, yet both are species of physical object. We should expect underlying basic laws governing the genus, of course, but then so should we, in the case of generic mass, specific schmass in w and specific $\text{mass}_@$ in the actual world, expect there to be a more basic law for the genus from which both the inverse square and inverse cube laws could be derived.

could go back in time—assuming time travel to be metaphysically impossible, as I do—they would choose a different career.

²² Such is the opinion of Sagan (2006, 55). It follows from Bertrand's Theorem that as a central force, gravity must depend on the square of the distance in order for the planets to have closed orbits (Goldstein 1980, 93).

²³ French 1971, 255. For simplicity I leave out consideration of general relativity and whether it provides a more basic explanation of the same phenomenon.

The upshot is that the inverse square/inverse cube putative counterexample to necessitarianism does not work. For the necessitarian, it provides no more than a template by which all supposed cases of fluctuables should be evaluated. We should expect the cross-world fluctuation of a this-worldly constant to result in radically different behaviour as between what the law requires in the actual world and what the analogous law requires in the possible world, sufficient for the necessitarian to insist plausibly that the laws govern essentially different kinds of object. If that expectation is not fulfilled, we should expect the behaviour to be different enough across worlds to underwrite the thesis of different laws governing different species of a common genus. This would be the case if, for example, there were an independently specifiable definition of generic mass as pure quantity of matter with some measure independent of appeal to the laws at issue. (Perhaps this can be done for mass but I leave that task to one side.) Needless to say, in all putative counterexamples to necessitarianism of the inverse-cube sort, the devil will be in the detail. While I do not think, for the reasons given, the necessitarian need be worried by fluctuables, I do not pretend he can avoid the need to evaluate cases on their individual merits. It is up to the contingentist, however, to come up with such cases.

6. No Room for a Hybrid View

The same line of argument applies to variants—constants that vary over time. Lowe asserts: ‘Indeed, we cannot even treat as sacrosanct the idea that the so-called “constants” of nature, such as G [the gravitational constant], really are *constant over time*. But if G were discovered to be gradually increasing over time, would it not be absurd to conclude that the property of mass which exists now is not the same property of “mass” which existed at some time in the past?’²⁴

This idea is picked up by Tahko, who takes the possible existence of variants (perhaps the fine-structure constant²⁵) to constitute ‘at least *prima facie* evidence for variation over metaphysically possible worlds’ (Tahko 2015, 520). Tahko embeds this idea within a ‘hybrid’ view of laws according to which some are necessary and others contingent. Here he agrees with Lowe, but where he differs is in holding that contingency may well derive from the fact that some, maybe most, laws do not ‘feature’ natural kinds at all, and so will not constitute essentialist truths about natural kinds. Since the fine-structure constant is involved in all laws concerning quantum electrodynamics,

²⁴ Lowe (2006, 151–2; emphasis in original), citing Milne’s theory of ‘kinematic relativity’ proposing temporal variation of the gravitational constant, which theory inspired Dirac’s cosmological work in the late 1930s (Kragh 2002, 529).

²⁵ The fine-structure constant characterizes the electromagnetic force between charged elementary particles and photons. Some physicists think there is evidence at least consistent with its variation over time and/or space: Webb, Flambaum, et al. 1999 (time); Webb, King, et al. 2011 (space). Tahko mistakenly cites Webb, King, et al. 2011 as evidence for time variation and gives the title as ‘Evidence for...’ rather than ‘Indications of...’, the former being the slightly more confident title given by the authors in their v.1 arXiv preprint at <https://arxiv.org/abs/1008.3907v1>, which Tahko cites (last accessed 16.5.18).

it does not look plausible to hold that there is a natural kind or class of natural kinds (*charged fundamental particle* does not seem to be one) underwriting the many laws that, due to the possible variance of the fine-structure constant, may well prove to be contingent.

Yet Tahko's case for contingency based on possible variation in the fine-structure constant (or perhaps other fundamental constants) is weak. He thinks that if there were variation over time this would be evidence even that there could be the same constants that in different worlds had different values, perhaps even 'radically different values', and did not 'give rise to the same laws' (Tahko 2015, 523). The use of the term 'constant' here is itself highly misleading. Nothing that varies is a constant. If what we call the 'fine-structure constant' (FSC) turns out to vary in this world, it is not a constant at all but, in my terminology, a variant—in effect, a physical variable much like mass or length, albeit perhaps constrained within a certain range. Why, then, mightn't a variant FSC simply be treated by physicists as akin to other physical variables, yielding different solutions for the same laws at different times or places? On the other hand, variation in the FSC might mean that what our best science currently takes to be laws governing the entire universe are more like 'local by-laws'²⁶ needing explanation by a deeper set of universal laws. In which case, what we took to be a single essential kind across the universe, such as the electron, would for the necessitarian be a genus containing as-yet-unknown species: the genus would be governed by the universal laws, the species by the 'local by-laws'. Yet what if there were local variation all the way down? Even if such a radical scenario as this were the case, the necessitarian would have to deny that either there was any single species (e.g., electrons) governed by fundamental laws or any plurality of congeneric species (e.g., species of electron) governed by those laws. Of course, such a view would be highly at odds with the current state of physics; but then so is the scenario itself.

Now Tahko would likely reply, as he does to Lange's rather less appealing 'eternal but time-dependent law' proposal,²⁷ that my approach is ad hoc. Why, though, should we

²⁶ As Webb himself puts it at http://www.science20.com/news_articles/if_finestructure_constant_varies_then_laws_physics_throughout_universe_do_too [last accessed 18.5.18].

²⁷ Lange 2008. It is less appealing than my approach inasmuch as Lange's idea seems to be that the same species of entity can be governed by an eternal, heterogeneous, time-dependent law. His example is an eternal law governing the repulsion force between electrons, where the force varies according as it occurs before or after a certain time in the age of the universe. Now this cannot be a *wholly general* solution to the problem of variants. It may be that, in a specific case such as the FSC, 'variation ... over time is part of what is in fact metaphysically necessary about the laws that it influences' (Tahko 2015, 521), but we cannot *assume* that this has to be the case. Presumably Lange's proposal would countenance an eternal time-dependent gravitational law varying as between an inverse square and an inverse cube, or between an inverse square and something even more radical such as my alien equation A. But we cannot know *in advance* whether there was a deeper explanation of the heterogeneity in the eternal, time-dependent law. Maybe if Lange's heterogeneous electron law obtained, we could find out something new about electrons—that *very same species of particle*—that explained the heterogeneity in terms of something fundamentally uniform about the nature of electrons. It might, however, also be that the heterogeneity indicated that we were dealing with species of a common genus—two different *kinds* of electrons. The more heterogeneous the variance, the more radical the ontological break underlying it, as far as necessitarianism is concerned. If A were an element

think this? For Tahko, it would be evidence of ‘a previous (ad hoc) commitment to the metaphysical necessity of the laws’ (Tahko 2015, 521). Well, if he means that the proposal is consistent with necessitarianism, that is fair enough: it is what you would expect the necessitarian to think (at least if they held Aristotelian sympathies), but this does not make it ad hoc. It does not introduce any radically new metaphysical idea, even though the physics underlying a given case might be radical.

To reinforce the point on behalf of the necessitarian, consider Tahko’s claim (held implicitly by Lowe) that variation over time would be evidence of modal variation. Yet why should we think this? A physical quantity may vary over time for reasons to do with the natures of the objects with which the quantity is related. Indeed, that’s the kind of explanation we should expect for the existence of variants. But it gives us no reason to think that in some other world the quantity had any particular value along the dimension of variation within the actual world. Perhaps variance in the actual world depends on the precise *life cycle*²⁸ of the objects concerned, such that the quantity only had a certain value in *w* if the objects in *w* were at the same stage of their life cycle as in the actual world. Or perhaps variance, if radical enough, indicated a transformation in ontological furniture from one congeneric species to another; in which case *w* would only contain the relevant quantity with the relevant value if one of those species, but not the other, existed at *w*. Again, if the variance were as radical as could be, then *w* would only contain that quantity with that value if, of a pair of species in the actual world that had no common genus, only one of those species existed in *w*. Not one of these scenarios contravenes necessitarianism; all are precisely what a necessitarian of Aristotelian leanings should say about the consequences of variation in putative ‘constants’. In general terms—the terms in which Lowe and Tahko speak—mere variants do not constitute even *prima facie* evidence of modal variation. Whether we have evidence in a *specific* case depends, quite obviously, on the specifics. As always at the interface between science and metaphysics, the devil is in the details. The Aristotelian necessitarian, moreover, has all the resources needed to interpret the details in a satisfying way.

Can contingentists hang their position on the idea that contingent laws are not grounded in, or explained by, or ‘feature’ (Tahko 2015, 517) essential kinds? One might appeal to the conservation laws as a case in point.²⁹ Only a few remarks are possible here; adequate discussion would require lengthy examination of various particular conservation laws. Without naming any of these, Tahko suggests that the kind *physical system* is all that either the necessitarian or the Lowe-style contingentist can appeal to in order to forge a connection between kinds and conservation laws. The implication is that *physical system* will not do, though we are not told why. One problem is that if

of a weird, supposedly eternal, time-dependent law, we would be justified in holding this to indicate the existence of two radically different kinds of object, perhaps with no generic overlap at all.

²⁸ Where by ‘life’ I do not mean anything essentially organic, of course.

²⁹ Tahko 2015, 517; Chalmers 1999 and 2006, 175–7. Chalmers is cited with approval by Fine 2001, 242, n. 13.

physical systems are defined as those obeying the laws of physics, then since the conservation laws are laws of physics we will have argued in a circle rather than giving a proper grounding of the conservation laws in the kind. This does not seem a fatal objection,³⁰ but suppose it is. And suppose the scientific essentialist proposal fails, namely that *world* is a natural kind, with a world such as ours, governed by the conservation laws, being one of that kind.³¹

Take, then, the conservation of energy: the total energy of an isolated system remains constant over time (Serway and Jewett 2008, 216). It seems on the face of it that this fundamental law governs *energy*, which looks as much like a natural kind as anything does. That is why stating the conservation of energy in explicit terms, rather than in terms of the impossibility of constructing a perpetual motion machine of the first kind, is metaphysically transparent.³² Now Feynman may be correct when he says that ‘in physics today, we have no knowledge of what energy *is*’ (Feynman, Leighton, and Sands 1964, 4–2) if he means that we do not have a complete definition of it, do not know all of its properties, or some such. But we know enough about energy to be certain that it is some *kind* of stuff—where ‘stuff’ is conceived as broadly as both physics and metaphysics will allow—that is fundamentally responsible for work, change, and causation. (Never mind whether any of these concepts are reducible to any of the others.) It has, as an essential property, its conservation, but to *define* it in terms of conservation would be as faulty as defining salt as something that dissolves in water, even if salt were the *only* thing that dissolved in water. The sub-kinds of energy—kinetic, potential, gravitational, thermal, and so on—do seem to be species of a common genus, as they all explain work, change, and/or causation in the local areas, or at the local levels, to which they apply. It would be surprising indeed were there not something *about energy as a kind* in virtue of which its conservation was a necessity.

Clearly a lot more needs to be said about particular cases, but the least we can be certain of is that conservation laws—assuming conservation of energy to be a typical example—do not obviously violate the metaphysical principle, central to Aristotelian necessitarianism, that laws are grounded in essential kinds. In fact, the conservation of energy looks to be just the sort of case in which some kind of stuff has properties whereby it is conserved in the way that the First Law states. That the law is not a causal one (assuming it not to be causal) is irrelevant, since the Aristotelian thesis is a wholly general thesis about the grounding of laws in the kinds they are about.

³⁰ One could just as easily reply that what makes a physical law *physical* is that it governs physical systems, so while it might not be obvious that the systems ground the laws, it can’t be ruled out that they do. Or perhaps there is some kind of virtuous interdefinability involved. Either way, the circularity charge is not clearly fatal. See also Chalmers 2006, 176 for a more general version of this familiar objection.

³¹ Ellis 2001, 249–53; Bigelow, Ellis, and Lierse 1992. The proposal is an odd one. After all, how can one even specify what *kind* of world ours is without appeal to the natural kinds of entity within it? In which case, it is hard to see what work the idea of *kinds of worlds* is doing in addition to the kinds within it, other than as a mere device to provide a kind for the conservation laws to work on. I cannot pursue the matter here.

³² Chalmers (2006, 176) focuses on the perpetual motion formulation, which naturally leads him to object that ‘[w]hen the law is so expressed, the difficulty of construing it as following from the essential property of anything becomes apparent’.

7. Conclusion

I do not pretend to have demonstrated that natural necessity, understood as a *sui generis* kind of necessity distinct from metaphysical necessity, is impossible. Such demonstrations are not easy to come by in philosophy, at least sufficiently to convince any opponent. What I do hope to have done is provide a cumulative case against natural necessity, and the strong contingentism characteristic of it, via a consideration of Lowe's position on the laws and that of some writers with similar views. What opponents of necessitarianism need to keep in mind is the central idea motivating it. If laws are about anything, they are about the *behaviour of things*. Moreover, the generality of laws means that they are about the *general* behaviour of things. Yet the general behaviour of things relevant to laws cannot, by definition, be accidental. For it not to be accidental, however, it must be explained, ontologically, by something that is itself non-accidental. The only plausible candidate is the *nature* of whatever it is that a given law is about. Laws, so understood, are *expressions* of the natures of what they are about. In other words, the laws of *nature* just are the laws of *natures*. Natures, at least as conceived by the Aristotelian, are not what things *happen* to have; they are what things *must* have. Hence the non-arbitrary division of reality into the so-called 'natural kinds', itself a term deserving abolition from philosophical vocabulary for all the harm it has done to clarity and understanding. Replacing it with the good old-fashioned term 'essence' would be a step forward.³³

8. References

- Armstrong, D. M. 1983. *What is a Law of Nature?* Cambridge: Cambridge University Press.
- Armstrong, D. M. 2010. *Sketch for a Systematic Metaphysics*. Oxford: Clarendon Press.
- Beebe, H. 2000. 'The Non-Governing Conception of Laws of Nature'. *Philosophy and Phenomenological Research* 61: 571–94.
- Bigelow, J., Ellis, B., and Lierse, C. 1992. 'The World as One of a Kind: Natural Necessity and Laws of Nature'. *The British Journal for the Philosophy of Science* 43: 371–88.
- Bird, A. 2001. 'Necessarily, Salt Dissolves in Water'. *Analysis* 61: 267–74.
- Bird, A. 2007. *Nature's Metaphysics*. Oxford: Clarendon Press.
- Chalmers, A. 1999. 'Making Sense of Laws of Physics', in *Causation and Laws of Nature* ed. H. Sankey, 3–16. Dordrecht: Springer Science.
- Chalmers, A. 2006. 'Why Alan Musgrave Should Become an Essentialist'. In *Rationality and Reality: Conversations with Alan Musgrave* ed. C. Cheyne and J. Worrall, 165–81. Dordrecht: Springer.
- Ellis, B. 2001. *Scientific Essentialism*. Cambridge: Cambridge University Press.
- Feynman, R. P., Leighton, R. B., and Sands, M. (eds). 1964. *The Feynman Lectures on Physics*, vol. 1. Reading, MA: Addison-Wesley.
- Fine, K. 1994. 'Essence and Modality'. *Philosophical Perspectives* 8. Atascadero, CA: Ridgeview Publishing Company.

³³ I am grateful to Tuomas Tahko and John Heil for helpful comments on an earlier draft.

- Fine, K. 1995. 'Senses of Essence'. In *Modality, Morality, and Belief: Essays in Honor of Ruth Barcan Marcus* ed. W. Sinnott-Armstrong, 53–73. Cambridge: Cambridge University Press.
- Fine, K. 2002. 'The Varieties of Necessity'. In *Conceivability and Possibility* ed. T. S. Gendler and J. Hawthorne, 252–81. Oxford: Clarendon Press.
- Foster, J. 2004. *The Divine Lawmaker*. Oxford: Clarendon Press.
- French, A. P. 1971. *Newtonian Mechanics*. New York: W. W. Norton.
- Goldstein, H. 1980. *Classical Mechanics*. Reading, MA: Addison-Wesley; 2nd edn.
- Kment, B. 2014. *Modality and Explanatory Reasoning*. Oxford: Oxford University Press.
- Kragh, H. 2002. 'Cosmologies and Cosmogonies of Space and Time'. In *The Cambridge History of Science*, vol. 5 ed. M. J. Nye, 533–7. Cambridge: Cambridge University Press.
- Lange, M. 2008. 'Could the Laws of Nature Change?'. *Philosophy of Science* 75: 69–92.
- Lange, M. 2009. *Laws and Lawmakers*. Oxford: Oxford University Press.
- Lewis, D. 1973. *Counterfactuals*. Oxford: Blackwell.
- Lewis, D. 1983. 'Scorekeeping in a Language Game', in his *Philosophical Papers*, vol. 1. Oxford: Oxford University Press: 233–49. (Orig. pub. *Journal of Philosophical Logic* 8 (1979): 339–59.)
- Loewer, B. 1996. 'Humean Supervenience'. *Philosophical Topics* 24: 101–27.
- Lowe, E. J. 1980. 'Sortal Terms and Natural Laws'. *American Philosophical Quarterly* 17: 253–60.
- Lowe, E. J. 1982. 'Laws, Dispositions and Sortal Logic'. *American Philosophical Quarterly* 19: 41–50.
- Lowe, E. J. 2006. *The Four-Category Ontology*. Oxford: Clarendon Press.
- Lowe, E. J. 2009. *More Kinds of Being*. Oxford: Wiley-Blackwell.
- Lowe, E. J. 2011. 'Locke on Real Essence and Water as a Natural Kind: A Qualified Defence'. *Proceedings of the Aristotelian Society Supplementary Volume* 85: 1–19.
- Oderberg, D. S. 2011. 'Essence and Properties', *Erkenntnis* 75: 85–111.
- Sagan, C. 2006. *The Varieties of Scientific Experience* (ed. A. Druyan). New York: Penguin.
- Serway, R. A. and Jewett, J. W., Jr. 2008. *Physics for Scientists and Engineers with Modern Physics*. Belmont, CA: Thomson; 7th edn.
- Shoemaker, S. 1980. 'Causality and Properties', in *Time and Cause*, ed. van Inwagen, 109–35. Dordrecht: Reidel.
- Shoemaker, S. 1998. 'Causal and Metaphysical Necessity'. *Pacific Philosophical Quarterly* 79: 59–77.
- Tahko, T. 2015. 'The Modal Status of Laws: In Defence of a Hybrid View'. *The Philosophical Quarterly* 65: 509–28.
- Tooley, M. 1987. *Causation: A Realist Approach*. Oxford: Clarendon Press.
- Van Inwagen, P. 1979. 'Laws and Counterfactuals'. *Noûs* 13: 439–53.
- Webb, J. K., Flambaum, V. V., et al. 1999. 'Search for Time Variation of the Fine Structure Constant'. *Physical Review Letters* 82: 884–7.
- Webb, J. K., King, J. A., et al. 2011. 'Indications of a Spatial Variation of the Fine Structure Constant'. *Physical Review Letters* 107, 191101: 1–5.
- Witte Jr., J. and Alexander, F. S. (eds). 2007. *The Teachings of Modern Protestantism on Law, Politics, and Human Nature*. New York: Columbia University Press.

6

The Epistemology of Essence

Tuomas E. Tahko

1. Defining the Notion of Essence

The notion of essence is notoriously mysterious: philosophers seem to use it in a number of different senses, and even if they *do* use the notion in the same sense, it is often not quite clear what that sense is. At the same time, essences, when they are invoked, are generally supposed to do a lot of explanatory work: natural kinds can be identified in terms of their essences, metaphysical modality may be reduced to essence,¹ the causal powers of various entities can be explained with the help of essences, and so on. In what follows, I will first attempt to offer a working definition of the notion and will then lay out the available options regarding the epistemology of essence. Following these introductory remarks, I will proceed to analyse the options.

As E. J. Lowe often puts it, perhaps the closest thing to a definition of essence that we may have is the familiar phrase from Locke: ‘the very being of any thing, whereby it is what it is’ (1975, III, III, §15). But this phrase is not particularly informative. What exactly is ‘the very being’ of a thing, and how is it that we come to know ‘the very being’ of things? Locke himself considered *real* essences (as opposed to merely *nominal* essences) to be unknowable to us, but in contemporary metaphysics, a modal interpretation of essence due to the work of Kripke and Putnam has been the standard.² In the tradition of ‘Kripke–Putnam essentialism’, essences are explained in terms of *de re* modality: an attribution of necessity to a proposition is *de dicto*, but when we attribute necessity to an object we are dealing with *de re* necessity, and hence essence.³ Another feature of the Kripke–Putnam tradition is that it is commonly thought that science discovers essences; that is, essences are discovered (at least for the most part) *a posteriori*, empirically. The apparent problem with this approach is that ‘little, if anything,

¹ Following Kit Fine (1994), more on this below.

² See Lowe (2011a) for an illuminating discussion of Locke’s views on essence, and for a comparison of Locke’s account with the Kripke–Putnam line.

³ I should note that by ‘Kripke–Putnam essentialism’ I refer mainly to the work of those philosophers who are broadly sympathetic to the idea that there are mind-independent (natural kind) essences and that they can be interpreted in terms of *de re* modal properties. My remarks should not be understood as exegetical; in fact, Putnam (1990) has distanced himself from this line of thought altogether, and it is not entirely clear what Kripke’s own position is. For further discussion of these issues, see Tahko (2015a).

is known about how or why objects have their *de re* modal properties,' as L. A. Paul (2006, 335) puts it. Accordingly, the epistemic problem has been merely postponed. This is certainly a pressing problem at least for those who consider the possession of *de re* modal properties to be primitive.⁴

Before we proceed any further, it should be made clear that the understanding of essence to be adopted in this chapter is not the one familiar from the Kripke–Putnam tradition, which represents what we might call the 'modalist' view of essence: an object has a property essentially if and only if it has that property necessarily. The alternative, broadly Aristotelian approach has been made popular by Kit Fine (1994, 1995a, 1995b) and E. J. Lowe (e.g., 2006, 2007, 2008a, 2011a). Other contemporary proponents of a broadly Aristotelian conception of essence include, for instance, David S. Oderberg (2007, 2011) and Kathrin Koslicki (e.g., 2012).⁵ This broadly Aristotelian view of essence—which I do not claim to be entirely faithful to Aristotle—suggests that not all necessary truths about a given entity *x* are essential truths about *x*, but all necessary truths are true in virtue of (or, one might say, *grounded in*) essential truths (about some entity or other). This implies that essential truths about *x* are a proper subset of the necessary truths about *x*, but even those necessary truths about *x* that are not essential truths *about x* are nevertheless essential truths about some entity or other. So, according to this view, essence is ontologically prior to modality in the sense that essential truths are more fundamental than modal truths. On this view we should not reduce essence to *de re* modal properties.⁶

Lowe sometimes refers to his own, broadly Aristotelian view of essence as 'serious essentialism' (e.g., Lowe 2013, 144). But as we saw, Lowe in fact takes his inspiration from Locke, suggesting simply that the essence of *x* is just the very *identity* of *x*. Moreover, it is important for Lowe that essences are not themselves further entities (in contrast to Locke). Since he takes it that all entities have an essence, there would seem to be a threat of infinite regress if essences themselves were entities. Indeed, why would we think that the *identity* of a thing would itself be an entity? More precisely, the conception of essence at hand suggests that once we know the identity and existence conditions of an entity, we know its essence; we may *express* this essence in terms of a set of these identity and existence conditions, or in terms of a proposition listing these conditions, but the essence itself is not a set or a proposition. I believe that this comes close to Lowe's view of the matter, but things are somewhat less clear with other broadly Aristotelian versions of essentialism, such as Kit Fine's, who sometimes writes as if

⁴ Paul (2004) develops an account of *de re* modal properties in terms of bundle theory, but if this means that essences are to be conceived as bundles of properties, then this view as well has its problems, as Oderberg (2007, 2011) has argued. These issues will be examined in detail in what follows.

⁵ Scott Shalkowski's work (e.g., 1997, 2004) in defence of and developing on a Finean conception of essence and modality should be mentioned in this connection as well, as should Fabrice Correia's (e.g., 2011). There are, of course, many others who have developed relevant accounts, such as Crawford Elder (2004), but it will not be possible to discuss all of them here. My focus will be on Lowe, for obvious reasons.

⁶ As Shalkowski (2008, 51) puts it, '*de dicto* necessity is a species of *de re* necessity', and hence the former is 'not free from essentialist implications'. For more critical remarks regarding the traditional approach, see Lowe 2013, ch. 8.

essences themselves *are* propositions: ‘we may identify the being or essence of *x* with the collection of propositions that are true in virtue of its identity’ (Fine 1995c, 275). I will leave this issue aside, interesting as it is—in what follows it is assumed that essences are not themselves entities.

Finally, the distinction between *general* and *individual* essences should be mentioned; or in other words, the distinction between *kind* essences and *particular* essences. This is an important distinction for Lowe (2013, 145), although personally I am somewhat sceptical about individual essences, such as the essence of an individual cat. General essences such as the essence of the kind ‘cat’ will be my main focus—although there may be reasons to be sceptical about some general essences as well, such as those of biological species.⁷ Let that pass for now. General essences are nevertheless somewhat less controversial than individual essences. In a classic paper, Baruch Brody (1973) considers it an advantage of ‘Aristotelian essentialism’ that it connects essentiality with what it is to be a natural kind. It follows that essences of artificial objects, such as tables and chairs, are also among the more controversial cases—I will largely omit discussion of them.

We think of our epistemic access to essence (and also modality) in terms of the *a priori* vs. *a posteriori* distinction, although I should immediately note that I consider this distinction to be somewhat vague.⁸ It should also be noted that even though essence and modality are undoubtedly linked in an important way, the initial assumption of this chapter is, following Lowe, that essence is *ontologically* and *epistemically* prior to modality. In what follows I will examine the *a priori* and the *a posteriori* routes to essentialist knowledge, before concluding with a brief discussion of a hybrid view, where each method is acknowledged. I am presently more interested in mapping our options regarding the epistemology of essence as well as clarifying Lowe’s view rather than defending a particular position. Another preliminary point worth noting is that in my analysis I will first explore the possibility of a unitary view of the epistemology of essence, even though I am doubtful that our epistemic access to essence is always via the same route (which leads to the hybrid view). However, for the sake of parsimony, a unitary account would be preferable, so I think that hybrid accounts should be considered only if all unitary accounts fail.

2. Epistemic Access to Essence

Given that essence is understood to be ontologically prior to modality, it may, at least initially, seem more promising to give an account of the epistemology of essence independently of the epistemology of modality. This would imply that the epistemology of modality is a special case of the epistemology of essence. Hence, if we could give a

⁷ Defenders of biological essentialism are few and far between, but see for instance Austin (2017) for a recent account.

⁸ See Tahko (2017) for discussion. See also Vaidya (2015) for a helpful classification of different approaches to modal epistemology. Vaidya also discusses, e.g., Bob Hale’s essentialist view, which I will not attempt to cover in this chapter.

plausible account of the epistemology of essence, we would also have the beginnings of an account of the epistemology of modality. Although explicit defences of this idea are relatively rare, it seems that this is a popular route for contemporary essentialists.⁹ However, there is certainly less agreement about whether our knowledge of essence is *a priori* or *a posteriori*. I will consider each option.

2.1 A posteriori access to essence

One of the supposed advantages of Kripke–Putnam essentialism is that our knowledge of essence, or essential properties, can be traced to scientific knowledge in a seemingly straightforward manner. Indeed, many contemporary essentialists continue to support this type of approach; it is familiar from the literature on ‘scientific’ or ‘dispositional’ essentialism (e.g., Ellis 2001; Bird 2007a). However, since the conception of essence taken for granted in much of this literature is that essence reduces to modality (rather than the other way around), it is not obvious that scientific essentialism is able to give us a sufficiently fine-grained account of the epistemology of essence understood in a broadly Aristotelian manner.¹⁰ A typical conviction of the scientific essentialist is that the laws of nature are metaphysically necessary, in which case our knowledge of the laws of nature is a direct route to substantial modal knowledge, with empirical science playing a key role. But while this conception may bear some resemblance to Aristotelian essentialism, it neglects a key feature of the Aristotelian ontology of essence, namely that essence is ontologically prior to modality.

Can we build on the work of scientific essentialists while adopting the Aristotelian ontology of essence? Oderberg’s essentialism is perhaps the most interesting attempt at this.¹¹ However, Oderberg (2007, 13) thinks, contrary to some versions of the Kripke–Putnam line and scientific essentialism, that discovering essences is not just the work of scientists. He would nevertheless insist that essentialism is a fallibilist position; that is, our knowledge of essence is subject to revision (*ibid.*, 48). Crucial to this line of thought is that although scientists play an important role in the discovery of essences, we cannot simply rely on experts to explain the epistemology of essence. Oderberg (*ibid.*, 13) argues, like Lowe, that *everyone* can have knowledge of essences. It is first and foremost the metaphysician’s task to explain essence, but scientific knowledge is indispensable for this task. So, Oderberg’s essentialism is of the *a posteriori* kind—and probably the best example of this approach combined with an Aristotelian ontology of essence. I will devote the rest of section 2.1 to a discussion of Oderberg’s account.

⁹ In discussion, both Kit Fine and E. J. Lowe have indicated to me that this option is more attractive to them. In Lowe’s case this is also clear from his written work. Note however that nothing here entails *conceptual* reductionism about modality; that is, the *concept* of modality is not necessarily analysable in terms of the concept of essence, even if the former does reduce to the latter on an ontological level, and even if essence is epistemically prior to modality. So, an understanding of the concept of modality may not be available simply via an understanding of the concept of essence. This is something that Lowe has stressed in discussion.

¹⁰ See Lowe (2006, chs 9–10) for discussion and criticism of scientific essentialism.

¹¹ Regarding laws of nature in particular, see Oderberg’s contribution to this volume.

2.1.1.1 ODERBERG'S ESSENTIALISM

Oderberg insists that no direct empirical test could enable us to discover essences, even though essences are tracked by the empirical sciences—the account is fallibilist. It also seems clear that Oderberg holds essence to be epistemically prior to modality. In particular, it is important for Oderberg that the essence of an object is not merely a bundle of the essential properties of that object. His main reason for resisting this type of 'bundle theory' is what he calls 'the unity problem'; namely, there should be something to hold a bundle of essential properties together in order to ensure that, say, the essential properties of a given kind are always featured in the members of that kind (Oderberg 2011, 90). The unity problem, sometimes also called 'the problem of complex essences' (Dumsday 2010), will turn out to be of great importance for the epistemology of essence. In fact, the problem goes back to the Aristotelian problem of *propria*. Here is a passage from Oderberg with a relevant example:

Having a capacity for humour is an essential property [...] of human beings, and in this sense we can say it flows *from the essence* of human beings to have a capacity for humour. But the *essence* of being human is to be a *rational animal*, and humans have a capacity for humour *only because* they are rational animals. (Oderberg 2007, 49)

This seems correct insofar as we need to distinguish between the *essence* of an entity and what that essence may entail (ignoring any problems with this particular example). But I do not regard the term 'flow' to be ideal. It is a historical notion, used by Locke, which Oderberg adopts into contemporary discussion for want of a better notion. Does the notion of 'flow' simply suggest that the essential properties of an entity are logically entailed by the essence of that entity? If this is the case, then a distinction that Fine (1995b, 56–8) draws between *constitutive* and *consequential* essence may be relevant here: a property is a constitutive part of the essence of a given object if it is 'directly definitive' of the object, and merely consequential if it is had in virtue of being a logical consequence of some 'more basic' essential property of the object. However, the 'basic' essential properties of objects entail all sorts of things and not all of these seem like very plausible essential properties; for example, the constitutive essential properties of humans entail any disjunction of an essential and a non-essential property of humans, such as humans having a capacity for humour *or* flying.¹² Oderberg is determined to settle this problem, but not in terms of entailment. Rather, he gives a more rigorous definition of 'flow': a set of properties of the objects belonging to a given kind with a particular essence are caused by and originate in the form of that kind (Oderberg 2011, 99–103). The idea is that *form*—a notion which is central to Aristotelian hylomorphic essentialism (which Oderberg is developing)—provides the essence and hence the properties that 'flow' from it.

As I understand it, then, 'flow' concerns the dependence between a set of essential properties and the essence which they are a part of. A concern regarding this solution

¹² See Gorman (2005), Koslicki (2012), and Correia (2011) for further discussion and some suggested solutions to the problems caused by the constitutive/consequential distinction.

is that it seems impossible for us to distinguish, epistemically, between the essence itself and an essential property that ‘flows’ from the essence. For instance, if we are looking for the essence of water, we might be able to point out a number of properties that seem essential but fail to be so. One example might be the diffraction of water waves. Diffraction, the bending of waves around obstacles, is a feature of any wave, but it will of course only be apparent when we have a body of water rather than just one water molecule. So, is diffraction a property of the water molecule or something that merely flows from the essence of water? One might think that the solution to this problem—how to distinguish essential properties from mere *propria*, properties that flow from the essence—could be addressed in the same way that we can distinguish essential properties from merely accidental properties. Oderberg is aware of this type of challenge and attempts to address each of these issues.

Regarding the problem of genuine essential properties and mere accidents, Oderberg suggests that we can use our reason and common sense to determine when a given property is genuine in the sense that it is caused by and originates in the essence. Crucial for this process is to consider whether the thing in question, say, the kind water, would continue to display ‘the characteristic properties, functions, operations and behaviour’ that it normally does even if a certain quality of it were to be removed (Oderberg 2007, 50–1). If this is the case, then the quality in question is not a part of the essence of the thing. But if removing the quality would cause ‘a general disturbance or radical change’ in the said functions of the thing, then it is a part of the thing’s essence (*ibid.*).¹³ However, on Oderberg’s position, it might seem that we have only our imagination to rely on to determine whether a given change is of the first or the second type. These issues prompt Oderberg to recognize an epistemological caveat concerning essence. Essential properties, including the ones that ‘flow’ from the essence, are presumably open to empirical research (given sufficient technical resources etc.). But since essences are not mere bundles of essential properties, we need something that unifies these properties to get to the essence—this role is played by the Aristotelian notion of *form*. Here is where Oderberg as well needs *a priori* input, because he thinks that the existence of such a unifier can only be deduced by metaphysical *a priori* reasoning, even though determining what the unifier is will require empirical investigation. As Oderberg (2011, 97) puts it: ‘That gold must have a principle of unity is not within the remit of observation; that gold is a metal whose atomic constituents have atomic number 79 is.’ This highlights the importance of the unity problem: *a priori* elements appear to be unavoidable in determining when a collection of essential properties constitutes an essence.

¹³ It is not entirely clear what counts as a general disturbance of the type that Oderberg has in mind. One might suggest that, say, in the case of water, these disturbances would include changes in the *microphysical* behaviour of water molecules. Yet, Needham (2011) and Lowe (2011a) have argued that the *macrophysical* properties of water are more plausible defining characteristics. For discussion, see Tahko 2015a. This issue is of course crucial for the previous example, because diffraction is a macrophysical phenomenon.

2.1.1.2 A PRIORI ELEMENTS IN A POSTERIORI ESSENTIALISM?

The problem emerging for the *a posteriori* route to essence is that we must already have grasped the essence we are looking for before we can identify the essential properties that ‘flow’ from it. It seems that our epistemic access to essence is often piecemeal: we wonder if a certain kind of entity, say, a certain subatomic particle, could possibly exist. We determine this by considering the essential properties that the particle would have, were it to exist: perhaps its mass etc. But, as we saw, it would appear that we can grasp an essence only *after* we have acquired sufficient knowledge about the essential properties associated with that essence. Or, to put it another way, we must have some prior conception of the essence of an object before we can recognize that it acts as a unifier for a given set of essential properties. If we need something like this in any case, then perhaps an account which takes us to have *a priori* access to essence would be better off to begin with?

Oderberg’s reply to this type of concern is that all we need to know *a priori*—at least in the case of natural kinds as opposed to abstract mathematical objects—is that a given kind, say, a subatomic particle, *has an essence*, not what that essence is.¹⁴ Further, we can know that a given property (such as mass, perhaps) is a good candidate for an essential property of a given kind because of things like universality in the kind, the way it characterizes the kind, and the difficulty or impossibility of removing that property from random members of the kind. It seems to me that these are indeed all good ways to identify properties that might be a part of some essence or other, but this may not remove the core of the difficulty. To know *which* kind a given essential property is associated with, some prior knowledge about the essence of that kind seems to be necessary. Consider the Higgs boson, which was finally discovered in 2012. This discovery was highly anticipated and physicists had a very good idea about the mass range of the predicted Higgs as well as its role in the Standard Model of particle physics well in advance of the discovery. But how did physicists know that the observed mass of the Higgs boson, now confirmed to be in the range of 125.09 ± 0.24 GeV/c is truly part of the essence of the *Higgs* kind, rather than, say, an emergent feature of some yet to be discovered natural kind? If it had turned out that the data from the Large Hadron Collider is not attributable to the Higgs boson, but rather to some emergent feature of a further kind of thing, then could it not also have turned out that *all* the data was due to some merely *accidental* properties of this further kind of thing?

Perhaps all this is something that the *a posteriori* essentialist can address with the built-in fallibilism of the account, but a friend of the *a priori* approach can continue to push for the importance of the *a priori* task of unifying the essential properties, which already seems to presuppose an understanding of what kind of thing we are dealing with. If it is only possible to engage in an analysis about what characterizes a given kind *after* we have at least a partial understanding of what that kind *is*, that is, of the essence

¹⁴ Based on an e-mail exchange with Oderberg in 2011.

of the kind, then we ought to consider the possibility of acquiring this type of *a priori* knowledge about essences. Let us now see how that approach fares.

2.2 *A priori access to essence*

The obvious challenge faced by any account of the epistemology of essence postulating direct *a priori* access to essence is that the cognitive faculty that enables this epistemic access calls for an explanation. There may be a temptation to take this epistemic access to be primitive. I do not consider this to be a good strategy, but there are very few attempts in the literature to provide a better explanation. However, the shortcomings of traditional Kripke–Putnam essentialism have prompted some philosophers to move towards *a priori* essentialism despite the epistemic challenge. Perhaps a negative argument is the best argument we can have? Lowe often motivates his view via such negative arguments (e.g., 2007, 2008a, 2008b, 2013), aimed towards the Kripke–Putnam type *a posteriori* essentialism of the likes of Alexander Bird.¹⁵ Lowe argues that since the Kripke–Putnam inference-pattern used to deduce knowledge of individual essences is suspect, perhaps all essential truths are *a priori*. But we ought to be able to say something positive as well. Note also that Lowe himself explicitly rules out conceivability and intuitions (e.g., Lowe 2014) as a potential *a priori* route to knowledge of essence. In what follows I will reconstruct Lowe’s *a priori* essentialism and propose some ways to develop it.

2.2.1 LOWE’S ESSENTIALISM

Lowe’s arguments for *a priori* essentialism rely on his more general metaphysical commitment according to which, in general, ‘essence precedes existence’ both ontologically and epistemologically. On the epistemic side, the idea is that we can—indeed we must—generally know the essence of a thing before we can know whether that thing exists. In other words, we have to know that it is possible for a thing of a particular kind to exist before we can determine whether a thing of that kind actually exists. One of Lowe’s best examples, in my view, is the case of transuranic elements: many of them were only synthesized after their *possible* existence was determined by non-empirical means. With the help of Mendeleev’s periodic table, chemists have been able to predict the existence of a number of yet to be discovered elements and to give highly accurate predictions of their properties. Later on they were able to synthesize these elements and verify that they indeed had the predicted properties. The idea that Lowe proposes is that this process would not have been possible without a prior grasp of the essences of transuranic elements. In fact, Lowe (2008a, 35) thinks that we cannot even ‘talk or think comprehensively’ about things whose essences we do not know.

There is something appealing about this story, and I will develop on it below, but there are some aspects that require further elaboration first. Perhaps the most pressing of these is Lowe’s (2008a, 36) acknowledgement that we only need to know a *part* of a

¹⁵ See also Bird 2008 and Lowe’s (2008b) reply to Bird.

thing's essence to be able to talk or think about it. As Bird (2007b) has pointed out, this appears to leave the possibility of *a posteriori* essentialism open, since it could be that we come to know most of a thing's essence by *a posteriori* means even if we must know a part of it *a priori*. Indeed, it is plausible that to be able to distinguish different kinds of things, and especially things like natural kinds, a considerable amount of empirical information is needed in addition to knowledge of general essentialist facts (such as chemical substances having their actual molecular composition essentially). This suggests that the Kripke–Putnam story about how we can derive knowledge of metaphysical necessities from general essentialist (*a priori*) facts with the help of particular empirical facts may have something going for it after all.

In various places, Lowe hints that *essential dependence* has a key role in our epistemic access to essences:¹⁶

Consider the following thing, for instance: the set of planets whose orbits lie within that of Jupiter. What kind of thing is that? Well, of course, it is a set, and as such an abstract entity that depends essentially for its existence and identity on the things that are its members—namely, Mercury, Venus, Earth, and Mars. Part of what it is to be a set is to be something that depends in these ways upon certain other things—the things that are its members. Someone who did not grasp that fact would not understand what a set is. (Lowe 2008a, 37)

So, Lowe thinks that in many cases knowing how a thing is related to other things is central to our knowledge of *what* a thing is. Even in the case of sets Lowe is careful to specify that depending essentially for its existence and identity on its members is only a *part* of what it is to be a set. In general, he seems to think that knowing the *complete* essence of a thing is not a very simple affair—perhaps almost impossible—but knowing a part of a thing's essence is often sufficient for talking or thinking about it comprehensively, and for being able to distinguish it from other things. Let me now briefly discuss the case of sets—Lowe's account is at its strongest when it comes to the essences of abstract objects, so sets make for a good case study. We will get back to the case of concrete objects below, in section 2.2.3.

2.2.2 BUILDING ON LOWE'S ESSENTIALISM: THE CASE OF SETS

Let us look at sets more closely, as they provide a fairly simple yet illuminating example. On the face of it, it would appear that one can work with sets without knowing very much, if anything, about their essence. Since set theories, like ZFC, are extensional, we can derive a number of theorems without going beyond extensionality. Extensionality guarantees that two sets are equivalent if and only if they share all their members, although this does not mean that the sets were formed in the same manner. Moreover,

¹⁶ See Tahko and Lowe 2015 for an analysis of different kinds of ontological dependence, including essential dependence. Lowe himself defines essential dependence as follows: 'To say that X depends essentially on Y for its existence and identity is just to say that it is part of the essence of X that X exists only if Y exists and part of the essence of X that X stands in some unique relation to Y' (Lowe 2008a, 38). See also Lowe 2013, 147.

working with sets on a purely extensional level would not seem to require a deeper comprehension of what the nature of sets is, in Lowe's sense, so the idea that one must know the existence and identity conditions of sets before one can understand what a set is becomes less plausible. But we must be careful here, for it is one thing to prove set-theoretical theorems on a purely extensional level according to the rules of some formal system, and quite another to prove that these theorems express truths about sets. In order to establish truths about *sets*, we also need some independent reasons to think that the purely extensional system accurately models the behaviour of sets, and here it seems that some further insight into what constitutes the essence of a set may be required. Some such reasons may very well be available—plausibly this will involve the role of the axiom of extensionality in set theory. But once it is acknowledged that such reasons are needed to express truths about sets, it is once again possible to defend the idea that a previous grasp of the essence of sets may be needed before it is possible to express truths about sets (rather than about purely extensional formal systems).

A possible way to defend this idea is to build on the classic iterative conception of set (Boolos 1971). For instance, Øystein Linnebo (2010) has suggested that the set-theoretical hierarchy and the quantifiers used in set theory should be interpreted in modal terms; that is, as having an implicit modal character.¹⁷ If this is correct, then it seems that the iterative conception of sets cannot be purely extensional, as the implicit modal character must be explained somehow. Linnebo develops one approach towards understanding the modality involved, but we do not need to go into the details here; all that is needed for the purposes of this example is a plausible case for a need to go beyond extensionality. Thus, one line of thought that would support the need for prior understanding of the essence of sets goes roughly as follows.¹⁸ Starting with the iterative conception of set we can determine a certain set-theoretical hierarchy, which has an implicit modal character. This conception captures the essence of sets, and it is only after one has this conception of sets that one can recognize the theorems of ZFC, for instance, as expressing truths about *sets*, rather than just about the elements of a purely extensional formal system. The most controversial part of this line of thought is perhaps the claim that the iterative conception captures the essence of sets. I will not attempt to support this claim here, but it is at least partly corroborated by the fact that many of the axioms of ZFC can be motivated by the iterative conception—this much is well documented in the literature (e.g., Boolos 1971; Parsons 1983; Yablo 2006; and Linnebo 2010).¹⁹

¹⁷ See also Parsons (1983) and Yablo (2006).

¹⁸ I would like to thank Ramiro Caso for suggesting this type of line of thought to me. See also Fine (2005) for a relevant analysis of sets.

¹⁹ In discussion (e-mail exchange in 2011) Oderberg has questioned the inference from our knowledge of the essence of sets to the belief that sets *exist*. I take it that in the case of abstract objects like sets, this inference is fairly straightforward: if it is part of the essence of a set that it has the very members that it has, then the *existence* of the set appears to be nothing over and above the existence of the members of that set. Of course, the strategy for establishing the existence of a given object depends on what the essence of the

2.2.3 LOWE'S ESSENTIALISM AND CONCRETE OBJECTS

The examples that we have considered so far have been fairly theoretical, such as trans-uranic elements and sets—the latter being of course abstract objects. But one objection that might be raised against Lowe's idea that essence precedes existence stems from more familiar everyday objects.²⁰ Consider cats, for instance. If Lowe is right, then we would have to know (a part of) the essence of the kind cat before we can talk or think about cats comprehendingly, or indeed to distinguish cats from other objects. Yet, it seems that we do not need to know very much at all about cats to be able to think about them, individuate them in thought, and refer to them quite comprehendingly.²¹ Perhaps we only need to know that they are living organisms. But even if one thinks that cats are robots or demons, it would still seem to be the case that when one sees a cat one is immediately capable of thinking about it (as a cat, albeit a robot-cat or a demon-cat), and successfully referring to it in conversation. While I think that there is much more to be said about the apparent difference between familiar everyday objects and abstract or theoretical objects, it may be that the apparent simplicity of thinking and talking about these familiar objects is exactly due to their being so *familiar* to us. That is, one might doubt that there is a genuine difference between familiar, concrete objects and more theoretical or abstract objects.

The key here is how we interpret Lowe's requirement for talking or thinking about an object *comprehendingly*. Lowe does not go into much detail about this requirement, but it appears to be doing a lot of work in his account; it is also a possible escape route from the objection under consideration—namely, that the familiarity of everyday objects is all that drives Lowe's idea that essence precedes existence. Perhaps it is just the inherited human knowledge about cats and other familiar objects that is doing the work here?²² By adopting a fairly strong reading of 'comprehendingly', it could be argued that one does indeed need to know a number of things about the essence of the kind 'cat' before it is possible to think or talk about cats comprehendingly.

According to Lowe (2008a, fn. 23; see also Lowe 2013, ch. 8), to be able to think or talk about cats comprehendingly, it may be sufficient to know that cats are animals or living organisms. This would at least rule out radical mistakes such as thinking that cats are robots. I take it that the underlying idea is as follows: someone who thinks that

object is, but I consider the case of sets to be among the easier cases. Things get much trickier with material, concrete objects.

²⁰ At this point it may be useful to note that our epistemic access to the essences of different kinds of objects, and perhaps abstract vs. concrete objects in particular, could plausibly differ even radically. In fact, this would suggest that we must resort to some sort of a hybrid view after all (i.e., combining *a priori* and *a posteriori* elements). On the face of it, it is difficult to see how *a posteriori* essentialism could account for the essences of abstract objects (although perhaps there is a way to do this), so unless we can explain our epistemic access to the essences of concrete objects in terms of *a priori* essentialism as well, then a hybrid view looks unavoidable. See Tahko (2017) for discussion; we will return to this in section 3.

²¹ The problem at hand does not concern our ability to *refer* to objects though; as Lowe (2008a, fn. 24) notes, he is not interested in semantic questions in his theory, but rather about our acquaintance with 'objects of thought'. Accordingly, I will also omit any discussion of semantic matters here.

²² Thanks to David Oderberg for suggesting this line of thought.

cats are robots or demons does not understand what it means to be a cat; that is, we would have to say that this person is not acquainted with cats (as an object of thought). Hence, such a person would not be able to pick out cats with any reliability, since they might for instance think that an actual robot is *also* a cat. A crucial element of ‘comprehension’ would thus appear to be a sufficient grasp of the identity and existence conditions of a given object. What constitutes a ‘sufficient’ grasp? We will get to this in a moment. In any case, there is always room for error, so the requirement should not be understood as the absence of any possibility to mistake one kind of thing for another kind of thing. An important specification of this idea concerns the relationship between general essences and categorial hierarchy—a thesis that Lowe calls ‘categorialism’ and which maintains that one necessary condition for a thinker’s ability to single out objects in thought is the thinker’s grasp of a *categorial concept* under which the thinker conceives the object to fall (Lowe 2013, 21). This grasp may be implicit. Lowe’s own example concerning Oscar, the cat, will help to illustrate the idea (Lowe 2013, 23ff.) (and we will get the answer to our question about a ‘sufficient’ grasp as well).

Let us suppose that the object of your singular thoughts is Oscar, the cat. You see Oscar and track his movements with your eyes. But suppose, for *reductio*, that you do not acknowledge categorialism and hence do not conceive Oscar to fall under any categorial concept, such as *being a cat*. Now, following the idea that statues and lumps of matter coincide, Lowe suggests that Oscar as well must coincide with a certain hunk of matter. If that’s right, then you will have presumably seen and tracked the hunk of matter as well. If you were an anti-categorialist, you would now have to explain why you would be able to have singular thoughts about *Oscar*, the cat, rather than the numerically distinct hunk of matter—which might indeed just as well be a robot, or a demon. Lowe of course insists that you cannot have genuine, *de re* singular thoughts about an object without at least an implicit grasp of the relevant categorial concept—a ‘sufficient’ grasp requires one to be able to place the entity in question under the relevant categorial concept.

The example concerning Oscar is still very crude, but Lowe does specify the case (2013, 27). Let us suppose that your neighbours talk about Oscar, saying that Oscar is white and beautiful, but never reveal that he is in fact a cat, rather than say, a vase (or a robot, or a demon...). As Lowe observes, according to the usual causal-historical theory of linguistic reference, you can refer to Oscar successfully. But Lowe protests that since Oscar could be any kind of *thing* whatsoever, you have not really managed to express singular thoughts about Oscar: ‘thing’ and ‘object’ are *transcategorial* terms, whereas Lowe’s categorialism requires that to have genuine singular thoughts about Oscar you would at least have to grasp that Oscar is a *living being*, an animal, because that would appear to be the narrowest categorial concept that Oscar could fall under. It *might* be sufficient for one to have singular thoughts about Oscar if one thinks that Oscar is a dog—another kind of living being—but not if one thinks that Oscar is an ornament of some kind. The details of such examples will of course partly depend on one’s view of the correct categorial hierarchy, but the picture that Lowe is developing

seems clear enough: to be able to think or talk about something comprehendingly, we must have grasped, at least implicitly, some subset of the relevant thing's *criteria of identity*. So it seems that a natural reading of Lowe's view is to take it that grasping the identity and existence conditions of an object just is grasping the essence of an object. This may put some fairly strong constraints on our ability to have *de re* singular thoughts concerning objects, but this is not the place to discuss that problem.

We now have a rough idea about Lowe's epistemology of essence, but some questions remain. In particular, it still seems that there must be something more to essence than just the bundle of the relevant identity and existence conditions, for how are we supposed to know, *a priori*, which object a given set of such conditions is associated with? Here we once again face the unity problem. As we saw in section 2.1.1, Oderberg introduces the idea of a unifying *a priori* principle in order to establish this link, but I expressed some doubts about this move. If each object is associated with a unique unifying principle, then the unifying principle itself is starting to look very much like the essence (if not a *haecceity*), and it is not clear that this will help us to address the original epistemic problem. For Oderberg, it is the Aristotelian notion of *form* that goes towards addressing this problem, but Lowe himself does not adopt Aristotelian hylomorphism or the form/matter distinction as it is usually applied in this tradition (see Lowe 1998).

One further worry that might be raised at this point concerns children's ability to talk about and reliably pick out, say, animals, without any deeper comprehension about the general essence of animals. Might children be able to have singular thoughts about animals without even knowing that animals are living organisms? I think that Lowe would answer negatively. As a first attempt to settle the matter, we could subject a group of young children to a test to determine whether they have grasped the essence of cats to a sufficient degree to talk and think about them comprehendingly. If they are able to pick out the cat from a sample space including a cat, a cat-like robot, and a dog, then they have at least a partial grasp of the essence of cats. However, this would surely beg the question, for what guarantees that it is *grasping the essence* that is doing the work here?²³ Lowe's requirements for grasping (a part of) an essence, it seems to me, are fairly modest. Everyone, even quite young children, is supposedly able to grasp partial essences. But something more needs to be said in support of the account to address the concerns raised above. I venture to suggest that the most promising reply available to Lowe is to introduce context as a variable in cases of 'comprehension'. In a situation where cats are the only animals about, it is sufficient to know that cats are animals to be able to think and talk about them comprehendingly, and to distinguish them from other objects. However, if there are other animals about, then more detailed information about the essence of cats would be required, such as cats being feline. Similarly, we would need to know even more about cats to distinguish them from other felines. Eventually, we might encounter difficult situations where we may not know

²³ For relevant discussion, see also Vaidya 2010 and Horvath 2014.

enough about the essences (and so the relevant criteria of identity) of a combination of objects to adequately distinguish them, and ‘species’ may very well be one such problematic case.

Perhaps one final example would be useful. Consider the classic case of a bronze statue and the lump of bronze that it is composed of. If the lump and the statue are indeed two different things, as Lowe (2008a, 46) thinks they are, then no amount of empirical information will help us to determine this. Empirically, the statue and the lump are indistinguishable. It is only because we know what kind of things bronze statues and lumps of bronze are—and specifically knowing what they essentially depend on for their existence and identity—that we can distinguish them. Here it may be helpful to recall Oderberg’s idea that if a removal of a certain quality that a thing has would produce a radical change in the properties, functions etc. of that thing, then we are most likely dealing with a quality that is part of the thing’s essence. We can for instance consider a change in the shape of the bronze statue and the lump of bronze: the lump could survive such a change without any radical consequences for its key properties and functions, whereas the statue could not, and hence the shape of a bronze statue is plausibly a part of its essence. In passing, Lowe seems to make reference to this type of idea, since he notes that we might be able to grasp the identity and existence conditions of entities with the help of their *causal powers* (Lowe 2013, 115). Accordingly, Lowe would likely attempt a strategy not unlike Oderberg’s in order to address the unity problem: we can determine whether an object continues to exist by observing whether its distinctive causal powers (which are grounded in its essential properties) continue to be manifested. But as we will see in the next section, this move may not be available for a pure *a priori* essentialist.

While this brief reconstruction of Lowe’s *a priori* essentialism leaves many questions open, I hope to have given a fair summary of the core ideas and some of the problems concerning the account. I will now conclude by outlining some reasons to consider a hybrid approach to the epistemology of essence.

3. A Hybrid Approach?

One key problem for a unitary account of the epistemology of essence concerns the distinction between concrete and abstract objects. We discussed the prospects of Lowe’s *a priori* essentialism regarding concrete objects and found some problems, although I think that Lowe’s *a priori* essentialism is very plausible in the case of abstract objects, such as geometrical objects (see Lowe 2012 for a discussion of geometrical objects). As I have argued elsewhere (Tahko 2017), it is much more difficult to adapt Lowe’s account to the general essences of natural kinds. I will not dwell on this issue here, but will instead focus on a passage from Lowe himself which, appropriately interpreted, may take us toward a hybrid view, despite the fact that he is typically quite explicitly in favour of *a priori* essentialism.

The passage I have in mind is the following (similar passages can be found elsewhere, e.g., Lowe 2014, 257):

The growth of objective knowledge consists [...] in a constant interplay between an *a priori* element—knowledge of essence—and an *a posteriori* element, the empirical testing of existential hypotheses whose possibility has already been anticipated *a priori*. (Lowe 2013, 156)

Lowe adds that this interplay does not have a foundational ‘starting point’ and it’s clear that he wants the process to be a fallibilist one. He also makes it clear that the empirical element in the process is not merely perceptual evidence, but rather ‘empirical evidence’ as it is understood in scientific practice. But it looks as if knowledge of essence is supposed to be contained entirely in the *a priori* element. The context of this passage is Lowe’s criticism of conceptualism; that is, the view that essentialist knowledge concerns concepts. The idea is that we can at least sometimes know *a priori* what it is to be an object of a certain kind even if no objects of that kind actually exist. Empirical evidence enters the picture when we want to determine whether any objects of that kind do in fact exist. As an example, recall the case of transuranic elements discussed in section 2.2.1.

There’s much that I find appealing in Lowe’s account, but I think that the mentioned interplay between the *a priori* element and the *a posteriori* element is more subtle than Lowe leads us to believe. In particular, I’m not fully convinced that we can, properly speaking, know what it is to be an object of a certain kind entirely *a priori* (see Tahko 2015a)—and this would suggest that knowledge of essence is the *result* of this interplay rather than present only in the *a priori* element. In fact, I think that there is a slight tension in Lowe’s own account in this regard, for recall what was said above about the role of causal powers in this process: it seems that in order to address the unity problem, Lowe has to resort to a strategy familiar from Oderberg; namely, to insist that we can determine the identity and existence conditions of objects (at least partly), by observing whether the object’s distinctive causal powers (which are grounded in its essential properties) continue to be manifested (Lowe 2013, 115). Now, it should be plain to see that if we must refer to the causal powers of objects in order to gain epistemic access to their essential properties (in which these causal powers are grounded), then we must at least sometimes rely on empirical evidence to accurately determine what it is to be an object of a certain kind. We may, of course, rely on previous empirical information when we speculate what kind of causal powers objects of a certain kind (such as some yet to be synthesized transuranic elements) might have, but how could we know that these causal powers arise from the essential properties of the very object that we had in mind? Only by rigorous empirical testing, it seems to me: we may have in mind several candidate objects that have very similar causal powers, but empirical testing will help us to single out which of these candidate objects we are in fact dealing with. This is how we can determine the existence and identity conditions of that object more accurately. Note that there is still room for an *a priori* element here, for in order to get this process started (even if there is no foundational ‘starting point’, as Lowe claims),

we must indeed have at least some *a priori* knowledge about what kinds of objects are metaphysically possible. This is especially striking in the case of transuranic elements, since many of them can only be created by our particle accelerators or by supernovae, but we have nevertheless been able to determine many of their essential properties prior to having the relevant empirical evidence. For Lowe, this is a crucial aspect of the relationship between metaphysics and natural science.²⁴

To conclude, it seems that Lowe has given us an intriguing picture of the role of *a priori* knowledge in the epistemology of essence, but this picture does need to be completed with the help of empirical elements. If this is right, then it would be a mistake to describe Lowe's account as purely rationalist, even in the sense that we first grasp the essences of possible entities *a priori* and then proceed to determine which ones of them actually exist with the help of empirical evidence. Instead, we can see that Lowe's account assumes a subtle interplay of *a priori* and *a posteriori* elements, which may, properly interpreted, come close to a type of hybrid view about the epistemology of essence.²⁵

4. References

- Austin, C. J. 2017. 'Aristotelian Essentialism: Essence in the Age of Evolution'. *Synthese* 194(7): 2539–56.
- Bird, A. 2007a. *Nature's Metaphysics*. Oxford: Clarendon Press.
- Bird, A. 2007b. 'A Posteriori Knowledge of Natural Kind Essences'. *Philosophical Topics* 35 (1/2): 293–312.
- Bird, A. 2008. 'Lowe on A Posteriori Essentialism'. *Analysis* 68: 336–44.
- Boolos, G. 1971. 'The Iterative Conception of Set'. *Journal of Philosophy* 68(8): 215–31.
- Brody, B. A. 1973. 'Why Settle for Anything Less than Good Old-Fashioned Aristotelian Essentialism'. *Noûs* 7(4): 351–65.
- Correia, F. 2011. 'On the Reduction of Necessity to Essence'. *Philosophy and Phenomenological Research* 84(3): 639–53.
- Dummsday, T. 2010. 'Natural Kinds and the Problem of Complex Essences'. *Australasian Journal of Philosophy* 88(4): 619–34.
- Elder, C. 2004. *Real Natures and Familiar Objects*. Cambridge, MA: MIT Press/Bradford Books.
- Ellis, B. 2001. *Scientific Essentialism*. Cambridge: Cambridge University Press.
- Fine, K. 1994. 'Essence and Modality'. In *Philosophical Perspectives 8: Logic and Language* ed. J. E. Tomberlin, 1–16. Atascadero, CA: Ridgeview.
- Fine, K. 1995a. 'The Logic of Essence'. *Journal of Philosophical Logic* 24(3): 241–73.
- Fine, K. 1995b. 'Senses of Essence'. In *Modality, Morality and Belief: Essays in Honor of Ruth Barcan Marcus* ed. W. Sinnott-Armstrong, D. Raffman, and N. Asher, 53–73. Cambridge: Cambridge University Press.

²⁴ For more details about this aspect of Lowe's work, see Lowe 2011b. For discussion, see Tahko 2015b and Morganti and Tahko 2017.

²⁵ I'd like to thank Ramiro Caso, Jonah Goldwater, Kathrin Koslicki, Laurie Paul, and David Oderberg for helpful comments on earlier versions of this chapter.

- Fine, K. 1995c. 'Ontological Dependence'. *Proceedings of the Aristotelian Society* 95: 269–90.
- Fine, K. 2005. 'Our Knowledge of Mathematical Objects'. In *Oxford Studies in Epistemology* ed. T. S. Gendler and J. Hawthorne, 89–109. Oxford: Oxford University Press.
- Gorman, M. 2005. 'The Essential and the Accidental'. *Ratio* 18: 276–89.
- Horvath, J. 2014. 'Lowe on Modal Knowledge'. *Thought* 3: 208–17.
- Koslicki, K. 2012. 'Essence, Necessity and Explanation'. In *Contemporary Aristotelian Metaphysics* ed. T. E. Tahko, 187–206. Cambridge: Cambridge University Press.
- Linnebo, Ø. 2010. 'Pluralities and Sets'. *The Journal of Philosophy* 107(3): 144–64.
- Locke, J. 1975. *An Essay Concerning Human Understanding*, ed. P. H. Nidditch. Oxford: Clarendon Press.
- Lowe, E. J. 1998. 'Form without Matter'. *Ratio* 11(3): 214–34.
- Lowe, E. J. 2006. *The Four-Category Ontology: A Metaphysical Foundation for Natural Science*. Oxford: Oxford University Press.
- Lowe, E. J. 2007. 'A Problem for A Posteriori Essentialism Concerning Natural Kinds'. *Analysis* 67: 286–92.
- Lowe, E. J. 2008a. 'Two Notions of Being: Entity and Essence'. *Royal Institute of Philosophy Supplements* 83(62): 23–48.
- Lowe, E. J. 2008b. 'Reply to Bird on a Posteriori Essentialism'. *Analysis* 68: 345–7.
- Lowe, E. J. 2011a. 'Locke on Real Essence and Water as a Natural Kind: A Qualified Defence'. *Aristotelian Society Supplementary Volume* 85(1): 1–19.
- Lowe, E. J. 2011b. 'The Rationality of Metaphysics'. *Synthese* 178(1): 99–109.
- Lowe, E. J. 2012. 'What is the Source of our Knowledge of Modal Truths?'. *Mind* 121: 919–50.
- Lowe, E. J. 2013. *Forms of Thought*. Cambridge: Cambridge University Press.
- Lowe, E. J. 2014. 'Grasp of Essences versus Intuitions: An Unequal Contest'. In *Intuitions* ed. T. Booth and D. Rowbottom, 256–68. Oxford: Oxford University Press.
- Morganti, M. and Tahko, T. E. (2017) 'Moderately Naturalistic Metaphysics'. *Synthese* 194(7): 2557–80.
- Needham, P. 2011. 'Microessentialism: What is the Argument?'. *Noûs* 45(1): 1–21.
- Oderberg, D. 2007. *Real Essentialism*. London and New York: Routledge.
- Oderberg, D. 2011. 'Essence and Properties'. *Erkenntnis* 75: 85–111.
- Parsons, C. 1983. 'What Is the Iterative Conception of Set?'. Reprinted in his *Mathematics in Philosophy*, 268–97. Ithaca: Cornell.
- Paul, L. A. 2004. 'The Context of Essence'. *Australasian Journal of Philosophy* 82: 170–84.
- Paul, L. A. 2006. 'In Defense of Essentialism'. *Philosophical Perspectives* 20(1): Metaphysics, 333–72.
- Putnam, H. 1990. 'Is Water Necessarily H₂O?'. In *Realism with a Human Face* ed. J. Conant, 54–79. Cambridge, MA: Harvard University Press.
- Shalkowski, S. 1997. 'Essentialism and Absolute Necessity'. *Acta Analytica* 12(2): 41–56.
- Shalkowski, S. 2004. 'Logic and Absolute Necessity'. *The Journal of Philosophy* 101(2): 55–82.
- Shalkowski, S. 2008. 'Essence and Being'. *Royal Institute of Philosophy Supplements* 83(62): 49–63.
- Tahko, T. E. 2015a. 'Natural Kind Essentialism Revisited'. *Mind* 124(495): 795–822.
- Tahko, T. E. 2015b. *An Introduction to Metametaphysics*. Cambridge: Cambridge University Press.
- Tahko, T. E. 2017. 'Empirically-Informed Modal Rationalism'. In *Modal Epistemology After Rationalism* (Synthese Library) ed. R. W. Fischer and F. Leon, 29–45. Dordrecht: Springer.

- Tahko, T. E. and Lowe, E. J. 2015. 'Ontological Dependence'. *The Stanford Encyclopedia of Philosophy* (Spring 2015 Edition), ed. E. N. Zalta, <<http://plato.stanford.edu/archives/spr2015/entries/dependence-ontological/>>.
- Vaidya, A. 2010. 'Understanding and Essence'. *Philosophia* 38(4): 811–33.
- Vaidya, A. 2015. 'The Epistemology of Modality'. *The Stanford Encyclopedia of Philosophy* (Summer 2015 Edition), ed. E. N. Zalta, <<http://plato.stanford.edu/archives/sum2015/entries/modality-epistemology/>>.
- Yablo, S. 2006. 'Circularity and Paradox'. In *Self-Reference* ed. T. Bolander et al., 139–57. Stanford: CSLI.

Essence and Necessity

The Case of Normative Nonnaturalism

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1. Introduction

There are at least two ways, in metaethics, to be a naturalist.¹ The first consists in denying that moral values and norms are properties inherent to the real world, which is a purely physical world. This position is traditionally labeled ‘non-cognitivism’—an umbrella term that covers a range of antirealist moral views.² The second form of naturalism, by contrast, accepts moral realism, but only on the condition that moral properties are somehow amenable to natural ones. It is true that the twentieth century witnessed a gradual liberalization of naturalism in metaethics, as well as in other domains, like the philosophy of mind. Philosophers moved from various identity theories to nonreductive naturalism. In the stricter—as in the more liberal—way, however, the majority of moral philosophers still adhere to the tenets of naturalism. That said, it must be noted that the quite recent renewal of interest in moral nonnaturalism reflects a certain degree of dissatisfaction with naturalism and its predominance in most fields of philosophy (Koons and Bealer 2010). Just as it is reasonable in the philosophy of mind to affirm that many aspects of the mental cannot be reduced to natural characteristics (such as the subjective character of conscious experiences or the intentionality of the mind), it is also plausible in metaethics to maintain that issues about the normative status of moral properties or the is–ought dichotomy

¹ At the beginning of this chapter, I would like to give the reader some linguistic clarifications. ‘Normative’ is a broader notion than ‘moral’: while normativity refers to a general ‘ought-to-be’, the moral ‘ought-to-be’ prescribes not only that one’s own good be realized, but also the good of others. ‘Normative’—and ‘moral’—belong to a different modality of being than ‘natural’ and ‘descriptive’. In Section 3 I shall introduce the notion of ‘end’, which will have either a normative or a moral meaning according to the context. I shall also introduce other meanings which that section will illustrate. On the difference between normativity and morality see Enoch (2011, 2 and 86ff).

² Moral realism consists of the following claims. 1. Moral facts exist independently of the subject which makes them objects of knowledge. 2. Sentences referring to moral facts are susceptible of being true. 3. At least some moral sentences are true. The moral realist accepts all three of these claims, the non-cognitivist does not accept any, and the *error theorist* accepts 1 and 2, but not 3. For an overview of non-cognitivism—currently known mainly as expressivism—and of other metaethical perspectives, see Shafer-Landau and Cuneo (2007) and (2014).

should hardly be reduced to issues of a mere natural character. The debate around nonnaturalist metaethics rages and its pluralistic nature makes it impossible to reach even a partial result. Thus, in my chapter, I will limit myself to analyzing a specific aspect of this debate—namely, the fact that there is still some reluctance to propose nonnaturalism as an autonomous ontological view. Some authors borrow too much from nonreductive naturalism; others think it more appropriate to conduct a defense at a merely epistemological level.

I shall try to overcome such reluctance and show what an ontology for normative and moral nonnaturalism could look like. As it is in the philosophy of mind that no progress can be made without taking into account the deep ontological roots of the discipline, so it is that the central issues of metaethics can only be properly addressed on the basis of a thorough investigation of their ontological grounding.

2. Ontological vs. Epistemological Moral Nonnaturalism

A brief survey of the recent literature reveals that many works on nonnaturalism have been published around the world in the last few years.³ What does contemporary nonnaturalism look like? Even though there are many different ways of defending nonnaturalism, it happens that the most traditional approach to nonnaturalism—that is, the ontological one—tends, nowadays, to be neglected in favor of a merely epistemological account. In this section, I shall illustrate the view of an author who supports nonnaturalism from a mainly epistemological point of view and I shall address some criticisms of his perspective.

Moral Realism. A Defense by Russ Shafer-Landau can be seen as a powerful and thoroughly argued nonnaturalist counterpoint to David Brink's *Moral Realism and the Foundations of Ethics*—the 'manifesto' of nonreductive moral naturalism. As Shafer-Landau himself admits, he owes a great deal to Brink's conception of moral ontology (76), to the point of declaring: 'Indeed, it is a picture which I, counting myself a non-naturalist, accept in all essentials' (64). At first glance this opening might disturb a sincere nonnaturalist and puzzle those who believe there to be a difference between nonreductive naturalism and nonnaturalism. But this concern is soon dispelled by Shafer-Landau, in whose view the true difference between the two conceptions is mainly epistemological, rather than ontological. From the ontological point of view, nonreductive naturalism and nonnaturalism are very similar, their only difference being that the latter does not conceive of moral properties as natural properties (64). But, according to Shafer-Landau, the natural/nonnatural distinction is not, contrary to tradition, a metaphysical question. Rather, this classification rests on a methodological and epistemological criterion: whether we are willing to consider ethics as a science. And, if so, whether we believe that ethical truths can be discovered in the

³ Among the book-length defenses of moral nonnaturalism, see Shafer-Landau (2003), Oddie (2005), Cuneo (2007), Wedgwood (2007), Enoch (2011).

same manner in which natural and social scientists discover truths in their domains; that is, *a posteriori*. Those who agree with this characterization are naturalistically oriented. The nonnaturalists maintain instead that ethics is not a science and that the knowledge of the fundamental moral axioms is a kind of *a priori* knowledge.

A first critical remark on Shafer-Landau's theses is that he appears to underestimate the natural/nonnatural distinction. Otherwise he would not affirm that he more or less shares the nonreductive naturalistic point of view, his *only* exception being the claim that moral properties are natural ones. But the natural/nonnatural distinction is so fundamental as to make up two radically different metaethical conceptions. Moreover, the relevance of this distinction is not undermined by the fact that Shafer-Landau's view bases it more on epistemological than on ontological reasons.

According to Shafer-Landau the nonreductive aspect of Brink's realism means that this kind of naturalism can countenance the existence of real properties over and above those of physics and the special sciences. This aspect of Brink's realism implies that both nonreductive naturalism and nonnaturalism give rise to a property dualism—or rather, to a property pluralism. In Shafer-Landau's view the natural/nonnatural distinction is not as important as the fact that there are many levels of reality, the properties of which are described by the corresponding special sciences and are not reducible to the properties of the other levels.

As to Shafer-Landau's claim that both nonreductive naturalism and nonnaturalism would imply a property dualism or pluralism, I would like to note critically that he seems seriously to misunderstand the role that special sciences play in nonreductive naturalism. This conception construes ethics as a second-order special science, whose properties cannot be deduced from laws belonging to the social or biological level. In the same way, the first-order special sciences cannot have their properties deduced from the laws of physics. This nonreductive point of view, however, in no way implies a property dualism or pluralism, since both the moral properties and the social and biological properties are natural properties, even though they are not reducible to the physical ones. What we have here is an ontological monism combined with an epistemological pluralism, a combination that would be more at home in an emergentist perspective.⁴ There is thus only one kind of genuine dualism; namely, the dualism between natural and nonnatural properties. It is a form of dualism because nonnaturalism countenances two different categories of reality, the natural and the normative, hence is pluralistic both from the ontological and the epistemological points of view. The consequence to be drawn is that nonreductive naturalism and nonnaturalism do not have 'identical ontological inventories' and, if they both share a nonreductive metaphysics, this metaphysics is not the same (64). Nonreductive naturalists and nonnaturalists do not have parity as regards their respective ontological commitments. It is therefore incorrect to maintain, as Shafer-Landau does, that the only genuine difference between the two views is an epistemological one. Epistemological inquiry is needed to discover whether and how nonnatural properties exist, but the

⁴ On this see a classic of British Emergentism, C. D. Broad (1925).

natural/nonnatural difference is an ontological one. Being aware of this allows us to re-establish a symmetry—which Shafer-Landau's approach does not maintain—between the ontological and the epistemological levels. In fact, one might ask how *a priori* knowledge of a certain domain is possible if that domain contains no nonnatural characteristics. Unless one abides by a Kantian perspective, it is quite unreasonable to postulate an *a priori* synthesis which is not correlated to the structure of reality.

A second aspect of Shafer-Landau's viewpoint that I intend to examine critically is the modality of the relationship posited between moral and natural properties. The parallel that the author draws between nonreductive naturalism and nonnaturalism extends to such a relationship. Indeed, he maintains that moral properties are *constituted* by natural properties. To be more precise, '... natural facts do exhaustively compose moral ones. There is nothing to a case of generosity, or viciousness, or dutiful action, other than the natural features that constitute such properties. Something exemplifies a moral property entirely in virtue of its possessing certain natural features' (2003, 75).⁵ Constitution explains the supervenience of moral properties on natural ones. 'Covariance is explained by means of constitution: something possesses the moral status it does in virtue of its descriptive constituents' (86). Moreover, a moral theory which is capable of explaining the supervenience of the moral on the natural is a better theory than one—such as non-cognitivism—that is not able to do so (89).

But why is the constitution thesis of such importance to this author's picture? On my understanding, it principally serves two aims, which are intertwined. The first aim consists in allaying any suspicion that the nonnaturalist is not ontologically parsimonious: he is not as such required to postulate extravagant entities or magic. In this respect, a property dualism according to which moral properties are exhaustively constituted by natural ones, is fully sufficient for the nonnaturalist's purposes. Shafer-Landau thinks in particular that a position similar to substance dualism should be avoided because it would multiply entities beyond necessity, and results in a bloated ontology (74). But, as the author himself admits, his ideas about constitution draw much of their inspiration from nonreductive physicalism, as the latter is understood in the philosophy of mind. In nonreductive physicalism, however, there is no question of genuine dualism. In fact, this view regards mental properties as situated at two different levels: the abstract and the concrete. At the abstract level, properties are exclusively defined by their formal role in producing behavioral output. This view is not committed to any ontological position, and is thus, in principle, also compatible with dualism. But, at the concrete level, mental properties are implemented and fixed by physical states. Mental properties are thus distinct from the physical ones only at the abstract level; once implemented, they are token-identical with them. On the other hand, if we had a genuine property dualism—that is, one that considers mental properties to be concrete properties, which were token-different from their physical base—we would see that the difference between property dualism and substance

⁵ For a treatment of constitution within nonreductive naturalism, see Brink (1989, 157ff).

dualism is not as pronounced as is often believed. In fact, both kinds of dualism countenance true ontological novelty—that is, either new domains of reality, such as the mental, or new categories of reality, such as the normative. Thus, a nonnaturalist à la Shafer-Landau faces a dilemma that, I submit, he cannot solve. Either the claim that the constitution thesis implies ontological parsimony is valid in virtue of a mistaken thesis—that is, that nonnaturalism and nonreductive physicalism share the same kind of property dualism. Or, if one switches to the—correct—thesis that only non-naturalism implies genuine property dualism, then the claim that the constitution thesis implies ontological parsimony is no longer valid. The reason for this is that substance dualism is not so ontologically different from property dualism as to be excluded from a nonnaturalist ontology, simply in virtue of the parsimonious nature of constitution. In conclusion, the constitution thesis does not offer good reasons for championing ontological parsimony.

Let us come now to the second main function that Shafer-Landau ascribes to constitution. In his opinion, the constitution thesis lends itself successfully to explaining the fundamental role played by supervenience in a nonnaturalist framework. The supervenience relation serves to ensure that the non-moral world keeps the moral one under control. ‘If the moral fails to supervene on the non-moral, then the non-moral world does not control the moral world. But if that world does not control the moral world, then the moral world is out of control. Moral assessments would be arbitrary’ (78). But supervenience is only able to exert such control if moral properties are exclusively constituted by the natural ones. In many places in his 2003 book Shafer-Landau emphasizes that natural facts have to constitute moral ones *exhaustively*. Otherwise it would be necessary to complement natural facts with something else, and Shafer-Landau thinks that that ‘something else’ can only be supernatural facts, like divine will. Without exhaustive constitution we could not explain the systematic covariance between the moral and the natural. This is similar to the mental/physical case, in which we could not explain the intimate connection between mind and body if we disavowed exhaustive constitution in favor of substance dualism. In such a case we should espouse occasionalism, positing that the actions of God explain the close dependence of the mental on the physical (78).

The aforementioned topics pave the way for a range of questions, which I shall try to answer in section 3, where I shall outline my own view on moral nonnaturalism. So, I shall close this section by highlighting the issues addressed in it.

A first fundamental question concerns the notion of constitution. Shafer-Landau does not carry out any in-depth investigation into the features that the constitution thesis should assume in metaethics, resorting merely to a thesis similar to that espoused in the philosophy of mind. But even in the philosophy of mind, the constitution thesis is controversial. The example that is usually employed to illustrate constitution is that of the statue constituted by the material of which it is made. However, when the constitution thesis is applied not to particulars, but to the relationship between properties, it is not crystal clear what it means to say that a physical property constitutes a mental or a

moral one.⁶ A second question is whether natural properties should constitute moral ones only in an exhaustive way. Exhaustiveness is both a necessary and a sufficient condition for supervenience to exert control on the moral. But is it really true that the apparent dependence of the moral on the natural is so close as only to be explicable through an appeal to exhaustive constitution? What, instead, if the moral and the natural had an intimate connection that did not require dependence, but reciprocal independence and interdependence? Furthermore, is it true that the only alternative to exhaustive constitution is recourse to positing that supernatural properties complement the natural ones? I do believe that this claim is incorrect, and, to be all the more so, if we introduce an issue that has remained behind the scenes so far—namely, the place of *normativity* in the relations of constitution and supervenience. If the moral supervenes on the descriptive *as a matter of conceptual truth* (Shafer-Landau, 78, my italics), the nonnaturalist faces the problem of avoiding a logical fallacy, consisting in the breaking of Hume's Law.⁷ This problem, moreover, paves the way for another big question: what is the modal nature of the supervenience within a nonnaturalist standpoint?

I shall stop here. As the reader can see, there is much material to be examined.

3. Outlines of an Ontological Foundation for Moral Nonnaturalism

We have just seen that, according to Shafer-Landau, supervenience is in need of explanation, and that, to him, this explanation can be provided by the constitution relation. Constitution explains the main role of supervenience—that is, the function of the control that, in virtue of supervenience, the natural world exerts on the moral one. But control implies dependence. In this part of my chapter, I shall try to show that constitution is useful for explaining supervenience, but that this does not allow us to draw the consequence that moral properties depend on natural ones.

3.1 Normative supervenience

Let us first try to clarify the notion of normative supervenience. Why are normative predicates supervenient? They are, because they are attributable to any object *x* (be it an individual, a state of affairs or an action) only if:

- (i) *x* is characterized by suitable predicates, called basic predicates. Basic predicates denote descriptive properties. Such properties can be either empirical properties or properties of a different ontic kind, such as certain essential properties of an object. Let us use the term *subvenient base* to denote a suitable set of descriptive

⁶ Among the authors who have leveled this criticism of constitution, see Crane (2001, 58).

⁷ Hume's Law reads that the supervenience between descriptive and normative properties cannot be of a logical kind. Therefore, it is important to determine what it means that the normative supervenes *conceptually* on the descriptive.

properties, and the term *basic predicate* as synonymous with *basic property*. Let us use D to denote any descriptive property.

- (ii) x is characterized by suitable normative predicates. These predicates denote either deontic or evaluative properties. Normative predicates are *supervenient* on a suitable subvenient base. Again, we will use the term *normative predicate* as synonymous with *normative property*. Let us use N to denote any normative property.
- (iii) A supervenience relation occurs between x's descriptive and normative predicates.

What is a supervenience relation? Usually it is illustrated in the following way:

For any y, if y has some normative property N, then y possesses some descriptive property such that any object x with this property has normative property N.

Formally:

$$(\forall N)(\forall y)[Ny \rightarrow \exists D(Dy \wedge (\forall x)(Dx \rightarrow Nx))]$$

The heart of this definition—and of the debate that has developed around it—is the second implication, and its interpretation. First of all, this implication is understood as necessary implication; for that reason it is said to represent a strong kind of supervenience:⁸

$$(\forall N)(\forall y)[Ny \rightarrow \exists D(Dy \wedge \Box(\forall x)(Dx \rightarrow Nx))]$$

The major problem, however, concerns the nature of such necessity. Is it an analytical, metaphysical, nomological necessity, or otherwise? To answer this question, it is necessary to have a basic idea of the various kinds of necessity.

3.2 Different notions of necessity

Currently, there are two main lines of thought on the nature of modalities. According to the traditional line, which is currently represented mainly by B. Hale (1996, 2013), the fundamental notion of necessity is the logical, while the others are definable by relativization. Simplifying a little, the situation can be formalized as follows: logical necessity (which is absolute, $\Box_p$) coincides with the truth in all of the logically possible worlds; mathematical necessity ($\Box_{m,p}$) coincides with the truth in all of the worlds where mathematical laws are true, besides the logical; metaphysical necessity ($\Box_{m,p}$) coincides with the truth in all of the worlds where metaphysical laws obtain, besides the logical and the mathematical; and natural necessities coincide with truth in all of the worlds where natural laws are true, besides the metaphysical, mathematical, and logical. Clearly the various kinds of necessities that are different from the logical are defined by relativization. This is determined by the relevant set of laws in respect of which the set of possible worlds is restricted. Among the various kinds of necessity, appropriate hierarchical relationships are given. Naturally possible worlds, for example,

⁸ Leary (2015, 4) and Hamilton (2016, 13–15).

are such from the mathematical and metaphysical point of view, too. For this reason, the logical modality is the fundamental and maximally general modality.

The traditional view of modality presents highly appreciable aspects—as the hierarchical structure just illustrated—but those that are open to criticism, too. The most criticizable aspect is that the set of the respectively mathematical, metaphysical, and natural laws is presupposed by the definition of the corresponding concept of necessity. This conflicts with the fact that the laws themselves are instances of their respective necessity.

This shortcoming is overcome by the new essentialist conception of modality, developed independently by E. J. Lowe and K. Fine. To Lowe and Fine, in fact, the concept of necessity—including the logical—is derived from the notion of essentiality. In short:

Essential necessity: $\Box p =_{\text{def}} p$ is true in virtue of the essence of the items—either objects or properties—that are involved in the state of affairs expressed by p .

Other forms of necessity are then traced to the essential necessity. The relationship between essential necessity and the other kinds of necessity is only apparently similar to that obtaining in Hale's view between logical necessity and the other kinds of necessity. While in the traditional conception, the various kinds of necessity are definable by relativization, in the essentialist conception, they are definable by restriction. Natural necessity, for example, is a particular kind of essential necessity and the same holds for all the other forms of necessity.

However, the concept of restriction is not completely unrelated to that of relativization. In fact, owing to the definition of essential necessity, we have:

p is true in virtue of the essence of the items—either objects or properties—that are involved in the state of affairs expressed by $p \Leftrightarrow p$ is true in all of the metaphysically possible worlds.

The fundamentality of metaphysical necessity follows immediately, therefore, from the fact that necessity depends on essentiality, and that essences define the space of metaphysical possibility.⁹ The claim I intend to make is that the concept of necessity expressed by supervenience is that of essential necessity, where the essences at play are those of the items involved in the supervenience sentences and, in particular, the essences of normative properties.¹⁰

3.3 Normative supervenience and essential necessity

Once the framework of the ontic modalities has been clarified, we want to deploy the notions of essence and essential necessity to tackle the analysis of the relationship $\Box(\forall x)(Dx \rightarrow Nx)$. Let us utilize the letter P to denote the normative property to be positive.

⁹ Finally, even if the essentialist position is characterized by the concept of restriction, and not by that of relativization, restriction is, at least partially, conservative towards the hierarchization by relativization. That the foundation of modal necessity is—according to the essentialists—the essential (or metaphysical) necessity, does not prevent us from distinguishing between the various forms of necessity, and restricting them to subsets of the totality of the essential necessities. We will make use of this technical aspect later on.

¹⁰ Lowe (1998, 2006, 2008, 2010, 2011, 2012); Fine (1994a, 1994b, 1995).

Let us then start from the following supervenience sentence:

$$\Box(Dx \rightarrow Px) \quad (1)$$

As is known, the nonreductivist naturalists like Brink (1989, 160) adhere to a natural nomological concept of supervenience, and claim that the positive value of x , given the D property, obtain in all the nomologically—that is, naturally—possible worlds that are similar to the actual world. In short, (1) is now restricted to:

$$\Box_n(Dx \rightarrow Px) \quad (2)$$

where $\Box_n$ is the operator of nomological necessity.

However, we are not forced to share the naturalistic point of view, and to maintain that the supervenience relation is only nomological. Let us consider the case of a morally good action, such as rescuing someone from drowning. Such an action is good because it results in saving the life of the person drowning—in other words, if an action x is characterized by the natural properties D which define it as an instantiation of rescuing, then x is good (P). Now, it is reasonable to think that the necessity connecting antecedent and consequent is a broader kind of necessity than the nomological. Is it not true that in any world in which a rescuing action occurs, such an action is intrinsically good, independent of any possible variations of the natural laws ruling the actual world? It is at this point that the essentialist theory of necessity can help. According to this theory, the essence of an object is the set of traits that cannot be absent—that is, are unconditionally necessary—if the object is to be that which it is. Among the properties characterizing the essence of an object susceptible of valuation (such as a human being, for example) there is also the property of possessing a certain *telic structure*. Therefore, an object's metaphysical component is not only defined by a set of descriptive properties (be they observational, theoretical, modal, or non-modal), but also by the *end* to which the object is directed. The relation between essence and end is a metaphysically necessary relation. This metaphysical necessary relation is clearly shown in the example of the rescuing action. The relationship between human agency and the end of the preservation of life is essential, and independent of any nomological circumstances. If this end is properly grounded in human agency, then it is so in every metaphysically possible world. We are now in a position to introduce the definition of normative supervenience by means of that of metaphysical necessity:

$$\Box_m(Dx \rightarrow Px) \quad (3)$$

where $\Box_m$ is the operator of metaphysical necessity.

In the remainder of my chapter, I will try to show why the relation of constitution is useful in explaining the relation of metaphysical supervenience. Constitution will be analyzed in light of the telic nature possessed by objects, actions or states of affairs. This analysis will highlight the fact that end-directedness is a component of constitution along with descriptive properties, and that constitution cannot therefore be exhaustive.

3.4 Normative constitution

On Baker's understanding, the basic idea of the notion of constitution is that '... When certain things with certain properties are in certain circumstances, new things with new properties come into existence' (2000, 32). Baker offers the example of a piece of marble which constitutes David: this example highlights the fundamental characteristics of constitution. First, the properties of the piece of marble are different from David's properties, since the constituted thing (David) has causal powers which the constituting thing (the piece of marble) would not have had if it had not constituted a statue, such as attracting visitors or arousing the pleasure of aesthetic contemplation (33). Second, constitution is an asymmetric relation, because the piece of marble constitutes David, while David does not constitute the piece of marble (33). Third, one thing constitutes another only in certain circumstances. This aspect of the constitution relation is particularly important and can be formally codified in the following way (41ff). Let F be the property which characterizes the constituting object (for example, the property of being that piece of marble), and let G be the property that characterizes the constituted object (David). Then: if x , which is F , is in G -favorable circumstances, then there is an entity y , which is G , such that it spatially coincides with but is not identical with x , and is constituted by x .

The G -favorable conditions are thus crucial for constitution. The piece of marble is David only insofar as it is perceived as artwork, and in that the conditions are in place for this to happen. The question is now: does the property of being G supervene on that of being F , if an x which is F constitutes a y which is G ? The answer is yes, if the G -favorable conditions occur. Formally: if an x which is F constitutes a y which is G , then the following corresponding supervenience sentence holds true:

$$\Box_{G\text{-conditions}} (Fx \rightarrow Gx) \quad (4)$$

In fact, although it is true—as constitution theory argues—that the F is not identical with the constituted G , it is also true that, once constituted, the G possesses all of the F 's properties and vice versa, enabling us to say:

$$\Box_1 (Fx \wedge G\text{-conditions} \rightarrow Gx) \quad (5)$$

where $\Box_1$ is the operator of logical necessity. But (4) is easily derivable from (5), owing to the restriction of the necessity domain.

The G -favorable conditions can be various. The piece of marble constitutes the statue because there are subjects capable of enjoying artwork; the piece of colored cloth which constitutes a flag represents the unity of a nation, which requires the existence of human subjects who take themselves to belong to this nation; the piece of paper that constitutes the dollar bill represents a certain amount of purchasing power in the exchange of goods; the strands of DNA that constitute genes perform specific informational functions in the living cell. And what about an object endowed with a telic structure? What kind of G -favorable conditions obtain so as to constitute such an object? This is the most interesting case for our purposes. As we know from the previous sections, in fact, Shafer-Landau argues that the relation of constitution should be deployed to explain the supervenience of moral facts on natural ones. We thus need to show that supervenience

follows from constitution in the moral case. I surmise that that can occur only if P-favorable conditions are given—that is, conditions suitable to establish the normative dimension of property P. These conditions, moreover, should be grounded in the essential structure of the telic object. If things were this way—as I shall try to show in the next subsections—then we could say that in the moral case (5) should be interpreted as:

$$\Box_1(Dx \wedge \text{P-conditions} \rightarrow Px) \quad (6)$$

from which it is easy to derive:

$$\Box_{\text{P-conditions}}(Dx \rightarrow Px) \quad (7)$$

(3) and (7) are closely related in meaning. But (7) derives from the constitution thesis as applied to the normative domain—certain descriptive properties constituting object *x* as a normatively positive object—whereas (3) is a direct expression of the supervenience of P over D. For this reason (7) is the explanation of (3). And this is not the only result. From (7), it also follows that the implication of *Px* by *Dx* is grounded in the essence of the P-conditions; that is, those conditions that ensure the emergence of the moral dimension over the non-normative.

So, the constitution relation allows us to clarify the conditions in virtue of which normative properties can be conceived of as supervenient on natural ones. While supervenience only establishes the logical relation of implication (according to the modality which is the most adequate to a given context), constitution also aims to identify what grounds supervenience. Of course, as just mentioned above, the constitution of moral properties by natural ones depends on the existence of conditions favorable to the emergence of the normative dimension.

The nature of these conditions is not rigidly defined. It depends on the chosen normative theory, and on the conception of the good that underlies it. In this chapter I shall adopt a particular way of conceiving the good—namely the Aristotelian—according to which the good is in the attainment of the agent's end. It should be emphasized, however, that the thesis of normative constitution does not depend on this particular theory of the good. The theses I will endorse might just as well be adapted to a utilitarian theory of the good, or to a contractualist theory of the right.

3.5 Two notions of 'end'

In the context of language—be it ordinary, philosophical or scientific—the term 'end' can be understood according to very different meanings. However, in this chapter, the only concept of 'end' that is of interest to us is the normative. For this reason, it is important to draw some demarcation lines in respect to other meanings of the term, such as, in particular, the explanatory meaning. For this purpose, it is appropriate to expand on our topic, and to start from the notion of *convenience*. The reasons of convenience (*rationes convenientiae*) are usually expressed by means of sentences employing relational concepts, such as 'to have the function of', 'to be directed to the end of', and 'to be made for'. Let us consider the following sentences: 'The arrow has

the function of hitting the target', 'The arrow is directed to the end of hitting the target', and 'The arrow is made for hitting the target'. All of these sentences, which—aside from semantically irrelevant nuances—are synonymous, express the fact that hitting the target convenes to the arrow. Such sentences pertain not only to objects but also to states of affairs. We may say, for example, 'The heartbeat has the function of circulating blood', 'The heartbeat is directed to the end of circulating blood', and 'The heartbeat occurs in order to circulate blood'. These sentences express a relation of convenience, too.

In both cases, however, the convenience relation can be reversed, so it is possible to exchange 'that which convenes' with 'that to which it convenes'. That is, the convenience relation is not only expressed by the sentence 'The circulation of the blood convenes to the heart's beating', but also by the reverse sentence 'The heartbeat convenes to blood circulation'. Likewise, both 'hitting the target convenes to the arrow' and 'the use of an arrow convenes to the hitting of the target' are meaningful. At first glance, it may seem that both readings are equivalent, but, on closer inspection, it appears that the second reading is weaker than the first, since it reveals a narrower understanding of the convenience relation. Indeed, that the heartbeat convenes to blood circulation means that, given a certain physiological situation, the heart's beating is a necessary condition for blood circulation, or, more rigorously, a necessary component of the entire sufficient condition guaranteeing blood circulation. On the first reading, however, that blood circulation convenes to the heart's beating means that blood circulation fulfills the function implicit in the heartbeat or, in different words, that blood circulation represents the fulfillment of the end for which the heartbeat is made.

The difference between the two meanings of the convenience relation is of fundamental importance. In fact, only the first relation is normative. This normativity is revealed by the relevance of the notion of *fulfillment* in both examples I have mentioned above. Blood circulation fulfills the function implicit in the heartbeat, and hitting the target fulfills the function of the arrow as a projectile. In general, that which convenes represents a fulfillment of that to which it convenes. Now, that which represents a fulfillment is *good* with respect to that which is fulfilled. Thus, that which convenes to a certain object or state of affairs is good with respect to that object or state of affairs.

The second convenience relation, by contrast, is not normative in character, since it simply expresses the *descriptive* relation between a state of affairs and the conditions that are required for it to occur. For this reason there is no difficulty in recognizing that the second convenience relation can be empirically attested. It is a theoretical relation, but it is confirmable by empirical evidence, in the form of either constant regularities or—owing to the complexity of phenomena—of statistical regularities. The same does not hold true for the normative meaning of the convenience relation, for which evidence of empirical regularities has no relevance. That hitting the target convenes to the arrow is true, because this is its function, not because it hits it regularly. Hitting the target always convenes to the arrow, even if it never hits the target.

This last remark shows that the normative kind of purposiveness does not coincide with the claim that the end-directedness of things—and, as a consequence, the moral end-directedness of the human being—is attainable by means of the observation of regularities, understood as the sign of the purposiveness of things. It is of fundamental importance to distinguish between two different kinds of purposiveness. The first one is a *de jure* purposiveness, which is based on the normative relation of convenience, and is thus able to ground moral truths. The second one, instead, is a *de facto* purposiveness. This is because it consists in an *explanatory hypothesis* concerning the fact that certain phenomena regularly occur in nature. An analysis of Aquinas's fifth proof of the existence of God will clarify this notion of *explanatory purposiveness*. Aquinas starts from the empirical premise that certain entities without consciousness act constantly—or at least mostly—in a manner such that they can achieve what is best for them. From this, it is evident that they accomplish their ends not by accident but because they have been directed towards them. On the other hand, because they are devoid of consciousness, they cannot pursue their ends by dint of their own capacities. Hence, there exists someone that pre-ordains them to their ends (Aquinas, *Summa Theologiae*, I, q.2, a.3).

My aim is not so much to establish the validity of this argument as to discuss two essential aspects of it. First, the argument as a whole presupposes that the good of the entities at stake is justified in a way that is autonomous from the explanatory purposiveness in which the argument results. In order to assert that there is a coincidence between the outcome and the *de jure* end of a natural process, one must presuppose that the notion of fulfillment is already meaningful. This means that *explanatory purposiveness* presupposes *normative purposiveness* as its necessary condition. There are thus two notions of purposiveness—*de jure* purposiveness and *de facto* purposiveness—but only *de facto* purposiveness derives from experience, since it derives from *de jure* purposiveness, together with observed regularities. This conclusion accords perfectly with the second aspect of the argument that is worth highlighting. If regularity of behavior were also evidence of *de jure* purposiveness, then we should ask ourselves why such a regularity is in need of explanation. A regularity can be explained by higher regularities (e.g., laws are explained by principles) but it is not necessary *per se* to find an explanation for certain events' regular occurrence. Instead, it is when something does not correspond to one's expectations that an explanation is needed. In Aquinas's fifth proof the fact to be explained is the regularity of behavior which, according to *de jure* purposiveness alone might not be such. In fact, it is typical for normative necessity (as opposed to ontic necessity) not to be regularly respected. It does not hold true in general that, if *p* is a positive state of affairs, then *p* is true—that is, the implication $Pp \rightarrow p$ is not a principle of the normative modalities. But then, if for a *p* of a certain kind not only *P* but also *p* holds true, this fact needs to be explained. This is exactly the case of the normatively convenient states of affairs that occur regularly. An explanation of the correspondence between the 'ought-to-be' and the 'is' must be given. This explanation is provided, according to Aquinas, by the *de facto* purposiveness that has been impressed on things by God's intervention. In conclusion, from this viewpoint, there

are also two distinct kinds of purposiveness, only one of which, the *de facto* kind, can be justified by empirical evidence.

Of course, the Darwinian view has no place for finalistic explanations of living phenomena. From my standpoint, however, that aspect of the Darwinian theory is not important. The crucial point is that the Darwinian understanding of end-directed phenomena does not cast *de jure* purposiveness into doubt. In fact, *de jure* purposiveness is not an empirical notion and thus cannot be falsified by any empirical theory. Thus, only *de facto* purposiveness is open to Darwinian criticism. If *de facto* purposiveness is rebutted, however, the reason is not that the domain of complex living phenomena requires no explanation of the coincidence between perfection and its *de facto* realization. The reason is rather that contemporary Darwinism, which is supported by the findings of molecular biology, often succeeds in giving a causal explanation for this coincidence. But Darwinism also accepts the principle that it is not possible to explain a complex fact like life on the basis of structurally less complex facts, without giving an account of how this can occur. Why is this so? The reason is that the complex is more perfect than the less complex, and the occurrence of the more complex is not normal but, rather, abnormal. If the more complex occurs—as it does in the domain of living phenomena, where myriad conveniences unfold—then an adequate explanation is called for.

A possible counter-criticism to these reflections could proceed as follows. It is true that from the simple regularity with which the arrow hits its target, it is not possible to deduce that hitting the target is the arrow's *de jure* end. It is also true, however, that the fact that this is the arrow's *de jure* end can be established by considering how the arrow is made: its weight is distributed in a balanced way, so to allow for regular trajectory which is conducive to hitting the target, its tip is sharp, and so on. Now, these are empirically observable properties, leading one to think that knowledge of *de jure* end can be gained from experience. The answer to this criticism is that knowledge of *de jure* end cannot be obtained from experience *alone*. That hitting the target is the arrow's *de jure* end cannot be known only on the basis of the empirically observable fact that the arrow is useful for hitting the target. Rather, it is known from that fact along with the further fact that the realization of the potentialities inherent in an object is good for that object, and that is thus good for the arrow to deploy its characteristics in order to hit the target. But the principle, according to which the realization of the potentialities inherent in an object is good, is a bridge-principle that cannot be derived from experience.

3.6 *Nonnaturalist conditions for normative supervenience*

The distinction between the two notions of an end, which were examined in the last subsection, will now be deployed to investigate the relationship between normative supervenience and constitution. Let us take again the example of the rescuing action. We saw above that the property of positivity P concerning action *x* supervenes on the descriptive property D, which describes the rescuing action as a modality of rescuing.

We also saw that this relation is explained by the fact that the positivity of the action is constituted by the properties defining its modality as a rescuing action. However, the constitution relation works only if the favorable conditions for establishing positivity are met. What are these conditions? How can we answer this question if we want to take into account that the conditions for positivity must be fixed in order to safeguard the principle of the invariance of supervenient normative properties? The favorable conditions consist in the accordance of the rescuing action with the agent's end. Since the agent's end is to realize the good, then to save the lives of others is, *ceteris paribus*, part of this. The action is positive, then, in virtue of the fact that it realizes the conditions of realization of the agent's end.

The rescue example has a clear parity of structure with the example of the arrow. The end of hitting the target is constitutive of the arrow (it is convenient to the arrow), in the same way that the end of helping others when they are in danger is constitutive of the human agent (it convenes to the human agent). The formalization of what has just been said will allow us to derive further relevant consequences. First, these two cases can be easily adapted to principle (6), which expresses the content of the constitution relation, and to principle (7)—the translation of (6) in terms of supervenience. Concerning (7) let us first discuss the case of the arrow. Let Ax mean that x is an arrow; let Tx mean that x hits the target and let $C(x, Tx)$ mean that realizing T convenes to x ; that is, that x is directed to the realization of T . Clearly, this indicates that it is part of x 's essence to possess a telic structure, and that x fulfills this *telos*. We then have:

$$\Box_1 (Ax \wedge (Ax \rightarrow C(x, Tx)) \rightarrow Px) \quad (6')$$

Prima facie one does not notice that (6') is a kind of logical necessity. But, in order to realize this, it suffices to introduce the following definition of normative positivity:

$$Px =_{\text{def}} \exists Q(Qx \wedge C(x, Qx)) \quad (8)$$

(8) affirms that an object is positive if being Q is its end, and x is Q —that is, if x realizes the end to which it is directed. By logical rules one then obtains:

$$Tx \wedge C(x, Tx) \vdash \exists Q(Qx \wedge C(x, Qx))$$

from which, by the definition of Px

$$Tx \wedge C(x, Tx) \vdash Px$$

thus

$$Ax \wedge (Ax \rightarrow C(x, Tx)) \wedge Tx \vdash Px$$

Clearly, this is equivalent to (6').

At this point, let us consider the nature of the implication

$$Ax \rightarrow C(x, Tx) \quad (9)$$

(9) is a sentence expressing the normative component of x 's essence. This component constitutes the condition according to which if x reaches the target, x is positive.

In other words, it is in the essence of x that, if x reaches the target, then x is a good arrow. It thus expresses a kind of metaphysical necessity. But then the relation between $Ax \wedge Tx$ and Px is also metaphysical. As a consequence, we can also write the constitution sentence corresponding to principle (7) as:

$$\Box_m (Ax \wedge Tx \rightarrow Px) \quad (7')$$

In conclusion, the supervenience sentence regarding the relation between 'being an arrow that hits the target' and 'being a good arrow' has been obtained by means of the constitution analysis of the arrow's telic structure. The analysis has highlighted the metaphysical nature—tied to the arrow's essence—of the relation. The arrow's essence has been understood in a broader sense, since the supervenience sentence contains—besides descriptive properties—a normatively mixed sentence like (9).

4. Conclusions

In the second section of my chapter I have addressed a number of questions, which I shall now try to answer.

Must constitution be exhaustive? Is a recourse to supernatural properties the only alternative to its exhaustiveness? The answer to both questions is negative. Constitution is not exhaustive, because the essence of an object or event includes not only descriptive properties, but their telic structure, too. Therefore, a non-exhaustive constitution relation must not necessarily rely on supernatural entities.

Does the non-exhaustiveness of constitution prevent natural properties from exercising control over the moral ones? Non-exhaustive constitution ensures the natural/moral covariance, and continues to be the foundation of supervenience, but without being controlled by the natural properties and without dependence of the moral properties on the natural. If we examine sentence (9), we see that it is a mixed normative sentence expressing the fundamentality—the originality—of the normative dimension. This explains the intimate connection between the moral and the natural, which is present in the antecedent of the constitution sentence. If things are this way, control is not necessary. Control, namely, is exercised over something external, not towards a co-original dimension, as normativity happens to be.

Does normative supervenience imply the breach of Hume's law? The answer is negative, again: as we know, the necessity typical of the relation of normative supervenience is metaphysical, not logical. This gives an answer to the last question too—that is, what is the modal nature of the supervenience relation within a nonnaturalist perspective? My guess is that only a metaphysical modality, based on the relationship between descriptive and normative properties in the essences of objects can express the nature of the constitution and supervenience relations from a nonnaturalist viewpoint.

5. References

- Baker, L. R. 2000. *Persons and Bodies. A Constitution View*. Cambridge: Cambridge University Press.
- Brink, D. O. 1989. *Moral Realism and the Foundations of Ethics*. Cambridge: Cambridge University Press.
- Broad, C. D. 1925. *The Mind and Its Place in Nature*. London: Routledge and Kegan Paul.
- Crane, T. 2001. *Elements of Mind. An Introduction to the Philosophy of Mind*. Oxford: Oxford University Press.
- Cuneo, T. 2007. *The Normative Web. An Argument for Moral Realism*. Oxford: Oxford University Press.
- Enoch, D. 2011. *Taking Morality Seriously. A Defense of Robust Realism*. Oxford: Oxford University Press.
- Fine, K. 1994a. 'Essence and modality'. *Philosophical Perspectives* 8: 1–16.
- Fine, K. 1994b. 'Ontological dependence'. *Proceedings of the Aristotelian Society* 95: 269–90.
- Fine, K. 1995. 'The logic of essence'. *Journal of Philosophical Logic* 24: 241–73.
- Hale, B. 1996. 'Absolute necessities'. *Philosophical Perspectives* 10: 93–117.
- Hale, B. 2013. *Necessary Beings: An Essay on Ontology, Modality, and the Relations Between Them*. Oxford: Oxford University Press.
- Hamilton, C. 2016. *Supervenience, Explanations, and Non-Naturalism*. Undergraduate Honors Thesis. Austin: Texas.
- Koons, R. C. and G. Bealer, eds. 2010. *The Waning of Materialism*. Oxford: Oxford University Press.
- Leary, S. 2017. 'Nonnaturalism and normative necessity'. Oxford: *Oxford Studies in Metaethics* 12: 76–105.
- Lowe, E. J. 1998. *The Possibility of Metaphysics*. Oxford: Oxford University Press.
- Lowe, E. J. 2006. *The Four-Category Ontology. A Metaphysical Foundation for Natural Science*. Oxford: Oxford University Press.
- Lowe, E. J. 2008. 'Two notions of being: entity and essence'. In *Being: Developments in Contemporary Metaphysics* ed. R. Le Poidevin, 23–48. Cambridge: Cambridge University Press.
- Lowe, E. J. 2010. 'Ontological Dependence'. *The Stanford Encyclopedia of Philosophy*, Edward N. Zalta ed. <http://plato.stanford.edu/archives/spr2010/entries/dependence-ontological/>
- Lowe, E. J. 2011. 'Locke on real essence and water as a natural kind: A qualified defense'. *Proceedings of the Aristotelian Society Supplementary Volume*, 85: 1–19.
- Lowe, E. J. 2012. 'What is the source of our knowledge of modal truth?'. *Mind* 121: 919–50.
- Oddie, G. 2005. *Value, Reality and Desire*. Oxford: Clarendon Press.
- Shafer-Landau, R. 2003. *Moral Realism. A Defence*. Oxford: Oxford University Press.
- Shafer-Landau, R. and T. Cuneo, eds. 2007. *Foundations of Ethics. An Anthology*. Malden, MA: Blackwell.
- Shafer-Landau, R. and T. Cuneo. 2014. 'The moral fixed points: new directions for moral non-naturalism'. *Philosophical Studies*, 171: 399–443.
- Smith, M. 2004. *Ethics and the A Priori. Selected Essays on Moral Psychology and Meta-Ethics*. Cambridge: Cambridge University Press.
- Wegdwood, R. 2007. *The Nature of Normativity*. Oxford: Clarendon Press.

8

Lowe's New Ontological Argument

Peter van Inwagen

1. Ontological Arguments

As every schoolchild knows, it was Immanuel Kant who coined the term *Ontologische Beweis*, which he applied to an argument that had been invented by Descartes and had later been refined in various ways by Leibniz and the members of the Wolff-Baumgarten school—the school in which Kant had for so many years slumbered dogmatically. At some point in the nineteenth century, the term began to be applied to an argument—or perhaps there were two of them—that had been devised by St Anselm over 500 years before Descartes wrote his *Meditations on First Philosophy*. Apparently the word *Beweis* was not regarded by Kant and his contemporaries, and is not regarded by present-day German speakers, as an ‘achievement term’—for Kant, there could be a *Beweis* that was incorrect or a failure, the *Ontologische Beweis* of course being a case in point. Although the usual English translation of *Beweis* is ‘proof’, Anglophone philosophers are very strongly inclined to hear ‘proof’ as an achievement term. For that reason, an *Ontologische Beweis* is usually called an ontological argument in English. One does, of course, frequently hear the phrase ‘the ontological argument’, but I don’t think that there is any useful sense in which Anselm’s argument (or his two arguments) and Descartes’s argument and Leibniz’s argument can be said to be the same argument or to be variants on or elaborations of some proto-argument.

It is pretty generally agreed amongst present-day analytical philosophers that the arguments of Anselm and Descartes and Leibniz are logically defective (in a broad or ‘postwar-Oxonian’ sense of ‘logically’: they can certainly be presented in forms that are valid according to ordinary textbook logic) and irremediably logically defective. I don’t mean to imply that *all* present-day analytical philosophers regard them as logically defective, but it’s a safe bet that those philosophers who don’t so regard them constitute a very small minority. Owing largely to the work of Charles Hartshorne and Alvin Plantinga, however, the second half of the twentieth century witnessed the discovery of some arguments of a kind that have come to be called *modal* ontological arguments.¹ And these arguments, or so most philosophers who have studied them

¹ This work of Hartshorne and Plantinga was to a certain extent anticipated by Leibniz, but imperfectly.

will agree, are not logically defective. Here is a typical modal ontological argument. (I choose this argument as an example because its conclusion is the same as that of the argument of E. J. Lowe's that is the topic of this chapter.) It is the argument that in my book *Metaphysics* I called by the rather unwieldy name 'the Minimal Modal Ontological Argument'.

It is possible for there to be something that is both concrete and necessarily existent.
Necessarily, whatever is concrete is essentially concrete

hence,

There is something that is both concrete and necessarily existent.

(Let us say, for the moment, that something is *concrete* if it is capable of entering into causal relations, if it is something that can be either an agent or a patient. Lowe has a somewhat different account of 'concrete', which I shall discuss presently.) This argument is valid in S5 and in no weaker modal system. I would not say that 'being invalid in all modal systems weaker than S5' was a logical defect in an argument—any more than I would say that 'depending on the Axiom of Choice' was a defect in a mathematical proof. J. N. Findlay once argued that any argument that purported to prove the existence of a necessarily existent thing must be logically defective (in the broad, postwar-Oxonian sense of 'logically') since the very concept of a necessarily existent thing was logically defective. But his argument for that conclusion depended on the premise that 'necessity in propositions merely reflects our use of words, the arbitrary conventions of our language'. He calls this 'the more modern view'. On the *still* more modern view, however, this thesis is false: 'Iron is composed of atoms with 26 protons in their nuclei' and 'Between every number greater than 2 and its double there is a prime number' are generally, if not universally, accepted as counterexamples to it.

If the Minimal Modal Ontological argument contains no logical defect, however, that does not imply that it is a demonstration of the truth of its conclusion. And the present-day consensus is that neither that argument nor any other modal ontological argument is a demonstration of the truth of its conclusion. And the reason why this is generally supposed to be so is that it is generally supposed that any such argument must have as a premise the thesis that the proposition that is its conclusion is possibly true. And that premise (the consensus holds) is something that no human being could know—or at any rate a human being could know it only if that human being (somehow) knew that the conclusion of the argument of which it was a premise was true. That is, a human being could know that the crucial modal premise of a modal ontological argument was true only by *first* knowing that its conclusion was true, *actually* true, and then deducing its possible truth from its actual truth. If that consensus is right, then, obviously, none of the many indisputably logically valid modal ontological arguments can serve as a vehicle by which an enquirer can pass from *not knowing whether* its conclusion is true to *knowing that* its conclusion is true.

2. E. J. Lowe's New Modal Argument

Lowe calls his argument a 'new modal version of the ontological argument' (2012). Perhaps the most interesting and important feature of this argument is that it does not employ a 'possibility' premise. Its conclusion is, as I said, the same as the conclusion of the Minimal Modal Ontological Argument, although it is phrased a bit differently: 'A necessary concrete being exists.'² He defines a 'concrete' being not as I did a moment ago, but rather as a being that exists in space and time—or at least in time. (A being that exists neither in space nor in time is an 'abstract' being.)

But if Lowe's argument for the existence of a necessary concrete being does not have as a premise the proposition that it is possible for a necessary concrete being to exist, why does he call his argument a 'modal ontological argument'?

He calls the argument an *ontological* argument because it is an argument *a priori* for the existence of a being that, if not *aliquid quo maius nihil cogitari potest* or *ens summe perfectum*, at any rate has at least two of the important characteristics of a being who answered to either of those two very demanding concepts would have: concreteness and necessity. He calls it a *modal* ontological argument because it involves modal reasoning. And I am perfectly happy with these reasons for giving his argument that title. It is not, after all, as if the term 'ontological argument' had a fixed and perfectly clear sense. If we do not restrict the application of 'ontological argument' to the arguments that Kant had in mind when he coined the term *Ontologische Beweis*—that is, arguments that proceed by attempting somehow to wring existence out of the concept *ens summe perfectum* by a process analogous to the process by which some Greek first wrung 'having interior angles whose sum is equal to the sum of two right angles' out of the concept 'triangle'—then I don't see why we shouldn't apply it to Lowe's argument. If both Anselm's argument and my Minimal Modal Ontological Argument can properly be called ontological arguments, I don't see why the term can't be applied with equal propriety to Lowe's argument. And, of course, there can be no objection to calling an argument that contains modal inferences a modal argument.

Let us now turn to the argument. It's rather complicated, and I will try to make it as easy to grasp in its entirety as possible by presenting it in the form of a sort of diagram, which is shown in Figure 8.1.

There are two definitions at the top of Figure 8.1. Below them there are some sentences in a roman font, and below *them* some sentences in italics—and there are also, you will observe, five crooked arrows (if arrows can have 'branches'), each using a different style of line. The roman sentences are the argument's premises. The final italic sentence is its conclusion. The other italic sentences are 'intermediate' conclusions: if we think of 'A necessary concrete being exists' as our 'theorem', the earlier italic sentences are 'lemmas'. And, as you will no doubt have deduced, the arrows display the

² Toward the end of the paper, he presents an argument for the conclusion that if there are necessary concrete beings, some of them must be rational beings. But this argument is best viewed as a separate argument—as opposed to a part of the New Modal Ontological Argument.

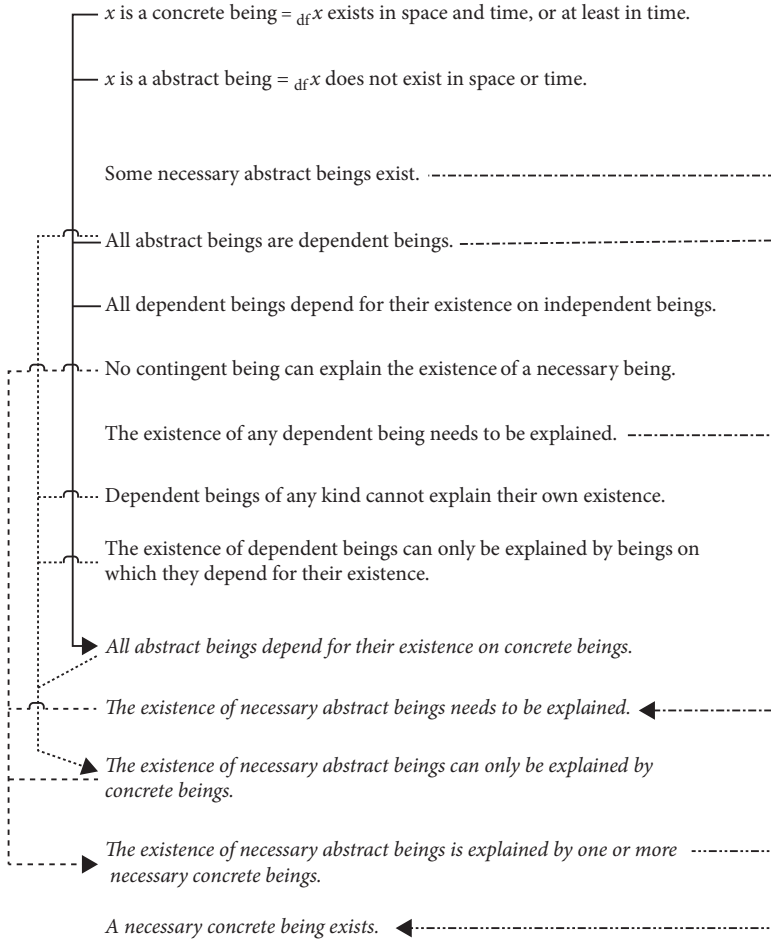


Figure 8.1. A diagrammatic representation of Jonathan Lowe's 'new modal ontological argument'

sentences—roman or italic—from which each of the italic sentences is supposed to follow. For example, the arrow starting on the upper right, using a dash-dot-dash-dot line, represents the assertion that the 'lemma'

The existence of necessary abstract beings needs to be explained

follows from the three premises

Some necessary abstract beings exist

All abstract beings are dependent beings

The existence of any dependent being needs to be explained.

Before examining Lowe's argument, I'll make two remarks about the definitions.

3. Lowe's Definitions

First, Lowe tells his readers that the exact content of his definitions of 'concrete' and 'abstract' is not really essential to the validity of his argument: the argument would be valid on any definition of those two terms that implied that 'concrete' and 'abstract' were exclusive and exhaustive categories. If the only thing we wanted to determine about the argument was whether it was logically valid, therefore, we could replace the two definitions with a premise:

Everything is concrete if and only if it is not abstract.

(Or, of course, with the logically equivalent 'Everything is abstract if and only if it is not concrete'.) Nevertheless, our interest is not only in the validity of the argument; we want also to discover what can be said for and against each of the premises, and doing that will very likely depend on what one means—perhaps on *exactly* what one means—by 'concrete' and 'abstract'. And that brings me to 'secondly'.

Secondly, I do not find the definitions entirely satisfactory, although my reservations about them (there are two) in the end turn out to have little relevance to the question of the persuasiveness of the argument. Still, I might as well mention them.

My first 'reservation' is this. Let us suppose that we regard some beings as, so to speak, *paradigmatically* beings that exist in space or (inclusive) time: Durham Cathedral for example. And suppose that we regard some beings as paradigmatically beings that do *not* exist either in space or in time: the sine function, for example, or the key of C-sharp major. And suppose we are mereological universalists of a really extreme stripe—like Nelson Goodman and David Lewis. What are we to say of the fusion *F* of Durham Cathedral and the sine function? The law of the excluded middle tells us that if existing in space or time is a real concept, and not simply words, words that represent a sort of vague gesture in the direction of a concept, then *F* either exists in space and/or time or it does not exist either in space or in time. Suppose it exists in space and/or time. Then it is a concrete being, and this seems problematical, to say the least. Suppose, then that *F* does not exist either in space or in time. Then it is an abstract being, and that seems equally problematical. (There is probably a parallel problem with 'dependent being' and 'independent being'. If God is an independent being and the Cathedral is a dependent being, is the fusion of God and the Cathedral a dependent being or an independent being?) It seems to me that if Lowe's argument is to have any chance of success, he must reject mereological universalism. I don't regard that fact as a flaw in the argument, of course, but there are metaphysicians who are wedded to mereological universalism, and in most of their cases, the marriage seems to be a pretty stable one.

I turn to another problem about the definition. Lowe considers the question whether God—if he exists at all—exists in space and time 'or at least in time'. Well, no one supposes that God exists in space (he is, as they say, without local presence). And plenty of people—Boethius, Augustine, and Aquinas, for example—have said that he is also wholly outside time, achronic. Lowe toys with this question, and in the course

of toying with it he makes, almost incidentally, a very interesting remark: that we *must* classify God as concrete because he has causal powers. But if having causal powers trumps not existing in space and time when it comes to classifying a being as concrete, why not define 'concrete' and 'abstract' in terms of having or not having causal powers rather than in terms of existing in space and time or not existing in space and time? Why not adopt the definition I mentioned in connection with the Minimal Modal Ontological Argument: Something is *concrete* if it is capable of entering into causal relations, if it is something that can be either an agent or a patient—and abstract otherwise? (Of course, if one accepts absolutely unrestricted mereological summation, one will find this definition no easier to apply than Lowe's: an RAF GR-4 Tornado all-weather attack aircraft obviously has causal powers; the sine function obviously doesn't. But what about their fusion? Causal powers or not? Abstract or concrete?)

I believe that, in response to my proposed definition, Lowe would say something along these lines:

You—I imagine him addressing me—you, Peter, seem to identify having causal powers with being capable of entering into causal relations. You have, in effect, defined a concrete being as a being that can be either an agent or (inclusive) a patient, and an abstract being as a being that can be neither an agent nor a patient. I cannot accept those definitions, since it is essential to my argument that abstract beings are, one and all, dependent on other beings, and ultimately on concrete beings, for their existence. And it is at best very hard to see how a being that cannot enter into causal relations could depend on a concrete being for its existence. (The relation between a set of material things and its members is certainly a relation of ontological dependency and it may not be a causal relation, but if I cause such a set to cease to exist by causing one of its members to cease to exist, that looks like a causal relation between me and the set.) I should, however, be willing to accept a modified version of your definition: A concrete being is a being that can be an agent, and an abstract being is a being that cannot be an agent. According to Aristotle and his medieval followers, God is an agent who cannot be a patient. In my view, it is, so to speak, the other way round ontologically with abstract beings: they can be patients (in the sense of depending on other things for their existence), but cannot be agents.

If that is indeed what Lowe would say, I am happy to accept it. Let us understand a concrete being as a being that has causal powers and an abstract being as a being that has no causal powers.

4. Three of Lowe's Premises

Let us now turn to the argument. I'm not going through it step by step. It's before you and you can examine its logical structure to your heart's content. (I will make one remark about it. It is not clear to me that its conclusion entails the existence of a necessary being that is *essentially* concrete. The conclusion of the argument, the proposition that a necessary concrete being exists, is at least formally consistent with the proposition that the only necessary concrete being, although it is a concrete being in the actual world, is an abstract being in some other possible worlds. If the argument does not rule

this possibility out, however, that problem would be easily solved: one only need add the following proposition to Lowe's premises: that whatever is concrete is essentially concrete. And that proposition is obviously true: nothing is a concrete being in one possible world and abstract being in another.) Rather than discuss the argument as a whole, I am going to examine the interplay between the first three of his premises, the Principle of Abstract Necessity (as I'll call it):

Some necessary abstract beings exist,

the Principle of Abstract Dependency:

All abstract beings are dependent beings,

and the Dependency-Independence Principle:

All dependent beings depend for their existence on independent beings.

(The Dependency-Independence Principle is not to be understood as stating that all dependent beings depend *immediately* on some independent being; it is rather the thesis that every dependent being depends transitively or 'at some remove' on the existence of independent beings; and it need depend on none of them *individually*: it may be that some dependent beings depend on the existence of *some independent beings or other* but not on the existence of any particular independent being. For example, it may be that the kind *horse* depends on the existence of horses, on there being at least one horse, but does not depend on the existence of any particular horse.)

Lowe explains 'existential dependency' as follows:

By a *dependent* being, in this context, I mean a being that *depends for its existence* on some other being or beings. This kind of dependence can be called *existential dependence* and may be defined (at least to a first approximation) as follows:

(D5) x depends for its existence on $y =_{df}$ necessarily, x exists only if y exists.

(D5), however, only defines the existential dependence of one *particular* entity on another. We need also to speak of the existential dependence of one *kind* of entity on another, which we may define (again to a first approximation) as follows, where F s and G s are entities of different kinds (for instance, *abstract* beings and *concrete* beings):

(D6) F s depend for their existence on G s $=_{df}$ necessarily, F s exist only if G s exist. (2012, 182)

I want to make two comments about this passage.

First, I do not see that Lowe's argument ever makes use of the definition (D6). What he does need is something like this:

(D6') x depends for its existence on F s $=_{df}$ necessarily, x exists only if F s exist (i.e., only if some F s or other exist).

(An instance—and perhaps a true one—of the definiendum of (D6') would be 'the kind *horse* depends for its existence on horses'.)

Secondly, these definitions do not seem to me to be satisfactory. Consider (D5). (Similar points apply to (D6) and (D6')). (D5) is certainly unsatisfactory if the Principle of Abstract Necessity is true—and, like Lowe, I accept that principle. For it immediately follows from (D5) and the Principle of Abstract Necessity both that there are beings and that every being, abstract or concrete, contingent or necessary, depends for its existence on any necessary abstract being. And, of course, it follows that there are beings and that all of them are dependent beings—that there are beings but no independent beings. If, for example, the sine function is a necessary abstract being, then God himself is a dependent being, since his existence strictly implies the existence of the sine function. (And this result—that there are beings but no independent beings—in conjunction with the Dependency-Independency Principle leads immediately to a contradiction.) Fortunately, however, Lowe is not committed to using these definitions—as is signalled by his parenthetical qualification 'at least to a first approximation'. In a footnote to the passage I have quoted he says,

Note, however, that the two definitions (D5) and (D6) ... are not in fact formally called upon in the version of the ontological argument that I am now developing, so that in the remainder of this essay the notion of existential dependence may, for all intents and purposes, be taken as primitive. There is an advantage in this, inasmuch as finding a perfectly apt definition of existential dependence is no easy task, as I explain in ['Ontological Dependence' in *The Stanford Encyclopedia of Philosophy* (Tahko and Lowe 2015)]. (Lowe, 2012, 182 n. 4)

For my part, I am happy to allow Lowe to use the notion 'x depends for its existence on y' as an undefined or primitive notion. (And to allow 'dependent being' and 'independent being' to be defined in the obvious way in terms of this primitive.)

The three premises I have singled out, the Principle of Abstract Necessity and the Principle of Abstract Dependency, and the Dependency-Independency Principle can be used to show that it is impossible for there to be nothing (that is, impossible for there to be nothing concrete, impossible for there to be only abstract beings). Lowe has in fact presented just such an argument in an essay called 'Metaphysical Nihilism Revisited' (2013). (In that essay, he uses a slightly different technical terminology.) That proposition, of course, is entailed by but does entail the proposition that a necessary concrete being exists—or at any rate the conjunction of the latter proposition and the proposition that whatever is concrete is essentially concrete entails the proposition that it is impossible for there to be nothing concrete.

We can deduce that conclusion—that it is impossible for there to be nothing concrete—from those three premises as follows. If the Principle of Abstract Necessity is true, there are abstract beings in every world. Therefore, a world in which there were no concrete beings would be a world in which there were abstract beings and only abstract beings, and (by the Principle of Abstract Dependency) that world would be a world in which there were dependent beings and only dependent beings—that is, a world that contains dependent beings and no independent beings. And, by the Dependency-Independency Principle, a world containing dependent beings and no independent beings is impossible.

When I first encountered this argument (on reading ‘Metaphysical Nihilism Revisited’), my reaction, written up a bit, was this:

The argument is certainly valid. One might, of course, question the Dependency-Independence Principle; certainly a lot of critics of the cosmological argument have questioned it. But what really puzzles me is this: Why would any philosopher find both the Principle of Abstract Necessity (hereinafter, ‘Necessity’) and the Principle of Abstract Dependency (hereinafter, ‘Dependency’) plausible? Platonists like myself will of course find Necessity plausible, but will have no time for Dependency, especially if, like me, they believe that abstract objects cannot enter into causal relations. Aristotelians—that is, believers in immanent universals—will find Dependency plausible, but will reject Necessity. What’s going on?

To find out what’s going on, let us examine Lowe’s arguments for Necessity and Dependence.

His argument for Necessity is essentially this. The truths of arithmetic are necessary, true in all possible worlds. Numbers and the relations among them are the truth-makers for arithmetical truths, and, therefore, numbers exist in all possible worlds. And, of course, numbers are abstract objects.

This seems to me to be a very reasonable argument, although it is enthymematic. Strict logical validity would require an additional premise, something along these lines:

For any proposition p , if p is a truth of arithmetic in the distinct possible worlds w_1 and w_2 , then p has the same truth-makers in w_1 and w_2 .

(If we do not appeal to this or some very similar premise, we shall not have ruled out the possibility of theses like this one:

Consider the arithmetical truth $1 \times 1 = 1$, which is, of course, true in every possible world. The number 1 is the truth-maker, or at least one of the truth-makers, of this proposition. But the phrase ‘the number 1’ is not a rigid designator. The office ‘the number 1’ is filled by different objects in different possible worlds—although, necessarily, some one object occupies that office in every possible world. And every object that occupies that office in any possible world is a contingent being.)

But that premise seems reasonable enough.

Although I find Lowe’s argument for Necessity convincing, I think that it can be improved—that is, that it can be replaced by a similar but better argument. This ‘similar but better argument’ is *better* in that does not require any appeal to the controversial notion of a proposition’s having truth-makers. (In my view, the *deservedly* controversial notion of a proposition’s having truth-makers.) We can dispense with Lowe’s appeal to truth-makers if we note that not all arithmetical propositions are truths about the relations among particular, specified numbers—propositions like ‘ $7 + 5 = 12$ ’ and ‘ $2e > \pi$ ’. Arithmetic also includes universal and existential quantifications. (We all know that arithmetic is incomplete, right? If ‘arithmetic’ did not include quantified statements, this would not be true.) Here is one of the existentially quantified truths:

There exists a natural number x such that none of the first billion natural numbers greater than x is a prime number.

This proposition is true in all possible worlds. Therefore, the open sentence ' x is a natural number and none of the first billion natural numbers greater than x is a prime number' is satisfied in every possible world. Now say that such open sentences (one-place open sentences such that their universal or existential closures are arithmetical propositions—propositions of *pure* arithmetic, unlike 'Some number greater than 3 is the number of the Stuart kings of England') express 'arithmetical properties'. Since our offset sentence expresses a necessarily true proposition, in every possible world at least one thing has the following arithmetical property: being an x such that x is a natural number and none of the first billion natural numbers greater than x is a prime number. Having obtained this result, we introduce a premise that does roughly the same work as the premise that I said needed to be added to Lowe's argument:

For every (pure) arithmetical property F , the *same* objects have F in every possible world.

And, just to be on the safe side, we also appeal to this premise:

For every property F and every object x and every world w , if x has F in w , x exists in w .

It follows that some natural number exists in every possible world. And if numbers are abstract objects, or abstract beings as Lowe calls them, it follows that some necessary abstract beings exist.

Let us now turn to Lowe's argument for Dependency, which I will quote rather than paraphrase:

Now I want to return, as promised earlier, to the question of whether all abstract beings are indeed *dependent* beings, since it is crucial to my argument that at least all *necessary* abstract beings have this status. A clue here, however, is provided by the very expression 'abstract'. An abstract being, it would seem, is one which, by its very nature, is in some sense *abstracted*—literally, 'drawn out of, or away from'—*something else*. To that extent, then, any such being may reasonably be supposed to *depend for its existence* on that *from* which it is 'abstracted'. (2012, 189)

I am not moved by this argument. It simply does not seem to me to be true that an abstract being (abstract object or entity I'd prefer to say) is in any sense abstracted or drawn out of or away from something else. The process of abstraction may have something to do with how one *knows about* abstract beings: it may be that we know about, say, shapes like triangularity and circularity by in some sense applying a mental operation called 'abstraction' to many observed triangles and many observed circles. And perhaps some sophisticated account of abstraction could even explain how Socrates was able to say that no one was wise if he was right and there was no one who was wise (and hence no wise people for him to abstract wisdom from). Perhaps this same sophisticated account could explain how we know about 'the missing shade of blue'. Still, it is very hard to see how, if Lowe is right, shapes (or if you like, *shape-properties*

like triangularity and circularity) could exist in a world in which there was nothing that had a shape. Consider a world w in which there are no solid objects at all, in which there exists nothing concrete but point-particles moving about in a (Euclidean) void. Could circularity exist in w if circularity—in all the worlds in which it does exist—depends for its existence on that from which it is abstracted? From *what* would circularity be ‘abstracted’ in w ? (By the way, the void is not one of the things that exist in w . The void is not a *thing*, not a fusion of really existing entities called spatial points. Speaking of the void is simply a picturesque way of saying that in w , there are point particles and nothing else concrete, and that two point particles are at some distance from each other; all true statements that make apparent reference to the void can be paraphrased in relational terms: as, for example, ‘At t , particles A, B, and C were co-linear’ and ‘At t , particle A was 4.6 times farther from particle B than particle C was from particle D. If you ask why I have bothered explicitly to deny real existence to the void, the answer to your question is: To prevent anyone from saying that there must be circular things in w , namely circular bits of the void.)

‘Well, perhaps Lowe could grant that circularity *wouldn’t* exist in w . After all, his premise is that *some* abstract objects exist necessarily, not that they all do’.

But consider: if circularity does not exist in w , then neither does the proposition that something is circular. If the proposition that something is circular does not exist in w , then it’s not possibly true in w , that something is circular. But the actual world is a world in which something is circular, and w is accessible from the actual world. Therefore, if circularity does not exist in w , the accessibility relation is not symmetrical. But that the accessibility relation is symmetrical is a *truth of reason*. Our actual state of affairs is essentially a *possible* state of affairs. It seems, therefore, that, while the concept of abstraction may have a role to play when we are addressing epistemological questions about abstract objects, it has no role to play when we are addressing ontological questions. And, if that is the case, then circularity must exist in all possible worlds, including worlds in which there’s nothing from which it could possibly be said to be an abstraction. Lowe’s argument for the dependency thesis therefore fails.

My critique of this argument is, of course, not correct beyond any possibility of dispute—for the critique itself involves arguments, arguments whose conclusions are positive and non-trivial philosophical propositions, and no argument for any such conclusion is ever correct beyond all possibility of dispute. My argument for the conditional thesis that if circularity does not exist in the world w , then the accessibility relation is not symmetrical, for example, depends on the premise that if the proposition that something is circular does not exist in w , then it is not possibly true in w that something is circular. And many philosophers have, in effect, disputed that proposition. And no doubt there are philosophers who are willing to say, ‘All right, so Dependency entails that the accessibility relation isn’t symmetrical. So what? The accessibility relation *isn’t* symmetrical. You can say that the symmetry of the accessibility relation is a truth of reason till you’re blue in the face, but that won’t make it so.’ I do not claim, therefore, to have *proved* that Lowe’s argument for Dependency fails. I have only given my own reasons for thinking that it fails. And, of course, even if Lowe’s argument for Dependency does fail, that doesn’t mean that Dependency is false.

5. Against the Principle of Abstract Dependency

What, then, can be said against Dependency? I will try at least to cast doubt on Dependency. I will try to describe a class of abstract objects that do not depend for their existence on anything else. These objects are, so to call them, modal objects, and I must say something about the kind of modality I mean to appeal to when I am describing them.

We are no doubt all familiar with the distinction between epistemic and alethic modality, and, in particular, epistemic and alethic *possibility*. We would all agree that the two sentences

It is possible that the number of primes between 10^{40} and 10^{41} is even

It is possible that the number of primes between 10^{40} and 10^{41} is odd

both express true propositions if 'possible that' means 'true for all I know or anyone else knows' (i.e., if 'possible' is understood in its epistemic sense) and that there's also a sense of 'possible' such that we know *a priori* that if the occurrences of that word in these two sentences are understood in that other sense, one of these sentences expresses a truth and the other a falsehood. That second meaning of 'possible' is its *alethic* sense.

There is, however, more than one kind of alethic modality. We may distinguish between, on the one hand, unrestricted or intrinsic or absolute or metaphysical modality, and on the other, the various restricted modalities: physical possibility and necessity, for example, or technological possibility and necessity. There are many interesting distinctions that can be made within each of these categories. For example, in one sense of 'physically possible' it is physically possible for me to be in Tokyo ten minutes from now and another it is not. But we can ignore these nice distinctions, because the kind of modality we shall be concerned with is unrestricted (etc.) alethic modality. Unrestricted alethic modality is what is restricted in various ways to obtain the restricted forms of alethic modality. To say, for example, that it is physically impossible for there to be a Christmas pudding the size of the moon is to say that it's impossible in the unrestricted sense, impossible *full stop*, that the laws of physics should be as they in fact are *and* there be a Christmas pudding the size of the moon. Unrestricted possibility and impossibility are not, incidentally, what is often called 'logical possibility' and 'logical impossibility' because the former phrase is perfectly meaningless and the latter is of very little philosophical interest or utility.

The objects I mean to discuss are *possibilities*—that is unrestricted or intrinsic or absolute or metaphysical possibilities.

Possibilities are denoted by phrases—I consider them to be definite descriptions—that consist of 'the possibility that' followed by a sentence in the subjunctive mood. For example:

The possibility that Socrates have been wise

The possibility that Durham shall be the capital of the UK at some point in the third millennium

The possibility that $7 + 5$ equal 12.

My 'subjunctive mood,' incidentally, is what Orwell called the American subjunctive. If I had used instead the 'classical subjunctive,' the first of my examples would have been phrased this way: 'The possibility that Socrates should have been wise'. These days, when the subjunctive mood in any form is moribund, I suppose that most people would prefer 'The possibility that Socrates was wise'—but that phrase, to my ear, denotes an *epistemic* possibility, and the possibilities I am talking about are alethic possibilities.

Such phrases as 'The possibility that $7 + 5$ equal 22' are improper descriptions, precisely analogous to 'The present king of France': there is no such man as the present king of France, and there is no such possibility as the possibility that $7 + 5$ equal 22.

Possibilities are *realized* or *unrealized*. For example, the possibility that Socrates have died in 399 BC is realized and the possibility that Plato have died in 399 BC is unrealized. (The phrase 'the possibility that $7 + 5$ equal 22' does not denote an unrealized possibility, since it denotes nothing at all.) We could take these two terms as primitive (i.e., take either as primitive and define the other as its negation) or we could define them in terms of truth and falsity. It is perhaps clear enough what we mean by the proposition 'associated with' a possibility (the proposition associated with the possibility that Socrates have been wise is the proposition that Socrates was wise). Using that notion, we can say that a possibility is realized just in the case that its associated proposition is true and unrealized just in the case that its associated proposition is false.

Like objects of any sort, possibilities can be 'quantified over': 'Two of the dire possibilities that the Office for Nuclear Regulation had called attention to in its report were realized before as many years had passed'. If you think that quantifying over possibilities requires a 'principle of individuation' for possibilities (I'm not very strongly inclined to think that myself), I offer the following: if x and y are possibilities, then $x = y$ if and only if the proposition that is associated with x = the proposition that is associated with y . If you protest that propositions themselves are in need of a principle of individuation, I present you with a dilemma. Either a principle of individuation can be provided for propositions or one cannot. In the former case, our problem is solved. And, if no principle of individuation can be provided for propositions... well, we must quantify over propositions if we're going to say much of anything—and certainly much of anything philosophical. (Show me any lengthy piece of philosophical text, and, to a very high probability, I shall be able to find in it sentences that can be understood only as saying either that all propositions have a certain feature or that some proposition has a certain feature.) And if that is so, then it must be permissible to quantify over objects for which we have no principle of individuation. (I should perhaps mention that in this argument—and in this paper as a whole—I do not use the word 'object' in the restricted sense in which Lowe used it in his 1995 paper 'The Metaphysics of Abstract Objects', or in the many books and essays he has written in which the word 'object' is assigned a special role in his 'ontological square'. In my usage, 'object' is simply a count-noun of maximum generality. Lowe often uses the word 'entity' in just that sense. I might just as well have used 'entity' for that purpose myself.) If we individuate possibilities according as their associated propositions are individuated, this will have the consequence

that there are distinct possibilities x and y that such that, necessarily, x and y are both realized or, necessarily, x and y are both unrealized. For example, the possibility that the number of solar planets be greater than the number of Platonic solids, and the possibility that the number of solar planets be greater than the number of primes smaller than 12 are numerically distinct, although 'necessarily co-realized' possibilities.

Possibilities are extremely useful objects. It seems to me, for example, to be useful to think of possible worlds as possibilities. We may say that a possibility x *includes* a possibility y just in the case that it is impossible for x to be realized and y unrealized. (For example, the possibility that Socrates have been a philosopher who died in 390 BC—which is, unfortunately, unrealized—includes the realized possibility that Socrates have been either a philosopher or a king.) And each of two possibilities *precludes* the other if it is impossible for them both to be realized. (The possibility that John F. Kennedy have been assassinated and the possibility that John F. Kennedy have died a natural death preclude each other.) A possibility x is *comprehensive* just in the case that, for every possibility y , either x includes y or x precludes y . A possible world is simply a comprehensive possibility, and an actual world is simply a comprehensive possibility that is realized. (If there are distinct but necessarily co-realized possibilities, it seems reasonable to suppose that, if there are comprehensive possibilities, there are distinct but necessarily co-realized comprehensive possibilities, and, in fact, that for every comprehensive possibility x , there are comprehensive possibilities distinct from x but necessarily co-realized with x . If these theses are indeed true, they imply that if there are any actual worlds there is more than one.)

I am, of course, aware that in his official philosophy of modality, Lowe has been extremely critical of the use of 'possible worlds' in philosophy and would no doubt be extremely critical of my allowing such objects as possibilities to play a serious role in a philosophical argument. I would note, however, that in the presentation of the new modal ontological argument, he does present his modal reasoning in terms of possible worlds. No doubt he would say that his use of possible worlds in that context is a mere heuristic device, a device that could in principle be dispensed with. Still, none of his reasons for rejecting possible worlds as suitable for being what Jonathan Bennett likes to call 'load-bearing members' in a philosophical argument seems to me to have any force. For example, he has charged that to analyse 'Socrates is essentially human' as 'Socrates is human in every possible world in which he exists' wrongly makes what is essential to Socrates depend on objects other than Socrates rather than on Socrates' own being or nature. I must insist that it does no such thing. To say that Socrates is human in all possible worlds is simply to say that his being or nature is such that he would have been human *no matter what*—as long as he managed to exist at all. To say that Socrates is human in all possible worlds in which he exists *is* therefore to make a statement about his being or nature; it is to say that humanity is, of metaphysical necessity, inseparable from that nature.

If I am right in my criticisms of Lowe's opposition to the use of the concept 'possible world' in the philosophy of modality, it does not, of course, follow that there are such

things as possible worlds or objects of the more comprehensive sort amongst which I have located them—that is, possibilities. It nevertheless seems to me to be extremely odd to say, for example, that there exist no unrealized possibilities. That statement seems to me to imply Spinozism, the doctrine that all truths are necessary truths. But this neat little argument, like all neat little arguments in philosophy, is far from decisive. I'll ask you, therefore, to think of my larger argument as having only a conditional conclusion: *if* there are such things as possibilities, then Dependency is false.

Let us then turn to the question: *are* possibilities (if such there be) dependent objects? If they exist, does this existence of theirs depend on anything else?

Suppose it is possible—possible in the unrestricted sense; absolutely possible, metaphysically possible, possible *full stop*—for there to be chunks of transparent iron, good-sized pieces of metal made of atoms with 26 protons in their nuclei through which electromagnetic radiation with wavelengths in the 390–700 nm range can pass as easily as (in actuality) it passes through flawless sheets of optical glass. It is no doubt *physically* impossible for there to be such things, but I'm asking you to consider the possibility that the laws of nature be different from what they actually are—perhaps in worlds containing such objects the fine-structure constant has a value different from its actual value, or some such jargon. And, of course, I'm asking you to suppose that it is possible for the laws of nature to be different in a certain way: a way that would permit the existence of transparent iron. If this is indeed possible, then the—of course unrealized—possibility that there be chunks of transparent iron exists. I should say that I have chosen to use this rather esoteric example because I thought I could best make the point I am trying to make by considering a metaphysical possibility that is a physical impossibility.

On what—other than itself and its own inner nature—does the existence of this possibility depend? In one way it depends on the fact—we are pretending that this is a fact for the sake of the example; it is certainly epistemically possible (for you and me, at least) that it is a fact—it depends on, I say, the fact that it is possible for there to be transparent iron. But this is what one might call 'in virtue of' dependence, not *ontological* dependence. This possibility of there being transparent iron does not depend for its existence on the fact that it is possible for there to be transparent iron in the way in which Socrates' unit set depends for its existence on Socrates—or in the way in which the set {the fact that it is possible for there to be transparent iron} depends for its existence on the fact that it is possible for there to be transparent iron. (If there are such objects as facts, then by Lowe's definition of ontological dependence, the possibility does depend ontologically on the fact—since the existence of the possibility strictly implies the existence of the fact. We have seen, however, that that definition is unsatisfactory.)

If the possibility of there being transparent iron depends on the fact that it is possible for there to be transparent iron only in the 'in virtue of' sense, and not in the ontological sense, does it then perhaps depend in the ontological sense on the *truth-makers* of the proposition that it is possible for there to be transparent iron? Well, what

are the truth-makers of that proposition? What *sort* of thing are they? Are they, for example, abstract or concrete? If they are the entities on which the possibility of there being transparent iron ultimately depends ontologically, then, if Lowe's ontological argument is sound, they must be concrete, owing to the fact that the premises of his argument entail

All abstract beings depend for their existence on concrete beings.

I am, as I have hinted, no friend of the undeservedly popular view that the idea of 'truth-makers' has some sort of role to play in metaphysics—or, truth be told, in any part of philosophy. And one of the two reasons I am no friend of this view is that I do not see what the truth-makers for *modal* propositions could be. Here is a slightly simplified presentation of a conversation I once had with the late David Armstrong (the old order passeth...).

PvI David, is there a possible world in which there's nothing but an iron ball (I mean, nothing but an iron ball and such parts and constituents as it may have)?

DA I'd suppose so, yes.

PvI And would it be possible in such a world for there to be *two* iron balls? I mean, in a world in which there was nothing but one iron ball, would modal propositions like 'It is possible that there are two iron balls' and 'There being two iron balls is a possible state of affairs' be true?

DA That seems the plausible thing to assume. I certainly wouldn't want to deny it.

PvI And what, in that world, would be the truth-maker or truth-makers for those propositions?

DA I suppose it would have to be the iron ball.

But if David was right, and I don't see what else a truth-maker theorist could say, I'm afraid I've lost whatever grip I had on the concept of a truth-maker. I can in a way see what is meant by saying that an iron ball is a truth-maker of such propositions as the proposition that there is at least one iron ball and the proposition that some things made of iron are balls, but I have no sense of what it means to say that an iron ball is the truth-maker for the proposition that there being two iron balls is a possible state of affairs—much less for the proposition that an Austrian logician's proving the incompleteness of arithmetic is a possible state of affairs (which would be one of that solitary iron ball's uncountably many feats of modal truth-making if David was right).

In the end, it seems to me that the possibility that there be chunks of transparent iron is just *there*. It exists necessarily, and, more to the point, depends on *nothing* for its existence—or on nothing besides itself and its own intrinsic nature. Although I count myself a very orthodox theist, I do not hesitate to say that it does not depend on God for its existence. God's business—I say—is with which possibilities are realized, not with which possibilities exist. We may, I think, imagine the act of Creation on this model: we may imagine God surveying every possibility (they are all on display in the

infinite gallery of the Divine Mind) and ‘then’ saying, ‘Let *these* possibilities be realized and let *those* possibilities go unrealized’. But in every possible world, it is the same possibilities that he surveys. And they in no way depend on him for their existence. Many present-day Christian philosophers insist that ‘If there are abstract objects, God has created them all’ is a logical consequence of the Creedal statement that God is the creator of all the *invisibilia*—for any abstract object there may be must of course be an *invisible*. And so it is, in a sense. In that same sense, however, it is a logical consequence of that statement that God created God, and hence that God is a created being. In that sense, the statement ‘God can create a cubical ball’ is a logical consequence of the biblical (and in fact Dominical) statement that with God all things are possible. In that sense of ‘logical consequence’, the statement ‘Beer is a myth or at least a thing of the past; if there ever was such a liquid as beer, there isn’t any such liquid today’ is a logical consequence of what you say when, having assiduously searched every corner of your house, you sadly utter the words ‘There’s no beer’. Enough, I hope, said.

I conclude that the Abstract Dependency Principle is false, and that Lowe’s new modal ontological argument has a false premise. Or, more modestly, I must conclude that *I* am convinced that Abstract Dependency is false, that *I* must regard that argument as unsound—and must also regard Lowe’s argument for the weaker conclusion that it is impossible for there to be nothing concrete as unsound.

6. Envoi

It will not astonish anyone who knows both Jonathan’s and my work in metaphysics to learn that there is an important metaphysical thesis—Abstract Dependency—that he accepts and that I reject. (In these closing remarks, I’m going to speak of him and me in the present tense. It seems right.) In point of fact, despite our warm personal friendship and our great respect for each other’s work—our respect for the philosophical craftsmanship of the other’s work, a thing in which we both delight—there is almost nothing in philosophy that we *agree* about, and this is particularly true of that part of philosophy that we both love above all others, metaphysics. In my own work, there are very few references to Jonathan’s work; the few such references that do exist, I believe, are all citations whose purpose was to establish that someone held a certain view, that a certain philosophical position was in fact occupied. In Jonathan’s book *Forms of Thought* there is no reference to my work at all, although I have written extensively on most of the topics he discusses. And this is how things should be, for our metaphysical views are so far apart that neither of us can make any real use of the other’s work. If, for example, Jonathan’s four-category ontology is anything like right, then all my work in metaphysics is fundamentally wrong, wrong from the ground up. And if the positions I defended in ‘A Theory of Properties’ (2004) are anything like right, then the four-category ontology is wrong from the ground up. Each of us would be hard put to it to find a single substantive sentence in the writings of the other that he agreed with. And yet, there is much more to our relationship than personal affection and the

respect for the other's intellectual honesty and philosophical craftsmanship. In philosophical *conversation*, the fact that there is little we agree about is no barrier to our exchange of ideas. For in conversation each of us is in a position to bring each other's presuppositions and unargued-for assumptions to the fore and subject them to the merciless dialectical pressure that is the lifeblood of philosophy. I have learned an immense amount about the foundations of my own thought from conversation with Jonathan—simply by being forced to think about the question, *Why* am I convinced that this, that, or the other fundamental pillar of my philosophy rests on firm ground? I hope he has learned something about the foundations of his own thought from talking with me. I don't *know* whether he has because we never discuss things of that nature: our serious talk is about philosophy and nothing else. We are both immensely shy introverts, and neither of us would ever dream of saying anything, well, *personal* to the other—anything like 'You know, even though you and I agree about hardly anything in philosophy, conversations with you have been of immense importance to my understanding of the foundations of my own philosophical positions'. I suppose that this is to be regretted—but then there probably isn't much point in wishing that two people weren't the kinds of people they are. Nevertheless, I can find it in myself to wish that I were a different sort of person and that I could have told him what I've just told you.

7. References

- Lowe, E. J. 1995. 'The metaphysics of abstract objects.' *Journal of Philosophy* 92: 509–24.
- Lowe, E. J. 2012. 'A New Modal Version of the Ontological Argument'. In *Ontological Proofs Today*, ed. M. Szatkowski, 179–91. Frankfurt: Ontos Verlag.
- Lowe, E. J. 2013. 'Metaphysical Nihilism Revisited'. In *The Puzzle of Existence* ed. T. Goldschmidt (London: Routledge): 182–96.
- Tahko, T. E. and Lowe, E. J. 2015. 'Ontological Dependence'. *The Stanford Encyclopedia of Philosophy*, Edward N. Zalta (ed.), <<https://plato.stanford.edu/archives/win2016/entries/dependence-ontological/>>.
- van Inwagen, P. 2004. 'A Theory of Properties'. *Oxford Studies in Metaphysics* 1: 107–38.

PART III

Mental Causation

9

The Ontology of E. J. Lowe's Substance Dualism

Alexander Carruth and Sophie Gibb

1. Introduction

A central area of concern in contemporary philosophy of mind is the debate surrounding mental causation. In particular, being able to account for the causal relevance of mental phenomena in the physical domain is usually taken to be a key desideratum for accounts of the metaphysics of mind and body. It has frequently been claimed that interactive substance dualism—according to which first, there is a genuine metaphysical distinction between mental and physical substance, and second, the mental is causally relevant within the physical domain—must be false because it fails to meet this desideratum in virtue of the supposed causal closure of the physical domain, according to which every physical event that has a cause has a sufficient physical cause. The problem is supposed to be that, given the causal closure of the physical domain, then there is no causal role left for the mental to fill. Thus, if substance dualism is maintained, interaction must be abandoned and epiphenomenalism embraced. Alternatively, interaction can be maintained, but at the cost of dualism: either way, the position is argued to be untenable.

Interactive substance dualists typically respond to this problem by rejecting the causal closure principle; that is, by denying that every physical event that has a cause has a sufficient physical cause. They maintain, instead, that some physical events have no sufficient *physical* cause, and in order to account for these events, mental causes must be cited. However, E. J. Lowe's model of psychophysical causation offers a way of reconciling interactive substance dualism with the causal closure principle. It does this by denying the homogeneity of the causal relata—more specifically, by invoking a distinction between 'fact causation' and 'event causation'. According to Lowe, purely physical causation is event causation, whereas psychophysical causation involves fact causation. Given this model of psychophysical causation, the dualist can therefore accept that every physical event that has a cause has a sufficient physical cause as, according to it, mental events do not cause physical events. Rather, the mental is causally efficacious in virtue of causing certain facts to obtain, about which the causal closure principle is silent. If this is correct, then contrary to the anti-dualist argument

sketched out above, there *is* a causal role for the mental to fill that is not filled by physical phenomena. Therefore, the causal closure of the physical does not force the substance dualist to reject either interaction or dualism, and so interactive substance dualism can be maintained.

However, Lowe's dualist model is only as plausible as the distinction between fact and event causation upon which it rests. In this chapter it is argued that a suitable distinction between fact and event causation is difficult to maintain within most common ontological systems. The chapter examines whether accepting the four-category ontology that Lowe defends can alleviate the aforementioned problem, but it is argued that it is not clear that it can.

The chapter begins by outlining the argument against interactive substance dualism from the causal closure of the physical. Following this, Lowe's response to the causal closure argument is explained and examined in detail. It is then argued that, at least assuming any of the standard accounts of the metaphysics of facts and events, Lowe's claim that the causal relata are heterogeneous cannot be maintained. The chapter goes on to examine in more detail what Lowe himself says about the metaphysics of facts and of events—which is surprisingly little for an author who is usually so ontologically candid. Lowe's suggestion that the account of the distinction between facts and events developed by Jonathan Bennett in *Events and their Names* can be adopted in order to secure the heterogeneity of the causal relata is rejected, as given a close reading of Bennett's work, it is clear that for him, event causation is a species of fact causation, and this is an outcome that Lowe cannot accept. Finally, the four-category ontology developed and defended by Lowe is briefly outlined, and it is argued that despite the richer categorial structure of this ontology compared to more common one- and two-category ontologies, the four-category ontology does not have sufficient resources to draw the distinction between facts and events in a manner suitable for grounding the heterogeneity of the causal relata. Thus, it is concluded that in the absence of a well-motivated distinction between facts and events, Lowe's response to the causal closure argument against interactive substance dualism fails, and that supporters of interactive substance dualism must find an alternative way in which to respond to that argument.

2. Responding to the Causal Closure Argument

The standard interactive dualist response to the causal closure argument is to deny the principle of the causal closure of the physical domain. However, an alternative response is to demonstrate that an interactive substance dualism can consistently accept the premises of the causal closure argument while denying its conclusion. One way of doing this is first to observe that the causal closure argument's conclusion that mental events are identical with physical events only follows if one is assuming the homogeneity of the causal relata—more specifically, if one is assuming that psychophysical causation and physical causation both provide examples of causation involving the

same categories of being. The second step is to argue that interactive substance dualism can offer a plausible denial of the homogeneity of the causal relata. This is Lowe's response to the causal closure argument (Lowe 1996, 2000, 2008, ch. 2).¹

The causal closure argument can be set out as follows:

- i. *Relevance*: Some mental events are causally relevant to physical effects.
- ii. *Closure*: Every physical event that has a cause has a sufficient physical cause.
- iii. *Non-Overdetermination*: There is no systematic causal overdetermination.

Therefore, mental events (that are causally relevant to physical effects) are identical with physical events.

Given *Closure*, the physical effects that physical events are causally relevant to are physical events. But, if one denies the homogeneity of the causal relata—if, more specifically, the physical effects that mental events are causally relevant to are not physical events, but some other category of physical effect, then the acceptance of *Relevance* requires neither the denial of *Closure* nor the denial of *Non-Overdetermination* nor the acceptance of event monism. This is Lowe's claim. According to him, while physical events cause physical events, mental events do *not* cause physical events but rather some other category of physical effect.

3. Lowe's Account of Mental Causation

Before assessing the plausibility of this claim, let us consider Lowe's account in more detail. Lowe maintains that the self is a simple substance that is distinct from the body or any part of it. Although maintaining an interactive dualism, he rejects the proposal that the self causes bodily behaviour by causing any physical event. According to the standard dualist model of psychophysical causal relevance, the causal role of a mental event in the physical domain is to cause some physical event or set of physical events—more specifically, some neural event or set of neural events—thereby initiating a causal chain of physical events that ultimately gives rise to some bodily movement. Lowe rejects this model because he does not think that there is any neural event or set of neural events that could be identified as being the immediate effect of a mental event, such that it would allow the claim that that mental event initiated a chain of neurological causes that gave rise to a particular bodily movement. Take any bodily movement that a mental event is assumed to have initiated. Lowe considers that if one were to trace this causal chain of events back from the bodily movement into the brain, the causal chain would display a complex, tree-like structure. This tree-like structure will be fractal—that is, the branches of the tree will be smaller trees, the branches of

¹ In places, Lowe has also sketched an alternative weaker dualist model of psychophysical causal relevance which appeals to the notion of simultaneous causation (Lowe 2000, 580; 2008, ch. 2). For a discussion of this model, see Gibb forthcoming and Robb (this volume). There is not space in this chapter to examine in detail a third dualist model of psychophysical causal relevance which Lowe sketches and which identifies intentional causation with fact causation (Lowe 2008, ch. 5). However, it should be noted that insofar as this third model relies on the distinction between facts and events, the arguments presented here will also apply to it.

which will be still smaller trees, and so on. Furthermore, the branches of the tree will be interwoven with other causal trees of neurological events that result in other distinct bodily movements. There is, Lowe claims, nowhere to locate a mental cause in this entangled causal web of physical events, such that it would be correct to say that it initiated a particular bodily movement. Although the

causal tree *mediates* the causal relation between my desire or volition and the movement of my arm, it does not do so by virtue of my volition or desire initiating the 'growth' of the tree from the 'tips' down: for the tree has *no* tips, and certainly none that it can call exclusively its own. (Lowe 1996, 65)

What, then, is the causal role of mental events in the physical domain? Consider the causal tree of neurological events that gives rise to some intentional movement of the limbs. From this purely physical perspective, the fact that this causal tree of neurological events converges on this bodily event is

a quite remarkable and inexplicable coincidence—rather like rings on the surface of a pond miraculously converging upon a central point, instead of spreading out from a disturbance there towards the pond's edges. (Or—to continue the tree metaphor—it is like supposing that a tree might grow from the tips of its branches down to its trunk, instead of vice versa).

(Lowe 1996, 68)

The reason why it seems a remarkable coincidence from this purely physical perspective is that as physical science 'traces back the physical causes of our bodily movements into the maze of antecedent neural events, it seems to lose sight of any unifying factor explaining why these apparently independent causal chains of neural events should have converged upon the bodily movements in question' (Lowe 2000, 581). Lowe's claim is that the causal role of mental events in the physical domain is to render the fact that a causal tree of neurological events converges on some bodily event non-coincidental. Mental events, unlike neural events, are suited to play this causal role because of their intentional nature—that is because a mental event, unlike any neural event, is directed upon the occurrence of a particular bodily movement.

Importantly, Lowe does not claim that a mental event plays this causal role by causing any neural event or set of neural events. Instead, a mental event plays this role by causing a physical causal fact—the fact that certain neural events, $N_1, N_2, \dots, N_n$, are causes of bodily event, P (Lowe 2000, 582). To demonstrate the intelligibility of the idea of a mental event causing a physical fact, Lowe provides a theological example. Imagine that the world is one in which every physical event has a sufficient physical cause and there is no simultaneous or backwards causation. As Lowe observes, such a world cannot have a beginning in time, because in that world there can be no first physical event. But, still, we might ask why *that* world, as opposed to some other possible world, is actual. One response is that it is because God had chosen it to be actual. This, Lowe claims, would be an example of a mental event causing physical facts. Note that God cannot cause these physical facts by causing some set of initial physical events, as this world has no beginning in time (Lowe 2000, 583).

4. Distinguishing Facts and Events

For Lowe's model of psychophysical causal relevance to be plausible, the distinction that he makes between event causation and fact causation must be metaphysically coherent. And this is precisely where a central problem with Lowe's account lies. It is not at all obvious how one should understand an 'event' and a 'fact' so that it is plausible to motivate a distinction between event causation and fact causation. Unfortunately, in his work on the mental causation debate, this is not an issue that Lowe explores. He does not set out an ontological account of events or facts, or hint in any detail at how one might understand either category in order to motivate the distinction between event causation and fact causation. In what follows, the motivation for thinking this distinction is problematic will be set out.

Lowe is denying the homogeneity of the causal relata. Physical events cause physical events. This is event causation. Mental events cause physical facts. This is fact causation. Fact causation is distinct from event causation, and hence psychophysical causation is distinct from physical causation. For Lowe's denial of the homogeneity of the causal relata to be plausible, Lowe's understanding of events and facts must meet three requirements. First, both events and facts must be suitable candidates for the causal relata. Secondly, events and facts must be distinct. If it were the case that events were just facts or complexes of facts, then event causation would be nothing over and above fact causation and, hence, a distinction between psychophysical causation and physical causation would not have been established. Thirdly, the understanding of events and facts that is being advanced must be combinable in a single ontological framework.

Lowe's account requires that while events are causes, both events and facts can be effects. This sets limits on what events and facts could be. Singular causation is a relation between entities existing at the level of particulars, where the notion of a 'particular' is to be contrasted with that of a 'kind'. It is particulars that act—this stone or, if modes are the causal relata, this stone's hardness. And, it is particulars that are acted upon—this window or, if modes are the causal relata, this window's fragility. Hence, given Lowe's account, both events and facts must be particulars. If, for example, events are literally repeatable, as Chisholm (1970) maintains, then events could not be the causal relata, although particular instantiations of such events could be.

Equally, the causal relata cannot be abstract entities. Causes and effects are concrete entities that occur in time (if not space). Consequently, if facts can be effects it follows that facts must be states of affairs. One can distinguish between three non-equivalent understandings of a fact. According to the first two propositional conceptions, facts are not parts of the world, but are descriptions of it: Ducasse (1940) and Carnap (1947) maintain that a fact is a true proposition and Moore (1953) holds that a fact is the truth of a proposition. According to the third position, facts or 'states of affairs' are not merely descriptive of the world, but are part of it. States of affairs are not true propositions; they underlie true propositions. Hence, according to David Armstrong—one of the central proponents of states of affairs—'A state of affairs exists if and only if a

particular has a property or, instead, a relation holds between two or more particulars' (1997, 1). Thus, for Armstrong, the instantiation of a property by a substance, that is, a property-instantiation, is a state of affairs. 'The redness of the apple' refers to a state of affairs, the apple being red. This state of affairs includes a substance (the apple) and a universal (redness) as constituents. While the first two interpretations of facts treat facts as wholly abstract entities, states of affairs are complexes that have concrete constituents and are therefore themselves concrete. To this extent, states of affairs are suitable candidates for the causal relata. If facts are instead interpreted merely as true propositions or the truth of a proposition then they are more suited to the role of causal explanantia, and Lowe clearly wants the mental to play more than an explanatory role. Hence, for there to be fact causation, facts must be concrete entities and, hence, one must identify facts with states of affairs. This is a point that Lowe is in agreement with—as he states in one of his discussions of his account of psychophysical causal relevance in 'a case of fact-causation, what is brought about is not an event, but a fact or state of affairs' (Lowe 2000, 582).

So let us assume that facts are states of affairs and accept Armstrong's account of states of affairs. According to Armstrong, the instantiation of a property by a substance, that is, a property-instantiation, is a state of affairs. Properties, according to Armstrong, are universals. Substances are thin particulars, where a thin particular is an entity that instantiates properties but is distinct from any property that it instantiates, only differing from other thin particulars numerically. Armstrong considers that unless one wishes to abandon ontological seriousness one should accept the truthmaker principle—the requirement that all truths have a truthmaker (Armstrong 1997, 135). But given this principle, if one accepts the existence of thin particulars and properties, one must accept monadic states of affairs as an ontologically additional category. Say that a thin particular *S1* instantiates a property *P1*. According to the truthmaker principle there must be something in the world that serves as the ontological ground for this relation—something is needed to weld substance and property together. This can be neither *S1* nor *P1* considered individually, nor can it be the pair *S1* and *P1*, for just because both exist does not mean that there need be an instantiation relation between them. Armstrong claims that the truthmaker must be the state of affairs '*S1* instantiating *P1*' for it is the state of affairs that brings the substance and property together in a non-mereological form of composition. This argument stands, Armstrong considers, regardless of whether the properties that thin particulars instantiate are universals or modes—if a thin particular instantiates a mode, something ontologically additional is still required to weld the two together (Armstrong 1997, §8.12).

Armstrong holds that singular causation is a relation between first-order states of affairs; where *S1* and *S2* are thin particulars and *P1* and *P2* are properties, *S1* instantiating *P1* brings it about that *S2* instantiates *P2*. But although Armstrong considers all causes and effects to be states of affairs, is there room, given his account of states of affairs, for event causation? That is, if facts are Armstrongian states of affairs, is there an understanding of an event that would allow one to motivate a distinction between event causation and fact causation?

Armstrong's own understanding of an event can be ruled out. According to Armstrong, an event is a 'succession of states of affairs'—a complex of states of affairs in a temporal zone (1997, 206). Events are not entities that exist over and above states of affairs—they just are complexes of states of affairs. Hence, given this understanding of an event, event causation is nothing over and above fact causation. Consequently, Armstrong's understanding of an event is of no help whatsoever in motivating Lowe's denial of the homogeneity of the causal relata. Indeed, clearly, Armstrong's distinction between states of affairs and events cannot be the one that Lowe is intending. According to Lowe, in the case of physical causation events cause events, in the case of psychophysical causation events cause facts. Given Armstrong's account of events, this is just to say that in the case of physical causation a complex of states of affairs in a temporal zone causes another complex of states of affairs in a temporal zone, while in the case of psychophysical causation a complex of states of affairs in a temporal zone causes a state of affairs at a time. In his articulation of the distinction between physical causation and psychophysical causation this is clearly not what Lowe is suggesting.

Could one hold on to Armstrongian states of affairs while advancing a different account of events, one that would allow one to motivate a distinction between event causation and fact causation? As the most popular understanding of an event in the mental causation debate is Kim's, the next obvious question is whether a combination of Armstrongian states of affairs and Kimean events allows one to motivate a distinction between fact causation and event causation. Indeed, in his discussions of the causal closure argument, Lowe does occasionally indicate that his understanding of events is precisely that of Kim's (see, e.g., Lowe 2008, 93).

According to Kim, each individual event has three constituents, a substance, a property it exemplifies, and a time. Event $(S1, P1, t1)$ exists just in case the substance $S1$ exemplifies the property $P1$ at time $t1$. Events $(S1, P1, t1)$ and $(S2, P2, t2)$ are identical if and only if $S1 = S2$, $P1 = P2$, and $t1 = t2$ (Kim 1993). Note that Kimean events do not include as simple events occurrences such as carnivals and weddings, because Kim purposefully underplays the difference between events and states. If one preserves the term 'event' for an entity that possesses a temporal dimension (as Armstrong does), hence identifying events with changes in the properties of a substance, Kimean events should be classified as states. But, then, are Kimean events anything other than Armstrongian states of affairs?

At first glance, Kim's events appear very similar, if not identical with, Armstrong's states of affairs. According to Kim, if the apple exemplifies the property of redness at time $t1$ this is an event. According to Armstrong, it is a state of affairs. Both Kimean events and Armstrongian states of affairs occur at a time, and are constituted by a substance and a property. Both Kim and Armstrong assume there to be an event/state of affairs if there is a relationship of exemplification between a substance and a property. (Note, that although Armstrong refers to the relationship as one of instantiation, he acknowledges that one could equally talk about exemplification (Armstrong 1997, 115).) Furthermore, the criteria of identity for Kimean events and Armstrongian states of

affairs are remarkably similar—both maintain that the identities between the respective entities depend on their internal constitution and structure; if they contain the same constituents and these are organized in the same way, then the entities are identical.²

Whether Kimean events are actually nothing other than Armstrongian states of affairs comes down to the question of how to understand the category of substance that partly constitutes Kimean events. Kim is neutral on this issue—he explains that by a ‘substance,’ he means ‘things like tables, chairs, atoms, living creatures, bits of stuff like water and bronze’—according to him, ‘there is no need to associate this notion with a particular philosophical doctrine about substance’ (1993, 33). But, if a Kimean event is the exemplification of a property by a *thin particular* at a time, then Kimean events must be Armstrongian states of affairs, as both will contain exactly the same constituents. There would be nothing to distinguish them. Event causation would be fact causation. Hence, this combination will not work to establish Lowe’s denial of the homogeneity of causation.

Now, of course, one might combine Kim’s notion of an event as a property-instantiation with some alternative philosophical doctrine about substances. Hence, for example, one might understand an event to be the instantiation of a property by an Aristotelian substance, rather than a thin particular.³ And, then, proceed to argue that, given this alternative understanding of the nature of a substance, events are not states of affairs, and certainly not Armstrongian states of affairs. Given this alternative Aristotelian account of substance, a substance is not something wholly separate from its properties in the sense that a thin particular is—and, for this reason, arguably states of affairs are not required to weld property and substance together. But nor should it therefore be concluded that an Aristotelian substance just is a state of affairs, for unlike states of affairs, Aristotelian substances are not complexes. The relationship between property and Aristotelian substance is not that of part to whole. Rather, Aristotelian substance comprises a basic irreducible category, which does not have constituents, but on which properties ontologically depend. However, a discussion of the merits of this alternative account of substance and whether it does indeed allow one to disassociate Kimean events from states of affairs is not one that we need to get into here, for it is of no use in grounding Lowe’s denial of the homogeneity of the causal relata. Armstrong’s account of substances as thin particulars and the Aristotelian account of substances provide competing accounts of the nature of an individual substance. Either individual substances are thin particulars or they are Aristotelian substances or they are something else. Both thin particulars and Aristotelian substances cannot be incorporated into a single ontological framework. The suggestion that a fact is the instantiation of a property by a thin particular on the one hand, while an event is the instantiation of a property by an Aristotelian substance on the

² For Armstrong’s criterion of identity for states of affairs, see Armstrong 1997, 131–2.

³ Proponents of the Aristotelian account of substance include Heil (2003), Martin (2008), and Lowe (2005) himself.

other, is untenable, because the two claims do not ontologically cohere—they cannot be embedded in a single ontological framework.

Consequently, if facts are Armstrongian, and events are Kimean, to make these two claims ontologically consistent, fact causation and event causation must ultimately be one and the same thing. More generally, given any two-category ontology of substance and property—a categorial framework that both Armstrong and Kim accept—it is very difficult to see how one might differentiate events from facts, and hence, event causation from fact causation in the way that Lowe requires. Say that facts are worldly entities, states of affairs—which they must be if they are to figure in causal relationships, as opposed to mere causal explanations. What could events be in a two-category system that accepts states of affairs, other than states of affairs, or complexes of states of affairs in a temporal zone? The alternative acceptance of a one-category ontology—a bundle theory of properties—clearly does nothing to alleviate the problem. But what if one instead embeds Lowe's denial of the homogeneity of the causal relata in his own four-category ontology? Does this allow one to distinguish event causation from fact causation in the way that Lowe wishes?

Finally, a brief note about one major approach to the metaphysics of events, namely Davidson's (see, for instance, 1970), and whether this could be more successfully combined with an Armstrongian account of facts than the Kimean approach. The short answer is that it cannot, and the reason for this is that the Armstrongian approach is committed to realism about properties (for Armstrong, properties are *de re* Aristotelian universals) whereas the Davidsonian approach to events assumes property nominalism. These two approaches to the nature of properties are inconsistent with one another, and therefore a combination of Armstrong's account of facts and Davidson's account of events violates the third criterion set out above, as they are not combinable in a single coherent ontological framework.

5. Lowe and Bennett on Facts and Events

Before going on to address whether adopting Lowe's own four-category ontology can alleviate the problem outlined, there is one brief suggestion he makes about understanding the distinction between events and facts that deserves attention. We have already seen that it is not immediately clear how Lowe's proposal—that a dualist model for mental causation can be developed by an appeal to heterogeneity of the causal relata—should be understood. Lowe suggests that two different classes of entities are capable of standing in the 'effect' place of the causal relation: facts and events. As has been argued, the plausibility of Lowe's account rests on whether or not an elaboration of the fact/event distinction can be given that meets the three desiderata outlined earlier in this chapter. However, appealing to the most common accounts of the metaphysical status of both facts and events do not allow for this. The first step in examining this issue further is to look closely at what Lowe has said about the nature of this distinction.

Lowe fails to give much detail on exactly what he took the distinction between facts and events to consist in. There are two points in *Personal Agency* where he does briefly elaborate on the distinction. At one stage, he explicitly discusses facts as states of affairs in the Armstrongian sense (Lowe 2008, 34–5). As has already been shown, this approach is of little help, however, for this account of facts fails to make them suitably distinct from events. A little later, when Lowe introduces the idea that there is a distinction to be made between fact causation and event causation, he directs the reader to Jonathan Bennett's *Events and Their Names* (2008, 42). The rest of this section will first offer a brief exposition of Bennett's account of the fact/event distinction, and will go on to argue that appealing to Bennett's way of drawing the distinction will not resolve the issues raised above.

Bennett begins with some observations about the ways in which events and facts are referred to in ordinary language. He draws on a number of features of the grammatical structure of phrases that name facts and events, the precise details of which need not detain us here, as nothing said below will be sensitive to these details. He concludes that events can be picked out by definite descriptions that are partially indeterminate, in that whilst they mention only intrinsic features of that event, they need not exhaustively list these features (Bennett 1988, 12). Contrastingly, when a fact is picked out by a definite description, it must either be referred to extrinsically, or *all* its intrinsic features must be mentioned in the relevant description. Thus, the following can both be names of the very same event:

'Mary's submission of her thesis.'

'Mary's submission of her thesis on causation last Thursday'

Whereas the following all pick out different facts:

'Mary's submitting her thesis.'

'Mary's submitting her thesis on causation.'

'Mary's submitting her thesis last Thursday.'

The key point here is that facts are much more determinate, fine-grained, and fragile than events, which admit of a certain degree of vagueness. Indeed, later, Bennett goes further, arguing that for any event we choose to name, the 'slack'-ness (1988, 64) of the event concept means that it is always indeterminate what property instances are actually picked out by that event name.

Following this linguistically focused discussion of how names for events and facts operate, Bennett goes on to give some further detail about the ontology of each category. In discussing concerns about the fragility of events understood in a Kimean sense, Bennett argues that Kimean 'event' semantics actually pick out facts: that is, facts are substances instantiating properties at a time. This should immediately give us pause, as we have just seen at length that conceiving of facts in this manner makes it very difficult to see how we can draw a distinction between facts and events that can support the

heterogeneity of the causal relata. We do not need to rehearse these arguments, but should proceed bearing in mind that there is already reason to suspect that adverting to Bennett's conception of the fact/event distinction may not be a fruitful move for Lowe.

Events, Bennett says, are complex tropes at a spatio-temporal zone (1988, ch. 4). Usually, as mentioned above, the relevant complex trope will not be fully specified by the name(s) we use to refer to the event. But neither, it should be emphasized, is the relevant complex trope composed of *all* the simple tropes that are at the zone. This implies that two distinct events can occupy the same zone. Bennett is clear that tropes are property instances (1988, 128), and so presumably the complex tropes-at-a-zone that constitute events are complexes of simpler property instances. But if this is the case, it seems hard to see how on Bennett's account one can draw a distinction between events and facts adequate to the task of hand: showing how the causal relata can be heterogeneous. This is because, as has just been discussed above, Bennett takes Kimean 'events'—that is, property instances at a time—to be facts; but if events are just complexes of facts, as on this account they seem to be, then events and facts are not suitably distinct.

Bennett himself eventually states that 'events are facts of a kind' (1988, 129), and goes on to claim that event causation is nothing over and above fact causation—we only maintain event-causal locutions for explanatory reasons. The indeterminacy of event names noted above accords with our typical epistemic situation regarding causes: that is to say, we are often relatively ignorant of the full details of causal situations, and so it suits us to have vague ways to talk about them (Bennett 1988, 135–42). This is a particularly infelicitous outcome for Lowe, for remember his suggestion is that all non-mental causation is event causation, with fact causation being invoked to account for the special role played by the mind. But on Bennett's account it turns out that *all* causation is fact causation: adopting this account for Lowe would lead to the surely unacceptable conclusion that there is *no* non-mental causation. Adopting Bennett's view will be of no help in explaining how a distinction between facts and events can be drawn that meets the requirements for being an adequate account of the heterogeneity of the causal relata.

6. The Four-category Ontology

At the end of section three it was suggested that ontologies that admit of no more than two fundamental categories of being simply will not have the resources to draw a distinction between facts and events that will be strong enough to support Lowe's dualist account of mental causation. In what follows we shall examine whether adopting a richer categorial system—the four-category ontology that Lowe developed and defended—can remedy the situation. Whilst there is not space here to examine the four-category ontology in detail or to outline the arguments in its favour, a brief outline of the main features of the view is provided below.

Lowe's four-category ontology is formed by cross-combining two categorial distinctions: that between substance and property—which forms the basis of most two-category ontologies—and that of particular and repeatable. Thus, Lowe recognizes the following categories of being: 'kinds' are repeatable substances; 'attributes' are repeatable properties conceived as *de re* Aristotelian universals; 'objects' are particular, individual substances, and 'modes' are particular property instances. On this schema, objects are instances of kinds: the particular glass I am drinking from now, say, is an instance of the kind 'glass' of which all and only glasses are members. Modes are instances of attributes: the particular fragility of this particular glass is an instance of the general property 'fragility' shared by other glasses and indeed by non-glasses—by vases, windows, eggshells, and so on. Modes characterize the objects by which they are had and attributes characterize the kinds they typify. Objects exemplify attributes in two ways: a particular glass may exemplify the attribute fragility in virtue of being a member of the kind 'glass' that is typified by that attribute, or in virtue of possessing a particular mode of fragility at some time: say, an actual instance of breaking when struck. Finally, it should be noted that there is no relation between repeatable substances—kinds—and particular properties—modes—analogue to the exemplification relation that obtains between objects and attributes. The categories of being and the relations they stand in which constitute the four-category ontology can be represented by what Lowe termed 'the ontological square', as depicted in Figure 9.1.

Lowe argues that in spite of its relative lack of parsimony compared with more popular one- and two-category ontologies, the four-category ontology is preferable due to its greater theoretical power: it alone has the necessary resources to provide a metaphysical foundation to the scientific enterprise. According to him, only the four-category ontology

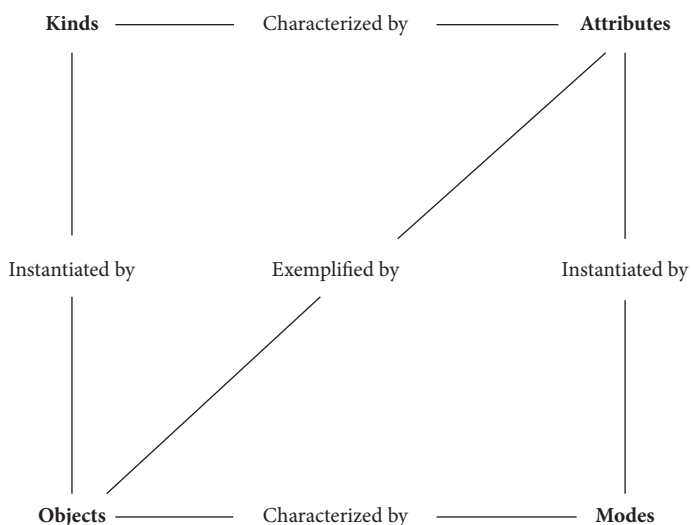


Figure 9.1. The four-category ontology (Lowe 2005, 19)

can provide a satisfactory account of all the following topics of interest in metaphysics: laws of nature; the distinction between dispositional and occurrent states of affairs; and modality.

Given its rich structure and potentially greater theoretical resources, it might be thought that the four-category ontology will fare better in being able to draw the distinction between facts and events than simpler, rival ontologies. However, it is not clear that this is the case, at least not with regard to appealing to the distinction in order to argue for the heterogeneity of the causal relata. To see why, we shall consider some potential ways the distinction might be drawn within the four-category ontology, and it will be shown that these cannot satisfy the three desiderata outlined earlier in this chapter.

Standard two-category ontologies admit of either particular substances and repeatable properties—universals—or particular substances and particular properties—tropes or modes. Lowe's four-category ontology admits of both particular and repeatable substance and also particular and repeatable properties. As has already been discussed, one criterion that must be met by candidate causal relata is particularity, for singular causation is a relation between particulars. Thus, if the four-category ontology has resources not had by two-category ontologies for grounding the heterogeneity of the causal relata, this can only be in virtue of its admitting more types of particular than rival two-category ontologies. At this point, therefore, it is worth noting that appeal to just those entities on the top row of the ontological square—kinds and attributes—will not be of help in grounding the distinction between facts and events, for these are both repeatable entities, and any complex of them, that is, any characterization of a kind by an attribute, will likewise be a repeatable.

Table 9.1 indicates all the types of particular that the four-category ontology contains. It should be clear that the simple particulars taken alone cannot ground the distinction between facts and event, as neither facts nor events are objects or modes taken in isolation. Thus, it is to the four different types of complex particular that we must look. Let us first consider what we have called the 'vertical' particulars—the instantiation of a kind by an object, or of an attribute by a mode. These are not suitable as candidate causal relata, although this is not due to their lacking complexity or particularity. Consider an example of each type of 'vertical' particular:

- (a) this glass's being an instance of the kind 'glass'
- (b) this patch of red's being an instance of the attribute 'redness'.

Table 9.1. The particulars of the four-category ontology

Simple Particulars	Complex 'Horizontal' Particulars	Complex 'Vertical' Particulars
Objects	Object characterized by a mode	Object instantiating a kind
Modes	Object exemplifying an attribute	Mode instantiating an attribute

Neither (a) nor (b) is suitable to stand in the effect-place of the causal relation because they are not the sort of thing to be brought about causally. Rather, an object's membership in a kind or a mode's being an instance of an attribute, stem from the nature or essence of that object or mode—and what nature or essence an object or mode has is not an *effect* of anything. Although we can cite causes for the presence, say, of an object belonging to some kind or other—why there is a *glass* here rather than an *avocado*—or the presence of a mode that is an instance of some attribute—why this object is *red* rather than *green*—these are *not* causes for a glass's membership in the kind 'glass', nor of this red patch's being an instance of 'redness'. If the four-category ontology can ground the distinction between fact and event in a manner suitable to uphold Lowe's response to the causal closure argument, it cannot do so by appealing to the 'vertical' complex particulars.

This leaves just the two 'horizontal' complex particulars—an object's being characterized by a mode and an object's exemplifying an attribute. The first thing to note is that both of these are property instantiations, which as we have already seen have been taken to be facts (by, for instance, Armstrong) and events (by, for instance, Kim). Within a two-category ontology, a distinction cannot be drawn between different types of property instantiation, as, depending on the system, either all properties are repeatable, or all properties are particulars. However, as the four-category ontology includes both particular and repeatable properties, it does have the resources to distinguish between two fundamentally different types of property instantiation. At first glance, this looks like a promising feature with regards to the prospects for drawing a suitable distinction between facts and events. Despite the initially promising outlook, however, we will argue that this approach is also inadequate.

Let us suppose, for the sake of argument, that an event is an object's being characterized by a mode, and that a fact is an object's exemplifying an attribute. If this is how the distinction between facts and events is drawn, then Lowe's claim is that whilst physical causation brings it about that objects are characterized by certain modes, mental phenomena can be causally relevant by bringing it about that objects exemplify certain attributes. As explained earlier, according to Lowe, there are two distinct ways in which an object can exemplify an attribute within the four-category ontology, either in virtue of being:

- (c) a member of a kind typified by that attribute, or
- (d) characterized by a mode that instantiates that attribute.

Thus, for the proposal that mental phenomena are causally relevant in virtue of being causally responsible for certain facts, where a fact is understood as the exemplification of an attribute by an object, to be a successful one, then it must be plausible that mental phenomena have as their effects either (c) or (d). However, we've already seen that (c) is not a suitable candidate effect: the kind-membership of an object stems from its essence or nature, and is not brought about causally—although there are causes for the presences and absences of objects of one kind or another, there are not causes for it

being the case that the object is of *this* rather than *that* kind. The other option, (d), is equally untenable, as it is precisely what we had, in attempting to draw the distinction in this manner, assumed that an event is: that is, an object's being characterized by a mode. (c) fails to be suitable as a candidate for standing in causal relations; (d) fails to make events and facts distinct: the distinction between facts and events cannot be grounded in that between the complex 'horizontal' particulars of the four-category ontology. It is important to recognize that nothing said above is contingent on the assumption that an object's being characterized by a mode is an event: we could have started with the assumption that this was a fact, and the same conclusions would follow.

It should be noted at this point that one final potential ground for the distinction remains: the two manners in which an attribute can be exemplified. However, it should be clear from what has been said already that this will not be adequate: one way in which an attribute can be exemplified, namely in virtue of an object's being an instance of a kind, is not a suitable candidate effect. Moreover, Lowe explains the distinction between occurrent and dispositional predication by appeal to the two different ways in which an object can exemplify an attribute, and it would be deeply confused to suggest that this distinction is the same one as that which obtains between facts and events.

7. Concluding Remarks

Lowe defends interactive substance dualism from the threat levelled by the causal closure argument by outlining an account of mental causation that is premised on the denial of the homogeneity of the causal relata. By drawing a distinction between event causation and fact causation, Lowe hopes to reconcile the causal relevance of the mental with the causal closure of the physical through claiming that the closure principle applies to event causation only, allowing for mental causes to cause physical facts without thereby violating the principle. For Lowe's account to be tenable, the distinction drawn between facts and events must be well motivated and plausible. We have argued that there are three criteria that must be met, whatever account of facts and events is given: both must be suited to occupying the effect place in singular causal relations; they must be genuinely distinct, and the accounts must be combinable within a single ontological framework.

As has been shown, on standard accounts of the metaphysics of facts and events, it is not possible to meet these three criteria, and we have suggested that any typical one- or two-category ontology will not have the resources to distinguish between facts and events in a satisfactory manner. It has also been argued that, whilst Lowe's richer four-category ontology may at first glance appear to have the theoretical power to solve this problem, under closer inspection it is not clear that it does. This was done by outlining several candidate ways in which the distinction could potentially be drawn utilizing the categorial structure of the four-category ontology, and showing that none of these ways of drawing the distinction could meet the three criteria outlined above. Whilst this does not prove conclusively that the distinction cannot be drawn, we contend that

it places the onus onto the defender of the position to show that it can be. Furthermore, it should be noted that if the distinction can only be drawn in a satisfactory manner within ontologies with higher numbers of categories, then this represents a previously unnoticed theoretical cost of adopting this account of mental causation.

Given this, defenders of interactive substance dualism who wish to accommodate, rather than deny, the causal closure principle cannot do so by appeal to Lowe's account of mental causation without first demonstrating how an account of the distinction between facts and events can be given that meets the criteria set out earlier in this chapter. They must either do this, or else abandon Lowe's account and seek out an alternative response to the causal closure argument.

8. References

- Armstrong, D. M. 1997. *A World of States of Affairs*. Cambridge: Cambridge University Press.
- Bennett, J. 1988. *Events and Their Names*. Oxford: Oxford University Press.
- Carnap, R. 1947. *Meaning and Necessity*. Chicago: University of Chicago Press.
- Chisholm, R. M. 1970. 'Events and Propositions'. *Noûs* 4: 15–24.
- Davidson, D. 1970. 'Mental Events'. In *Experience and Theory* ed. L. Foster and J. W. Swanson, 79–101. Humanities Press.
- Ducasse, C. J. 1940. 'Propositions, Opinions, Sentences, and Facts'. *Journal of Philosophy* 33: 701–11.
- Heil, J. 2003. *From an Ontological Point of View*. Oxford: Oxford University Press.
- Kim, J. 1993. 'Events as property exemplifications'. In J. Kim, *Supervenience and Mind: Selected Philosophical Essays*. Cambridge: Cambridge University Press.
- Lowe, E. J. 1996. *Subjects of Experience*. Cambridge: Cambridge University Press.
- Lowe, E. J. 2000. 'Causal Closure Principles and Emergentism'. *Philosophy* 75(294): 571–86.
- Lowe, E. J. 2005. *The Four-Category Ontology: A Metaphysical Foundation for Natural Science*. Oxford: Oxford University Press.
- Lowe, E. J. 2008. *Personal Agency: The Metaphysics of Mind and Action*. Oxford: Oxford University Press.
- Martin, 2008. *The Mind in Nature*. Oxford: Oxford University Press.
- Moore, G. E. 1953. *Some Main Problems in Philosophy*. London: George Allen and Unwin, Ltd.

10

Could Mental Causation be Invisible?

David Robb

1. Emergentism

My topic is an emergentist model of mental causation recently proposed by Jonathan Lowe (2003). Weak forms of emergentism are limited to epistemic or explanatory claims, but the form I'll consider here, following Lowe, is of an especially strong sort.¹ It consists of three theses, the first of which is:

- (1) Mental properties are not physical.

Lowe's emergentism goes beyond (1) to event and substance dualism. I believe that the main points of this chapter are compatible with these more ambitious forms of emergentism, but in what follows, the official emergentist position will be limited to property dualism.

The second thesis is:

- (2) Mental properties depend on physical properties.

Emergent properties, while non-physical, cannot float free of the physical. Emergentists differ on the strength of this dependence. Lowe takes it to be causal. Adding more detail to (2), then, an emergentist says that when (and only when) physical systems achieve a certain level of complexity, they cause to be instantiated non-physical properties of those same physical systems.

The final emergentist thesis is:

- (3) Mental properties exert a novel (non-redundant), downward causal influence on the physical systems from which they emerge.

Emergent properties make a causal difference to what happens physically, and in particular to (intentional) human behavior. In producing such behavior, emergents do

¹ A version of the weak/strong distinction is in Bedau 2002. On the varieties of emergentism, see McLaughlin 1992, Kim 1999, and O'Connor and Wong 2015.

causal work that the physical on its own cannot. This third doctrine distinguishes emergentism from more moderate brands of property dualism on which mental properties are realized in, and so share causal powers with, their physical base properties.² Emergent properties aren't like this. There are new causal powers in the world—extra causal 'oomph'—when an emergent comes on the scene.

So-understood, emergentism has the scientific virtue of making a bold empirical claim about the long-term results of human physiology. As we study the structure and dynamics of the human body, emergentism predicts we will eventually find that physical properties do not have the causal resources to produce all of what we observe. But with emergentism's prediction comes risk, for empirical research may instead reveal that everything physical—and in particular, whatever happens in and to the human body—can be fully accounted for by physical causes. In that case, we would have discovered that the physical world (at least our part of it) is 'causally closed' or 'causally complete'. Following Lowe and others, we can formulate the resulting Closure principle like this:

Closure: Everything physical that has a cause at time t has a sufficient physical cause at t .³

If Closure were empirically confirmed, this would mean trouble for emergentism. Indeed, we'd have the state of affairs envisioned by those who have advanced a 'causal argument' for physicalism, an argument using Closure as a premise.⁴ Closure ensures that any mental cause of a physical effect will be accompanied by a sufficient physical cause of that effect. If the mental cause is included in (reducible to) the physical cause, then emergentism is false, as thesis (1) says mental properties aren't physical. On the other hand, if the mental cause is not included in the physical cause, then any efficacy of the former is redundant, as the physical cause is (by hypothesis) sufficient for the effect. But this too is incompatible with emergentism, as thesis (3) says the efficacy of mental properties is non-redundant. And to abandon mental causes altogether—the epiphenomenalist option—is also incompatible with thesis (3). Closure seems to leave the emergentist with no options, leaving some form of physicalism as the best alternative.

That said, I won't here develop or defend the causal argument for physicalism, nor will I weigh in on whether science has vindicated (or will vindicate) Closure. Instead, my more modest aim is to argue that—*contra* Lowe—the case for or against emergentism can be, as it appears, empirical. Along the way, I will use the issue to touch on some

² One prominent version of such a view has been worked out in detail in Shoemaker 2007 and Wilson 2011.

³ The principle is stronger than needed in this context: a local version, restricted to the human body, would suffice. But I'll follow Lowe in using the universal version.

⁴ They include Lewis 1966 and Papineau 2001. Presentations of the argument often include a 'no over-determination' premise, but the version in the text can dispense with this, as emergent properties by definition exert non-redundant efficacy.

broader topics in the metaphysics of mental causation, including the proper formulation of Closure and the nature of causal powers.

2. Lowe's Model of Mental Causation

Turn then to Lowe's model of mental causation. While Lowe is an emergentist, he argues that emergent mental causation could be empirically undetectable, and in this sense 'invisible'.⁵ This seems to me unwelcome news, for it means that what we thought was a chance for progress on the mind-body problem—namely, that a significant portion of it can be resolved empirically—may not be in the cards after all. Indeed, Lowe suggests that if his 'invisibility' model is correct, the dispute between physicalists and emergentists may have to remain largely within philosophy, where Lowe thinks emergentists have the stronger case.

I'll build Lowe's model by starting with the physical effect, *B*. This could be an instance of *behavior*, say my raising my arm, but given other aspects of the model, it's better to think of *B* as the *beginning* of a physical process that leads to (or partly constitutes) that behavior. Ignoring such niceties, I'll refer to *B* as the *behavioral effect*. *B* is physical, and occurs at time t_2 . Now suppose *B* is initiated by a mental cause, *M*, at t_1 , the time immediately preceding t_2 . (Here time is discrete, a simplifying but non-essential aspect of the model.) Given Closure, *B* also has at t_1 a sufficient physical cause, *P*. Lowe has doubts about the proper formulation, not to mention the truth, of Closure (see his 2000; 2008, ch. 2). But here he grants the principle as formulated; indeed, this is central to his case for invisibility. So now the materials for the causal argument are in place: *M* must either be included within *P* or causally redundant, and neither option is compatible with emergentism.

But in an ingenious block of this argument, Lowe says the mental cause could be emergent, but in a way that respects Closure. Suppose that at t_1 , *P* causes emergent property *M* to be instantiated at that very time. Moreover, *M* then aids non-redundantly in the production of *B*. That is, if (counter-nomically) *P* were present without *M*, *P* would not be enough to cause *B*. *M*'s added oomph is needed. But this does not impugn *P*'s sufficiency for *B*, since *P* is itself sufficient for *M*, its needed helper. The resulting model can be represented by Figure 10.1, where each arrow stands for causal sufficiency:

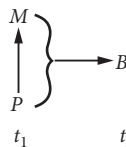


Figure 10.1. Lowe's model of mental causation

⁵ In a number of publications, Lowe explores several models of dualist interactionism: Lowe 1992; 1996, ch. 3; 2003. My remarks here are limited to the model defended in Lowe 2003 (and his 2008, ch. 3). In what follows I will simplify Lowe's presentation, but in a way that I hope preserves its essential details.

If this is how we're to understand emergent mental causation, is M 's efficacy empirically detectable? Except for a hedge at one place,⁶ Lowe says No: M 's efficacy is invisible to empirical investigation, however sophisticated.⁷ Central to the point is that Closure is not violated, so that anyone tracking the causal history of B will find a sufficient physical cause at t_1 , and so will see no reason to postulate a non-physical cause, this in spite of the fact that a non-physical cause is required for B .

Note that there's not going to be a way to test for M 's non-redundant efficacy by removing the mental state (by, say, distracting or anesthetizing the subject) while holding P fixed. From our God's-eye view of the model, we see that the counter-nomic conditional mentioned earlier is true, but our hypothetical scientists can't see this, at least not in the way envisaged. Any removal of M would also involve tampering with P , thus altering B 's physical causal history. It will continue to appear as if P is doing all of the work.

It's important for Lowe's model that P 's causation of M be simultaneous. For suppose instead that t_1 and t_2 are not adjacent times, and that there is some time between them so that P causes M to be instantiated, not at t_1 , but at this intermediate time.⁸ In that case there would be a detectable violation of Closure at the intermediate time: a cause of B (namely, M) but no sufficient physical cause of B . We might try to save Closure here by postulating a sufficient physical cause of B at this intermediate time. Such an intermediary would be either sufficient on its own for B , or sufficient for B by producing the needed helper M . But in the former case, M 's efficacy becomes redundant, contrary to emergentism. And in the latter case, we are back to Lowe's model with just some re-labeling. The upshot is that an emergentist trying to respect Closure should require, as Lowe does, that M be simultaneously caused by its physical base P .

Having sketched Lowe's model, I now want to explore his invisibility claim in more detail. This will involve some back-and-forth between imagining M 's efficacy exposed, then wondering whether it might be hidden after all. However, I will eventually conclude with an optimistic hypothesis, a potential outcome of empirical investigation on which M 's efficacy will be visible.

3. Powers and Causal Sufficiency

It's because Closure is allegedly respected that M 's causal role is invisible. But there nevertheless seems to be a clear sense in which the model violates the principle. After all, something non-physical is required for the production of a physical effect. Closure says the physical world contains the full resources for the production of any physical

⁶ 'While this is not to say that there is no kind of empirical evidence that could have a bearing on this debate, the implication would seem to be that purely metaphysical arguments must have a larger role to play than is commonly supposed in determining which of [interactive dualism or one of its physicalist rivals] is superior' (Lowe 2008, 58).

⁷ The question here and below is not whether M itself is visible (Lowe 2003, 151), but instead whether M 's efficacy—and in particular, its efficacy with respect to B —is visible to empirical investigators.

⁸ I take it such is the picture endorsed by, for example, O'Connor and Wong 2005, who thereby reject Closure.

effect, yet on Lowe's model, the physical world *doesn't* contain the full resources for the production of *B*, as the non-physical *M* is required. Granted, *P* is sufficient for *B* on a nomic understanding of causal sufficiency: *P* and the relevant laws of nature ensure that *B* occurs. But since the model doesn't respect Closure, it looks as if nomic sufficiency isn't the appropriate way to understand causal sufficiency in this context.

An alternative way to understand causal sufficiency is first to think of *M*, *P*, and *B*, not just as properties, but more specifically as *powers* and causation as the manifestation of powers (see, e.g., Molnar 2003, Mumford and Anjum 2010, Heil 2012). On Lowe's model, *M* is the manifestation of *P*, and *B* is the manifestation of *P* 'partnered' with *M*. Switching to powers-talk might be optional here, but it provides a useful way to discuss Lowe's model, and in particular why Closure does in fact appear to be violated in it, in spite of Lowe's intentions.

On the powers account of causation, power *y*'s being a manifestation of powers $x_1 \dots x_n$ (equivalently, $x_1 \dots x_n$'s producing *y*) is taken as a primitive,⁹ but one of its important features can be mentioned here: If *y* is a manifestation of $x_1 \dots x_n$, then $x_1 \dots x_n$ are *complete* in the sense that no other power needs to be partnered with them to produce *y*. It is in this sense that $x_1 \dots x_n$ are causally sufficient for *y*, and it's this productive sense of causal sufficiency, not the nomic sense, that, it seems to me, should appear in Closure. (Another reason causal sufficiency in this powers sense isn't necessitation is that it's usually possible for a complete set to be present while some other power on the scene prevents the manifestation.¹⁰ One may be tempted to say in that case that the original set isn't complete after all: the *absence* of the preventing power must be present as well. But even if there are absences, the absence of a power isn't a power, any more than the absence of a chair is a chair.)

Return then to Closure, which currently reads:

Closure: Everything physical that has a cause at time *t* has a sufficient physical cause at *t*.

This is compatible with the powers account of causation, so long as 'sufficient physical cause' is understood in terms of causal production (manifestation), as above. But it's worth reformulating the principle to make this explicit:

Closure (revised): Every physical manifestation is produced by physical powers.

A few notes on this new version:

First, it allows the principle to be a bit more compact. Since manifestations are by definition caused, 'every physical manifestation that has a cause' would be redundant.

Second, a temporal variable no longer needs to appear in the principle. Such a variable was originally required to rule out violations of Closure in which a sufficient physical

⁹ I take this conceptual point to be compatible with manifestation's having a discoverable 'deep essence'. For example, it could turn out that manifestation essentially involves the persistence of some quantity, such as energy. But I'm neutral on such matters here.

¹⁰ See Mumford and Anjum 2010.

cause of a physical effect worked only through some non-physical cause at an intermediate time. But powers and their manifestations are never mediated. Any would-be production requiring causal mediation at other times would not be complete, as the intermediate power(s) would have to be added to produce the manifestation.

Third, this modified principle, like its predecessor, is not so strong as to preclude a role for non-physical powers in producing physical manifestations. For all the revised principle tells us, a given physical effect might be produced by physical as well as non-physical powers. Closure requires only that every physical manifestation is produced at least by physical powers, whether or not it happens to be produced (that is, over-produced) by other sorts of powers as well.

Fourth and finally, this rephrasing has the virtue of explaining why Lowe's model is, as I believe it should be, a violation of Closure. *B* on Lowe's model is not produced by *P*, for *P*, although complete for *M*, is not complete for *B*. Rather, *B* is produced by the partnering of *P* and *M*. Furthermore, the invisibility claim starts to look suspect in a way that it might not have on the nomic understanding of causal sufficiency. After all, if *P* doesn't produce *B*, this should be detectable. With respect to producing *B*, *P* should look as odd as a gear with some missing teeth that nevertheless smoothly turns another gear. This is in fact what I will propose, though getting there will require some additional remarks on powers.

4. Principles of Combination

Consider a bit more concretely how we might expose the efficacy of an emergent property given the powers understanding of causation. Thesis (2) of emergentism, when elaborated, says that mental properties emerge only in physical systems of a certain degree of complexity. This means *M* emerges from a complex combination of physical powers, so that *P* is really a structural property composed of properties of the system's physical parts (perhaps combined with properties of the immediate environment). Call these the 'micro-powers' composing *P*.

I assume these micro-powers can be studied in non-mental contexts—that is, when they do not compose *P* or any other psychological base, so that the activities of non-physical powers are not in question. Such a study will let us construct an increasingly detailed *causal profile* for each micro-power, one that specifies the contribution the power makes when a partner in purely physical production.¹¹ The general strategy for exposing *M*'s efficacy is then unsurprising: with these empirically discovered causal profiles, predict how the micro-powers will partner in the mental context. If *B* is what's predicted, we have evidence that they are the powers producing this behavior. If

¹¹ A causal profile may perhaps be represented with (some fragment of) a 'Shoemaker sentence': see Hawthorne (2001, 376).

not—if *B* is anomalous with respect to these profiles—we have evidence for the contribution of a non-physical power, and *M*'s efficacy is no longer hidden.

But exposing *M*'s causal role cannot be this simple. The worry here is not inductive skepticism about causal profiles. (If skepticism were in play, Lowe's invisibility claim would be more secure, but less interesting, perhaps trivial.) The worry, rather, is that the causal profiles gleaned from non-mental contexts aren't useful for this project without some *principles of combination* saying how micro-powers with these profiles combine when partnered in a new (in this case, mental) context. Such principles cannot be known *a priori*, nor should we expect micro-powers of all types to combine in the same ways (see, e.g., McLaughlin 1997). But empirical study could reveal the relevant principles of combination. For example, some powers will combine additively, say by simple vector addition. Others may fail to combine additively, yet will combine continuously, so that similar powers in combination with similar partners produce proportionally similar manifestations. Still other micro-powers may not obey this principle, and will combine with one another in a radically discontinuous manner. I take it all these principles of combination, and more, are empirically possible for *P*'s micro-powers.

Suppose then that we have confirmed a range of such principles in non-mental contexts. Armed with the causal profiles of our micro-powers, we should be able to predict how they will jointly manifest when composing *P*. If something other than *B* is predicted, then when *B* occurs, this is evidence of a non-physical power at work. (Might this simply be evidence for a previously undetected *physical* power? No, for skeptical worries about physical properties are here set aside. Lowe's invisibility thesis is interesting because it's claimed to hold when empirical information about all of the relevant physical properties is already in.) Note that the epistemic bar here is low. Empirical evidence for some conclusion needn't be infallible, and inductive skepticism is off the table. We just need good empirical reasons for thinking that a non-physical property is efficacious, at it looks as if this qualifies.

5. The Micro-latency Hypothesis and Intelligibility

The upshot so far is that the efficacy of *M* in Lowe's model will not be invisible so long as the manifestation *B* is not what we would predict from the micro-powers composing *P* and the governing principles of combination, all of which can be discovered empirically. There is, however, more one might say here to defend Lowe's invisibility thesis.

Even when some confirmed principles of combination are apparently violated in mental contexts, emergence is not the only viable hypothesis. Physicalists and emergentists alike agree that mentality is special, found only in a small part of the universe (perhaps only this planet), and only in a small sliver of time. Moreover, there are well-known idiosyncrasies of mental contexts: only in them do we find phenomenal consciousness, self-awareness, and other unique capacities. It would not be too

surprising, then, if a principle of combination confirmed in non-mental contexts were violated in mental contexts, not because of non-physical powers on the scene, but because mental contexts—unique on everyone's lights—bring out novel manifestations of physical causal powers. One does not need to be an inductive skeptic to consider this a live possibility.

Call this the micro-latency hypothesis (Shoemaker 2002). This hypothesis is not emergentist in our sense, for there's nothing in physicalism prohibiting physical powers from having novel manifestations in unique contexts. The micro-latency hypothesis proposes that this is what's going on when *P* apparently produces the surprising result *B*. Even though *M* is at work, empirical investigators will have no reason to postulate *M* rather than what has previously been latent in the ordinary micro-powers composing *P*. *M*'s efficacy will continue to be invisible.

(In defense of this hypothesis, I should also note that it needn't say the micro-powers composing *P* are transformed into different powers—*pace* O'Connor who, unlike Lowe, is a 'visible' emergentist. As O'Connor reads the micro-latency hypothesis, it requires that the micro-powers composing *P* are locally responsive to their macro-level circumstances (O'Connor 2000, 113). He then objects that it's mysterious how a power's nature could be affected non-locally according to the broader structure of which it is a part. How does a micro-power 'know' that it's now included in such a larger, psychological structure? But the hypothesis doesn't have to be understood in this way. The micro-powers composing *P* don't *change* in a mental context. Rather, in this context, they maintain their intrinsic natures and simply compose in novel ways.)

If the micro-latency hypothesis is always available as a (non-skeptical) alternative to emergentism, Lowe's invisibility claim would seem to be vindicated. But continuing to press, could it turn out that there is a principle of combination whose violation could not be explained by this hypothesis? Suppose we were to empirically confirm a *meta-principle* of combination, one that governs the way powers combine no matter what first-order principle (additivity, continuity, or whatnot) they obey in any particular situation. If powers are observed to combine in various first-order ways in non-mental contexts, we might be able to discern such a meta-principle.

What could a meta-principle look like? One idea—call it the Principle of Intelligibility—is that powers are always manifested *intelligibly*. As a first pass, think of the principle like this: when powers are manifested, we can see that the powers and manifestation fit together, much like pieces of a jigsaw puzzle. Knowing (by prior inspection) how the pieces are shaped, we can see that they should interlock. Similarly, knowing (empirically) the causal profiles of various powers, we can see how they and their manifestations fit together. A version of the Principle of Intelligibility is expressed by Philo in a well-known passage of Hume's *Dialogues*:

It is observed by arithmeticians that the products of 9 compose always either 9 or some lesser product of 9 if you add together all the characters of which any of the former products is composed.... To a superficial observer so wonderful a regularity may be admired as the effect either

of chance or design; but a skilful algebraist immediately concludes it to be the work of necessity, and demonstrates that it must forever result from the nature of these numbers. Is it not probable, I ask, that the whole economy of the universe is conducted by a like necessity, though no human algebra can furnish a key which solves the difficulty? And instead of admiring the order of natural beings, may it not happen that, could we penetrate into the intimate nature of bodies, we should clearly see why it was absolutely impossible they could ever admit of any other disposition? (Hume 1779/1998, Part IX)¹²

Suppose we've found substantial support for Philo's vision by studying the ways physical powers combine in non-mental contexts, yielding a variety of first-order principles of combination, all of them intelligible. And suppose, then, that the micro-powers composing *P* are *not* manifested intelligibly by *B* in the mental context. That is, empirical investigators can't see how *P* and *B* fit together. This looks like good evidence for a non-physical power partnering with *P* to produce *B*. Now imagine the defender of invisibility appealing again to the micro-latency hypothesis: for all our scientists know, the anomalous manifestation *B* is a result, not of a non-physical power partnering with *P*, but of *P*'s ordinary micro-powers, which combine in new, unexpected ways in mental contexts. The Principle of Intelligibility is supposed to block this move. Novel first-order principles of combination are fine, but the meta-principle requires that we see *how* the novel manifestation arises. If we can't see this, we again have evidence for an emergent power on the scene.

The Principle of Intelligibility is very schematic, and will no doubt raise eyebrows. For starters, what exactly does it mean to be intelligible? The example of the jigsaw puzzle is merely an analogy, not a definition. And doesn't the principle add an epistemic (and possibly subjective) element to what should be a metaphysical issue? And in any case, didn't Hume show that causal powers are *never* manifested intelligibly? I can't answer all these questions here, but I'll make a more limited response in conclusion.¹³

First, to lessen the worries of Humeans: the principle doesn't require that we can know the causal profile of a property *a priori*. We can agree with Hume that Adam, newly created, couldn't simply intuit that fire would burn him, or that bread would nourish him. What it does require is that once we empirically discover the causal profiles of micro-properties, their subsequent manifestations will be intelligible to us. This may involve *a priori* inference, but as Sanford (1994) has argued, it's one that could come only after substantial empirical input. Consider two gears that are meshed. Only empirically do we know the natures of the gears' properties—shape, rigidity, and so on. But given this knowledge, we can see—*a priori* infer—that one gear's turning clockwise will cause the other to turn counterclockwise. It's this last bit of insight that Sanford calls 'intelligibility'.

¹² Whether Hume himself endorsed the Principle of Intelligibility is controversial: see the papers collected in Read and Richman 2007.

¹³ I have explored this in more detail in Robb 2007, 2015.

Second, the Principle of Intelligibility as formulated is stronger than needed to answer Lowe's invisibility claim. I've been speaking of 'micro-powers', but a structural property such as *P* can be broken down into properties at various mereological levels. For simplicity, suppose there are two such levels: the level of basic physics (the micro-level), and above it the level of neuroscience. Each level has its characteristic properties (powers). Now the Principle of Intelligibility as understood so far requires that powers manifest intelligibly at all levels, but this is more than is needed in the present context. Suppose it turns out that powers needn't manifest intelligibly at the level of basic physics. Perhaps it just looks brute that (e.g.) like charges repel. It's compatible with this that neuroscientific powers *do* manifest intelligibly.¹⁴ (Returning to the puzzle analogy, it would be like discovering that a puzzle is ultimately composed of tiny, colored pixels. Why one pixel should be next to another may look brute. But *given* that pixels are arranged to form certain shapes at higher mereological levels, their subsequent ability to interlock is intelligible.) So if we were to discover that this more limited Principle of Neuro-Intelligibility holds, our scientists could use it to look at what *P* produces in the mental context. If *B* is an unintelligible result at the neuroscientific level, this again is evidence for the efficacy of an emergent, evidence that cannot be explained away with the micro-latency hypothesis.

The Principle of Intelligibility is an ambitious hypothesis, and so may obscure my aim here, which is modest: find some way, at least in principle, to reveal *M*'s efficacy on Lowe's model of mental causation. Empirical confirmation of the Principle of Intelligibility (or its weaker cousin) looks like it would be enough. If this is right, then it has a lesson for those bold emergentists who take their view to make empirical predictions: they should hope for confirmation of the Principle of Intelligibility or some similar meta-principle of combination. Otherwise, the debate between physicalists and emergentists may remain with philosophy, and there the evidence is that the debate is likely to remain unresolved.¹⁵

6. References

- Bedau, M. 2002. 'Downward Causation and the Autonomy of Weak Emergence'. *Principia* 6: 5–50.
- Craver, C. F. 2007. *Explaining the Brain: Mechanisms and the Mosaic Unity of Neuroscience*. Oxford: Oxford University Press.
- Hawthorne, J. 2001. 'Causal Structuralism'. *Philosophical Perspectives* 15: 361–78.
- Heil, J. 2012. *The Universe as We Find It*. Oxford: Oxford University Press.
- Hume, D. 1779/1998. *Dialogues Concerning Natural Religion*, ed. R. H. Popkin, 2nd edn. Cambridge, MA: Hackett Publishing.

¹⁴ Mill (1843/2011, III.vi.2) makes a similar point, though it's put in terms of laws. The Principle of Neuro-Intelligibility is closely related to the mechanistic style of explanation defended in Craver (2007).

¹⁵ For helpful discussion of this chapter, I thank audiences at Washington University, St. Louis in 2011; the Workshop on Dispositions and Mind, Seoul in 2012; the Triangle Area Philosophy Symposium (2012); and The Philosophy of E. J. Lowe: A Memorial Conference, Durham University (2014). I also thank James Van Cleve for his questions and suggestions on a number of issues, including the formulation of Closure and the nature of a causal profile.

- Kim, J. 1999. 'Making Sense of Emergence'. *Philosophical Studies* 95: 3–36.
- Lewis, D. K. 1966. 'An Argument for the Identity Theory'. *Journal of Philosophy* 63: 17–25.
- Lowe, E. J. 1992. 'The Problem of Psychophysical Causation'. *Australasian Journal of Philosophy* 70: 263–76.
- Lowe, E. J. 1996. *Subjects of Experience*. Cambridge: Cambridge University Press.
- Lowe, E. J. 2000. 'Causal Closure Principles and Emergentism'. *Philosophy* 75: 571–85.
- Lowe, E. J. 2003. 'Physical Causal Closure and the Invisibility of Mental Causation'. In *Physicalism and Mental Causation* ed. S. Walter and H. Heckmann, 137–54. Exeter: Imprint Academic.
- Lowe, E. J. 2008. *Personal Agency*. Oxford: Oxford University Press.
- McLaughlin, B. 1992. 'The Rise and Fall of British Emergentism'. In *Emergence or Reduction?* ed. A. Beckermann, H. Flohr, and J. Kim, 49–93. Walter de Gruyter.
- McLaughlin, B. 1997. 'Emergence and Supervenience'. *Intellectica* 25: 25–43. Portions reprinted in M. Bedau and P. Humphreys, eds, *Emergence*. MIT Press, 2008.
- Mill, J. S. 1843/2011. *System of Logic*. In *Collected Works of John Stuart Mill*, Vol. VIIA ed. J. M. Robson, Reprint edn. Routledge.
- Molnar, G. 2003. *Powers*. Oxford: Oxford University Press.
- Mumford, S. and R. Anjum. 2010. 'A Powerful Theory of Causation'. In *The Metaphysics of Powers* ed. A. Marmodoro, 143–59. Routledge.
- O'Connor, T. 2000. *Persons and Causes*. Oxford: Oxford University Press.
- O'Connor, T. and H. Y. Wong. 2005. 'The Metaphysics of Emergence'. *Noûs* 39: 658–78.
- O'Connor, T. and H. Y. Wong. 2015. 'Emergent Properties'. In E. Zalta, ed., *The Stanford Encyclopedia of Philosophy*. Stanford, CA.: Metaphysics Research Lab, Stanford University.
- Papineau, D. 2001. 'The Rise of Physicalism'. In *Physicalism and Its Discontents* ed. C. Gillett and B. Loewer, 3–36. Cambridge: Cambridge University Press.
- Read, R. and K. A. Richman, eds. 2007. *The New Hume Debate*. Revised edn. Routledge.
- Robb, D. 2007. 'Power Essentialism'. *Philosophical Topics* 35: 343–58.
- Robb, D. 2015. 'Mental Causation and Intelligibility'. In *Causation and Mental Causation, Humana.Mente Journal of Philosophical Studies* 29 ed. R. Campaner and C. Gabbani, 213–26.
- Sanford, D. H. 1994. 'Causation and Intelligibility'. *Philosophy* 69: 55–67.
- Shoemaker, S. 2002. 'Kim on Emergence'. *Philosophical Studies* 108: 53–63.
- Shoemaker, S. 2007. *Physical Realization*. Oxford: Oxford University Press.
- Wilson, J. 2011. 'Non-reductive Realization and the Powers-based Subset Strategy'. *The Monist* 94: 121–54.

Bibliography of the works of E. J. Lowe

Books

1. ***Kinds of Being: A Study of Individuation, Identity and the Logic of Sortal Terms*** (Aristotelian Monograph Series, No. 10), Oxford & New York: Basil Blackwell, 1989, pp. vi + 210. *Reviews*: P. F. Snowdon, *Philosophical Books* 32, 1991, 37–9; H. W. Noonan, *Philosophy* 66, 1991, 248–9; G. Engelbrechtsen, *Iyyun* 40, 1991, 100–5; P. Simons, *Mind* 101, 1992, 581–2; M. Baur, *Review of Metaphysics* 46, 1992, 166–8; F. J. Pelletier, *History and Philosophy of Logic* 13, 1992, 125–8
2. ***Locke on Human Understanding***, London & New York: Routledge, 1995, pp. x+203. *Reviews*: A. Miller, *Locke Newsletter* 26, 1995, 141–55; P. Snowdon, *Mind* 105, 1996, 348–51; B. Maund, *Australasian Journal of Philosophy* 74, 1996, 528–30; N. Unwin, *Philosophical Books* 38, 1997, 97–8; E. Barbanell, *Canadian Philosophical Reviews* 15, 1995, 410–12; D. W. Hamlyn, *Philosophical Investigations* 20, 1997, 155–9; D. Doering, *Jahrbuch zur Liberalismus-Forschung* 8, 1996, 251–2; A. Nelson & L. Nolan, *Philosophia* 27, 1999, 665–8
3. ***Subjects of Experience***, Cambridge: Cambridge University Press, 1996, pp. x + 209. *Reviews*: K. Sutherland, *Times Higher Education Supplement*, Oct. 11, 1996, 21; J. Heil, *Times Literary Supplement*, Jan. 10, 1997, 27; G. Madell, *Philosophy* 72, 1997, 147–50; J. Brown, *Philosophical Books* 39, 1998, 56–8; K. Paprzycka, *Canadian Philosophical Reviews* 17, 1997, 45–7; J. L. Bermudez, *Philosophical Quarterly* 49, 1999, 272–5; U. Uus, *Journal of Consciousness Studies* 6, 1999, 90–1; C. Macdonald, *Philosophy and Phenomenological Research* 60, 2000, 224–8; C. J. L. Talmage, *Dialogue* 37, 1998, 631–3
4. ***The Possibility of Metaphysics: Substance, Identity, and Time***, Oxford: Oxford University Press, 1998, pp. viii + 275. *Reviews*: F. Jackson, *Times Literary Supplement*, Apr. 9, 1999, 33; K. Hawley, *British Journal for the Philosophy of Science* 50, 1999, 478–82; S. McLeod, *History and Philosophy of Logic* 20, 1999, 141–3; E. Olson, *Mind* 109, 2000, 967–9; M. D. Gossiaux, *Review of Metaphysics* 54, 2000, 159–60; S. Shalkowski, *Philosophical Books* 41, 2000, 275–8; J. Heil, *Philosophical Review* 110, 2001, 91–4; G. Rosenkrantz, *Philosophy and Phenomenological Research* 64, 2002, 728–36; R. Le Poidevin, *International Philosophical Quarterly* 42, 2002, 546–7. Italian language version: ***La possibilità della metafisica: sostanza, identità, tempo***, ed. & trans. S. Galvan, A. Corradini & C. L. De Florio, Soveria Mannelli: Rubbettino, 2009
5. ***An Introduction to the Philosophy of Mind***, Cambridge: Cambridge University Press, 2000, pp. xiii + 318. *Reviews*: M. di Francesco, *Il Sole 24 Ore*, 292, Oct. 29, 2000, 34; B. Benham, *Metapsychology Online Reviews*, Oct. 11, 2000; St. E. Cuypers, *Tijdschrift voor Filosofie* 63, 2001; J. Gosling, *International Philosophical Quarterly* 4, 2001, 253–5; H. Langsam, *Philosophical Books* 42, 2001, 229–31; S. Crawford, *Times Literary Supplement*, 5167, Apr. 12, 2002, 29; B. Majors, *Review of Metaphysics* 56, 2002, 438–9; D. Heider, *Filosoficky-Casopis* 51, 2003, 1035–44. Spanish language version: ***Filosofía de la mente***, trans. O. Fernandez Prat, Idea Books, 2000, pp. 281. Estonian language version: ***Sissejuhatus vaimufilosoofiasse***, trans. T. Hallap, Tartu: Tartu Ülikooli Kirjastus, 2008, pp. 331. Persian language version: Nashr-e Markaz Publishing Company, 2010

6. **A Survey of Metaphysics**, Oxford: Oxford University Press, 2002, pp. xiii + 402. *Reviews*: T. Crane, *Times Higher Education Supplement*, May 31, 2002; G. Keil, *Zeitschrift für Philosophische Forschung* 57, 2003, 320–4; H. Clement, *Tijdschrift voor Filosofie* 67, 2005, 347–8; A. M. Nguyen, *Teaching Philosophy* 29, 2006, 384–7
7. **Locke**, London & New York: Routledge, 2005, pp. xiii + 220. *Reviews*: J. Hill, *Locke Studies* 6, 2006, 211–18; M. Stuart, *Notre Dame Philosophical Reviews*, 13/6/2006; W. E. Morris, *Philosophical Books* 49, 2008, 48–50
8. **The Four-Category Ontology: A Metaphysical Foundation for Natural Science**, Oxford: Oxford University Press, 2006, pp. xiv + 222. *Reviews*: I. Johansson, *Dialectica* 60, 2006, 513–18; R. Wasserman, *Notre Dame Philosophical Reviews*, 11/4/2006; D. W. Mertz, *The Modern Schoolman* 84, 2006, 79–81; P. van Inwagen, *Times Literary Supplement*, 5415, Jan. 12, 2007, 22; S. Bostock, *Philosophical Books* 48, 2007, 274–7; J. Westerhoff, *Mind* 116, 2007, 759–62; B. C. Look, *Review of Metaphysics* 60, 2007, 666–8; P. Bricker, *Australasian Journal of Philosophy* 87, 2009, 675–8; A. R. Thompson, *Polish Journal of Philosophy* 3, 2009, 143–7
9. **Personal Agency: The Metaphysics of Mind and Action**, Oxford: Oxford University Press, 2008, pp. xvi + 222. *Reviews*: M. Griffith, *Notre Dame Philosophical Reviews*, 15/6/2009; I. Persson, *Times Literary Supplement*, 5559, Oct. 16, 2009, 9; A. Gyr, *Metaphysica* 10, 2009, 215–21; B. P. Göcke, *Philosophy* 85, 2010, 302–6; R. Clarke, *Mind* 119, 2010, 820–3; A. Buckareff, *Philosophy in Review* 30, 2010, 276–9; L. O'Brien, *Australasian Journal of Philosophy* 89, 2011, 172–4
10. **More Kinds of Being: A Further Study of Individuation, Identity, and the Logic of Sortal Terms**, Malden, MA & Oxford: Wiley-Blackwell, 2009, pp. xii + 227. *Reviews*: G. S. Rosenkrantz, *Notre Dame Philosophical Reviews*, 12/8/2010
11. **Forms of Thought: A Study in Philosophical Logic**, Cambridge: Cambridge University Press, 2013, pp. xii + 213. *Reviews*: C. Hermes, *Philosophical Quarterly*, 64, 2014, 352–4; S. Gross, *Analysis*, 75, 2015, 165–7; P. Washburn, *Review of Metaphysics*, 68, 2015, 669–70
12. **Locke's Essay Concerning Human Understanding**, London & New York: Routledge, 2013, pp. x + 207. [Revised and expanded edn of 2 above]

Edited volumes

1. **Analytic Philosophy Without Naturalism**, ed. A. Corradini, S. Galvan, & E. J. Lowe, London & New York: Routledge, 2006, pp. xiii + 266
2. **Psycho-Physical Dualism Today: An Interdisciplinary Approach**, ed. A. Antonietti, A. Corradini, & E. J. Lowe, Lanham, MD: Rowman & Littlefield (Lexington Books), 2008, pp. xv + 241
3. **Truth and Truth-Making**, ed. E. J. Lowe & A. Rami, Stocksfield: Acumen, 2009, pp. x + 262. *Reviews*: D. Eford, *Notre Dame Philosophical Reviews*, 17/8/2009; J. Dodd, *Analysis Reviews* 70, 2010, 567–71
4. **Mental Causation and Ontology**, ed. S. C. Gibb, E. J. Lowe, & R. D. Ingthorsson, Oxford: Oxford University Press, 2013, pp. viii + 272

Articles in journals and contributions to edited volumes

1. Neither intentional nor unintentional, *Analysis* 38, 1978, 117–18
2. Indicative and counterfactual conditionals, *Analysis* 39, 1979, 139–41
3. Active and passive euthanasia: an objection, *Philosophy* 55, 1980, 550–1

4. An analysis of intentionality, *Philosophical Quarterly* 30, 1980, 294–304
5. For want of a nail, *Analysis* 40, 1980, 50–2
6. Peacocke and Kraemer on Butler's Problem, *Analysis* 40, 1980, 113–18
7. Reply to Davis, *Analysis* 40, 1980, 187–90
8. Sortal terms and natural laws, *American Philosophical Quarterly* 17, 1980, 253–60 [Prize Essay]
9. Against an argument for token identity, *Mind* 90, 1981, 120–1
10. 'All actions occur inside the body', *Analysis* 41, 1981, 126–9
11. Indirect perception and sense data, *Philosophical Quarterly* 31, 1981, 330–42
12. Intentionality: a reply to Stiffler, *Philosophical Quarterly* 32, 1982, 354–7
13. Intentionality and intuition: a reply to Davies, *Analysis* 42, 1982, 85
14. Laws, dispositions and sortal logic, *American Philosophical Quarterly* 19, 1982, 41–50
15. On being a cat, *Analysis* 42, 1982, 174–7
16. On the alleged necessity of true identity statements, *Mind* 91, 1982, 579–84
17. Reply to Geach, *Analysis* 42, 1982, 31
18. The paradox of the 1,001 cats, *Analysis* 42, 1982, 27–30
19. A simplification of the logic of conditionals, *Notre Dame Journal of Formal Logic* 24, 1983, 357–66
20. Instantiation, identity and constitution, *Philosophical Studies* 44, 1983, 45–59
21. On the identity of artifacts, *Journal of Philosophy* 80, 1983, 220–32
22. Reply to Hornsby on actions, *Analysis* 43, 1983, 140–1
23. A note on a response of Hornsby's, *Analysis* 44, 1984, 196–7
24. Wright versus Lewis on the transitivity of counterfactuals, *Analysis* 44, 1984, 180–3
25. 'If A and B, then A', *Analysis* 45, 1985, 93–8
26. Reply to Baldwin on *de re* modalities, *Mind* 94, 1985, 101–3
27. Reply to Wright on conditionals and transitivity, *Analysis* 45, 1985, 200–2
28. On a supposed temporal/modal parallel, *Analysis* 46, 1986, 195–7
29. Necessity and the will in Locke's theory of action, *History of Philosophy Quarterly* 3, 1986, 149–63 [reprinted in U. Thiel (ed.), *Locke: Metaphysics*, Aldershot: Ashgate, 2002, 247–61]
30. Noonan on naming and predicating, *Analysis* 46, 1986, 159
31. Reply to Dale, *Analysis* 46, 1986, 83–5
32. Reply to Over, *Analysis* 46, 1986, 200
33. Sortal terms and absolute identity, *Australasian Journal of Philosophy* 64, 1986, 64–71
34. The topology of visual appearance, *Erkenntnis* 25, 1986, 271–4
35. What do we see directly?, *American Philosophical Quarterly* 23, 1986, 277–85
36. Miracles and laws of nature, *Religious Studies* 23, 1987, 263–78
37. Lewis on perdurance versus endurance, *Analysis* 47, 1987, 152–4 [reprinted in H. Noonan (ed.), *Identity*, Aldershot: Dartmouth, 1993, 311–13]
38. Not a counterexample to modus ponens, *Analysis* 47, 1987, 44–7
39. Reply to Le Poidevin and Mellor, *Mind* 96, 1987, 539–42
40. Reply to Noonan, *Analysis* 47, 1987, 201–3
41. The indexical fallacy in McTaggart's proof of the unreality of time, *Mind* 96, 1987, 62–70
42. What is the 'problem of induction?', *Philosophy* 62, 1987, 325–40
43. Substance, *An Encyclopaedia of Philosophy*, ed. G. H. R. Parkinson, London: Routledge, 1988, 255–78
44. Substance, identity and time, *Aristotelian Society Supplementary Volume* 62, 1988, 61–78

45. The problems of intrinsic change: rejoinder to Lewis, *Analysis* 48, 1988, 72–7
46. Impredicative identity criteria and Davidson's criterion of event identity, *Analysis* 49, 1989, 178–81
47. What is a criterion of identity?, *Philosophical Quarterly* 39, 1989, 1–21 [reprinted in H. Noonan (ed.), *Identity*, Aldershot: Dartmouth, 1993, 435–55]
48. Conditionals, context and transitivity, *Analysis* 50, 1990, 80–7
49. David Lewis, *Handbook of Metaphysics and Ontology*, ed. H. Burkhardt & B. Smith, Munich: Philosophia, 1991, 453–4
50. Jackson on classifying conditionals, *Analysis* 51, 1991, 126–30
51. Noun phrases, quantifiers, and generic names, *Philosophical Quarterly* 41, 1991, 287–300
52. One-level versus two-level identity criteria, *Analysis* 51, 1991, 192–4
53. Real selves: persons as a substantial kind, *Human Beings*, ed. D. Cockburn, Cambridge: Cambridge University Press, 1991, 87–107 [Review: C. Gill, *Philosophical Quarterly* 42, 1992, 502–4]
54. Substance and selfhood, *Philosophy* 66, 1991, 81–99
55. Experience and its objects, *The Contents of Experience*, ed. T. Crane, Cambridge: Cambridge University Press, 1992 [Reviews: T. Sorell, *New Scientist*, 14 Nov. 1992, 45–6; A. Millar, *Mind* 102, 1993, 362–6]
56. McTaggart's paradox revisited, *Mind* 101, 1992, 323–6
57. Reply to Ramachandran on conditionals and transitivity, *Analysis* 52, 1992, 77–80
58. The problem of psychophysical causation, *Australasian Journal of Philosophy* 70, 1992, 263–76 [reprinted in T. O'Connor & D. Robb (eds), *Philosophy of Mind: Contemporary Readings*, London: Routledge, 2003, 49–64]
59. Are the natural numbers individuals or sorts?, *Analysis* 53, 1993, 142–6
60. Comment on Le Poidevin, *Mind* 102, 1993, 171–3
61. Contributions to *Encyclopedia of Language and Linguistics*, ed. R. E. Asher et al., Oxford & Aberdeen: Pergamon Press & Aberdeen University Press, 1993: *a priori*; abstract ideas; epistemology; natural kinds; sortal terms [reprinted in *Concise Encyclopedia of the Philosophy of Language*, ed. P. V. Lamarque, Oxford: Elsevier Science, 1997]
62. Perception: a causal representative theory, *New Representationalisms: Essays in the Philosophy of Perception*, ed. E. Wright, Aldershot: Avebury, 1993, 136–52
63. Rationality, deduction and mental models, *Rationality*, ed. K. Manktelow & D. Over, London: Routledge, 1993, 211–30
64. Self, reference and self-reference, *Philosophy* 68, 1993, 15–33
65. The causal autonomy of the mental, *Mind* 102, 1993, 629–44
66. Ontological dependency, *Philosophical Papers* 23, 1994, 31–48
67. Primitive substances, *Philosophy and Phenomenological Research* 54, 1994, 531–52
68. Vague identity and quantum indeterminacy, *Analysis* 54, 1994, 110–14
69. Coinciding objects: in defence of the 'standard account', *Analysis* 55, 1995, 171–8
70. Contributions to *The Oxford Companion to Philosophy*, ed. T. Honderich, Oxford: Oxford University Press, 1995: *a priori* and *a posteriori*; A-series and B-series; affirmative and negative propositions; analytic and synthetic statements; axiom; being; Bennett; capacity; Carnap; cat, Schroedinger's; class; contingent and necessary statements; Dummett; existence; existential proposition; fact; generalization; identity, criterion of; Jackson; kind, natural; Kripke; Mellor; mereology; mode; monism and pluralism; names; necessary and sufficient

- conditions; necessity, epistemic; necessity, logical; necessity, metaphysical; nominalism; ontology; particulars and non-particulars; philosophical logic; potentiality; propensity; Ramsey; realization; recursion, definition by; redundancy theory of truth; reflective equilibrium; rigid designator; self; semantic theory of truth; specious present; Stalnaker; Strawson; things; thought experiments; time; time preference; time travel; truth; universals; Wiggins
71. Die Metaphysik und ihre Möglichkeit, *Metaphysik—Neue Zugänge zu alten Fragen*, ed. J. Brandl, A. Hieke, & P. Simons, St. Augustin: Academia Verlag, 1995, 11–32
 72. The metaphysics of abstract objects, *Journal of Philosophy* 92, 1995, 509–24
 73. The problem of the many and the vagueness of constitution, *Analysis* 55, 1995, 179–82
 74. The truth about counterfactuals, *Philosophical Quarterly* 45, 1995, 41–59
 75. There are no easy problems of consciousness, *Journal of Consciousness Studies* 2, 1995, 266–71 [reprinted in *Explaining Consciousness: The Hard Problem*, ed. J. Shear, Cambridge, MA: MIT Press, 1997, 117–23]
 76. Conditional probability and conditional beliefs, *Mind* 105, 1996, 603–15
 77. Why is there anything at all?, *Aristotelian Society Supplementary Volume* 70, 1996, 111–20
 78. Why there are no easy problems of consciousness [abstract], *History and Philosophy of Psychology Newsletter* 22, Spring 1996, 17–18
 79. John Locke; Identity, *Encyclopedia of Empiricism*, ed. D. Garrett & E. Barbanell, Westport, CT: Greenwood Press, 1997, 172–7, 206–15
 80. Ontological categories and natural kinds, *Philosophical Papers* 26, 1997, 29–46
 81. Reply to Noonan on vague identity, *Analysis* 57, 1997, 88–91
 82. Whose rationality? Logical theory and the problem of deductive competence, *Cahiers de Psychologie Cognitive/Current Psychology of Cognition* 16, 1997, 140–6
 83. Commentary on A. J. Hamilton's 'False Memory Syndrome and the Authority of Personal Memory Claims', *Philosophy, Psychiatry and Psychology* 5, 1998, 309–10
 84. Entity, Identity and Unity, *Erkenntnis* 48, 1998, 191–208
 85. Form without Matter, *Ratio* 11, 1998, 214–34 [reprinted in *Form and Matter: Themes in Contemporary Metaphysics*, ed. D. S. Oderberg, Oxford: Blackwell, 1999]
 86. Personal experience and belief: the significance of external symbolic storage for the emergence of modern human cognition, *Cognition and Material Culture: The Archaeology of Symbolic Storage*, ed. C. Scarre & C. Renfrew, Cambridge: McDonald Institute for Archaeological Research, 1998, 89–96
 87. Tense and persistence, *Questions of Time and Tense*, ed. R. Le Poidevin, Oxford: Clarendon Press, 1998, 43–59
 88. Abstraction, properties and immanent realism, *Proceedings of the Twentieth World Congress of Philosophy, Volume 2: Metaphysics*, ed. T. Rockmore, Bowling Green, OH: Philosophy Documentation Center, 1999, 195–205
 89. Concreta: substance—introduction, *Metaphysics: Contemporary Readings*, ed. S. D. Hales, Belmont, CA: Wadsworth, 1999, 371–6
 90. Contributions to *The Cambridge Dictionary of Philosophy*, ed. R. Audi, 2nd edn, Cambridge: Cambridge University Press, 1999: Armstrong, Dummett
 91. Contributions to *The Fontana/Norton Dictionary of Modern Thought*, ed. A. Bullock and S. Trombley, 3rd edn, London: Harper Collins, 1999: behaviourism; conventionalism; meaning; nominalism; realism; relativism

92. John Locke, *Companion to the Philosophers*, ed. R. L. Arrington, Oxford: Blackwell, 1999, 369–75 [reprinted in *The World's Great Philosophers*, ed. R. L. Arrington, Oxford: Blackwell, 2003, 195–202]
93. Self, agency and mental causation, *Journal of Consciousness Studies* 6, 1999, 225–39 [reprinted in *The Volitional Brain: Towards a Neuroscience of Free Will*, ed. B. Libet, A. Freeman, & K. Sutherland, Thorverton: Imprint Academic, 1999]
94. Vague identity and quantum indeterminacy: further reflections, *Analysis* 59, 1999, 328–30
95. Causal closure principles and emergentism, *Philosophy* 75, 2000, 571–85
96. In defence of the simplicity argument, *Australasian Journal of Philosophy* 78, 2000, 105–12
97. Locke, Martin and substance, *Philosophical Quarterly* 50, 2000, 499–514 [reprinted in *John Locke: Critical Assessments of Leading Political Philosophers, 2nd Series, Volume III: Metaphysics*, ed. P. Anstey, London & New York: Routledge, 2006]
98. Dispositions and laws, *Metaphysica* 2, 2001, 5–23
99. Event causation and agent causation, *Grazer Philosophische Studien* 61, 2001, 1–20
100. Identity, composition, and the simplicity of the self, *Soul, Body and Survival*, ed. K. J. Corcoran, Ithaca, NY: Cornell University Press, 2001, 139–58
101. Ontic indeterminacy of identity unscathed, *Analysis* 61, 2001, 241–5
102. Recent advances in metaphysics [abstract], *Formal Ontology in Information Systems: Collected Papers from the Second International Conference*, New York: ACM Press, 2001, 1
103. A defence of the four-category ontology, *Argument und Analyse*, ed. C. U. Moulines & K. G. Niebergall, Paderborn: Mentis, 2002, 225–40
104. A serious look at serious naturalism: review essay on W. L. Craig & J. P. Moreland (eds), *Naturalism: A Critical Analysis, Philosophia Christi* 4, 2002, 197–2002
105. Is knowing a state of mind? Critical notice of T. Williamson's *Knowledge and its Limits*, *International Journal of Philosophical Studies* 10, 2002, 483–9
106. Kinds, essence and natural necessity, *Individuals, Essence and Identity: Themes of Analytic Metaphysics*, ed. A. Bottani, M. Carrara, & P. Giaretta, Dordrecht: Kluwer, 2002, 189–206
107. La connaissance metaphysique, *Revue de metaphysique et morale*, 2002, 453–71
108. Material coincidence and the cinematographic fallacy: a response to Olson, *Philosophical Quarterly* 52, 2002, 369–72
109. Metaphysical nihilism and the subtraction argument, *Analysis* 62, 2002, 62–73
110. Properties, modes and universals, *The Modern Schoolman* 74, 2002, 137–50
111. The rational and the real: some doubts about the programme of 'rational analysis', *Reason and Nature: Essays in the Theory of Rationality*, ed. J. L. Bermudez & A. Millar, Oxford: Clarendon Press, 2002, 175–85
112. 3D/4D equivalence, the twins paradox, and absolute time, *Analysis* 63, 2003, 114–23 [co-author Storrs McCall]
113. Identity, Individuality and Unity, *Philosophy* 78, 2003, 321–36
114. In defense of medium-sized specimens of dry goods, *Philosophy and Phenomenological Research* 68, 2003, 704–10
115. Individuation, *Oxford Handbook of Metaphysics*, ed. M. J. Loux & D. W. Zimmerman, Oxford: Oxford University Press, 2003, 75–95
116. Metaphysical realism and the unity of truth, *Monism*, ed. A. Bachli & K. Petrus, Frankfurt: Ontos Verlag, 2003, 109–23

117. Personal agency, *Minds and Persons*, ed. A. O'Hear, Cambridge: Cambridge University Press, 2003, 211–27
118. Physical causal closure and the invisibility of mental causation, *Physicalism and Mental Causation: The Metaphysics of Mind and Action*, ed. S. Walter & H.-D. Heckmann, Exeter: Imprint Academic, 2003, 137–54
119. Recent advances in metaphysics, *Facta Philosophica* 5, 2003, 3–24
120. Substance causation, persons, and free will, *Persons: An Interdisciplinary Approach*, ed. C. Kanzian, J. Quitterer, & E. Runggaldier, Vienna, 2003, 76–88
121. Substantial change and spatiotemporal coincidence, *Ratio* 16, 2003, 140–60
122. Locke: compatibilist event-causalist or libertarian substance-causalist? Critical study of G. Yaffe's *Liberty Worth the Name: Locke on Free Agency*, *Philosophy and Phenomenological Research* 68, 2004, 688–701
123. Lois et dispositions, *La structure du monde: objets, proprietes, etats de chose: Renouveau de la metaphysique dans l'ecole australienne de philosophie*, ed. J.-M. Monnoyer, Paris: J. Vrin, 2004, 299–313
124. Non-Cartesian dualism, *Philosophy of Mind: A Guide and Anthology*, ed. J. Heil, Oxford: Oxford University Press, 2004, 851–65
125. Some formal ontological relations, *Dialectica* 58, 2004, 297–316
126. The four-category ontology: reply to Kistler, *Analysis* 64, 2004, 152–7
127. Endurance versus perdurance and the nature of time, *Philosophical Writings* 10, 2005, 45–58
128. Entries in *The Oxford Companion to Philosophy*, 2nd edn, ed. T. Honderich, Oxford: Oxford University Press, 2005: animalism in personal identity; causal asymmetry or direction; counterpart theory; disquotatation; emergence; essentialism; frame problem; modal realism; modality and metaphysics; presentism; probabilistic causality; reasoning, psychology of; slingshot arguments; tense; time's arrow; truth (revised); zombies
129. How are ordinary objects possible? *The Monist* 88, 2005, 510–33
130. Identity, vagueness and modality, *Thought, Reference, and Experience: Themes from the Philosophy of Gareth Evans*, ed. J. L. Bermudez, Oxford: Oxford University Press, 2005, 290–310
131. Indeterminist free will, *Philosophy and Phenomenological Research* 70, 2005, 681–90 [co-author Storrs McCall]
132. Interview with Jonathan Lowe, *Discourse: Learning and Teaching in Philosophical and Religious Studies* 5, 2005, 17–28
133. Needs, facts, goodness and truth, *The Philosophy of Need*, ed. S. Reader, Cambridge: Cambridge University Press, 2005, 161–73
134. Obituary of W. von Leyden, *Locke Studies* 5, 2005, 17–18
135. Ontological dependence, *The Stanford Encyclopedia of Philosophy*, ed. E. N. Zalta, 2005 [revised 2009 and again in 2016, when additional material was added by Tuomas Tahko], <http://plato.stanford.edu>
136. Philosophical logic, *The Map of Contemporary British and American Philosophy*, ed. Ouyang Kang, People's Press [China], 2005, 479–519 [in Chinese]
137. Substance and identity, *Substanz: Neue Überlegungen zu einer klassischen Kategorie des Seienden*, ed. K. Trettin, Frankfurt: Klostermann Verlag, 2005, 33–51
138. Syntax and ontology: reflections on three logical systems, *The Old New Logic: Essays on the Philosophy of Fred Sommers*, ed. D. S. Oderberg, Cambridge, MA: MIT Press, 2005, 49–65

139. The particular–universal distinction: a reply to MacBride, *Dialectica* 58, 2005, 335–40
140. Vagueness and endurance, *Analysis* 65, 2005, 104–112
141. Agent causation/Jonathan Bennett, *Encyclopedia of Philosophy*, 2nd edn, ed. D. M. Borchert, Detroit: Macmillan, 2006, vol. 1, 88–90, 549–50
142. Can the self disintegrate? Personal identity, psychopathology, and disunities of consciousness, *Dementia: Mind, Meaning and the Person*, ed. J. Hughes, S. Louw, & S. Sabat, Oxford: Oxford University Press, 2006, 89–103
143. Critical notice of Fabrice Correia's *Existential Dependence and Cognate Notions*, *Grazer Philosophische Studien* 73, 2006, 255–8
144. Endurantism versus perdurantism and the nature of time, *Revista di Filosofia Neoscolastica* 4, 2006, 713–727 [revised and expanded version of 127]
145. How real is substantial change? *The Monist* 89, 2006, 275–93
146. Ideational theories of meaning, *Encyclopedia of Language and Linguistics, Second Edition*, volume 5, ed. K. Brown, Oxford: Elsevier, 2006, 483–5
147. In defence of the autonomous mind, *SWIF Philosophy of Mind Review* 5, 3, 2006, 31–8, <http://lgxserver.uniba.it/lei/mind/swifpmr/0520063.pdf>
148. Is conceptualist realism a stable position? *Philosophy and Phenomenological Research* 71, 2006, 456–61
149. Non-Cartesian substance dualism and the problem of mental causation, *Erkenntnis* 65, 2006, 5–23
150. Powerful particulars: review essay on J. Heil's *From an Ontological Point of View*, *Philosophy and Phenomenological Research* 72, 2006, 466–79
151. Radical externalism or Berkeley revisited?, *Journal of Consciousness Studies* 13, 2006, 78–87 [reprinted in A. Freeman (ed.), *Radical Externalism: Honderich's Theory of Consciousness Discussed*, Exeter: Imprint Academic, 2006, 78–87]
152. Rational selves and freedom of action, *Analytic Philosophy Without Naturalism*, ed. A. Corradini, S. Galvan, & E. J. Lowe, London & New York: Routledge, 2006, 163–77
153. The 3D/4D controversy: a storm in a teacup, *Nous* 40, 2006, 570–8 [co-author Storrs McCall]
154. A defence of anti-conceptualist realism, *Belief and Metaphysics*, ed. C. Cunningham & P. Candler, London: SCM-Canterbury Press, 2007, 291–322
155. A problem for a posteriori essentialism concerning natural kinds, *Analysis* 67, 2007, 286–92
156. Between agent causationism and volitionism: a middle path, *Agency and Causation in the Human Sciences*, ed. F. Castellani & J. Quitterer, Paderborn: Mentis, 2007, 77–88
157. Does the descriptivist/anti-descriptivist debate have any philosophical significance? *Philosophical Books* 48, 2007, 27–33
158. Formal ontology and the revival of metaphysics, *Metaphysics Today: Problems and Prospects*, ed. M. Lutz-Bachmann & T. Schmidt, Freiburg: Verlag Karl Alber, 2007, 76–93
159. Immanent Universals, *Documenti e Studi sulla Tradizione Filosofica Medievale* 18, 2007, 623–36
160. La métaphysique comme science de l'essence, *Métaphysique contemporaine: propriétés, mondes possibles, et personnes*, ed. E. Garcia & F. Nef, Paris: J. Vrin, 2007, 85–117
161. Sortals and the individuation of objects, *Mind and Language* 22, 5, 2007, 514–33

162. The ontological argument, *The Routledge Companion to Philosophy of Religion*, ed. C. Meister & P. Copan, London & New York: Routledge, 2007, 331–40
163. Truthmaking as essential dependence, *Metaphysics and Truthmakers*, ed. J.-M. Monnoyer, Frankfurt: Ontos Verlag, 2007, 237–59
164. A defence of non-Cartesian substance dualism, *Psycho-Physical Dualism Today: An Interdisciplinary Approach*, ed. A. Antonietti, A. Corradini, & E. J. Lowe, Lanham, MD: Rowman and Littlefield (Lexington Books), 2008, 167–83
165. Against disjunctivism, *Disjunctivism: Perception, Action, Knowledge*, ed. A. Haddock & F. Macpherson, Oxford: Clarendon Press, 2008, 95–111
166. Dualism, *Encyclopedia of Neuroscience*, ed. M. D. Binder, N. Hirokawa, & U. Windhorst, Berlin: Springer, 2008
167. Dualismo delle sostanze non cartesiano, *L'uomo a due dimensioni: il dualismo mente-corpo oggi*, ed. A. Lavazza, Milano: Bruno Mondadori, 2008, 185–207
168. Essentialism, metaphysical realism, and the errors of conceptualism, *Philosophia Scientiae* 12, 2008, 9–33
169. How are identity conditions grounded?, *Persistence*, ed. C. Kanzian, Frankfurt: Ontos Verlag, 2008, 73–89
170. 'If $2 = 3$, then $2 + 1 = 3 + 1$ ': Reply to Heylen and Horsten. *Philosophical Quarterly* 58, 2008, 528–31
171. Illusions and hallucinations as evidence for sense data, *The Case for Qualia*, ed. E. Wright, Cambridge, MA: MIT Press, 2008, 59–72
172. Metaphysics, *Routledge Companion to Twentieth-Century Philosophy*, ed. D. Moran, Abingdon & New York: Routledge, 2008, 438–68
173. New directions in metaphysics and ontology, *Axiomathes* 18, 2008, 273–88
174. Reply to Bird on a posteriori essentialism, *Analysis* 68, 2008, 345–7
175. The determinists have run out of luck—for a good reason, *Philosophy and Phenomenological Research* 77, 2008, 745–8 [co-author Storrs McCall]
176. The metaphysical foundations of natural science, *Norsk Filosofisk Tidsskrift* 43, 2008, 6–22
177. Tropes and perception, *Tropes, Universals and the Philosophy of Mind: Essays at the Boundary of Ontology and Philosophical Psychology*, ed. S. Gozzano & F. Orilia, Frankfurt: Ontos Verlag, 2008, 175–92
178. Two notions of being: entity and essence, *Being: Developments in Contemporary Metaphysics*, ed. R. Le Poidevin, Cambridge: Cambridge University Press, 2008, 23–48
179. What is 'conditional probability'?, *Analysis* 63, 2008, 218–23
180. Dualism, *Oxford Handbook of Philosophy of Mind*, ed. B. McLaughlin, A. Beckermann, & S. Walter, Oxford: Oxford University Press, 2009, 66–84
181. An essentialist approach to truth-making, *Truth and Truth-Making*, ed. E. J. Lowe & A. Rami, Stocksfield: Acumen, 2009, 201–16
182. David Wiggins, *A Companion to Metaphysics*, 2nd edn, ed. J. Kim, E. Sosa, & G. Rosenkrantz, Malden, MA & Oxford: Wiley-Blackwell, 2009, 625–7
183. Dualism, *The Oxford Companion to Consciousness*, ed. T. Bayne, A. Cleeremans, & P. Wilken, Oxford: Oxford University Press, 2009, 244–8
184. Free agency, causation, and action explanation, *New Essays on the Explanation of Action*, ed. C. Sandis, Basingstoke: Palgrave Macmillan, 2009, 338–55

185. Individuation, *A Companion to Metaphysics*, 2nd edn, ed. J. Kim, E. Sosa, & G. Rosenkrantz, Malden, MA & Oxford: Wiley-Blackwell, 2009, 28–36
186. Modes of exemplification, *Gustav Bergmann: Phenomenological Realism and Dialectical Ontology*, ed. B. Langlet & J.-M. Monnoyer, Frankfurt: Ontos Verlag, 2009, 173–91
187. Serious endurantism and the strong unity of human persons, *Unity and Time in Metaphysics*, ed. L. Honnefelder, E. Runggaldier, & B. Schick, Berlin: Walter de Gruyter, 2009, 67–82
188. The definition of endurance, *Analysis* 69, 2009, 277–80 [co-author Storrs McCall]
189. Action theory and ontology, *A Companion to the Philosophy of Action*, ed. T. O'Connor & C. Sandis, Malden, MA & Oxford: Wiley-Blackwell, 2010, 3–9
190. Another dubious counterexample to conditional transitivity, *Analysis* 70, 2010, 286–9
191. On the individuation of powers, *The Metaphysics of Powers: Their Grounding and their Manifestations*, ed. A. Marmodoro, London & New York: Routledge, 2010, 8–26
192. Substance dualism: a non-Cartesian approach, *The Waning of Materialism: New Essays*, ed. R. C. Koons & G. Bealer, Oxford: Oxford University Press, 2010, 439–61
193. Why my body is not me: the unity argument for emergentist self-body dualism, *Emergence in Science and Philosophy*, ed. A. Corradini & T. O'Connor, New York & London: Routledge, 2010, 127–48 [reprinted in 234]
194. Body, soul and self, *Rivista di Filosofia Neo-Scolastica* CIII, 2, 2011, 201–15
195. Critical notice of G. Strawson's *Selves: An Essay in Revisionary Metaphysics*, *Analysis Reviews* 71, 2011, 587–92
196. Experience of change and change of experience, *The Ways Things Are: Studies in Ontology*, ed. C. Kanzian, W. Löffler, & J. Quitterer, Frankfurt: Ontos Verlag, 2011, 121–30
197. How not to think of powers: a deconstruction of the 'dispositions and conditionals' debate, *The Monist* 94, 2011, 20–34
198. John Locke, *The Routledge Companion to Epistemology*, ed. S. Bernecker & D. Pritchard, London & New York: Routledge, 2011, 687–96
199. Locke on real essence and water as a natural kind: a qualified defence, *Aristotelian Society Supplementary Volume* 85, 2011, 1–19
200. Naturalism, imagination, and the scientific worldview, *Turning Images in Philosophy, Science, and Religion: A New Book of Nature*, ed. C. Taliaferro & J. Evans, New York: Oxford University Press, 2011, 91–113
201. Personal identity, *The Continuum Companion to Philosophy of Mind*, ed. J. Garvey, London & New York: Continuum, 2011, 203–19
202. Ontological categories: why four are better than two, *Ontological Categories*, ed. J. Cumpa & E. Tegtmeier, Frankfurt: Ontos Verlag, 2011, 109–26
203. The rationality of metaphysics, *Stance and Rationality*, ed. O. Bueno & D. P. Rowbottom, special issue of *Synthese* 178, 2011, 99–109
204. Vagueness and metaphysics, *Vagueness: A Guide*, ed. G. Ronzitti, Dordrecht: Springer, 2011, 19–53
205. A neo-Aristotelian substance ontology: neither relational nor constituent, *Contemporary Aristotelian Metaphysics*, ed. T. E. Tahko, Cambridge: Cambridge University Press, 2012, 229–48
206. A new modal version of the ontological argument, *Ontological Proofs Today*, ed. M. Szatkowski, Frankfurt: Ontos Verlag, 2012, 179–91

207. Against monism, *Spinoza on Monism*, ed. P. Goff, Basingstoke: Palgrave Macmillan, 2012, 92–112
208. Asymmetrical dependence in individuation, *Metaphysical Grounding: Understanding the Structure of Reality*, ed. F. Correia & B. Schnieder, Cambridge: Cambridge University Press, 2012, pp. 214–33
209. Athroniaeth meddwl: teuddiadau a themâu diweddar [Philosophy of mind: recent trends and themes], *Cred, llên a diwylliant: cyfrol deyrnged Dewi Z. Phillips*, ed. E. G. Matthews, Talybont: Y Lolfa, 2012, 67–96 [in Welsh]
210. Categorical Predication, *Ratio* 25, 2012, 369–86 [reprinted in 220]
211. Essence and ontology, *Metaphysics: Aristotelian, Scholastic, Analytic*, ed. L. Novak, D. D. Novotny, P. Sousedik, & D. Svoboda, Frankfurt: Ontos Verlag, 2012, 93–111
212. Free will and rational choice, *The Blackwell Companion to Science and Christianity*, ed. J. B. Stump & Alan G. Padgett, Malden, MA & Oxford: Wiley-Blackwell, 2012, 421–9
213. Individuation, reference, and sortal terms, *Perception, Realism, and the Problem of Reference*, ed. A. Raftopoulos & P. Machamer, Cambridge: Cambridge University Press, 2012, pp. 123–41
214. Mumford and Anjum on causal necessitarianism and antecedent strengthening, *Analysis* 72, 2012, pp. 731–5
215. Non-Cartesian substance dualism, *After Physicalism*, ed. B. P. Göcke, Notre Dame, IN: Notre Dame University Press, 2012, 48–71
216. The probable simplicity of personal identity, *Personal Identity: Complex or Simple?*, ed. G. Gasser & M. Stefan, Cambridge: Cambridge University Press, 2012, pp. 137–55
217. What is the source of our knowledge of modal truths?, *Mind* 121, 2012, 919–50
218. Body, soul and self, *Dualistic Ontology and the Human Person*, ed. M. Szatkowski, Munich: Philosophia Verlag, 2013, 161–79
219. Branching time and temporal unity, *Around the Tree: Semantic and Metaphysical Issues Concerning Branching and the Open Future*, ed. F. Correia & A. Iacona, Dordrecht: Springer, 2013, 73–80
220. Categorical predication, *Classifying Reality*, ed. D. S. Oderberg, Malder, MA and Oxford: Wiley-Blackwell, 2013, 5–22 [reprint of 210]
221. Complex reality: unity, simplicity and complexity in a substance ontology, *Johanssonian Investigations: Essays in Honour of Ingvar Johansson on his Seventieth Birthday*, ed. C. Svennerlind, J. Almäng, & R. Ingthorsson, Frankfurt: Ontos Verlag, 2013, 338–57
222. Mereological extensionality, supplementation, and material constitution, *The Monist* 96, 2013, 131–48
223. Metaphysical foundations for science (interview by R. Marshall), *3:AM Magazine*, 18/04/2013
224. Metaphysical nihilism revisited, *The Puzzle of Existence*, ed. T. Goldschmidt, New York: Routledge, 2013, 182–96
225. Naturalism, theism, and objects of reason, *Philosophia Christi* 15 (1), 2013, 35–45
226. Neo-Aristotelian metaphysics: a brief exposition and defence, *Aristotle on Method and Metaphysics*, ed. E. Feser, Basingstoke: Palgrave MacMillan, 2013, 196–205
227. Presentism and relativity: No conflict, *New Papers on the Present*, ed. R. Ciuni, K. Miller & G. Torrenço, Munich: Philosophia Verlag, 2013, 133–52

228. Some varieties of metaphysical dependence, *Varieties of Dependence: Ontological Dependence, Grounding, Supervenience, Response-Dependence*, ed. M. Hoeltje, B. Schneider, & A. Steinberg, Munich: Philosophia Verla, 2013, 193–210
229. Substance causation, powers, and human agency, *Mental Causation and Ontology*, ed. S. C. Gibb, E. J. Lowe, & R. D. Ingthorsson, Oxford: Oxford University Press, 2013, 153–72
230. The will as a rational free power, *Powers and Capacities in Philosophy: The New Aristotelianism*, ed. R. Groff & J. Greco, New York: Routledge, 2013, 172–85
231. How real are artefacts and artefact kinds?, *Artefact Kinds: Ontology and the Human-Made World*, ed. M. Franssen, P. Kroes, T. Reydon, & P. Vermaas, Switzerland: Springer, 2014, 17–26
232. Metaphysical knowledge, *The Armchair or the Laboratory*, ed. M. Haug, London: Routledge, 2014, 126–44
233. Metaphysics, *The Bloomsbury Companion to Analytic Philosophy*, ed. B. Dainton & H. Robinson, London/New York: Bloomsbury, 2014, 336–54
234. Why my body is not me: The unity argument for emergentist self-body dualism, *Contemporary Dualism: A Defense*, ed. A. Lavazza & H. Robinson, New York: Routledge, 2014, 245–65 [re-print of 193]
235. A new argument for realism from perceptual content, *Phenomenal Qualities—Sense, Perception and Consciousness*, ed. P. Coates & S. Coleman, Oxford: Oxford University Press, 2015, 257–73
236. Locke on language and meaning, *A Companion to Locke*, ed. M. Stuart, Oxford: Blackwell, 2015, 279–95
237. There are (probably) no relations, *The Metaphysics of Relations*, eds. A. Marmodoro & D. Yates, Oxford: Oxford University Press, 2016, 100–12
238. Coincidence, *The Handbook of Mereology*, ed. H. Burkhardt, J. Seibt, & G. Imaguire, Munich: Philosophia Verlag, 2017
239. Metaphysics as the science of essence, *Ontology, Modality, and Mind: Themes from the Metaphysics of E. J. Lowe*, ed. A. D. Carruth, S. C. Gibb & J. Heil, Oxford: Oxford University Press, 2018, [re-print in English of 160]

Book reviews

1. J. O. Urmson's *Berkeley* and G. J. Warnock's *Berkeley*, *British Journal of 18th Century Studies* 6, 1983, 89–90
2. P. Jones's *Hume's Sentiments*, *British Journal of 18th Century Studies* 7, 1984, 139–40
3. T. V. Morris's *Understanding Identity Statements*, *Philosophical Books* 26, 1985, 252–4
4. J. Bennett's *A Study of Spinoza's Ethics*, *British Journal of 18th Century Studies* 9, 1986, 116–17
5. R. J. Delahunty's *Spinoza*, *Durham University Journal* 78, 1986, 392–3
6. G. Forbes' *The Metaphysics of Mind*, *Mind* 95, 1986, 135–8
7. G. Schlesinger's *The Intelligibility of Nature*, *Philosophical Books* 27, 1986, 234–6
8. R. C. Stalnaker's *Inquiry*, *Philosophical Books* 27, 1986, 101–3
9. D. Berman (ed.), *George Berkeley: Essays and Replies*, *Durham University Journal* 80, 1987, 164–5
10. K. Konyndyk's *Introductory Modal Logic*, *Philosophical Books* 28, 1987, 165–6

11. W. Garnett's *The Springs of Consciousness*, *Durham University Journal* 80, 1988, 366
12. D. Lewis's *Philosophical Papers Volume II*, *Mind* 97, 1988, 484–7
13. S. G. Shanker (ed.), *Philosophy in Britain Today, History of the Human Sciences* 1, 1988, 132–4
14. D. Berman's *A History of Atheism in Britain*, *Durham University Journal* 81, 1989, 333–4
15. A. Brennan's *Conditions of Identity*, *Philosophical Books* 30, 1989, 103–6
16. F. L. Will's *Beyond Deduction*, *Philosophy* 64, 1989, 424–5
17. H. Noonan's *Personal Identity*, *Mind* 99, 1990, 477–9
18. S. Wolfram's *Philosophical Logic: An Introduction*, *Philosophical Books* 31, 1990, 34–5
19. M. C. Banner's *The Justification of Science and the Rationality of Religious Belief*, *Religious Studies* 27, 1991, 421–2
20. M. A. Box's *The Suasive Art of David Hume*, *Durham University Journal* 83, 1991, 316–17
21. K. Campbell's *Abstract Particulars*, *Philosophical Quarterly* 41, 1991, 104–6
22. P. Humphreys' *The Chances of Explanation*, *Isis* 82, 1991, 783–4
23. D. H. Sanford's *If P then Q: Conditionals and the Foundations of Reasoning*, *Philosophical Books* 32, 1991, 31–2
24. G. Strawson's *The Secret Connexion*, M. Tooley's *Causation*, and E. Fales' *Causation and Universals*, *Philosophical Quarterly* 41, 1991, 494–8
25. R. Coburn's *The Strangeness of the Ordinary*, *Mind* 101, 1992, 151–3
26. K. Lambert (ed.), *Philosophical Applications of Free Logic, History and Philosophy of Logic* 13, 1992, 246–7
27. D. H. Mellor's *Matters of Metaphysics*, *Philosophy* 67, 1992, 268–70
28. S. Stich's *The Fragmentation of Reason*, *Philosophical Quarterly* 42, 1992, 98–101
29. P. Bailhache's *Essai de Logique Deontique*, *History and Philosophy of Logic* 14, 1993, 249–50
30. C. A. J. Coady's *Testimony: A Philosophical Study*, *Philosophy* 68, 1993, 413–15
31. D. Hodgson's *The Mind Matters: Consciousness and Choice in a Quantum World*, *Philosophical Books* 34, 1993, 33–4
32. A. Phillips Griffiths (ed.), A. J. Ayer: *Memorial Essays*, *Philosophy* 69, 1993, 107–8
33. W. Spohn et al. (eds), *Existence and Explanation, History and Philosophy of Logic* 14, 1993, 130–1
34. T. Williamson's *Identity and Discrimination*, *Mind* 102, 1993, 210–12
35. D. Braine's *The Human Person*, *Philosophy* 69, 1994, 244–6
36. M. Dummett's *Origins of Analytical Philosophy*, *Philosophy* 69, 1994, 246–8
37. J. Heil's *The Nature of True Minds*, *Philosophical Books* 35, 1994, 56–7
38. R. I. G. Hughes (ed.), *A Philosophical Companion to First-Order Logic, History and Philosophy of Logic* 15, 1994, 255–6
39. A. Newman's *The Physical Basis of Predication*, *Philosophy and Phenomenological Research* 55, 1995, 490–2
40. R. Nozick's *The Nature of Rationality*, *Philosophical Quarterly* 45, 1995, 397–9
41. T. C. Potts' *Structures and Categories for the Representation of Meaning, History and Philosophy of Logic* 16, 1995, 140–1
42. H. Robinson's *Perception*, *Philosophy* 70, 1995, 463–6
43. G. Rosenkrantz's *Haecceity*, *Mind* 104, 1995, 202–5
44. H. Harris (ed.), *Identity*, *Philosophical Quarterly* 47, 1997, 395–7
45. *Journal of Consciousness Studies*, Vols 1 & 2, *Philosophical Books* 38, 1997, 30–1

46. F. P. Ramsey's *Notes on Philosophy, Probability and Mathematics*, ed. M. C. Galavotti, *British Journal for the Philosophy of Science* 48, 1997, 300–1
47. J. Copeland (ed.), *Logic and Reality, History and Philosophy of Logic* 19, 1998, 179–81
48. M. Jubien's *Contemporary Metaphysics* and M. J. Loux's *Metaphysics: A Contemporary Introduction*, *European Journal of Philosophy* 6, 1998, 365–8
49. T. Pink's *The Psychology of Freedom*, *Philosophy* 73, 1998, 305–7
50. R. A. Wilson's *Cartesian Psychology and Physical Minds*, *History and Philosophy of Psychology Newsletter* 26, 1998, 19–20
51. A. Biletzki & A. Matar (eds), *The Story of Analytic Philosophy: Plot and Heroes*, *History and Philosophy of Logic* 20, 1999, 64–5
52. N. Jolley's *Locke: His Philosophical Thought*, *Locke Newsletter* 30, 1999, 145–8
53. D. Lewis's *Papers in Metaphysics and Epistemology*, *History and Philosophy of Logic* 20, 1999, 138–9
54. M. Tooley's *Time, Tense, and Causation*, *Philosophical Books* 40, 1999, 45–7
55. E. Castellani (ed.), *Interpreting Bodies: Classical and Quantum Objects in Modern Physics*, *British Journal for the Philosophy of Science* 51, 2000, 353–5
56. J. Etchemendy's *The Concept of Logical Consequence*, *History and Philosophy of Logic* 21, 2000, 236–7
57. A. Gallois's *Occasions of Identity*, *Mind* 109, 2000, 354–7
58. U. Meixner's *Axiomatic Formal Ontology*, *Studia Logica* 64, 2000, 137–40
59. J. Foster's *The Nature of Perception*, *Times Literary Supplement* 5151, Dec. 21, 2001, 24
60. F. Jackson's *Mind, Method and Conditionals: Selected Essays*, *Mind* 110, 2001, 211–15
61. J. Kim's *Mind in a Physical World*, *American Journal of Psychology* 114, 2001, 303–8
62. D. Cockburn's *An Introduction to the Philosophy of Mind*, T. Crane's *Elements of Mind* and K. T. Maslin's *An Introduction to the Philosophy of Mind*, *Times Higher Education Supplement*, May 31, 2002, iv
63. R. L. Epstein's *Five Ways of Saying 'Therefore'*, *History and Philosophy of Logic* 23, 2002, 147–8
64. R. Mason's *Before Logic*, *Philosophical Books* 43, 2002, 41–3
65. B. O'Shaughnessy's *Consciousness and the World*, *Philosophy* 77, 2002, 283–7
66. D. Owens's *Reason without Freedom*, *Ratio* 15, 2002, 309–11
67. J. Baggini & P. Fosl's *The Philosopher's Toolkit*, *Times Higher Education Supplement*, Oct. 24, 2003, 31
68. D. Davidson's *Subjective, Intersubjective, Objective*, *Philosophy* 78, 2003, 555–8
69. K. Hawley's *How Things Persist*, *Philosophical Quarterly* 53, 2003, 613–16
70. W. G. Lycan's *Real Conditionals*, *Philosophical Books* 44, 2003, 177–8
71. D. Wiggins's *Sameness and Substance Renewed*, *Mind* 112, 2003, 816–20
72. J. C. Beall & B. C. van Fraassen's *Possibilities and Paradox: An Introduction to Modal and Many-Valued Logic*, *History and Philosophy of Logic* 25, 2004, 329–30
73. C. H. Conn's *Locke on Essence and Identity*, *Locke Studies* 4, 2004, 243–53
74. G. Molnar's *Powers: A Study in Metaphysics*, *British Journal for the Philosophy of Science* 55, 2004, 817–22
75. J. Bennett's *A Philosophical Guide to Conditionals*, *Philosophical Books* 46, 2005, 158–60
76. T. Crane & K. Farkas (eds), *Metaphysics: A Guide and Anthology*, *Times Higher Education Supplement*, May 27, 2005, xi
77. U. Meixner's *The Two Sides of Being*, *Erkenntnis* 62, 2005, 290–4

78. J. W. Yolton's *The Two Intellectual Worlds of John Locke*, *Locke Studies* 5, 2005, 223–6
79. D. Davidson's *Truth, Language, and History* and *Truth and Predication*, *Times Higher Education Supplement*, April 7, 2006, 26
80. U. Meixner's *The Two Sides of Being*, *Journal of Consciousness Studies* 13, 2006, 115–18
81. B. Greetham's *Philosophy*, *Times Higher Education Supplement*, May 25, 2007, xix
82. J. Hawthorne's *Metaphysical Essays*, *Notre Dame Philosophical Reviews*, 15/1/2007
83. P. Mackie's *How Things Might Have Been*, *Mind* 116, 2007, 762–6
84. R. Martin & J. Barresi's *The Rise and Fall of Soul and Self*, *Journal of Consciousness Studies* 14, 2007, 125–7
85. L. Newman (ed.), *The Cambridge Companion to Locke's 'Essay Concerning Human Understanding'*, *Locke Studies* 7, 2007, 213–23
86. J. C. Beall & B. Armour-Garb (eds), *Deflationism and Paradox*, *History and Philosophy of Logic* 29, 2008, 279–80
87. H. Lillehammer & G. Rodriguez-Pereyra (eds), *Real Metaphysics*, *European Journal of Philosophy* 16, 2008, 134–8
88. R. Sorensen's *Seeing Dark Things: The Philosophy of Shadows*, *Times Higher Education magazine*, March 20, 2008, 46–7
89. Peter Unger's *All the Power in the World*, *Philosophical Quarterly* 58, 2008, 745–7
90. C. B. Martin's *The Mind in Nature*, *International Journal of Philosophical Studies* 17, 2009, 609–11
91. E. Olson's *What Are We? A Study in Personal Ontology*, *Analysis Reviews* 69, 2009, 388–90
92. M. E. Reicher (ed.), *States of Affairs*, *Notre Dame Philosophical Reviews*, 15/10/2009
93. R. Sorensen's *Seeing Dark Things: The Philosophy of Shadows*, *Philosophy* 84, 2009, 615–19
94. D. Oderberg's *Real Essentialism*, *Philosophical Quarterly* 60, 2010, 648–52
95. D. M. Armstrong's *Sketch for a Systematic Metaphysics*, *Notre Dame Philosophical Reviews*, 20/1/2011 (reprinted in French in *Revue Étudiante de Philosophie Analytique* 3, 2011, 65–72)
96. S.-J. Savonius-Wroth, P. Schuurman & J. Walmsley (eds), *The Continuum Companion to Locke*, *Locke Studies* 11, 2011, 173–5
97. J. Schloss & M. Murray (eds), *The Believing Primate, Faith and Philosophy* 29, 2012, 243–7

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